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**Education India Journal: A Quarterly Refereed
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1. Teacher Educators' Perceptions about the Effectiveness of Social Networking Sites and Online Teaching Tools

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Abstract

A large number of students and teachers in higher educational institutions as well as in teacher education institutions are using mobile devices for connecting. Social networking sites and online teaching tools are playing a significant role in teaching-learning process. The most common social networking sites are YouTube, Instagram, WhatsApp, Facebook, Twitter, Snapchat, Skype, LinkedIn etc. There are two major approaches of online teaching: synchronous teaching (like audio-video conferencing tools, live chat, shared whiteboard, online slide shows etc.) and asynchronous teaching (e-mail, file attachments etc.). The present study adopted survey methodology focusing on sample from teacher educators working in teacher education institutions. The objective of the study was to investigate the teacher educators' perception towards effectiveness of social networking sites and online teaching tools. The results indicated that WhatsApp and YouTube were the two most preferred social networking sites used by teacher educators. Google Classroom, WhatsApp and YouTube were the three most preferred online teaching tools used by teacher educators. It was concluded that social networking sites and online teaching tools provide platform for organizing and structuring instructional content, prompt feedback. Further, it was found that the most important skills required for efficient use of social networking sites and online teaching tools are technical know-how and classroom management.

Keywords: Social Networking sites, Online Teaching Tools, Teacher Educators, Perceptions and Teacher Education Institutions

Introduction

Social networking sites are quite popular among students, teachers and teacher educators. It has changed the way of communication among humans. Social Networking sites have revolutionized the way of interaction, communication and thinking of the masses. (Weisgerber & Butler, 2010). In online

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social networks, the users (referred to as Nodes) are connected and interact with other users. The network topology has a great influence on the nodes (Lee & Oh, 2015). The services of social networks is dependent upon the accuracy of the reputation (user's real behaviour) information of the nodes. Such reputation (it is often subjective) can be calculated for websites through centralised (being used by commercial applications such as eBay and Amazon) and decentralised approaches (Szabo & Fath, 2007; Galeotti et al., 2010). The emergence of social networks is attributed to the power of Web 2.0 (Ellison, 2007) which allowed users to write on the web alongwith reading it and also the feature of wisdom of masses. One of the uses of social networking sites people make for news content (Newman, 2011), and accessing user generated content (Schifferes et al. 2014) like photos or comments posted by them. News professionals also make use of social networking sites (Newman, 2010) for collaborative / networked journalism. Thurman & Walters (2013) found the use of live blogging sites by newspapers. Phuvipadawat and Murata (2010) analysed twitter to determine the popularity of messages. Social media sites have also been used for reporting on sporting events (Cunningham, 2003).

The social networking sites in education enable the students to interact with the teachers and their peer learning groups so that they may get more useful information. (Bozkurt, Karadeniz, & Kocdar, 2017). There are varieties of social networking sites (Adamic & Adar, 2005) which are used by teachers and teacher educators for educational purposes (Burkhardt, 1994; Considine, Krahel, Lenk & Janvrin, 2016; Cross, Nohria & Parker, 2002; Debreceeny, 2015). The most common social networking sites are YouTube, Instagram, WhatsApp, Facebook, Twitter, Snapchat, Skype, LinkedIn etc. A large number of students in higher educational institutions as well as in teacher education institutions are using mobile devices for connecting. The economic recession and lockdown measure due to COVID-19 make online teaching learning very interesting (Bozkurt & Sharma, 2020). Internet and world wide web are main information highway for online teaching and learning. Online teaching is an option for teachers who wish to teach in their own environment using technology. There are two major approaches of online teaching synchronous teaching (like live chat, audio and video conferencing,

shared whiteboard, online slide shows etc.) and asynchronous teaching (e-mail, file attachments etc.) (Sharma, 2018b; Sharma & Mishra, 2007).

Now-a-days, teacher educators are also using social networking sites as well as utilizing different tools for online teaching. For proper development of a learner, it is pertinent that all three domains of personality, viz, cognitive, affective and psychomotor are given due importance. Emotion plays a deep role in effective learning. Plutchik (1980) proposed a wheel of emotions which are culturally independent. This factor makes this model of 'Wheel of Emotions' a strong one as it can be applied to any given language. Plutchik suggested basic emotions that are Joy- Sadness, Trust-Disgust, Fear-Anger & Surprise-Anticipation. Social media networking sites use algorithms to detect the positive or neutral or negative polarity of communications or messages. The emotions are usually subjective and thus supervised learning methods are good measures for such polarity detection (Paltoglou & Thelwall, 2012). A large number of teacher educators, teachers and students in higher educational institutions as well as in teacher education institutions are using mobile devices for connecting (Galego, Ferreira Dias, Amorim, Madaleno, Jimenez, & Osorio, 2020). So, it is important to conduct study to discover teacher educators' perceptions about the effectiveness of Social Networking Sites and Online Teaching tools. That effort will be carried out to enhance the online learning environment in teacher education colleges and universities.

Literature Review

In this section, we analyse and review some of the studies related to educational technologies, social networking sites and online teaching tools.

Dogan (2010) investigated the impact of educational technology on school teachers and its effect on learning process. The results indicated no significant difference in the experience of teachers whereas a significant difference was reported related to their perspective of educational technology being used at the institutions they are employed at and by their education level. Further, No contradiction between gender and technology was found. Galy, Downey & Johnson (2011) examined the differences in the perceptions of the student towards e-learning tools and how the performance of students is affected by using these tools. The findings indicated a differential value between online courses and campus-based

courses. The students preferred assignments, quizzes on a week basis, prompt feedback, strict schedule for online classes and meeting with the instructor once a month. Mundy, Kupczynski & Kee (2012) analyzed teacher's perceptions about the use of technology in the classroom. The study reported enhanced student engagement, and proficiency. The students experienced an acceleration in their learning and were excited about the use of technology in classroom. Hodge (2015) conducted a study on the perceptions of teachers towards online social network as an instructional platform. This study explored the effectiveness of online social network as an e-learning tool for teaching and learning. The major findings indicated that the Edmodo promotes continual interaction, provides a space for student-centric activities. Alshammari, Ali & Rosli (2015) explored the effectiveness and multiple benefits associated with the usage of social network sites as an effective learning tool for students. The study reported that social network sites improves collaborative environment, enhancing learning environment and peer engagements. Zade, Jadhav, Bhattacharjee & Kulkarni (2018) summarized the effectiveness of social networking in education. It was found that social networking is a useful tool in education and provides interactive services, blogging services, group services and helpful in searching for colleges and career opportunities to the students.

Rationale of the study

Above researches explored the use of social networking sites in teaching-learning process. National Education Policy 2020 of the Govt of India lays a greater focus on the development of digital competencies of teachers. Teacher education programmes in the country offer training and courses towards creation of teacher workforce. Suitable digital technologies and approaches are recommended for teacher educators. Therefore, this study aimed at investigating the teacher educators' perception towards effectiveness of social networking sites and online teaching tools. Further, this study provided the teacher educators an opportunity to express their opinions regarding the adoption of the social networking sites and online teaching tools.

Research Questions

- I. To identify preferred social networking sites used by the teacher educators and the reasons why they

use them.

- II. To identify preferred online teaching tools used by the teacher educators and the reasons why they use them.
- III. To investigate teacher educators' perception about the effectiveness of the social networking sites and online teaching tools.
- IV. To examine required skills for effective use of social networking sites and online teaching tools among teacher educators.

Research Methodology

This study used a mixed approach i.e. quantitative as well as qualitative research design. Mixed method research is an important research methodology in social and behavioural sciences (Teddlie & Tashakkori, 2003; Gorard & Taylor, 2004; Johnson & Onwuegbuzie, 2004). Mixed methods research involves qualitative and quantitative approaches of research (Tashakkori & Teddlie, 2003a, p. 711). The mixed methods research approach is useful to explain discrepancies arising between the qualitative and quantitative results. Mixed methods give a voice to each and every participant and ensure that study findings are grounded in participants' experiences. This study used survey method and follow up interviews and discussions to collect information about the perceptions about the effectiveness of social networking sites and online teaching tools.

Sample

The participants of this research were 50 teacher educators' working in teacher education institutions and department of education in one of the State University. Total number of teacher education institutions covered was 3. Teacher educators were randomly selected. Random selection ensures that results obtained from the sample as well as from entire population would have been approximately same (Shadish, Cook, & Campbell, 2002).

Research Instrument and Data Collection

A Questionnaire was prepared as the main research instrument for data collection. The questionnaire was named as 'Perceptions on Effectiveness of Social Networking Sites and Online Teaching Scale'. It included four aspects of effectiveness of social networking sites and online teaching tools, which are

as follows:

- Organizing and structuring instructional content
- Prompt feedback
- Online classroom management
- Assessment of learning

Quality of the survey instrument was established through face validity and content validity. This questionnaire as a digital survey was distributed to the teacher educators in the form of a Google Form. The collection period was 10 days. Follow up interviews and discussions took place on campus as well as through telephonic conversations.

Results and Findings

Responses given by the teacher educators on the survey on social networking sites and online teaching tools were examined and are presented in the following section:

I. Preferred Social Networking Sites by the teacher educators

WhatsApp and YouTube were the two most preferred social networking sites as being used by all the teacher educators. Telegram was also preferred by 70 per cent of the teacher educators. Twitter and LinkedIn were the least preferred social networking sites.

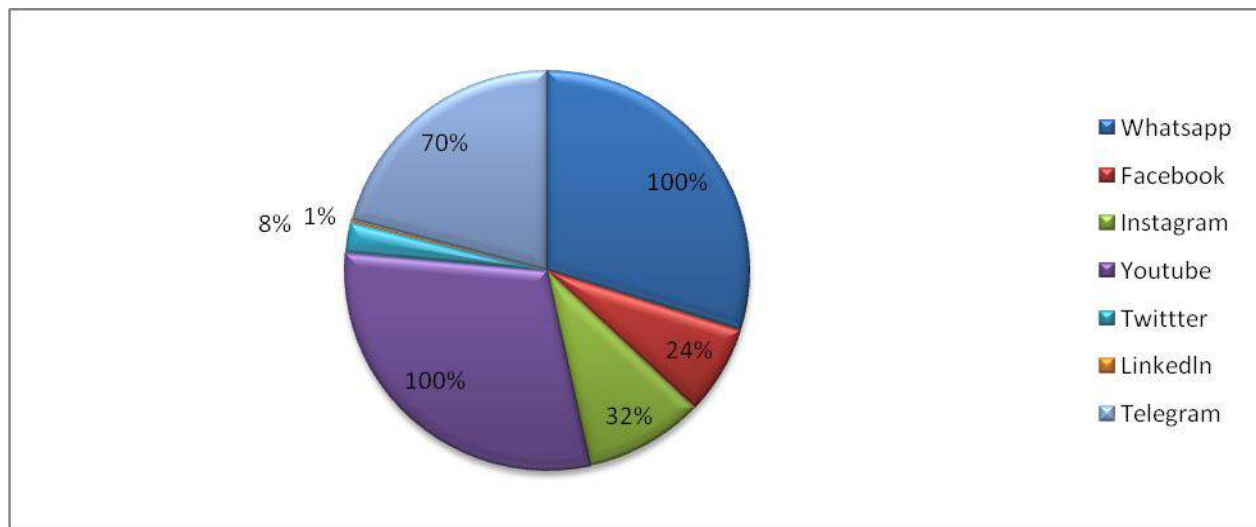


Figure 1: Social Networking Sites used by Teacher Educators (in Percentage)

II. Online Teaching Tools preferred by the teacher educators

Google Classroom, WhatsApp and YouTube were the three most preferred online teaching tools used by teacher educators. Zoom and Google Meet were also preferred by the teacher educators. Edmodo and Skype were the least preferred online teaching tools.

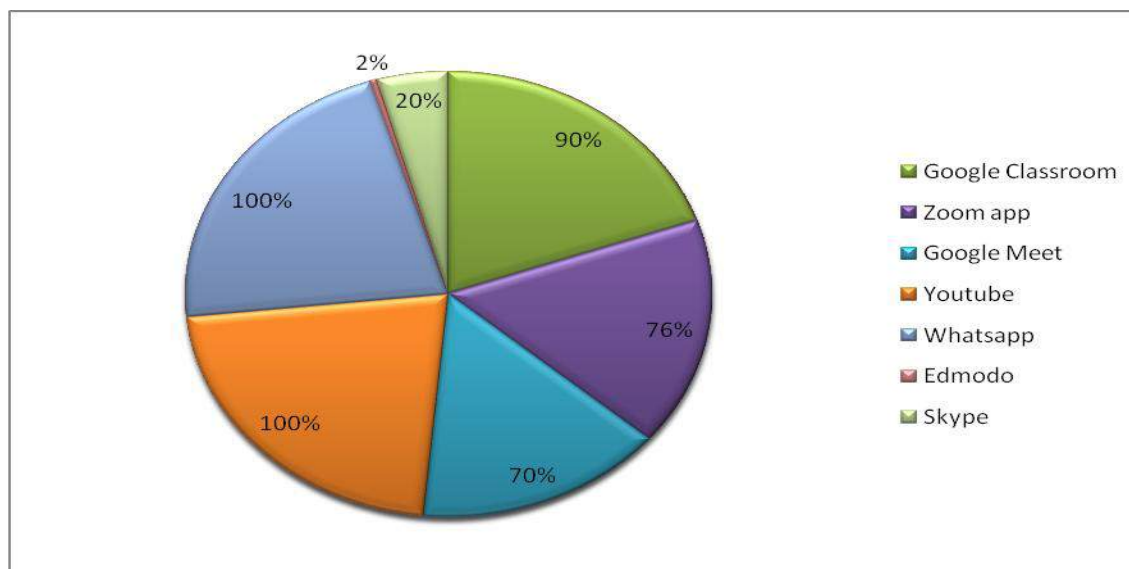


Figure 2: Online Teaching Tools used by Teacher Educators (in Percentage)

III. Teacher Educators' perception about the effectiveness of the social networking sites and online teaching tools

To fulfil the objective of this study, a questionnaire 'Perception on Effectiveness of Social Networking Sites and Online Teaching Scale' was prepared involving these four aspects:

- Organizing and structuring instructional content.
- Prompt feedback.
- Online classroom management.
- Assessment of learning.

Items	Description	Total Respondents	Strongly Agree	Agree	Indecisive	Disagree	Strongly Disagree
Item (i)	Organizing and structuring instructional content	50	4 (8%)	11 (38%)	10 (20%)	19 (22%)	6 (12%)
Item (ii)	Prompt feedback	50	4 (8%)	39 (78%)	3 (6%)	2 (4%)	2 (4%)
Item (iii)	Online classroom management	50	9 (18%)	23 (46%)	5 (10%)	11 (22%)	2 (4%)
Item (iv)	Assessment of learning	50	7 (14%)	6 (12%)	29 (58%)	7 (14%)	1 (2%)

Table 1: Responses of Teacher Educators on four aspects

(i) Organizing and structuring instructional content.

The responses of teacher educators to the question on effectiveness of social networking sites and online teaching tools for organizing and structuring instructional content are presented in figure 3.

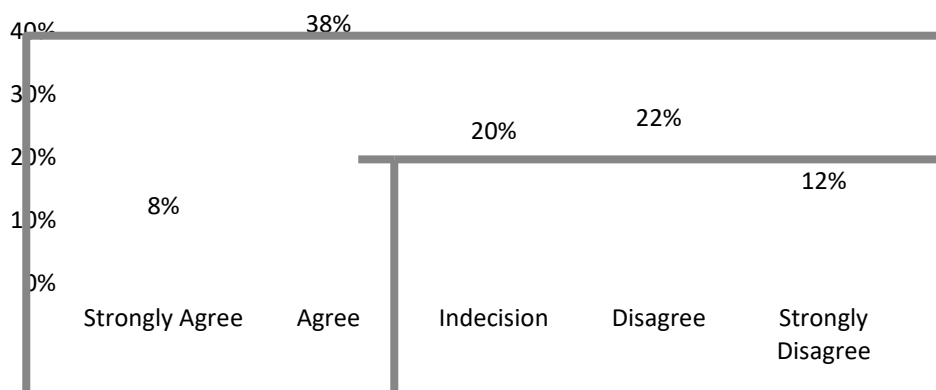


Figure 3: Responses related to Organizing and structuring instructional content

The findings indicate that out of total sample, 8 percent strongly agreed, 38 percent agreed, 20 percent were indecisive, 22 percent disagreed and only 12 percent strongly disagreed about the effectiveness of social networking sites and online teaching tools for organizing and structuring instructional content. Ignoring the 20 percent indecisive cases, it can be inferred that 46 percent of teacher educators find social networking sites and online teaching tools as an effective platform for organizing and structuring instructional content. Teacher Educators these days are making productive use of their social networking sites and online teaching tools for academic purposes. Social networking sites and online teaching tools help them gain knowledge and access to various resources that aid their studies.

(ii) Prompt feedback

The responses of teacher educators to the question on effectiveness of social networking sites and online teaching tools for prompt feedback are presented in figure 4.

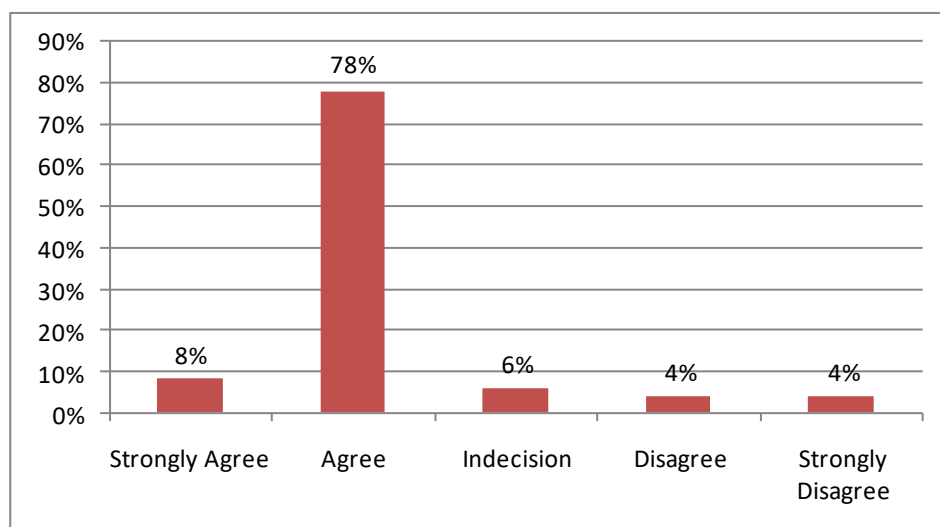


Figure 4: Responses related to prompt feedback

The findings indicate that 8 percent of respondents strongly agreed, 78 percent agreed, 6 percent were indecisive, 4 percent disagreed and only 4 percent strongly disagreed. It thus, shows that according to majority of respondents the social networking sites and online teaching tools provide a good platform for prompt feedback. Teacher Educators these days are making productive use of their social

networking sites and online teaching tools for the purpose of connection, collaboration and quick feedback. Feedback is important for personal and professional growth of the students and helps to improve quality of their work. Teacher educators can create their own user accounts to conduct discussions, posts or comments to provide feedback.

(iii) Online classroom management

The responses of teacher educators to the question on effectiveness of social networking sites and online teaching tools for online classroom management are presented in figure 5.

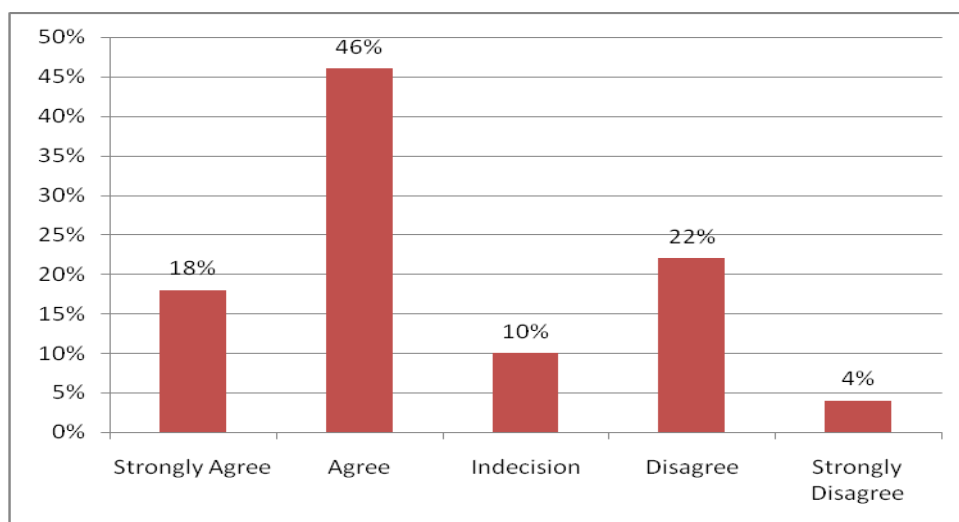


Figure 5: Responses of teacher educators on online classroom management

The findings indicate that when asked about their view on effectiveness of social networking sites and online teaching tools for online classroom management, 18 percent of respondents strongly agreed, 46 percent agreed, 10 percent were indecisive, 22 percent disagreed and only 4 percent strongly disagreed. It thus, indicates that social networking sites and online teaching tools provide a good platform to manage online classrooms. These platforms encourage students to ask questions and to answer each other. Teacher can allot teamwork projects and divide the students into groups to work in different breakout rooms.

(iv) Social Networking sites and online teaching tools are effective for assessment of learning

The responses of teacher educators to the question on effectiveness of social networking sites and

online teaching tools for assessment of learning are presented in figure 6.

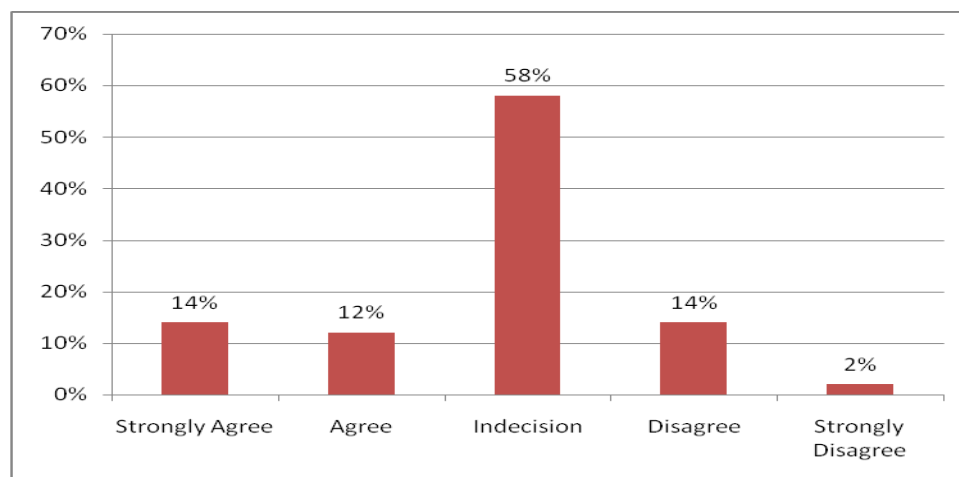


Figure 6: Responses of teacher educators on assessment of learning

The findings indicate that when asked about their view on effectiveness of social networking sites and online teaching tools for assessment of learning, 14 percent of the respondents strongly agreed, 12 percent agreed, 58 percent were indecisive, 14 percent disagreed and only 2 percent strongly disagreed. It thus, shows that social networking sites and online teaching tools are not providing effective platform for assessment of learning. Teacher educators are indecisive about the effectiveness of social networking sites and online teaching tools. The reason may be that teacher educators are not well trained and aware about the online assessment tools.

IV. To identify required skills for effective use of social networking sites and online teaching tools among teacher educators

Another objective of this study was to identify skills required by the teacher educators for the effective use of social networking sites and online teaching tools. The results are shown below in table 2.

Table 2: Responses of teacher educators (in percentages) with respect to skills required for effective use of social networking sites and online teaching tools

Skills	Total Respondents	Strongly Agree	Agree	Indecisive	Disagree	Strongly Disagree
Communication Skills	50	52%	36%	10%	1%	1%
Time Management	50	56%	36%	8%	--	--
Patience	50	56%	36%	8%	--	--
Compassion and Empathy	50	50%	34%	12%	2%	2%
Classroom Management	50	62%	12%	12%	6%	8%
Technical know-how	50	80%	18%	2%	--	--
Motivation and Independence	50	56%	36%	8%	--	--

From table 2, it can be concluded that the most important skills required for effective use of social networking sites and online teaching tools are technical know- how and classroom management. The other important skills are time management, communication skills, patience, compassion & motivation and independence empathy.

Discussions

From the above finding we can say that WhatsApp and YouTube are the two most preferred social networking sites used by teacher educators. Telegram is also very popular among the teacher educators. Twitter and LinkedIn are the least preferred social networking sites. These findings are in agreement with Patmanthara, Febiharsa & Dwiyanto (2019) who also reported YouTube, Instagram, WhatsApp, and Facebook to be most preferred social networking sites.

Google Classroom, WhatsApp and YouTube are the three most preferred online teaching tools used by teacher educators. Zoom and Google Meet are also a strong choice by the teacher educators. Edmodo and Skype were the least preferred online teaching tools. Our results match the findings of Mishra, Gupta and Shree (2020) who examined teaching-learning activities through online mode during the lockdown period and reported that the most widely used platforms are WhatsApp, Telegram and Email for active interactions, submission of quiz, tests and assignments.

Social networking sites and online teaching tools provide platform for organizing and structuring instructional content (Sharma, 2018a). Teacher educators these days are making effective use of social networking sites and online teaching tools for academic purposes, for example, Google Classroom,

Edmodo, YouTube etc. provides adequate storage space to organize the content matter and mounting other educational resources. Azhar & Iqbal (2018) examined the perceptions of teachers towards the effectiveness of Google Classroom. The study concluded that teachers found the reason of inefficiency of Google Classroom is its lack of user-friendly interface. Our study focused on how these sites can be used to manage the content.

Social networking sites and online teaching tools allow teachers to provide prompt feedback and manage the online classrooms. Teacher educators use these tools for connection, collaboration and quick feedback. Teacher educators can create their own user accounts to conduct discussions, posts or comments to provide feedback. Teacher can allot teamwork projects and divide the students into groups to work in different breakout rooms. Saini & Abraham (2015) found that the learners of all levels, be it the school level or the higher education level are using social media tools like Face book, YouTube, Twitter etc. to communicate with their peers and teachers. The study concluded that people now-a-days prefer to interact through social media rather than physical mode.

Our results indicate that the social networking sites and online teaching tools are not providing effective platform for assessment of learning. The reason may be that teacher educators are not well trained and aware about the online assessment tools. Kearns (2012) looked into the challenges and effective practices for student assessment in online learning and reported various assessment methods being used by teachers.

The most important skills required for effective use of social networking sites and online teaching tools are technical know-how and classroom management. The other important skills are time management, communication skills, patience, compassion & motivation and independence empathy. Hardy (2016) in her article discussed important skills like communication skills, compassion and empathy, patience, subject expertise and time management that online teachers are developing every day. Her study supports our findings.

Recommendations

- (i) Teacher educator's professional training should be planned around practical use of social networking sites so that they can handle online classes effectively.

- (ii) Teacher educators must use various online tools so that all teacher trainees can access the resource material.
- (iii) Teacher should be trained to record online attendance and assessments.
- (iv) Adequate infrastructural facilities need to be provided to both teacher educators and teacher trainees.
- (v) There should be adequate provision of internet connectivity at the teacher training institutions.

Conclusion

Teachers and teacher educators are connecting with their students and colleagues through social networking sites. They are well aware about their social presence; they like to participate in discussions. Online teaching is an exciting way to learn about almost anything. It has positive impact on the lives of students as well as on the teachers.

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2. Adoption of Leadership practices by Leaders in High Performing Schools.

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Abstract

Leadership plays an important role in every field of life whether it is political, social, cultural managerial or educational field. The present paper deals with leadership practices adopted by the school principals of the high performing schools. The study is based on the meta-analysis and uses research papers and articles from the EBSCO and Google scholar databases. The key terms used for searching are school leadership, leadership practices and leadership styles. The results of the study are presented in form of various leadership practices adopted by school leaders which are categorized under five main headings such as vision and mission, practices related to leaders, teacher's development, involvement and relationship.

Key words: leadership practices, school leaders, successful schools, leadership styles.

Introduction

A country aims for quality education for achieving multidimensional growth. The growth may be in the economic, political, social or educational field. In all these fields of life, leadership plays an important role for taking a nation to new heights. The successful school leadership is not only attaining the position but also possessing the knowledge and understanding about the leadership qualities and the ability for implementing these qualities (Pashiards & Savvides, 2011; Sharma, Sun & Kannan, 2012). The effectiveness of a school is determined by its leadership (Hallinger, 2005; Xhomara, 2018). The effectiveness of the school is defined as the realization of the cognitive, affective and behavioral development of the students (Kocak, & Bozkut, 2020). However it was observed that there is a significant relationship between school leadership and students development (Hallinger & Heck, 2009). There is significant and positive relationship between school leadership and student-teacher interaction (Xhomara, 2018). The role of leadership in success of schools in terms of enhancing teaching and learning, building relationships, improving students learning is well identified in related

research literature (Day and Sammons, 2014; Leithwood, Seashore, Anderson, Wahlstrom, 2004). There are many factors affecting the student's outcome such as enriched curriculum, effective teaching and learning practices, cooperative school environment and school leadership. Leadership is one of the tangible and indiscriminate resource without which fruitful results cannot be seen in the schools. The concept of leadership was developed in the 1960's to determine the ability of leaders to influence people working in the institutions for achieving the school goals. Leadership is a process in which a person influences a group of persons for achieving a common goal. Therefore, it is a process but not a trait or characteristic that exists in a person. It is a transactional event ("quid pro qui" which means exchange of something for something) which means that it is a bipolar process in which a leader affect and is affected by the followers. It involves influence which means how a leader influences his followers. Leadership does not exist without influence as it occurs in groups. The group may be a community or an organization. Leadership involves a common goal, which means that the leader and the followers have common purpose(Northouse, 2016, p.6).

School Effectiveness and Leadership

School leadership is important in the educational field because most of the functions in the schools are performed by the leaders. Firstly, the academic achievement of the students is determined by the school leadership (Boampong et. al. 2016; Kythreotis et. al., 2012; Tatlah & Iqbal, 2012). School leadership is considered as second important factor after classroom instruction that affect the learning outcomes of the students (Liethwood et. al., 2008). Secondly, it also impacts the job satisfaction of the teachers. The teachers in schools having good leadership were found highly satisfied with their job while as the teachers in schools where the leadership is not good are not satisfied with their job. It shows a positive relationship between school leadership and job satisfaction. (Atik & Celik, 2020; Bhatti et. al., 2011; Machumu and Kaitila, 2014; Naureen, Awan & Noshaba, 2015; Shah and Khan, 2018). Trust and psychological environment plays the role of mediator on the relation between job satisfaction school leadership behavior (Atik & Celik, 2020). It was observed that distributed leadership effects the job self efficacy and academic optimism of teachers (Zare & Nastiezaie, 2019).Thirdly, school culture is also affected by the leadership qualities. (Quin et. al., 2015; Niemann

and Kotze 2006; Sahin, 2004; Sahin, 2011; Turan et. al., 2013; Yilmaz & Jafarova 2019). It was also found that charismatic leadership and different types of school culture have positive and significant relationship (Ozgenel, 2020). Furthermore, school leadership is important in the sense that it monitors the daily working of the schools including the classroom supervision and other academic and non academic activities of the school. Moreover, leadership in schools has been playing a very crucial role for bringing changes and in up grading the school curriculum.

School Leaders and their Leadership Styles

The school leader's possess various types of behaviours of leadership such as task oriented, people oriented and change oriented behaviours. These different behaviours of leadership lead to the performance of the schools (Bahadur et. al, 2017). The theories of behavioural approach tried to explain the successful leadership styles which they used for making the institution successful (Naureen, Awan & Noshaba, 2015). However, Tatlah and Iqbal in 2012 used four leadership styles in his study such as delegating style, participating style, telling style and selling style. The delegating style is that in which the followers have high competence and high commitment. The second one, participating or supporting style is that type of style in which the followers have high competence and variable commitment. The telling or directing style is that type of style in which the followers have low competence and high commitment. The selling or coaching style is that in which the followers have low commitment and some competence. These four styles were further categorized into two main styles of leadership such as task oriented or autocratic and people oriented or democratic styles of leadership. In delegating style the attention of the school leader is on little task and little relationship. In participating or supporting style the leaders focus on low task and high relationship. In telling or directing style the leaders attention is on high task and little relationship. In selling or coaching style the leaders attention is on high task and high relationship. (Naureen, Awan, & Noshaba, 2015; Tatlah and Iqbal, 2012). There is a significant relationship between the styles of leadership and effectiveness of the school (Tatlah and Iqbal, 2012). This depicts that good leadership practices in school plays an important role for bringing effectiveness in the schools where as the schools with poor leadership practices leads to low performance of the schools. It was observed that authoritarian leadership is the

dominant style of leadership. Achimugu and Obaka found that students of chemistry performed better under democratic leadership style as compared to their counter parts under other styles of leadership. Effective leadership practices have significant and positive influence on the effectiveness of the students learning outcomes (Aslan, 2020; Cheung et. al; 2018; Korkmaz, 2006).

The Leadership styles of the principals have direct effect on the academic achievement of the students. It was found that five factors were associated with the academic achievement of the students. One such factor was leadership style of the principals and the other four factors are related to organizational culture. (Kythreotis et. al., 2012). It has been seen that 60 % of the head teacher leadership styles have influenced the academic achievement of the students. It was observed that for improving the performance of the students, it is better for leaders to adopt the democratic style of leadership and should avoid dictating his subordinates and should welcome teachers, students and parents for their opinions. (Boampong et. al., 2016). The democratic style of leadership is the most widely used styles of leadership than any other styles of leadership. The gender to a greater extent affects the effect of leadership styles on the teaching learning system. The female or women teachers are more democratic in their leadership style than their male counterparts. The male teachers were more autocratic in their style of leadership. So far as the democratic leadership style is concerned, there is good relationship between the teachers and the students. (Okoroji et. al., 2014). However, the laissez-faire style of leadership gives complete freedom to the participants for taking decision without the participation of the leader. In democratic leadership, leaders shares the decision making process with their sub-ordinates. Hoy in 2006 and Bass and Riggo in 2006 has given the concept of four I'S in transformational leadership such as inspirational motivation, idealized influence, intellectual stimulation, and individualized consideration. The mostly used styles of leadership in the schools is the democratic and the transformational style. (Omar and Kavale, 2016). The performance of the staff and students were better in the schools that were using the transformational and the democratic style of leadership as compared to the schools in which other styles of leadership are used. (Omar & Kavale, 2016). The transformational style of leadership affects positively job satisfaction and organizational commitment of teachers (Aydin, Sareer, & Uysal, 2013). As the leadership style of principals changes

from other styles of leadership to transformational leadership style, the satisfaction of job and commitment of organization of teacher's increases (Aydin, Sareer, & Uysal, 2013). The leadership of students showed more preference to authoritarian leadership style. The female students were more authoritative in leadership style than the male students. (Najar & Dar, 2017). However, it has been found that the students belonging to nuclear families were more authoritative in leadership style than the students belonging to joint families. (Najar & Dar, 2017). It was also observed that there is no significant difference in the leadership preferences of students with reference to rural urban dichotomy. (Najar & Dar, 2017). The leadership practices adopted by school leaders vary according to their leadership style and significantly affect the school success. The purpose of the present paper is to provide an overview of the types of leadership practices adopted by the leaders of high performing schools.

Research question

What type of leadership practices are adopted by the leaders in high performing schools?

Methodology

Review of the related research studies is done in order to gain an understanding of the leadership practices adopted by leaders of academically high performing schools. Semi-systematic review method is adopted in the present study to map the field of research in the area of school leadership and provide an overview of leadership practices of leaders of high performing schools. In the present study the EBSCO and Google scholar databases are used. The key terms used for search are: school leadership, school leadership practices and leadership styles are used for searching. The research papers and the research articles included in this study are downloaded from these databases.

Analysis

School leadership practices: Leadership practices refer to the habitual or the customary way of doing things. A combination of various techniques which the leader uses for the expected outcomes or it may be defined as the way of pursuing some work or the course of action. The principals adopt different type of leadership practices in different schools and different contexts. The practices of school leadership vary from principal to principal, place to place and context to context. The practices

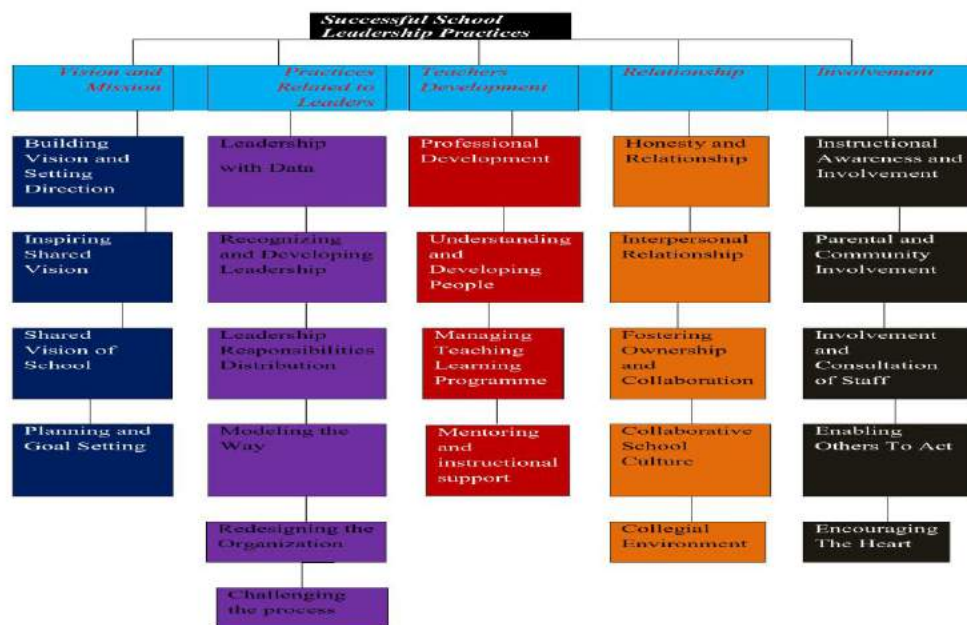
of leadership adopted by the school principals are monitoring the time of learning, improving the teachers welfare, focusing on building of character of student's, encouraging participation of student's in extracurricular activities, enhancing standards, and partnership building with subordinates (Jawas, 2017). The five transformational leadership practices which are adopted by the principals include challenging the process, modeling the way, enabling others to act, inspiring a shared vision, and encouraging the heart (Quin et. al., 2015). The dimension of school leadership practices such as developing people in schools has a very low impact on the learning of the children. Another dimension, setting direction for curriculum design has significant and positive impact on two domains of learning such as Attitudes-active learning and Skills-expression and communication. The dimension of school leadership practices, setting a direction for curriculum development has significant relation with three domains of children's learning such as knowledge acquisition and knowledge acquisition learning, Skills cognitive thinking and skills-Creativity and coordination. (Cheung et. al., 2018). However, it was also observed that 20 % variance in school culture scores is explained by the sub-dimensions of leadership practices such as creating a vision, guidance, encouraging personnel, questioning the process and encouraging audience. (Turan et. al., 2013). It was found that the relation between school culture and school leadership practices is significant. (Quin et. al., 2015; Niemann and Kotze, 2006; Tarun & Bektas, 2013). This means that the relation between school culture and school leadership practices is highly efficient. Therefore school culture is a broad area and it includes various dimensions and all these are affected by practices of school leadership.

The successful leadership practices adopted by the school principals are leadership with data; relationship and honesty; fostering collaboration and ownership; developing and recognizing leadership; and instructional involvement and awareness. (Crum, Sherman & Maren, 2014). The other studies showed that successful school leadership practices adopted by the principals are shared vision of school, collaborative culture of school, leadership responsibilities distribution, involvement and consultation of staff, community and parental involvement, mentoring and instructional support, interpersonal relationships and professional development (Salfi, 2010). Some of the leadership practices adopted in the different types of Indonesian schools are monitoring learning hours, focusing

on students' character building, improving the welfare of teachers, building partnership with stakeholders, encouraging students' participation in extracurricular activities and increasing standards (Jawas, 2017). These practices of leadership are in one way or the other way the responses of the government regulations. The leadership practices adopted by the successful principals in the high poverty and the challenging schools are setting direction, developing people and the organization redesigning. (Jacobson, 2011). The principals from high performing schools said that planning and goal setting is the important leadership practice while as some told for developing their own plan. However the principals from low performing schools did not make any statement like the principals of successful schools. The principals of successful schools considered collegial environment as the key factor for the success of the schools. The principals from low performing schools also considered school as a family but they did not consider it for having positive effect on teaching and learning. The principals from high performing schools were proactive in establishing the policies and practices. (Singh and Allison, 2016). The high performing schools use the practices such as modeling the way, challenging the process, inspiring a shared vision, encouraging the heart and enabling others to act more regularly than the low performing schools. However, the major difference was seen in challenging the process and inspiring the shared vision (Quin et. al., 2015). All successful leaders use the same basic leadership practices (Liethwood, Harris and Hopkins, 2008).

Categories of School Leadership Practices

The core practices of school leadership are divided into four categories such as building vision and setting direction, redesigning the organization, understanding and developing people, and managing the teaching learning program. All these leadership practices were categorized under the five main headings. The five main headings of leadership practices are given as vision and mission, practices related to leaders, teacher's development, involvement and relationship. The practices related to five main categories are shown in the table below:



The table shows that the first type of practice is vision and mission which provides a way or direction for the school leader and his subordinates. It helps them to avoid wrong paths and work according to a particular correct chosen path. It acts as a type of blue print and the leader and subordinates work according to the vision and mission of the school. The second leadership practice relates to leaders, which provides what type of behavior and strategies they are using while acting as a leader. It gives the data about the changes the leaders have brought in the institution. The third type of leadership practice is teacher's development. This type of practice deals with developing the skills and abilities of the school teachers and organizing seminars, workshops and in-service training programs for the teachers in schools. The fourth type of leadership practice is involvement which means involving the staff, parents and the members of the community in the school activities. The fifth category of leadership practices is relationship of the leader with his subordinates, parents and the members of community.

Conclusion

The leadership practices act as a catalyst for improving the learning of the students and their personality development as well as developing the competencies of the teachers. The study shows the various leadership practices that are adopted by the leaders of successful schools. It is observed that the schools perform successfully because of their leadership practices. It is recommended that low performing schools should also adopt these practices that are adopted by the leaders of successful schools. It was found that democratic leadership style is the most effective style of leadership in terms of academic performance and other school outcomes. The democratic and transformational leadership styles are the most used styles of school leadership. Both students and teachers are highly satisfied by these types of leadership styles. On the other hand, it was found that the autocratic style of leadership is not an effective style of school leadership. It was found that female leaders in schools use the democratic style of leadership as compared to their male counterparts. As per leadership among the school students is concerned, it is found that there is use of autocratic style of leadership as compared to other styles of leadership which means that student leaders are using the autocratic style of leadership. However, the practices adopted by the leaders of successful schools are leadership with data; fostering collaboration and ownership; relationships and honesty; developing and recognizing leadership; instructional involvement and awareness, planning and goal setting, collegial environment, modeling the way, challenging the process, inspiring a shared vision, encouraging the heart and enabling others to act; building vision and setting direction, redesigning the organization, developing and understanding people, and managing the teaching learning program. Therefore, the practices of leadership plays a significant role for taking the school to the new heights. The leadership practices affect every aspect of the school from student's academic achievement to job satisfaction of teachers. These school leadership were categorized under the five main headings as vision and mission, practices related to leaders, teacher's development, involvement and relationship. To conclude we can say that leadership practices will help the educational organizations to achieve the goals of the education which in turn will help in the building of the nation.

Educational Implications

The school leadership training programs are conducted at the National, State and District level by different agencies such as NCERT, NIEPA, SCERT, SIE's and DIET's. These training programs are conducted by keeping the different aspects of school education and leadership into consideration. It is recommended that while organizing the school leadership training programs, the educational planners should take into consideration the above mentioned successful school leadership practices in order to boost up the school education and to run the educational institutions in a systematic, planned and effective manner.

The study will be beneficial to the educational practitioners, principals, teachers and research scholars to have an understanding of the successful school leadership practices adopted in the schools and to use these practices in daily routine when they will become a school leader.

The study will be beneficial to the curriculum framers to keep some lessons on school leadership in the text books so that the students should develop an understanding on school leadership, leadership practices, leadership styles and other aspects of school leadership.

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3. Exploration of Teaching Competency among Secondary School Teachers of Sonitpur District of Assam

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Abstract

It is rightly stated by the Kothari Commission the destiny of our country is being shaped inside the classrooms. So teachers are considered to be the backbone of any nation. Therefore, teachers should be competent enough to perform this responsibility with full justice. Competency over this task of teaching is the essence of a successful education system. The present study was designed to know the level of teaching competency among secondary school teachers of Sonitpur district. Further, it also studied teaching competency with regard to gender, type of institutions and Educational qualifications. The sample for the present study consisted of 50 secondary school teachers selected through simple random sampling technique from the different schools of Sonitpur district. General Teaching Competency Scale by Prof. B.K Passi and Prof. M.S.Lalitha was used in this study. The results revealed that the level of teaching competency of secondary school teachers is average. It also revealed that there is no gender-wise difference however there is a significant difference with regard to the type of educational institutions and professional qualifications.

Keywords- Teaching Competency, Secondary School Teachers.

INTRODUCTION

As Rabindranath Tagore has quoted that “A teacher can never truly teach unless he is still learning himself. A Lamp can never light another lamp unless it continuous to burn its own flame”. The role of the teacher is very unique and important in any society. Teachers are the most important factor in educating the future generation. Teaching is considered the noblest profession. A teacher leads a child from the path of darkness to light. He is the one upon whom the responsibility of shaping the future of the nation is vested. Therefore, teachers should be competent enough to perform this responsibility with full justice. Dr. A.P.J. Abdul Kalam stated that “teaching is a very noble profession that shapes

the character, caliber, and future of an individual. If the people remember me as a good teacher that will be the biggest honour for me". Teaching is considered one of the major tasks of a teacher. Competency over this task of teaching is the essence of a successful educational system. The role of qualified and competent teachers for national development is of immense importance. Since the quality of the system of education is determined by the professional competency of the teachers, hence any investment in the field of education will be of little or no value if the teachers are ineffective and incompetent. The teaching competency is an overall assessment of the performance of the teacher in a classroom situation on the criteria of knowledge of the subject matter, methods and techniques of teaching, questioning, and use of teaching aids, pupil's participation and teachers' personality, rapport with the class, classroom management and clarity of objectives. (Ali, 2012, p.190). Teaching competency of teachers is the ability to promote qualitative and effective teaching-learning process and accomplishment of the goals of education through desirable skills and aptitudes. Competent teachers can contribute more in the classroom and in improving the whole system of education. For the attainment of teaching goals and making learning effective teachers' competence is very essential. There is a great importance of teachers' competency in the educational scenario. Dr. S. Radhakrishnan has remarked, "Teachers place in society is of vital importance. He acts as the point of transmission of intellectual tradition and technical skill from generation to generation and helps to keep the lamp of civilization burning". He even aptly said that "Until and unless we have dedicated and committed teachers, who can take teaching as a mission in their lives, we cannot have a good educational system. Teacher should be the best minds of the country". On the importance of teaching competency, the Kothari Commission (1964-66) stated that "of all different factors which influence the quality of education and its contribution to national development, the quality, competence and character of teachers are undoubtedly the most significant." The commission also states that "the teacher is one of the most important factors contributing to the national development. The best system of education may fail to achieve the desired ends in the absence of sincere, competent and professionally aware teachers." The National Policy on Education (1968) stated, "Of all the factors which determine the quality of education and its contribution to national development, the teacher is undoubtedly the most

important. It is on his personal qualities and character, his educational qualifications and professional competence that the success of all educational endeavor must ultimately depend.” The National Commission on Teachers (1983-85) noted, “In the eyes of the community recognition of professional competency and increase in emoluments and benefits were the most salient factors that were expected to enhance the teacher’s status.” The commission further noted, “Professional competence having an edge over the economic factor” was necessary to improve the status of the teachers”. Thus, there is a great need for teaching competencies for the teachers as they can improve and fulfilled the educational achievements of students, their needs and expectations as well as of the society. It is believed that a competent teacher can perform his duty in a better way and has the capacity of moulding the future generations. Secondary education is the instrument for national development that fosters the worth and development of the individual for further education and development, general development of the society. Secondary education is a crucial stage in the educational hierarchy as it prepares the students for the higher education and also for the world of work. Therefore, at this stage, teachers should be competent to provide knowledge that will help them in the preparation of their future life. Success of teaching entirely depends on the level of teaching competence of a teacher. There is a positive relationship between teaching competence and students’ academic achievements. A competent teacher teaches his/her students in a meaningful way. A teacher can never do justice to his/her teaching profession, until and unless, he/she is competent in the dimensions of planning, presentation, closing, evaluation and managerial. With this regard, the researcher conducted a study on the secondary school teachers of the Tezpur area to know their teaching competencies.

REVIEW OF RELATED LITERATURE

Bhattacharjee, D. & Carri, R.G. (2020) conducted a study to identify teaching competency of English teachers at secondary level of Tripura Board of Secondary Education (T.B.S.E.). The study found that there is no association between locality and gender with teaching competency of T.B.S.E. teachers. However, age of teachers is associated with teaching competency’ of T.B.S.E. teachers. **Praveen, S. & Srivastava, N (2020)** studied the teaching competency of upper primary school teachers of central and state government schools with reference to their gender and years of teaching

experience. The study revealed that upper primary school teachers of central and state government differ significantly in teaching competency on the basis of gender and teaching experience. Central school teachers were found more competent as compared to state government school teachers. **Rana, N. & Shivani (2019)** conducted a study with the purpose to investigate the differences in the teaching competency of secondary school teachers in relation to their gender, location of school, type of school, qualifications, subject background and teaching experience. The findings of the study revealed that there is no significant difference in the teaching competencies of male and female teachers. However, significant differences have been found in the teaching competencies of secondary school teachers in relation to their location of school, type of school, subject background, qualifications and teaching experience. **Mishra, S (2017)** revealed that there was significant difference in teaching competencies among secondary school teachers in relation to gender and teaching experience, whereas difference in teaching competencies among such teachers teaching science and non-science subjects was not significant. Similarly, there was no significant difference between teachers who had B.Ed. general and B.Ed. secondary qualification.

OBJECTIVES OF THE STUDY

- (1) To determine the level of teaching competency of secondary school teachers.
- (2) To find out the significant difference between the level of teaching competency with regard to gender, type of schools and professional qualification.

HYPOTHESES

(H_01): There is no significant difference between the level of teaching competency of male and female secondary school teachers.

(H_02): There is no significant difference in the level of teaching competency between the private and government secondary school teachers.

(H_03): There is no significant difference in the level of teaching competency between the secondary school teachers in terms of professional qualification.

DELIMITATION OF THE STUDY

1. The present study has been delimited to only Sonitpur district.

2. The study has delimited to 50 samples of teachers only.

RESEARCH METHODOLOGY

The researcher has adopted a descriptive survey method for the study. The population of the study consists of all the secondary school teachers of Sonitpur district of North Assam. A total sample of 50 teachers representing secondary school teachers of Sonitpur district was selected through Simple Random Sampling technique. General Teaching Competency Scale by Prof. B.K.Passi and Prof. M.S.Lalitha which is an observation schedule consisting of 21 items related to 21 teaching skills that encompass the entire teaching-learning process in the classroom. They are related to 5 major aspects of classroom teaching namely Planning, Presentation, Closing, Evaluation and Managerial. It is a 7-point rating scale measuring the use of the skill by the teacher in the classroom corresponding to each item ranging from '1' for 'Not at all' to '7' for 'very much'. The researcher has collected primary data by personally visiting different sample schools with prior permission for collecting the data. The tool that has been used is an observation schedule; therefore, the researcher has adopted the participatory observation method and has herself filled the questionnaire to know the teaching competencies of secondary school teachers. Percentage, Mean, Standard deviation, t-test was used for analysis and interpretation of results.

DATA ANALYSIS AND INTERPRETATION

The data gathered by the researcher was analyzed based on the objectives by using appropriate statistical techniques.

Analysis of Objective No. 1

To determine the level of teaching competency of secondary school teachers.

Teaching Competency and skill level	No. of sample	Percentage
Superior	3	6 %
High	3	6 %

Above Average	9	18 %
Average/Moderate	18	36 %
Below Average	13	26 %
Low	4	8 %
Inferior	0	0 %
Total	50	100 %

Table 9.1 Level of Teaching Competency of Secondary school teachers

It is understood from the **table no. 9.1** that 36% of secondary school teachers have average, 26% have below average, 18% of them have above average, 8% of them have low and 6% of them have both superior and high level of teaching competency. From the table, it can be inferred that none of the secondary school teachers have inferior level of teaching competency.

DISCUSSION

The study reveals that secondary school teachers possess a moderate (average) level of teaching competency. The result of this study is similar with the studies of Choudhury and Chowdhury (2015) and Akram, et.al (2015) in their studies also found that a strongly positive and significant relationship exists between teacher competence and job satisfaction. According to Bhargava & Pathy (2014), a positive attitude towards teaching profession can bring the desired quality in the education sector by developing sense of duty, professional competence and by giving them an insight of the student's needs and problems.

Analysis of Objective No. 2: *To find out the significant difference between the level of teaching competency with regard to gender, type of schools and professional qualification.*

Table 9.2: Significance of difference in the level of Teaching Competency with regard to gender, type of institutions and educational qualification

Selected variables	Group	No. of Samples	Mean	S.D.	d.f.	Level of Significance	Cal.Value	Critical Value	Remarks
Gender	Male	19	66.31	14.32	48	0.05	0.016	2.01	Accepted
	Female	31	66.38	14.17					
Type of schools	Private	26	73.84	14.12	48	0.05	4.53	2.01	Rejected
	Government	24	58.25	8.90					
Professional Qualification	B. Ed.	36	69.19	15.15	48	0.05	2.33	1.67	Rejected
	Non-B.Ed.	14	59.07	12.70					

It is evident from the **table no 9.2** that Mean and Standard Deviation of male teachers in teaching competency are 66.31 and 14.32 respectively. Similarly the same for female teachers are 66.38 and 14.17 respectively. However, the mean score of female secondary teachers is slightly higher than the male secondary teachers. It indicates that there is no significant difference in the level of teaching competency of the male and female secondary school teachers. Mean and Standard Deviation of private teachers in teaching competency are 73.84 and 14.12 respectively. Similarly the same for government teachers are 58.25 and 8.90 respectively. It indicates that there is significant difference in the level of teaching competency of the private and government secondary school teachers. Hence, the null hypothesis is rejected. Again, the Mean Score and Standard Deviation of teachers with B.Ed. training in teaching competency are 69.19 and 15.15 respectively. Similarly the same for teachers who are without B.Ed. training are 59.07 and 12.70 respectively. The obtained cal.value 2.33 is more than the critical value 1.67 at 0.05 level of significance. It indicates that there is significant difference in the level of teaching competency of secondary school teachers with and without B.Ed. training qualification. Hence, the null hypothesis is rejected.

DISCUSSION

The mean score of female secondary school teachers (66.38) is higher than the mean score of male secondary school teachers (66.31) but statistically there is no significant difference between the two.

The 'no significant difference' in teaching competency with regard to gender reveals that teaching competency is independent of male and female. So, male and female secondary school teachers were equally competent in their teaching competence. This finding is supported by the findings of the studies by Passi and Sharma (1982), Prakasham (1986), Kaur and Talwar (2014). Gender can never determine the quality and competency of a teacher. We know that quality teaching depends on the caliber of a teacher, and not in gender of a teacher. The findings of the present study reveal that private secondary school teachers are more competent than government secondary school teachers. The results are in line with the investigation carried out by Rajammal (2015), Quari and Bhat (2015). Himabindu (2012) in his study explored that the teaching competency aspect is not influenced in respect of variables like sex and age but is influenced in respect of variables like locality, experience, marital status and type of institution. (Pratibha, 2017). The study also found that teachers with B.Ed. teachers are more competent than the teachers who do not possess the B.Ed. training experience. The result of the study is in line with the study of Pratibha (2016), Aziz and Akhtar (2014).

SUGGESTIONS AND RECOMMENDATIONS

- (1) For increasing the teachers' competence, school authorities and government should look after it. Steps like increase of pay-scale, job security and even appreciating their hard-work by announcing different rewards and incentives should be taken into consideration.
- (2) At the time of recruiting teachers for secondary level, it is important that teachers with professional qualification like B.Ed. or M.Ed. should be taken into concerned. It should be made as an indispensable and compulsory qualification at the time of selection of teachers at the school level.
- (3) Teacher education institutions should give more importance on practice teaching than on theory. As sometimes, it can be seen that teachers might have mastery over that subject but is not able to communicate his/her knowledge to the students due to lack of training in real classroom scenario.
- (4) The educational institutions should organize seminars, workshops, refreshers courses on a timely basis, even the school management committee should invite resource persons to their institutions to motivate and enrich the teachers.

(5) All possible efforts should be taken by all concerns to enhance teaching competency of teachers through necessary trainings and orientation.

(6) The use of modern information and technological advancement should be included in the teacher education curriculum so as to prepare teachers who are competent to use it effectively in the classroom teaching in the future.

(7) The NCERT, SCERT, NCTE and the Department of Education of different universities should organize a series of workshops and seminars to re-orient the teachers especially of the school level.

CONCLUSION

National Knowledge Commission(2007) and National Curriculum Framework for Teacher Education (2009) have suggested that a teacher needs to be prepared in relation to the needs and demands arising in the school context, to engage with questions of school knowledge, the learner and the learning process. Teachers' competence can raise the standard and quality of the educational system. Education at secondary stage is considered to be the backbone and foundation for higher education. It is an investment as well as an instrument to prepare such students who will lead towards the achievement a more rapid economic, social, political, technological, scientific and cultural development in the country. Therefore, it is very essential that the teachers at the secondary level should be competent in their teaching profession. A competent teacher has the capacity to bring the whole world into the classroom through his/her teaching process. In a developing country like India where even the target of full literacy is not achieved and where country has to continue steady progress of school children if the teachers are not competent then there is little hope in the bright future of the coming generation. With incompetent teachers, it will be a difficult task to continue the progress of the whole nation. Therefore, at present it is not enough only to have an adequate number of teachers but at the same time, it is very important that the teachers who are appointed for secondary education in particular and for the general education should be competent in their teaching, only then it is possible to have an effective and improved standard of education system.

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4. Education in India: A Historical Perspective

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Abstract

Indian education system is strongly influenced by the religious systems and few have their own indigenous approach which serve as a foundation for secondary and higher education. The current paper attempts to comprehend how these institutions have played a role in formalisation of education at secondary school system through an overview ancient, medieval, colonial and post independent periods. The present paper aims to bring out common thread as well as differences in approach towards secondary and higher education and how over a period, formalisation of education have subjected to changes in responses to merits and demerits, respectively. The institutional differences of these religious systems from ancient to post independent periods are discussed in terms of (a) Objective (b) Pedagogy and (iii) Outcomes of Education.

Key words: Educational Philosophy & Pedagogy

1. Introduction

Education in India since time immemorial is influenced by religious systems and have been found to have indigenous approach. The long history in terms of the timeline has led to several multicultural and multireligious perspectives to education which is predominantly determined by the need of the hour. Thus, the objectives of education, the pedagogical approach and the expected outcome has changed over a period. irrespective of the differences the roots of any education system lie in its philosophical foundation. ‘Philosophy’ says Kautilya, “is the lamp of all the sciences, the means of performing all the works, and the support of all the duties” (Radhakrishnan, S., 1924). In sociological perspective ancient education is considered as the tool for cultural preservation, cultural transmission, acculturation, and enculturation (Rathi, K. N., & Ampili Aravind., 2015). Education is not a word confined rigidly to specified occupations of man but refers vaguely to more or less of human activity in the production and prevention of changes in other human beings (Thorndike, 1920).

A. Ancient System of Education (1500 BCE –300 BCE)

Two major streams of education were followed in the ancient Hindu system namely Aparavidya (material knowledge) and Paravidya (spiritual knowledge). It is interesting to observe from a historical and sociological perspective that over a period educationist emphasized on the apravidya vidya that led to emergence of universal education, streams of education based on the levels in terms of primary, secondary, higher, technical education etc, Religious institutions are left with propagating paravidya that is spiritual knowledge. Aims of Education were to provide good training to young men and women in the performance of their social, economic, and religious duties. Also, preservation and enrichment of culture, character and personality development and cultivation of noble ideals.

Two major types of educational systems namely gurukulas i.e., house of the teachers where learners resided till their education is completed and vidyapeetas the institutions for higher spiritual learning. Gurukulas could be considered as the centres for basic learning that advocated sravana, manana and nididhyasana as the three steps (Radhakrishnan, S., 1914) These three components in other words refer to listening to the readings of the scriptures by the teacher (sravana), attempting to memorise and arriving at subjective interpretation (manana) and the comprehension of the fact and applying to their life (nidhidhyasana) (Pandya, 2014; Yadav, U., 2018). Thus, in the ancient Hindu system of education Individual was of the prime concern and the centre of education. The pedagogy of oral tradition that facilitated sravana, manana and nidhidhyasana was followed in the ancient gurukulas as well as some of the prevailing gurukulas in the modern times.

Termination of learning was by the teacher who gives testimony for each one of his students about the knowledge that one has acquired through sravana, manana and nidhidhyasana or in other words the accuracy of his perception, ability to draw inferences and the capacity for valid comparisons. Based on the individual differences of the learners they were subjected to apara and paravidya/apadravya focused on the material knowledge through which one could earn one's livelihood ie. having to study subjects like elementary philosophy, elementary grammar, elementary arithmetic, along with subjects like Astrology, Medicine, Chemistry and Physics etc. On the other hand, subjects

under paravidya were elementary philosophy, elementary grammar, elementary arithmetic, along with subjects like Vedanga, Upanishads and Purnas.

Students of paravidya also mastered in the science of logic that helped in arriving at individual interpretations of the scriptures (writing explanations for the principles laid down in scriptures in others writing bhashyas to the sutras found in the scriptures. Centres for higher education namely vidyapeetas and parishads facilitated the process of learning by periodically organizing debating circles, goshtis. Goshtis and Parishads provided platforms for the testimony of their learning. Further divisions within the ancient education systems

The Chief aims and ideals of ancient Indian education are highlighted by Gupta in the following manner: a) Infusion of a spirit of piety and righteousness, b) Formation of high character, c) Development of personality, d) Inculcation of civic and social values, e) Promotion of social efficiency and f) Preservation and spread of national culture (Pandey, S. K., 2019).

B. Buddhist Education (300 BCE to 100 BCE)

The advent of Buddhism and its propagation during in the 6th Century BC brought a new perspective to understand the philosophy of mankind. The socio political system that was prevailing during the 6th century promoted the spread of this religion It satisfied the political and economic needs of the rulers and the merchants who found the principles of Hinduism as not approving their non-religious practices (Darian,1977) Ancient education system based on Hinduism was considered to be focused more on ritualistic modalities towards preserving the oral tradition while Buddhism education system was considered to be pragmatic (Rahdhakrishnan, 1924). There is no fundamental difference between Vedic and Buddhist approach towards the general educational theory or practice was concerned.

The aim of both the systems of education is the liberation or emancipation. The aim of Buddhist Education was to make a free man, a wise, intelligent, moral, non-violent & secular man. Students will turn out to be judicious, humanist, logical and free from superstitious. They also would be free from greed, lust, and ignorance. The Core of Buddhist philosophy of education focuses on three major points namely discipline, meditation, and wisdom (Meshram, M., 2013). Monasteries

were established to inculcate Buddhist principles of education. The primary goals of Buddhist education as practiced in monasteries is to achieve moral perfection through following eight-fold paths of life. The curriculum included Sastras of the five Sciences” namely grammar, the science of Arts & Crafts, Science of Medicine, Science of Reasoning and Inner Science.

In addition, prominence of logic, tours, conferences, and meditation in solitude were the additional pedagogical aspects of Buddhist education. The curriculum was chiefly spiritual in nature. It was because the chief aim of education was to attain salvation. So, the study of the religious books was most important. This type of curriculum was meant only for the monks. Besides these spinning, weaving, printing of the clothes, tailoring, sketching, accountancy, medicines, surgery, and coinage were the other subjects of Buddhist education. In the early period Buddhist Education was limited within the monasteries and only for the members of the monastery. But later, it was open to the mass, even lay people got scope to have education in those institutions.

In modern days Buddhist Education became wide open and embraced people of all walks of life. The aim of Buddhist Education is to change an unwise to wise, beast hood to Buddha hood. Buddhist Education made revolutionary change in the society. The Buddhists in the world first made education open to all. Students irrespective of caste, creed, religion got opportunity to have education which was denied by the superior class in the society (Masih, A., 2018). However, **Buddhist education could not give the proper attention to the occupational, industrial, and technical education. But in return, it gave severe blow to the social development because its derided family ties.** Leaving their family life Buddha Bhikshus devoted their whole lives to sangh and Buddhism.

C. Islamic Education (Post AD 1200)

Islam arrived in the **Indian** subcontinent in the 7th century when the Arabs conquered Sindh and later arrived in North **India** in the 12th century via the Ghurids conquest and has since become a part of **India's** religious and cultural heritage. Muhammad Ghauri in A.D. 1175, paved ways for the development of the 100 Estelar Muslim education in India. The Muslim educational institutions are attached to the Mosques. Mohammad Ghor replaced temples with mosques, and the pathasalas with

Maktabas and Madrasahs. Feroz Tughlak built many colleges and founded a residential University at Delhi and provided it with scholarships and endowments (Krishnamoorthy, 2019). Aim of Islamic education was Emancipation from dogmatic and ideological understanding of Islam. This view of human nature entails that Islamic education should develop the physical, spiritual, emotional, and intellectual aspects of a Muslim (Hussien, S., 2007). The programme of education and training proposed by Ibn Sina included study of the Quran, metaphysics, language, adab (belles-lettres), ethics and manual skills.

Islamic Education was offered at two stages Stage 1: Study of the Quran, metaphysics, language, adab, ethics & manual skills (curriculum different for female child). In the second stage of schooling, which Ibn Sina called the period of specialization, pupils should, in his view, begin to acquire manual skills, irrespective of their social status (Mirbabaev, A. K., Zieme, P., & Furen, W., 2000). The instructive way of thinking that was received from the Islamic training framework and comprehend talks from the Quran were the centre of school education. (Arockiasamy, T., & Fernandes, S., 2020).

In traditional Islamic pedagogy, authoritative acceptance of knowledge is stressed, with learning often based on listening, memorization, and regurgitation. Islamic pedagogical knowledge in to two divisions **revealed** (wahy) and **acquired** (iktisabi) knowledge. Revealed knowledge was sacred and given to the chosen few, namely, prophets. Acquired knowledge included transmitted traditions (naqli) and rational knowledge ('aqli). Gurukulas, Monasteries, and Madararas all had a strong philosophical perspective that focused on the spiritual dimension to life though the curriculum also contained subjects necessary for earning a livelihood.

D. British Colonial Education in India (1800-1940)

The Paradigm shift towards to the objectives, pedagogy and the outcome of Indian Education System could be traced to colonial period or British Rule in India. Under the British India, Christian missionaries were making efforts to introduce western education by opening elementary schools and providing education to the marginalized, deprived and economically weaker sections of the society. This was a period when the symmetry between Christianity and western scholarship was championed

by missionaries such as John Murdoch and Alexander Duff (Bellenoit, H. J., 2007). This period was characterised by educational reforms, acts and policy framework for Indian education towards modernisation industrialisation and building labour workforce for modern era. A committee were formed to take stock of systems and polices which recommended to revive secondary school system in India. Anglican mission schools run my Missionaries hoped to make their schools centres of spiritual and religious dialogue. They were engaged in the imparting of both useful and Godly knowledge (Bellenoit, H. J., 2007). Hunter Commission report (1852) highlighted the position of secondary schools at that time, suggested to withdraw as early as possible from the direct management of Secondary Schools, emphasising diversified courses. The management of primary education was transferred to the local bodies' setup under the Local Self-Government Acts passed during the period (Report of Secondary Education Commission, 1952). Indians usurped the educational enterprise from within to contest the 'superiority' of both western knowledge and Christianity. In doing so they located traditions of reason and logic in Indian thought. Rather than serving as a hegemonic colonial undertaking, mission schools were subject to Indian agency and contestation. Indians were alert to the advantages of western educational institutions introduced under British rule and appropriated them to reproduce their own social and intellectual foundations (Bellenoit, H. J., 2007).

On the other hand, Indian Education Policy drafted in 1904 introduced a new pattern of education both at the primary and higher level of learning to cater to the needs of the rural section of India as well as to train the people was to protect Britishers in their business transaction with landlords and grain dealers. The historical event of partition of Bengal led to national education movement in the year 1905 spearheaded by Sri Aurobindo, Rabindranath Tagore, and others. They focused and emphasized on reviving the past glory of India, to recreate a sense of pride among Indians by learning the traditions, culture, and religion of India, to rediscover the value of using Indian languages as the medium of education (Banerjee, D., 2015).

To promote learning procedure of Indian traditions along with modern Western values, Tagore formed the Visva-Bharati University, Satish Chandra formed Dawn Society and Sri Aurobindo established an ashram in Pondicherry. As the active members of National Council of Education, both Sri Aurobindo

and Tagore developed certain ideals regarding the teaching procedure, which helped reform of the traditional system of education (Banerjee, D., 2015). Changes in the Indian education system both at the primary, secondary and higher levels were periodically altered based on the recommendations by several commissions like National Council 1910, Primary Education Bill 1910, Hartog Committee 1929, Wardha Scheme of Education 1937 etc.

Maria Montessori's 1939 visit to India laid foundations of preschool education in India and teachers were trained in Montessori's method of education. Montessori's made an influence on educators in creating child centred learning environment. Though this method originally developed in Italy, this method focuses on developing children's independence, building collaborative relationships with teacher, and using hand on materials (Mira Debs, 2019). The primary objectives of Western education during this period were Universalisation of Education in terms of access and equality. Focus on formalisation with universal access in large numbers, and Establishment of Educational Infrastructure like council bodies (National council of education) National Laboratories, Universities, Colleges, Secondary School, Hostels. The expected outcomes of this education Establishment of Training and Development centres, setting up the minimum qualification of a teacher & review boards, Recognition for State Language & National language.

Though the outcomes national education movement has laid foundations to develop an education policy for Independent India, the negative impacts were on native learning systems, especially resulting in marginalization of institutions of religious education, ignoring the indigenous literature and thoughts. In colonial India, the job of deciding, selecting, and shaping school knowledge was performed initially by 'enlightened outsiders' and later by 'educated Indians'. The table 1 below summarises education systems with respect to the objective, pedagogy and outcomes. During this period, Indigenous knowledge and pedagogic traditions were seen to be of extraordinarily little use for educational purposes. Rather, indigenous knowledge was 'deficient' and was claimed to have several elements that would not qualify it as 'modern'. The new content and curricula were not challenged by 'educated Indians'. Eventually indigenous knowledge and cultural forms got excluded from the curricula of schools (Kumar, 1991).

Table 1: Showing the summary of education systems with respect to the objective, pedagogy, and outcomes

Education System	Objective of Education	Pedagogy	Outcomes
Vedic Education	1. Inhibition of those activities of the mind by which it gets connected with the world of matter. 2. Individual's supreme duty is thus to achieve his expansion into the Absolute, his <i>self-fulfilment</i>. 3. The <i>individual is the chief concern and centre of the Education</i>	1. Oral Tradition 2. Four reliable devices gaining knowledge <i>Perception, Inference, Comparison</i> and <i>Testimony</i> of experts. 3. Three steps of education process <i>Listen, Comprehend and Reflect</i>. 4. Elementary arithmetic and grammar, phonology & metrics also formed part of the primary curriculum.	Composition of Sutras & Bashyas. Skill Based education Sanskrit as Medium of Instruction
Buddhist Education	1. To achieve <i>moral perfection</i> and to develop a morally perfect personality. 2. The desire can only be removed, & moral development can be possible through <i>Eight-Fold Noble path</i>	1. A child is first introduced to a "Sidham" and then great Sastras of the five Sciences" 2. Aksharasvkarana (5 or 6 age) - Formal Introduction	Establishment of Monasteries & Universities Translation of Vedas, Vedangas & Upanishads in Tibet language
Islamist Education	1. Emancipation from Dogmatic and ideological understanding of Islam. 2. Islamic education should develop the <i>physical, spiritual, emotional, and intellectual</i>	1. Authoritative acceptance of knowledge is stressed, with learning based on listening, memorization, & regurgitation. 2. Listening to a teacher, who is active as a transmitter of knowledge, while the student is passive.	Advanced studies in various branches of sciences. Music, Painting & Philosophy

	aspects of a Muslim	3. Pedagogical knowledge into two divisions revealed & acquired knowledge	
British Colonial Education	Providing education to the marginalized, deprived economically weaker sections of India. 1. Anglican missionary schools as centers of spiritual and religious dialogue 2. Promote Technical and Women education	1. Reading and writing of vernacular language and arithmetic. 2. Introduction of Rural education - half time school tailored for children of agriculturists 3. Excessive prominence is given to examination 4. Training college for teachers 5. Medium of Language for Higher primary and Missionaries run school was English 6. Introduction of Kindergarten methods and Physical in the primary schools.	Establishment Industrial, Technical & Commercial Schools National education movement - promote technical education in local languages. Medium of Language - English

2. Modern Education System

Recognizing the critical importance of science and technology for post independent India, GOI brought out many National Education Policy reforms. Like every society in pursuit of knowledge, technology, and innovations, which defined the desired path, science was denoted as a precursor to economic growth globally. The emphasis was to reform the scientific temperament universally, so was technology, global leaders such as America and with other European nations took the lead. As a result, in 1953 the secondary education commission recommended the teaching of general science as a compulsory subject in high schools and higher secondary schools. The first form of science education post-independent India witnessed the establishment of universities and Institutes.

Education which was hitherto was in service sector today is more related to the business sector (Ganalakshmi N., 2015). Revolutions in the field of science and technology, the impact of globalization have resulted in redefining the goals of education. The modern education system is geared at developing a strong foundation to acquire knowledge and specialised skill set for a livelihood. The business orientation to education has brought in significant changes in its infrastructure, pedagogy, as well as in its goals. Indian education system is complex as it is governed by Central, State as well as by Private organizations. Separate agencies regulate different streams of education namely Primary & Secondary, Higher Education, Technical and Medical education etc. Each one of these define its own goals and pedagogy based on the ownership namely whether it is run by the Centre, State, or the Private organizations. The primary concern of the Government is to increase literacy and thus prioritizes on Accessibility and Equity. On the other hand, the priorities of private organizations are driven by the market and thus create and cater to the Demand.

The resultant of these is seen in measures like opening of more and more schools and colleges, incentives for private institutions. Unfortunately, Macaulay's ideas still dominate education in India. Elements of colonialism are evident in the authoritarian position of teachers in the class, control of student's responses, employing teacher-centred methods, which lead to passive acquisition of knowledge. Teaching of values, ethics, and morality informally at the family has taken the back seat and have become a subject matter in schools which on the one hand is taught with secondary importance by the school/teacher and on the other hand is learnt by the students a subject that need to be passed in the examination failing to adopt them in their personal life. The ancient system of

education that was based on strong cultural base and that recognized the needs of society and the individual differences among students is being replaced by universalization of education.

Majority of primary/secondary education systems in India still follow Macaulay perspective and western education frameworks, which are structured, content-based, reductionist and materialistic in nature. Modern Education became the priority of those who could afford it and hence creating a divide between educated and uneducated; affordable and poor; accessible and remote. Though Indian education benefited from influence of Self-imposed British Colonial education system reforming higher education for the modern or industrial world, but it was alien to secondary the education system and has not been aligned to the native mindset.

3. Merits and Demerits of Modern Education

Table 2: Showing the merits and demerits of Modern education system with respect to objective, pedagogy, and outcomes.

Objective	Merits	Demerits
	Foundation to acquire knowledge and specialised skill set for a livelihood – Emphasises Individual learning principle	Myopic view on purpose of education and meaning of education. Ignores collective learning principle
	Science and technology development for the nation	indigenous knowledge and cultural forms excluded from the curricula of mainstream of schools. Importance of Literature, Arts and craft were side lined. Subjects like metaphysics & philosophy were considered as unnecessary.
	Universalisation of Education in terms of access and equality. Focus on formalisation with universal access in large numbers.	Other alternative educational models, vocational education systems and experimental schools were not part of formalisation.
	Establishment of Educational Infrastructure like council bodies (National council of education) National Laboratories, Universities, Colleges, Secondary School, Hostel.	The original inspiration of national education movement was misled, and key stake holders took it only as an academic activity.

Pedagogy	Medium of Instruction – English preschool with Three Language formula	Academic problem of teaching many subjects at one time
	Predetermined set of curriculums and teaching methodology resulting mugging up information for exam.	Limited attention towards internalisation process of learning
	System of education became more responsive towards fill the gap of Industry and Global needs	Ethical and Value based education took a back seat
Outcomes	Established Training and Development centres, setting up the minimum qualification of a teacher & review boards	Dearth of properly qualified and trained teachers
	Recognition for State Language & National language	English language became compulsory for clearing matriculation and other competitive exams
	Three are tier school systems, two Central schools (CBSE and ICSE) and state schools	Central schools are not affordable for common. Divide between Public and Private schools

The fundamental purpose of education is filled with **resource myopia**. The early preschool programs and schemes in nation range from being affluent to being poor; from remote rural to urban metropolitan; from westernized nursery classrooms housed in large comprehensive private schools to independent nursery schools and day-care centres, to mobile crèches that serve as temporary classrooms in vans/buses at labour sites for the care of children of migrant construction workers; and from government/private or non-profit sponsored centres in affluent neighbourhoods to small, crowded, one-room schools in urban slum settlements (Gupta, 2006).

Hence resulting in a deep divide between Public and Private Centres/Schools is visible since Independent India. There are serious concerns regarding the quality of pre-school education offered by Anganwadis and Balwadis – specifically: lack of resources; lack of awareness of quality indicators; distortions in the curriculum; poor training; lack of institutional capacity; and inadequate advocacy about the need and significance of early care and education programs (NCERT, 2002).

Primary education institutions in India, especially in urban India is influenced by the Global standards of America, British and European countries. Lucrative offers (both teachers and Students) by education institutions towards a promising future of the child, filled with the road of success and industrious life. Even parents from rural India, farmers take a loan from the bank to send their

children to residential schools. This has become ascribed status in the family. Primary schools/ Government schools from rural India are perceived low in standards due to lack of basic facility, infrastructure, and trained teachers (Gupta, 2006). New models, innovation, and experiments in classroom environment are geared towards only cognitive learning a concept. Here learner acquires knowledge and skill by higher mental processes by manipulating the information based on mental representations of physical objects and events. There is a pressing need for the Development of experiential learning based approach through self reflection, self evaluation and self discovery, here the learner is directly involved in the experience of new knowledge the retrieval of that new information remain for longer period of time. Development of scientific rigor in absence of right context is the common sight.

The curriculum is being designed both at the secondary and higher grades of education caters to the needs and demands of global market. As a result of which educational institutions are turning out with large number of graduates trained with skills that were supposed to facilitate the need of global market. It is interesting to note how the expected outcome of the education has changed over a period from preparing an individual to be a self-fulfilment, and not in the acquisition of mere objective knowledge to shift from Manual work to Mechanical, Energy centric to Information-centric and now Automation technologies to Learning Machines.

Despite all these so-called advancement in the field of education brought out in the form of teacher training adopting technology, encouraging, and supporting establishment of centres of education the outcome of education is found to be far beyond the expectations of educationists, employers as well as by the public. The goals of ancient system of education that focused on the holistic development of the students both inner and outer self that prepared him for life is found to be missing.

Today educationists recognise the role and importance of multilingual and multicultural education, thereby connecting the ancient and the traditional knowledge with contemporary learning. Today people have realized that Technology is not a solution to a problem. It is just an instrument! Any knowledge in absence of **‘Ethical values’, ‘Morality’ and ‘Right Spirit’, it is a cultivation of dogmatic prepositions.**

NCERT (2005) in the National curriculum framework for school education highlights “Ethics is concerned with all human values and with rules, principles, standards, and ideals which give them expression. In relation to action and choice, therefore, ethics must be conceded primacy

over each of the forms of understanding; it also stimulates the ethical development in students (NCERT, 2005). In this context ethical development “calls for devising means and way of helping children to make choices and decide, what is right, what is kind and what is best for the common good, keeping in view the border implication for personal and social values”. Several of the issues recognized by educationists as the resultant of current education system like fear of failure, academic stress, lack of motivation, conflicting goals, etc., were found to be due to depleting importance given to morality and ethical dimension right from the primary education (Kumaraswamy N., 2013; Deb, S., Strodl, E., & Sun, J., 2015).

On the other side “*rote learning*” have been viewed as ineffective method of learning in NCF (2005). Turning away from rote learning solely towards cognitive based approach has its own profound concerns, which calls for future research attention and empirical evidence. Our Ancient education systems like Vedic and Buddhist traditions are fundamentally based on rote learning and experiential learning methods. Both are coupled with Sharavana, manana and nidhnyasa, question arises do we need to treat learning process independently with instrument of the learner?

By looking at the demerits of modern education system, there is a pressing need for hybrid content of Indian early childhood education to demand a sphere of value-based education for a sustainable livelihood. The paradigm shift is essential to revisit the purpose of education (secondary education) and need for alternative education systems along with the mainstream of education which is inclusive and collective in nature. An inclusive education not only for a student it includes the family as a unit. Family or Parent education aligning with the purpose of child education is also essential. The students should not get into a conflict zone of parents, teachers, and peers' demands or expectation. He or She should be able to stand for “*oneself*” it is possible only when one has realized oneself, this cannot be achieved in absence of spiritual, moral, ethical, and value-based education systems.

Modern education includes heavy study material on modern science and technology into the syllabus, resulting overemphasis courses like classical language and literature have dropped in Higher education institutions. There are various emerging trends which effect higher education system in India. New technological changes have become essential parts of one's life and due to modern education, its positive effects can be seen in terms breaking down the physical barriers, distance education and easy access etc. Despite merits of current education system is also leading to negative aspects are like self- centism, digital divide, weakening of Self structure due to depletion

of faith in indigenous approach to one's life. The major benefits of ethical students were found to be highly that they are committed, taking responsibility, giving respect, high on tolerance, non-discrimination, compassion, better understanding and being a good citizen (Reena Sahota, 2017; Shobana, S., & Kanakarathinam, R., 2017; Sharma, S., 2020).

As stated earlier in table 1, the common thread observed across religious institutions is riddance of the learner's ignorance as the key objective of education. Second major observation was that across these religious institutions emphasized on lifestyle, family education and development of right habits in early preschool education is prerequisite to achieve the purpose of education for Ancient, Buddhist and Islam education systems. Whereas in the modern education system developing a strong foundation to acquire knowledge and skill set for a livelihood have become critical.

It is also interesting to note that the ancient educational institutions adopted a pragmatic approach which was advocated by different religious systems. It is interesting to notice how these religious institutions catered (Ancient, Buddhist and Islamic education systems) to the learner's overall development of personality, morality, ethics, and values at the early childhood foundation. These systems are indigenous in nature, teacher-directed, child-centred, humanistic, and experiential.

This reflective knowledge, which value-based, ethics and moral education have been embedded by the ancient religious institutions itself. These indigenous systems have played a key role in building "*knowledge on oneself and collective knowledge of mankind*", at foundational level of a child before getting into mainstream education. The Ancient education system of India which includes both Vedic and Buddhism era. Institutions like Gurukula Education systems, Nalanda, takshila were built on basis of this collective knowledge.

4. Conclusion

The need of the hour is to objectively analyse how far the principles of Ancient education can be blended into the current/modern education system at both secondary and higher education level. There is a spurt of growth in establishing alternative educations centres based on religious/philosophical/spiritual foundation in totality or partially. This paper calls out for the emergence of alternative education systems, which aims to *facilitate knowledge of oneself and collective knowledge of mankind*. Educators are keen towards bringing a paradigm shift from cognitive based learning to other forms pedagogical approaches which is collective, holistic, and experiential in nature. however, the claims of these institution as finding solution of to the problems of modern education need to be empirically tested and validated.

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5. Case study as an Educational Research perspective for social justice

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Abstract

Research in itself is the part of life engaging us for collecting and processing of information. Unknowingly or knowingly people have participated in different pre-conditioned and after-conditioned researches in their daily lives. Like a person standing in a shop or mall and thinking about brands, trends of fashions, quality of products and finding out their results. In similar ways one can view other academic researches with regard to need and demand of specific research grounds; whether it is scientific, sociological, historical, cultural or any literary researches. Mostly, every research is done with the hope to explore their established uses and limitations, and also to examine some of the new possibilities that are opening up in the studies of education and society. So, as the case study researches important in its own ways of providing some sources of data about a person, case, individual; or a group, organization, or school. This paper is written with the view of understanding case-study researches from different dimensions as a social analyst what provides us a clear view of people's life issues and realities as a critical approach of doing educational research.

Key words: Why and how case study, types, considering points, ethical issues and research perspective for social justice.

What is case study?

In educational research there are cases to deal with, whether quantitative or qualitative. Somewhere the entire culture shaping group in ethnography is a case (Creswell & Poth, 2018); individual's personal experiences about family, society and culture in narratives are cases; participant's lived experiences about a concept or phenomenon is a case; participant's stories, histories, auto-biographies in grounded theory are also cases in larger scenario. It is all for developing an in-depth and critical understanding of single case or multiple cases. According to Simon (2009) the case study is an in-depth investigation of a specific real life 'project', policy, institution, program or system' from multiple perspectives in order to catch its 'complexity and uniqueness' (Cohen et al, 2018, p 375). Creswell and Poth (2018, p 153) have taken reference from Yin (2014) that the case may be a concrete entity, such as an individual, a small group, an organization, or a partnership. At a less concrete level, it may be a community, a relationship, a decision process, or a specific project. The

case study is not exactly a methodology, but by and large it is an intended choice of understanding someone's life or an instance from diverse angles.

Though a genuine case study is bounded or trapped within the context of phenomenon, so its demand for a more, deeper and deeper level of analysis and re-thinking on the known and un-known features of case and its context. Actually the boundary line between the case and its context is very thin, need to be re-allocated without any manipulation by the researcher to get a rich and thick description of the case. But how it is possible? When there are lots of biases, vulnerable situations and subjectivity of researcher are mean to obstruct him/her from neutrally judging the situation. Yin (2009) has rightly argued regarding this situation that the boundary line between the phenomenon and its context is blurred, as a case study is a study of a case in a context and it is important to set the case within its context (and rich descriptions and details are often a feature of a case study) (Cohen et al, 2018, p 375). Basically case study has a long history of origin, mainly known for its use first time by Sigmund Freud, father of psychoanalytic theorist; then in medicines as a problem case, then law for solving criminal purposes and political sciences as case reporting and lastly, or presently spontaneously used in the fields of Educational researches as school or classroom researches. The case study research is not a simple case understanding, rather it intended to view towards it with an Russian Doll approach, i.e. a nested approach where to understand a micro-level case involves understanding and including meso- and macro- contextual levels in which it is nested (Cohen et al, 2018, p 375).

Case study research has many unique characteristic features, what make it very different from other researches, such as real presentation of participants in their real context, widen the boundaries of units of analysis, identifying the units and level of units from the context, blending both numeric and qualitative data, go with 'holism' than 'reductionism', establishes cause and effect (how and why) relationship, etc. Mostly, the research design of case studies can be changeable, semi-structured or is emergent in nature. While explaining qualitative enquiry Cypress (2017) has rightly stated that the steps of qualitative inquiry are also repeated multiple times during the process. As the design unfolds, the elements of this design are put into place, and the enquirer has minimal control and should be flexible. There is continuous reassessment and reinteraction (Cypress, 2017, p 254). So, in the summary, case study has several highlighting points as per Cohen's (2018, p 376) analysis-

- It is concerned with a rich and vivid description of events relevant to the case.
- It provides a chronological narrative of events relevant to the case.
- It blends description with analysis of events.

- It focuses on individual actors or group of actors, and seeks to understand their perceptions of events.
- It highlights specific events that are relevant to the case.

Some other defining characteristics of case study according to Creswell and Poth (2018, p 115) are-

- Case study research begins with identification of a specific case, multiple cases, community, decision process or an event that will be described and analyzed.
- The case identification is bounded; meaning can be defined or described within certain parameters. Focus on procedures for particular type of case study.
- Relying on one source is not enough for developing in-depth understanding.
- How to approach the data analysis in a case study always differs.
- A key to generating the description of case involves identifying case themes. These themes may also represent issues or special situations to study each case. A complete findings section of the case study would then involve both description of the case and themes or issues that the researcher has uncovered in studying the case.
- Case studies often end with conclusions formed by the researcher about the overall meanings delivering from the case.

Why is case study?

The epistemological rigor of any research is the base for answering ‘why’ behind any research. According to Lincoln and Guba (2000) knowledge construction is an initial step before starting of every work, it asks us a very important epistemological question or the question of knowledge as “what is the nature of knowledge and relationship between the knower and the would-be known?”(Mertens, 2005). For understanding this important constructs of epistemology of doing research we must need to know about the four major paradigms of researches used in social science researches, such as

1. **Positivist/ post-positivist paradigm:** mostly quantitative, objectivity is important, one reality, manipulable variables, knowable within probability.
2. **Constructivist paradigm:** multiple socially constructed realities, mostly qualitative, interactive link.

3. **Pragmatic paradigm:** what is useful determines what are true, mostly use mixed methods as qualitative and quantitative both.
4. **Transformative paradigm:** multiple realities shaped by social, cultural, economic, ethnic, gender and disability values. It is mainly following Critical theory, Critical race theory, Neo-Marxist theory, Feminist theory, Freirean theory, etc in analyzing the texts.

According to Denzine and Lincoln (2018, p 196) each of the paradigms are having certain basic, critical issues to be addressed, such as;

1. Axiology (ethics and values),
2. Accommodation and commensurability (can paradigms be fitted into one another?),
3. Action (what the researcher does in the world?),
4. Control (who initiates inquiry, who asks questions?),
5. Foundations of truth (foundation verses anti-and non-foundation),
6. Validity (traditional positivist models vs. post structural-constructivist criteria); and
7. Voice, reflexivity and post-modern representation (single vs. multivoiced).

As this paper is basically talking about case study research, whose nature is exploratory, explanatory, inquiry-based, critical and interpretive for understanding a person, group or social phenomenon; hence the suitable paradigm to tackle with the issues is transformative paradigm. Every paradigms of research have several values, concerns, approach of enquiry, aspects of reliability, validity, prediction and control; and diverse stance points of looking towards the research topics. So, as the transformative research what works rigorously for uncovering for uncovering and analyzing the social and cultural ethos of cases. Now the question raises that why the Transformative Paradigm emerges? Do the other paradigms are in capable of analyzing the social issues or the Transformative research paradigm has four following characteristics which are common to the diverse perspectives represented within it and serve to distinguish it from the positivist and constructivist paradigms, (Mertenes, 2005)

1. It places central importance on the lives and experiences of the diverse groups that traditionally have been marginalized i.e. women, minorities, tribals and person with disabilities. Researchers must focus on how oppressed groups' lives are constrained by the actions of oppressors, individually and collectively and on the strategies that oppressed groups use to resist, challenge and subvert. Therefore, studying oppressed people's lives also means that study of the oppressors' means of domination must be included.

2. It analyzes how and why inequalities based on gender, race or ethnicity, disability, sexual orientation, and socio-economic classes are reflected in asymmetric power relationships.
3. It examines how results of social inquiry on inequalities are linked to political and social actions.
4. It uses a transformative theory to develop the program theory and the research approach. A program theory is the set of beliefs about the way a program works on the issue that how such problems occur.

Transformative researchers are pluralistic and evolving in their methodologies. The researchers who are working within the transformative tradition tend to use qualitative methods; however they emphasize a need for more care and rigor in the existing methods. Such researchers use a wide variety of qualitative methods, such as **interviews, observations, document review and narrative analysis** within that transformative framework (Reinharz, 1992 as cited in Mertens, 2005). In transformative research what comes from the **participatory action research** tradition, it is viewed as essential to involve the people who are the research participants in the planning, conduct, analysis, interpretation and use of the research. Now the question arises that how the researcher's view of the world is? What he/she wants to explore?

If a researcher is working with person with disability then how they understand participant's world; Peck and Furman (1992) noted that researchers have used qualitative methods to identify the fundamental roles of ideology, organizational dynamics, and socio-political process in shaping policy and practice in special education. They suggest that the most fundamental problems constraining the inclusion of children with disabilities in regular classes are less technical than political in nature. Therefore, focusing on the institutional process affecting the children's behavior is less appropriate than addressing political issues (as cited in Mertenes, 2005, p 231). It of course leads the researcher in asking some critical questions about why children with disabilities are not succeeding in school.

If a person is working with feminist perspective then how understand the world? This ability to provide insight into the socio-political process has been emphasized by feminists as well, because Fine in 1992 believes that qualitative methods are the only truly ethnographic data collection methods to focus on interpretation, allow the immersion of the researcher in the social setting and facilitate inter subjective understanding between the researcher and participants (Mertenes, 2005, p 232).

By and large all the issues are a part of critical race theory presently used by qualitative researches influences social sciences and other educational researches. The inter-disciplinary nature of education

is heavily believed on the aspects of race, class, caste, gender and disability as a meaning making category that directly or indirectly informs educational researches along with the fields of psychology, sociology and anthropology. The Critical Race Theory is broadly used as ‘critical pedagogy’ in the fields of educational research from different perspectives of curriculum, textbooks, syllabus and whole schooling process, etc; from Antonio Gramsci’s, Paulo Freire’s and Henry Giroux’s view points, started from a part of Frankfurt School, known as the school of Social Theory and Critical Philosophy.

How is case study?

The ‘how’ question in case study research is systematically gather enough information about a setup, phenomenon, organization or subject as per different methodological approaches. The nature of case study research is in-depth, enquiry based, able to adopt any method and approach of the study according to the nature of cases.

Lune and Berg (2017, p 171) have rightly stated about the use and meaning of case study approaches, what reveals about two essential elements, such as- first, case studies require multiple methods and sources of data through which we create a full and deep examination of the cases. Exactly which method was used and how to combine depends on the case itself, although the need for depth and context in on setting certainly favors qualitative research over quantitative research. Second, to call certain research a case study means that there is some broader category of events (or settings, groups, subjects, etc) of which the present study is one case. Some case studies are often adopted for post-facto (after the event) studies, rather than continuing issues. We need to understand this type of study from a study of Sociologist Kai Erikson, after an event taken place on a mining community, i.e. a massive flood attack and the survivors hired the researcher for making a quick assessment of the situation. For this Mr. Erikson has spent almost 5 years and published his groundbreaking work as “Everything in its path” for larger generalization (Lune and Berg, 2017).

Now the question raises that how the researcher spent such an extended long period and what is its utility? This technique of spending time in the field is called as ‘participant’s observation’ where an ‘observer’ becomes a member of his or her subject’s community. An advantage of this method of research is the opportunity it provides to study what actually occurs within a community, and then consider that information within the political, economic, social, and religious systems of that community. It also provides a sensitivity of understanding the norms, folkways, values, mores, attitudes, customs, and practices of other societies and cultures. Narrative analysis is another process

of gathering information of case study research by inquiring the facts for the purpose of research through storytelling. The researcher then writes a narrative of the experience. Connelly and Clandinin (1990) note that, "Humans are storytelling organisms who, individually and collectively, lead storied lives. Thus, the study of narrative is the study of the ways humans experience the world." (Retrieved online 1) In other words, people's lives consist of their stories or the journey of their social and cultural living and life.

Field notes, interviews, journals, letters, autobiographies, and orally told stories are all tools of case study research. Such research techniques illustrate how life of a person can be understood, revealed and transformed in stories and by very act of telling stories. Narrative research is a sort of autobiographical genre of writing and doing research that displays multiple layers of consciousness, connecting the personal to cultural or insider to outsider world. The entire field of study is often used in disciplines such as literary theory, history, anthropology, drama, art, film, theology, philosophy, psychology, linguistics, education, and politics and also in branches of sciences. Narratives has been studied from variety of perspectives, Mishler (1986) summarized the analysis of spoken narratives by emphasizing;

1. Textual matters or the syntactic and semantic or words and sentences as devises internally connecting parts of text,
2. Ideational matters or the referential meaning of what is said and
3. Interpersonal matters or the role relationship between the teller and listener as reflected in the speech (Margarete, 1990).

The considering points in case study

According to Creswell and Poth (2018, p 159) the most considering points are-

- Determine if a case study approach is appropriate for studying the research problem: A case study is a good approach when the inquirer has clearly identifiable cases with boundaries and seeks to provide an in-depth understanding of the cases or a comparison of several cases.
- Identifying the intent of the study and select the case (or cases): for constructing case study the intent of the researcher must be very clear about the case- whether single, multisite, within-site, focused on a case, individual program or event depends on purposeful sampling (Yin, 2009).

- Develop procedure for conducting the extensive data collection drawing on multiple data sources: there are six types of data collection sources- documents, archival records, interviews, direct observation, participant's observation, and physical artifacts (Yin, 2014).
- Specify the analysis approach on which the case description integrates analysis themes and contextual information: the type of analysis of data either can be holistic analysis of the entire case or an embedded analysis of a specific aspect of the case. The analysis is always rich in the context of the case or setting in which the case presents itself (Merriam, 1988). When multiple cases are chosen, a typical format is to provide first a detailed description of each case and themes within the case, called as within-case-analysis, followed by a thematic analysis across the cases, called as cross-case-analysis.
- Report the case-study and lessons learned by using case assertions in written form- like writing, describing, analyzing and interpreting cases, all involves reflecting processes.

Types of case study researches

There are several types of case study researches according to Yin (2009) there are three types in terms of outcomes;

1. Exploratory (it can be used for generating hypotheses that are tested in large scale surveys or experiments).
2. Descriptive (providing narrative accounts).
3. Explanatory (testing theories) (Cohen et al, 2018, p 377)

The nature of data analysis in case study researches are very contextual, sometimes depends on **holistic analysis** of the entire case or an **embedded analysis** of specific aspect of the case. However, in case studies the individual's subjective experiences or within case variables are more considered for de-constructing the social realities of the cases; focuses on ideographic approach for understanding changing human behavior in its social context. Creswell and Poth (2018, p157) have suggested some specific types of case study researches as per the focus of analysis for embedded cases, such as- 1. The single instrumental case study; 2. The collective or multiple case study; 3. The intrinsic case study.

1. The single instrumental case study (It is for exploring an issue, of a single case, through different sources of information).
2. The multiple case studies (Here, different representative case are selected for illustrating the issues and find out the inaccessible conclusions).

3. The intrinsic case study (Here, the whole focus is on the case itself what presents an unusual or unique situation; it focuses more on narrative researches).

Now the question rises whether issue of finding a case through inquiry of situations or vice versa, is the only way of categorizing cases or it depends on process? However the real fact in case studies is that the process and methods depends on nature and context of cases. So, Hamilton and Cobett-Whittier (2013) have suggested the following types-

1. Reflexive case study (which includes the personal reflections of the researcher as the case/practitioner in question),
2. Longitudinal case study (to catch changes over time, the dynamics of evolving situations, a sense of the history of events and work with different or same cases),
3. Cumulative case study (case studies which provide a cumulative body of data about a topic, phenomenon or situation),
4. Collective case study (Working separately and sometimes asynchronously to gather data about a particular phenomenon or a topic, etc),
5. Collaborative case study (working with others within and across institutions, to gather multiple perspectives and contexts).

Ethical issues and challenges

Social activists and case study researchers all have recognized the power of thick qualitative description for promoting change and social justice in the fields of educational research. The social or educational researches which are equally critical and justified as per the determined criteria and for maintaining the quality and rigor; is really a great question in front of researchers. In qualitative researches the criteria for attaining the worthiness of research is somehow different from quantitative researches in terms of internal and external validity, reliability and objectivity replaced to credibility, transferability, dependability and conformability (Lincoln and Guba, 1991). Rigor and trustworthiness of research are like pillars for improving the utility of any qualitative research and justifying the inherent subjectiveness of the qualitative researches.

According to Lincoln and Guba (1985) we may understand the following four identified criteria used for maintaining the quality of qualitative research designs;

1. Credibility (the accurate and truthful depiction of participant's lived experiences, achieved through prolonged engagement and observation of the context and environment),

2. Transferability (it is done through adopting purposive sampling method and providing a thick description of robust data with a wide possible range of information through the accurate description of participants and people related with them),
 3. Dependability (it is closely related with reliability in quantitative researches, based on expert's cross validation for reviewing and validating the themes and descriptors came out from the fields),
 4. Conformability (it is met by maintaining a reflexive journal by the researcher during the research process for keeping notes and do introspection daily. Its main work is to inspect or do an audit trial whether data are collected, analyzed and interpreted properly or not).
- (Cypress, 2017, p 257-258)

In qualitative researches, while designing the study, collecting the data and analyzing it, the criteria of validity and reliability used for evaluating rigor of the study, is really a point of concern for social and educational researchers. Because it follows a positivistic and scientific outline to interpret data and never reached to the statistical means of quantification of data. On the basis of above mentioned arguments it need to follow certain research ethics-cum-challenges to do planning for qualitative researches. The United Kingdom's Economic and Social research council in (2015) has broadly given an outline of six principles of research ethics;

- Research participants should take part voluntarily, free from any coercion or undue influence, and their rights, dignity and (when possible) autonomy should be respected and appropriately protected.
- Research should be worthwhile and provide value that outweighs any risk or harm. Researchers should aim to maximize the benefit of the research and minimize potential risks of harm to participants and researchers. All potential risk and harm should be mitigated by robust precautions.
- Research staff and participants should be given appropriate information about the purpose, methods and intended uses of the research, what their participation in the research entails and what risks, benefits, if any, are involved.

- Individual research participant and group preferences regarding anonymity should be respected and participant requirements concerning the confidential nature of information and personal data should be respected.
- Research should be designed, reviewed and undertaken to ensure recognized standards of integrity are met, and quality and transparency are assured.
- The independence of research should be clear, and any conflicts of interest or partiality should be explicit. (Cohen et al, 2018, p. 117)

According to Creswell and Poth (2018, p 161) interest in case study researches is growing spontaneously in the fields of educational and social researches along with some **challenges** to be addressed by every fields of researchers ;

1. **Case-selection** (the researcher need to decide, what case he/she wanted to select single/multiple; broad/narrow or specific in scope, etc. Choosing and loosening who is a big question for the researcher, where as large number of cases have very little meaning in comparison to small number of specific cases.)
2. **Resource limitation** (it is for both time and financial issues. It makes the researcher very judgmental to spend how much time in each case dilutes the study and reduces the in-deftness on single cases.)
3. **Cross-case analysis** (it follows the principle of implicit purpose for single case study and explicit purpose for multiple case studies. It also necessary to decide units of analysis for studying such as, object, phenomenon, event or organization, etc.)
4. Deciding the “boundaries” of case (it deals with constraints of research in terms of time, event and processes. Some case studies have clear beginning and end points and some don’t have. So, researcher need to be judicial in setting the boundaries that adequately surrounded the case.)

Conclusion

Use of case studies in educational researches is simply not a mere introduction of cases but an emerging trend for addressing different social issues related to children, their childhood and child’s psychology, their stress and adjustment; and no doubt the implementation of different methods and techniques of teaching and pedagogy in school and classrooms in larger scenario. In

the fields of Educational Research, case study as a truthful and rigorous methodology can able to address several issues of Qualitative Research, like natural setting, researcher as a instrument or tool, multiple methods of collecting data, complex reasoning through bottom up or top down methods, considering participant's multiple perspectives and meanings, context dependent research, emergent design and researcher's stance points etc. like all other qualitative researches Case Study also demands and involves more than simply going out into the field and observing a given group, individual, instance or phenomenon within their culture and context. While working with case study the educational and social researchers must understand trust worthiness, credibility, transferability and confirmability of as a necessary research aspect for attaining the quality and rigor in qualitative researches. Though the Case study research designs are well framed according to all the quantitative, mixed, as well as qualitative research formats as per the nature of the study, but in this paper it is taken as a transformative research paradigm to address different social issues of caste, class, gender, third gender, disability and other disadvantaged groups of society from their respective voices for maintaining equity and social justice.

Example of ethnographic case study in Andre Beteille's book Caste, Class and Power: Changing Patterns of Stratification in a Tanjore village. Here he studied the changing patterns of stratification in South Indian village in an intensive field study and acting as a participant observer which spread over ten months. The district in particular has been known for the complexity and rigidity of its caste structure. The structure not only divided the population of village into sections of unequal ritual status but also the settlement patterns of the village continue to reflect the basic cleavages of the traditional structure. The main concern of the writer to write this book is to analyze the changing relations of caste, class and power of Sripuram and to examine their interrelations in the very context. But today there are many areas which are becoming progressively caste-free in terms of landownership, occupation and even education. Education is no longer a monopoly of a single group of castes and the educational system is far more open, both in principle and practice for non-Brahmins of Sripuram. Here we have also observed the transformative approach what the author has adopted in doing his extensive research on the people of Sripuram village irrespective of their caste and class by taking education as a tool of 'sanskritization' (Shrinivas, 1987) or upliftment for the lower castes of the village.

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6. A Study on Problems Faced by Students of Rural College During Online Classes in The Time of Covid -19

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Abstract

Covid-19 has affected the education system all over the world. India is not untouched by it. During the period of Covid-19, there was a shift from the traditional mode of teaching to online or digital mode of teaching. This transition in the education system was all of a sudden and the major stakeholders associated with the education system were not fully equipped with the methods adopted for online teaching. One of the major stakeholders among them is the student group. This group faced problems during the online delivery of the teaching and learning process. In this article, the researcher has found problems faced by undergraduate students during their online classes in this pandemic period which is still going on. An online survey was conducted through a questionnaire designed in google forms. This study was a micro-level study and respondents were the students of Govt. College Sarkaghat, which is situated in a rural area.

Key Words: Online Classes, Student Problems, Education, Covid-19

1. Introduction

The coronavirus disease (Covid-19) outbreaked from Hubei province of China in December 2019 (Thakur, P., 2020). It spread all over the world in few months and it was declared as a global pandemic by World Health Organization on 11th March 2020. It shattered the global economy and there was the closure of many organizations. Education institutions all over the world have to close down their campuses and there was a shift from offline mode to online mode of education (Bao, 2020). The educational institutions and stakeholders associated with them were not ready to accept this transition as it was all of a sudden. There was a complete transition from classroom teaching to a digital model of teaching. However, this shift was a necessity as the spread of disease was on hype and the safety of lives was a major concern. The higher educational institutions initially lacked proper facilities to deliver online education to students and it was a bit tough for the educational institution situated in rural areas. In the rural areas, both teachers and students lacked the minimum required facilities to get engaged with digital teaching and learning processes. The teachers were expecting that the situation will get normalized and they will be back to the traditional classroom-based teaching process. But in the meantime, it was realized that they have to switch to an

online mode of teaching. Students also adopted this scenario. Teachers started taking classes on digital platforms like Google Class Room, Google Meet, Zoom and other online platforms. The teachers who are teaching science subjects and practical subjects faced more challenges than the teachers who are teaching non-practical or subjective subjects. Some of them uploaded videos on students' groups and also shared them on YouTube. Few teachers also used Pen Tab to demonstrate the figures and graphs to the students. But lack of data plans and low internet connectivity issue was claimed by students as belonging to a low-income family and some living in remote areas. However, the online mode of teaching emerged as a panacea in a time of crisis. This also led to adding a flexibility aspect to the teaching-learning process. Teachers can deliver their content at any time and students can learn anywhere across the globe if they are fully equipped with the resources and minimum required facilities time **(Dhawan, S. 2020)**. The students have the advantage that online classes have saved their travel time. The students in the rural areas are facing challenges due to limited resources and a lack of digital literacy. This pandemic has posed unsurprisingly many challenges to the students who are actively involved in online learning through synchronous or asynchronous mode. The technological gap between the student and teacher has made a digital divide. All the students are not digitally literate and they do not have equal access to digital technology due to financial and social problems **(Chakraborty et al.,2020)**. The continued lockdown and pandemic has affected human mental health across the globe. Many students are facing stress and anxiety **(Cao et. al,2020)**. Some are also facing eye-related problems as they have to continuously stare at the mobile screen for hours and it is very tough to watch the content on a small screen of mobile phones. The skills gap, as well as a noisy home atmosphere, such as multiple family members sharing a common room, divert students' attention away from their classes, where lectures are delivered in synchronous mode **(Zhang et al., 2020)**. In this article only undergraduate students of Govt. College Sarkaghat, a rural area college; was taken as respondents to conduct the study. A structured questionnaire was made through google forms and an online survey was conducted as the covid pandemic is still on a surge. The study revealed the various problems and issues they have faced and still facing during their online classes.

2. Literature Review

This pandemic made many organizations rethink their work plans and strategies and adopt new methods and techniques. But it was not easy for most of the organizations to introduce and integrate new strategies with existing ones. The technological advancements in every field have made

tremendous changes. The field of education is one of them. Now the education is being provided in online mode. However, this method was not so popular in institutions situated in rural areas of India but during this tough time of the Covid-19 pandemic, it became a necessity. In the pandemic, e-learning became the only way to transmit knowledge and to reduce the spread of Covid-19 disease (**Rahman et al.,2021**). Online Learning or e-learning is blended learning that can be given via synchronous and asynchronous mode. ICT plays a significant role in it. Asynchronous mode provides students with a flexible platform in which they can access online content when it is convenient for them. The problem with online teaching is that no one size fits all approach because it is dependent on the type of technology in use at that time as well as the curriculum material being taught. Though the advantages of online education are acknowledged but during online teaching students lack face-to-face interaction, group interaction in which they learn belongingness, active participation and contribution to learning experiences. The facilitator found it difficult to build up confidence and reduce the anxieties during online classes (**Gillett-Swan, J.2017**). Students believed they learn more in physical classrooms than they do online, according to empirical studies and they would like to join physical classrooms when the crisis; even their digital competencies are better (**Bojovik et al. 2020**). The students attending online classes at their respective homes have to face a wide range of distractions at home. They couldn't find a suitable space while sharing a common room and they face noise distraction. This was also constrained by a lack of personal devices and an unstable mobile network (**Zhang et al.,2020, N.Kapasia et al., 2020**). The students have failed to make a balance between their home life with their studies (**Dhawan, S. 2020**). Students were found to low in their morale for e-learning as they were deficient in e-learning competencies and they were not fully prepared to adopt an online learning management system (**Parkes et al.,2014**). The closure of institutions has elevated the stress and anxiety level of students (**N.Kapasia et al., 2020; Sahu P. 2020**). The students reported the problem of lower internet connectivity, unstable home environment while attending online classes especially in remote areas, and students belonging to marginalized class faced the problem of recharging data plans for their devices (**N.Kapasia et al., 2020**).

3. Data and Methods

3.1 Subjects: It is an online survey-based study of 112 college students of Govt. Degree College Sarkaghat, a rural area college in Himachal Pradesh, India.

3.2 Data Collection and Procedure

An online survey was conducted in the first week of May 2021 to collect the information. A structured questionnaire link was sent to students through WhatsApp. A total of 112 students provided the complete information regarding the survey.

3.3 Data Analysis

The data analysis was carried through descriptive statistics to study the demographic variables and other variables of the study regarding problems faced by students during online classes. All the analyses were done through Statistical Packages for Social Sciences (SPSS Version: 26).

4. Results and Discussion

4.1 Demographic Profile

Table 1.

Gender of Respondents				
	Frequency	Percent	Valid Percent	Cumulative Percent
Male	38	33.9	33.9	33.9
Female	74	66.1	66.1	100.0
Total	112	100.0	100.0	

Among the students surveyed who faced problems during online classes (n= 112); female respondents constituted 66.1% of the population while the male respondents constituted 33.9 % of the population as shown in Table 1.

Table 2.

Course of Respondents				
	Frequency	Percent	Valid Percent	Cumulative Percent
B.A.	60	53.6	53.6	53.6
BSc.	3	2.7	2.7	56.3
B.Com	22	19.6	19.6	75.9
Professional Courses	27	24.1	24.1	100.0
Total	112	100.0	100.0	

The majority of students who participated in this survey belong to BA course followed by professional courses. Students from the Science course least participated in the survey and constituted only 2.7% of the whole population as clear from Table 2.

Table 3.

		Religion			
		Frequency	Percent	Valid Percent	Cumulative Percent
	Hindu	106	94.6	94.6	94.6
	Muslim	6	5.4	5.4	100.0
	Total	112	100.0	100.0	

In this survey, the respondents who responded comes from two main religion Hindu & Muslim. The majority lies with Hindus. The Hindu constituted 94.6 % of the total population surveyed as depicted in Table 3

4.2 Variables of Study

Table 4.

It was not easy to make a shift from offline classes to online classes

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	4	3.6	3.6	3.6
	Disagree	11	9.8	9.8	13.4
	Neutral	12	10.7	10.7	24.1
	Agree	61	54.5	54.5	78.6
	Strongly Agree	24	21.4	21.4	100.0
	Total	112	100.0	100.0	

Table 4 shows that majority of students agreed (54.5%), followed by strongly agree (21.4%), neutral (10.7%), disagree (9.8%) and strongly disagree (3.6%) that they find it difficult to switch from offline classes to online classes.

Table 5.

It is not easy to maintain a schedule as online classes offer a flexi-time schedule

		Frequency	Percent	Valid Percent	Cumulative Percent
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Valid	Disagree	12	10.7	10.7	10.7
	Neutral	8	7.1	7.1	17.9
	Agree	62	55.4	55.4	73.2
	Strongly Agree	30	26.8	26.8	100.0
	Total	112	100.0	100.0	

The agreed students constituted 55.4% of the population followed by strongly agreed (26.8%), disagree (10.7%) and neutral (7.1%). No student strongly disagreed. The students agreed that it was a tough task for them to make a schedule as online classes offered a flexi-time schedule.

Table 6.

Technological issues like internet connectivity and login processes are creating problems during online classes

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	2	1.8	1.8	1.8
	Disagree	8	7.1	7.1	8.9
	Neutral	6	5.4	5.4	14.3
	Agree	56	50.0	50.0	64.3
	Strongly Agree	40	35.7	35.7	100.0
	Total	112	100.0	100.0	

The 50% of students agreed that they faced technical issues during their online classes, 35.7% of students strongly agreed, 5.4% were neutral, 7.1% disagreed and 1.8% students strongly disagreed on this variable as reported in Table 6.

Table 7.

The lack of computer literacy and how to use various apps

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	4	3.6	3.6	3.6
	Disagree	21	18.8	18.8	22.3
	Neutral	24	21.4	21.4	43.8
	Agree	44	39.3	39.3	83.0
	Strongly Agree	19	17.0	17.0	100.0

Total	112	100.0	100.0
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Table 7 depicts that out of total students surveyed (n=112) 39.3 % students agreed followed by 21.4% neutral, 18.8% disagree, 17% strongly agree and 3.6% strongly disagreed that they lack computer literacy and knowledge of using various apps.

Table 8.

Distraction due to sound and other barriers as confined to a room with family members around

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	3	2.7	2.7	2.7
	Disagree	11	9.8	9.8	12.5
	Neutral	6	5.4	5.4	17.9
	Agree	55	49.1	49.1	67.0
	Strongly Agree	37	33.0	33.0	100.0
	Total	112	100.0	100.0	

The 49.1% students reported agreed, 33% strongly agreed, 9.8 % reported disagree, 5.4 neutral and 2.7 reported strongly disagree out of the total students surveyed. It was confirmed that students get distracted during their online classes due to sound and other barriers as shown in Table 8.

Table 9.

Lack of self-motivation when you find it difficult to engage with online class

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	2	1.8	1.8	1.8
	Disagree	17	15.2	15.2	17.0
	Neutral	11	9.8	9.8	26.8
	Agree	61	54.5	54.5	81.3
	Strongly Agree	21	18.8	18.8	100.0
	Total	112	100.0	100.0	

The self-motivation decreased when students found it difficult to engage with online class. It was found 54.5% of students agreed followed by 18.8% strongly agreed, 15.2% disagreed, 9.8% were found to be neutral and 1.8 % were found to be strongly disagreed as per the results in Table 9.

Table 10.

Learning adaptability vary from student to student so faced with lack of concentration, inability to understand in live classes

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	3	2.7	2.7	2.7
	Disagree	15	13.4	13.4	16.1
	Neutral	12	10.7	10.7	26.8
	Agree	55	49.1	49.1	75.9
	Strongly Agree	27	24.1	24.1	100.0
	Total	112	100.0	100.0	

49.1% of students agreed and 24.1% of students strongly agreed that they lacked concentration when was not easy to understand during online classes. The 10.7% students were neutral, 13.4% disagreed and 2.7% students were found strongly disagree from the total population surveyed as presented in Table 10.

Table 11.

Fail to communicate their learning problems with teachers/ educators to clear their doubts

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	3	2.7	2.7	2.7
	Disagree	24	21.4	21.4	24.1
	Neutral	9	8.0	8.0	32.1
	Agree	46	41.1	41.1	73.2
	Strongly Agree	30	26.8	26.8	100.0

Total	112	100.0	100.0
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The sample statistics reported that 41.1% of students were found to agree, followed by 26.8% strongly agreed, 21.4% disagreed, 8% neutral and 2.7% strongly disagreed from the whole population under the survey. It was found that they failed to communicate their learning problems with their educators during online classes as shown in Table 11.

Table 12.

Feel mental pressure as their parents are acting as a watchdog at one end (Continuously watching their performance) and teachers on the other end during online classes

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	5	4.5	4.5	4.5
	Disagree	15	13.4	13.4	17.9
	Neutral	13	11.6	11.6	29.5
	Agree	56	50.0	50.0	79.5
	Strongly Agree	23	20.5	20.5	100.0
	Total	112	100.0	100.0	

The majority of students reported that they faced mental pressure during online classes. 50% of students agreed and 20.5% of students strongly agreed that they felt mental pressure. The 11.6% students were found to be neutral while 13.4% students disagreed and 4.5% students reported strongly disagreed on this variable as presented in Table 12.

Table 13.

Emotionally disconnected; the emotional connection indeed is beneficial for learning and development

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	5	4.5	4.5	4.5
	Disagree	8	7.1	7.1	11.6
	Neutral	12	10.7	10.7	22.3
	Agree	65	58.0	58.0	80.4
	Strongly Agree	22	19.6	19.6	100.0
	Total	112	100.0	100.0	

58% of the students were found to agree and 19.6% were strongly agreed showing that the majority of students feel emotionally disconnected. The results reported that 10.7% of students were found neutral, 7.1% disagreed and 4.5% were found strongly disagreed on the variable emotional disconnection under study which is depicted in Table 13.

Table 14.

Lack of infrastructure (such as laptops and smartphones) at homes; as parents have limited resources and students have to borrow their parent's mobile

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	4	3.6	3.6	3.6
	Disagree	16	14.3	14.3	17.9
	Neutral	8	7.1	7.1	25.0
	Agree	54	48.2	48.2	73.2
	Strongly Agree	30	26.8	26.8	100.0
	Total	112	100.0	100.0	

The 48.2% students agreed and 26.8% students from the whole population strongly agreed that they lack basic infrastructure to attend the online classes while 7.1% of students were found to be neutral, 14.3% disagree and 3.6% were found to strongly disagree as description shown in Table 14.

Table 15.

**Facing health issues like headaches and eye problems as have to stare on mobile screens/
laptop screens for an extended period**

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly Disagree	2	1.8	1.8	1.8
	Disagree	16	14.3	14.3	16.1
	Neutral	7	6.3	6.3	22.3
	Agree	45	40.2	40.2	62.5
	Strongly Agree	42	37.5	37.5	100.0
	Total	112	100.0	100.0	

The majority of students faced health issue. 40.2% of students were found to be agreed followed by 37.5% strongly agree, 14.3% disagree, 6.3% neutral and 1.2% students were found strongly disagree from the total population under study as shown in Table 15. The students were found to have eye weakness and headache problems.

4.3 Discussion:

The population surveyed was undergraduate students. The results of the variables under study have shown that the majority of students faced problems during their online classes. It was a bit difficult for the students to switch from traditional offline classes to online classes. The students failed to maintain the schedule as it gave flexibility for both educators and students to have online classes. This study revealed that due to lack of infrastructure required to attend online classes and lack of digital literacy among students posed a challenge for them. The financial constraint can be attributed to this. It was found in the study students lacked self-motivation and failed to communicate their problems with their teachers during online classes. They were under mental pressure as their performance was continuously monitored by their parents and educators and they developed health issues like eye problems and headaches. Moreover, their learning process and development was affected due to emotional disconnection from their peer group.

5. Conclusions, Limitations and Further Research

Covid-19 pandemic has led to the acceptance of virtual mode of classes by the educators and students despite their readiness. The present study has assessed the learning problems faced by students during

online classes. Although the students have switched from offline classes to virtual classes they are struggling with the challenges faced by them. The higher education authorities must provide financial assistance to develop infrastructure for digital learning and to ensure learning in future during health emergencies. The state govt. must make intervention strategies to address the challenges faced by students. Multi-prolonged strategies must be developed and adopted to ensure a high level of interaction during online studies so that an online education culture can be continued. Finally, there is an urgent need to make a resilient education system to train the young minds and to acquainted with the skills needed for employability.

Some gaps in the research should be discussed in future research studies. First, the study has been focused on undergraduate students of a rural area college. The findings of this research cannot be generalized to other courses such as school learning, master courses and Ph.D. courses. Second, the researcher was only able to meet a limited number of undergraduate students, implying that the results might not be indicative of their experiences.

To overcome these limitations and to generalize the context, further research initiatives could be taken, which consider school students, postgraduate students and Ph.D. students. A comparative study between offline mode and online study mode could be made.

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7. Unlocking Learners' Potentiality Through Adaptive Learning and Testing Systems

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Abstract

In the era of globalisation, postmodernism, industrialisation, integrating adaptive learning in the curriculum holds the interest of academics, writers, and educators. Adaptive learning, particularly over the last decade, has been a focal issue for educational institutions. When Artificial Intelligence (AI) became mainstream in the 1970s, adaptive learning became popular. Its core principle was to tailor the learning experience to the abilities and shortcomings of the students. Adaptive learning, in a sense, personalises learning strategies after a differentiating phase to determine the student's individual interests, while including a variety of options. The accessibility and scope of technology, as well as online instructional modelling, have increased the benefits of adaptive testing. Improvements in educational experience and student success, as well as real-time, personalised feedback, are among the benefits of this approach for validating learning. It allows for one-on-one mentoring, which boosts student engagement and lowers dropout rates. This article critically examines different issues like differentiated, personalised, and adaptive learning; Adaptive Testing, adopting the Learning Trend, Adaptive Learning Models, Adaptive Learning and Testing Systems, Adaptive Learning in Corporate Training and its benefits. The main objective of the essay is to motivate academics, writers, and educators in a new way of exploration in the traditional teaching-learning process, to impart critical thinking of the paradigm shift in the theoretical, ideological spheres in the light of postmodern, globalised popular culture perspectives.

Key words: Adaptive learning, personalised feedback, Adaptive Testing, Adaptive Learning Models, Adaptive Testing Systems.

Introduction:

Adaptive learning is an approach that takes into account and analyses important aspects of the instructional process, such as learning style, level of comprehension, input, communication abilities, and other data that make up the profile of a student's unique needs. This knowledge aids in the creation of a customised approach that provides many opportunities for great success. The concept

was first proposed in the 1970s, but it couldn't be implemented on a large scale until better computing equipment, advanced artificial intelligence techniques, and improved online virtual environments were created. Since assessing adaptive learning requires measuring improvement, assessment has become one of the most important tools for doing so. Alfred Binet's model is the foundation for today's computerised adaptive assessment. Adaptive experiments are made up of objects or components chosen from a collection (item bank). Elements are chosen to correspond to the individual's approximate competence (or skill) level: If someone succeed in one aspect, the next one will be slightly more difficult; if he/she fail, the next one will be slightly easier. When the individual's competence or skill exceeds the specified goal or when a certain number of elements have been presented, the evaluation is completed.

There is no such thing as a one-size-fits-all solution. Adaptive learning is a technique for implementing learning that is tailored to the needs of a learner, or, in other words, adapts to the learner's trust and understanding. Take a look at a typical e-learning course, which includes course material and elements. In a typical learning scenario, students log into the Learning Management System (LMS), complete the course, and then answer the questions that accompany the concepts. They move on to the next subject if they answer the question correctly. They will be asked to retake the quiz if they are unable to do so.

In a traditional learning route, every student has the same course experience. They answer the same questions on the same screens. Adaptive learning, on the other hand, does not follow the same pattern as traditional learning. Learners in an adaptive learning course can begin with the same material or video, but the platform feeds them only the appropriate content they need to master, rather than all of it, based on their answers and trust levels when answering the questions. Adaptive learning is a technology-based framework that analyses a learner's success in real time and adapts teaching methods accordingly.

Differentiated, personalised, and adaptive learning:

Adaptive learning, in a sense, personalises learning strategies after a differentiating phase to determine the student's individual interests, while including a variety of options. This has led to a misunderstanding of the differences between adaptive and customised learning. It's a simple blunder to use them interchangeably. As a result, it's important to understand that customised learning is an 'umbrella' term that encompasses a variety of methods and models, including ability-based learning, integrated instruction, lesson models, and adaptive learning.

Personalization goes beyond the ‘one size fits all’ method on a simple level, and computer driven tutorials on a more advanced level. Personalization, on the other hand, is not dependent on adaptability in and of itself. Adaptive learning is described as a strategy for creating a customised learning environment for students through the use of a sophisticated computer system based on data. Since its self-adjustment is based on the student’s experiences and at his or her demonstrated success level, this learning has a non-linear approximation when teaching, receiving feedback, and correction. As a result, the framework adapts and anticipates the form of material and tools that students may require to progress through the course at any given time.

Differentiated, personalised, and adaptive learning have similarities and variations. Differentiated learning is a type of personalization that entails the creation of several paths for students to gain knowledge; personalised learning, on the other hand, entails the use of diagnostics to determine students’ needs in order to have a customised solution; and adaptive learning, on the other hand, necessitates the use of data analyses, psychometric tests, and algorithms, among other things, in order to reach the desired outcome.

Adopting the Learning Trend:

Adaptive learning isn’t a novel concept. Its origins can be traced back to B. F. Skinner’s teaching computer and his Theory of Programmed Learning in the 1950s, which was followed by the Artificial Intelligence revolution in the 1970s.

Adaptive learning is now applicable to classroom education, distance learning, and tutorials as a result of the advancement of Information and Communication Technologies (ICT) and as a result of computers being smaller, more efficient, and more accessible. Adaptive learning systems are also commonly used in a variety of contexts to educate and practise more effectively. Adaptive learning has been linked to large-scale data collection in recent years. It is referred to as customised learning using affective computation methods. It must be noted that learning adaptability is possible. This achievement was made in the education market, where companies like ‘Sherton Software’, ‘Carnegie Learning’, and ‘Knewton’ have been designing adaptive learning technologies for years. By using a network that every institution can buy, ‘Knewton’ has been able to build on the principle of adaptive learning. It is maintaining vast data sets and educational materials as part of its new collaboration with Pearson (one of the world’s leading publishing houses and educational companies), allowing them to introduce this technology to the masses.

Gartner (2013), estimated that the cycle had just barely risen above the abyss of disillusionment. This suggests that, on the one hand, there is a lot of room for development in the coming years, and in the other hand, we will start to see more and more educational implementations. MOOCs, Big Data, and adaptive learning are called revolutionary in higher education because of their ability to include education in a variety of ways for new students, allowing for the collection of vast amounts of data and potentially improving the educational ecosystem (Gartner, 2013).

ADAPTIVE LEARNING AND TESTING	
1905	1ST Adaptive Test ‘Binet IQ Test’ by Alfred Binet
1950	Skinner’s machine
1959	Programmed Logic for Automated Teaching Operations (PLATO). System developed at the University of Illinois
1970	➤ SCHOLAR : First intelligent tutorial system created by Jaime Carbonell. ➤ Computer Aided Instruction (CAI)
1980	➤ Computer Based Testing (CBT) ➤ LOGO Programming language
1993	Internacional Conference on Artificial Intelligence in Education (AiED). Edinburgh, United kingdom.
1998	Carnegie Learning
2000	Learning Management System (LMS)
2006	ADAPTIVE TUTORIALS Graphic Simulators
2009	MOST POPULAR PLATFORMS ➤ KNEWTON ➤ LEARN SMART
2010	SMART SPARROW
2011	MOOC

2012	EMERGING TRENDS Adaptive learning reaches its highest peak of expectations
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Adaptive learning models:

Adaptive learning solution providers often operate in a variety of empirical study fields, such as intelligent tutoring programmes, automated learning, memory and cognitive theories, and so on. Furthermore, adaptive learning systems have been classified into various groups or types and given different names. Regardless of these differences, there are two basic versions that are not mutually exclusive. Researchers from EGA (2013, pp. 5-6) agree with this viewpoint:

- The focus is on the content. This model is focused on tracking student results, interactions, and metadata created by student-content interaction. This data, along with its relationship to the learning goals, is compiled on a dashboard that the professor uses to determine what instruction, material, or path improvements or changes are required.
- It's all about the tests. This is the paradigm that people think about as they think about adaptive learning. In this approach, the method adjusts instruction, learning materials, and course paths almost in real time and automatically, based on constant testing of the learner's success and order. It is not appropriate for the professor to intervene.

Adaptive Learning and Testing Systems:

Adaptive learning applications range from the most common, such as audience response systems (which enable a speaker to change his presentation in real time based on input from the audience), to more advanced systems, such as the computer-based TOEFL, which adjust the form of questions provided to students based on their previous responses.

Adaptive tests (computer-based):

These assessments are based on Computerized Adaptive Testing (CAT), which uses sophisticated algorithms to evolve and generate the best tests for each student (Fast Test, 2013). The test is designed to include elements that are most suitable for a given person's standard. According to Fast Test, these measures have the following advantages:

- Assessments that are 50 to 90 percent shorter
- Scores that are more reliable
- Higher evaluation reliability
- Increased personal encouragement

Adaptive tutorials:

These are Intelligent Tutoring Systems (ITS), which enable students to communicate with each other through a task-objective simulation while being directed and remediated. Adaptive tutorials can provide students and teachers with a variety of reviews. To encourage contemplation and material adaptation, students receive input based on their interactions, while teachers receive input based on their own authoring choices (Marcus et al. 2011, p.626). Adaptive tutorials are intended to allow teachers to keep track of large numbers of students' total responses, change the instruction and input given by the tutorials, and respond to common problems. We can see how tutorials involve students in working around conceptual problems by evaluating student input and success in evaluation tasks (Prusty, et al. 2011, p.2). Adaptation happens on three occasions in these systems:

- Information gleaned from tasks, answers, test reports, and mistakes.
- Activity sequence is dynamically determined by the students' results.
- Reflection (from teachers): Based on the wishes and success of the students.

Cognitive (intelligent) tutors:

Artificial Intelligence is used to mimic the actions of a human teacher in this form of ITS. They do so by relying on AI science, which is still in its early stages. Intelligent tutors provide lessons before students have learned the subject; after that, they provide customised guidance and input, enabling students to learn at their own pace. Finally, they collect comprehensive information that allows teachers to track and participate in the learning process if needed. Adaptive learning and testing have the ability to enhance the learning environment by motivating and engaging pupils, personalising pathways within courses and syllabi, and allowing teachers to make better use of lecture time. They may also play an important role in promoting student retention, assessing their skills, and improving their academic performance. Our students would be well suited for career achievement as a result of it. It's important to note that Mastery Learning is a pedagogy that integrates adaptive learning and testing effectively. According to Bloom's Taxonomy (1968), each student is given the sum and kind of guidance needed to attain a higher or total degree of achievement, and all students' success is matched (Kulik, et al. 1990). It also has advantages such as ensuring that the course's end purpose is met, that students understand and master the material, that each student receives direct and tailored input, and that students with exceptional achievement are supported in reaching their full potential by new experiences (Mazarin, 2014).

Adaptive technologies, on the other hand, can be used in blended learning environments to achieve greater personalization by encouraging students to monitor their own progress. In the personal learning process, this promotes self-control and participation (Dream Box Learning, 2014). Teachers can improve the instructional style they use to engage with students one-on-one by combining adaptive learning systems with a blended model.

Adaptive learning and assessment platforms actively track and review answers, allowing teachers to more accurately determine whether students are absorbing the materials and provide assistance to those who need it (Kerns, 2013). Furthermore, it makes for a proper compromise of online content and academic events, allowing for more time to delve further into advanced topics and engage in higher-level discussions. As a result, the professor takes on a more involved facilitator or mentor role.

Adaptive learning and testing, meanwhile, will improve the learning environment for both teachers and students by providing immediate guidance on student abilities and shortcomings while concentrating on specific needs. Furthermore, from the student's viewpoint (CTU, 2013), it has the following advantages:

- More power over their education: Students may step easily towards a field of mastery and then explore the areas they need to build upon when confronted with difficulties and issues that fit their level of comprehension and success on a specific subject.
- Improved results: Students can do better because integrated learning provides extra help in the situations where they struggle the most.
- More self-assurance: Students gain more self-assurance as they proactively resolve learning disparities.

Adaptive programmes, according to Christina Yu (2014), encourage students to have a greater mastery of their academic success by taking the following four steps:

- By making mistakes. A crucial component in improving literacy. The key factor adaptive systems grade is errors, which are used to create a learning path that is tailored to the needs of the students and the curriculum they need. It is the most important aspect of the procedure. The goal is to dynamically plan and adjust the learning path based on the progress of the students at each level.
- By providing immediate input. For attempting, struggling, and then attempting again. Real-time reviews will help students feel less anxious about their schoolwork and encourage

iterative testing. As a result, rather than a test-based ranking, the learning process relies on long-term discovery and skills creation.

- By reflecting on the interests of the students. To build in the student a sense of their own talents and ideals of hard work and perseverance, provide relevant knowledge to improve student skills or competences, answer recurrent questions, provide assistance for learning ‘gaps’, and provide positive input. Furthermore, this promotes the growth of intrinsic motivation.
- By self-awareness and meditation. Students may be able to recognise trends of their own learning processes, such as their most common mistakes, effective research behaviours, and the kinds of tasks they like the most, in order to improve or amend them. Adaptive systems can produce reports that recognise those patterns, allowing them to assist students in leaving and achieving successful outcomes.

The data embedded into each learning object to determine its importance for students when understanding and mastering a subject is the value of adaptive learning and research in reality. A significant amount of knowledge provided by educational institutions can yield tangible benefits (Gartner, 2013). Data management and analysis will aid in the development of quantitative techniques that teachers will use to enhance each student’s individual learning by introducing innovative strategies for monitoring and adapting to their needs that are difficult to do through conventional approaches.

Adaptive Learning in Corporate Training and its Benefits:

The future in employee training is adaptive learning in business training. Adaptive learning offers strategies and learning paths that are tailored to the individual needs of each student. Adaptive learning also aids companies in meeting their market and training objectives by built-in efficiencies. Here are six benefits of adaptive learning in business training, along with examples of how it can improve corporate learning.

1. One-To-One Instruction:

Adaptive learning aids in focusing emphasis on a single individual. When it comes to delivering personalised reviews, current e-Learning approaches fall short. Adaptive learning technology employs algorithms to adapt to the demands of the learner based on assignments and answers. This, in particular, resembles one-on-one training, which is essential in today’s business world.

2. Greater Time Efficiency:

If learners have a firm grasp of the basics, adaptive learning will help them invest half the time they would on a traditional course. Learners can obtain the same results with the aid of a “personalised strategy”. The key thing to remember is that learners should not need to go over material that they are already familiar with. Instead, they concentrate on material that can help them develop their skills. As a result, the staff will be able to save a significant amount of time off the field for preparation. As a result, adaptive learning in business training aids in the creation of quality training.

3. Confidence-Based Approach:

Adaptive learning employs a confidence-based evaluation approach to measure the learners’ current comprehension. Learners are asked to respond to questions about material about which they appear to be knowledgeable. This technique is effective in determining what the learners are aware of, whether knowingly or unconsciously, and therefore is an indirect method of delivering personalised guidance and learning.

4. Create Individualized Learning Paths:

When students advance through the courses, adaptive learning frameworks capture data. This information is then used to help each learner personalise their expectations, content, and learning journey. The information gathered also aids in the regulation of instruction to ensure that it meets the needs of students. The capacity of the adaptive learning platform to monitor an employee’s success in his or her ‘skill’ often provides HR managers with a scale for determining when the employee is ready for a promotion. As a result, adaptive learning in business training aids graduates in completing their objectives more quickly.

5. Personalized Learning for a Heterogeneous Group:

Adaptive learning is perfect for a diverse community of students. Adaptive learning caters to all groups of students, regardless of whether they are beginners, intermediate, or mature in their knowledge of the principles.

6. Provides Focused Remediation:

Adaptive learning in business training lets learners concentrate on points of vulnerability for effective remediation, rather than spending time over-studying subjects they have learned or under-studying things they haven’t yet understood.

Criticism:

There are two major obstacles to overcome before incorporating integrated learning. The first is collecting statistically accurate data sets in order to give tailored learning guidance to pupils, taking into account the time required for teachers or specialists to plan the contents, as well as the instructional style. The second issue, on the other hand, is the growing number of stakeholders. This could cause some frustration, which could have a negative effect on students when they switch between adaptive learning platforms. It's critical to address the problem of metadata standardisation when gathering the large data sets needed to ensure a smooth transfer between existing platforms and those that could emerge. It's also important to consider how students' privacy laws and their refusal to have their communications tracked would affect the data collection. Finally, we must show that adaptive learning platforms work with any subject, not just those that are reasonably formal, such as mathematics (Gartner, 2013).

Conclusion:

Adaptive learning is a technique for implementing learning that is tailored to the needs of a learner. Adaptive learning caters to all groups of students, regardless of whether they are beginners, intermediate, or mature in their knowledge of the principles. Alfred Binet's model is the foundation for today's computerised adaptive assessment. B. F. Skinner's Theory of Programmed Learning (1950s) is now applicable to classroom education, distance learning, and tutorials as a result of the advancement of Information and Communication Technologies (ICT). Adaptive learning has been linked to large-scale data collection in recent years, referred to as customised learning using affective computation methods. It has been used in the education market, where companies like 'Sherton Software', 'Carnegie Learning', and 'Knewton' have been designing adaptive learning technologies for years. Gartner (2013), estimated that MOOCs, Big Data, and adaptive learning are called revolutionary in higher education because of their ability to include education in a variety of ways for new students, allowing for the collection of vast amounts of data and potentially improving the educational ecosystem. Adaptive learning and testing have the ability to enhance the learning environment by motivating and engaging pupils. Adaptive learning in business training offers strategies and learning paths that are tailored to the individual needs of each student. Adaptive learning technology employs algorithms to adapt to the demands of the learner based on assignments and answers. It also aids companies in meeting their market and training objectives by built-in efficiencies. It will help staff to save a significant amount of time off the field for preparation.

Through constantly customising and designing individualised learning routes, adaptive learning in organisational training addresses pressing problems that organisations and learners face. Adaptive learning is here to remain, and it will become more widely used to improve learning performance, success, and career implementation.

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8. Trade Union Attitude and Reaction towards the Vigorous Political and Economical Changes in India

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Abstract

Today it is very alarming that gradually trade union membership is decreasing in Bengal as well as in India also, mostly this happens due to the frequent movement of labour force, capital and production in era of globalization. However, unions have rejuvenated themselves by shielding their evolutionary character of conflict between labour and capital and also embracing numerous approaches to improve themselves. The strategic rejoinder embraces to form original membership, enhance collaboration, aiming on to continuous enhancing of the technical and non-technical abilities of the workers by proper training, safeguarding production along with shielding the attentiveness of consumer. In contradiction of the backdrop, this article attempts to assess the roles of the principle trade unions, operating predominantly in India and working towards the modifications on the structure of employment, especially in West Bengal and Largely in India as well.

Keywords: Trade Union, Labour Force, Capital, Productivity, Globalisation

2. INTRODUCTION

In recent time, entire world is going through the fourth Industrial relation system which has a particular target of making a new Socio-Cyber Physical System, a few steps ahead of so called digitised automated system. 'IR', belongs to the transition phase along with the world economic progression, and frequent movement of labour and capital mostly in sub-continent country like India, Sri Lanka, Bangladesh etc., this makes a prominent changes, especially in the system of production, occupation and the structure of organization. By analysing secondary data from earlier literature and others relevant data base, this article attempts to evaluate whether the trade union belongs to Bengal and largely in India are incorporating adequate essential changes in their methods of operations to withstand in the troublesome social, financial and politically aware wisdom. Payne (2001) states, the contemporary atmosphere, where worldwide capitalist transformation enforced the worker's unions to invent mutual cause along with the central and state government to stop the changes from one state to another. Due to the impact of globalization, the demand of productivity has enhanced which the

trade unions or worker's union truly find unmanageable. Blyton et al. (2001) emphasizing on aviation industry and concluded that disintegration is shaped by the decentralization and therefore the detachment which directly encompasses the shifting responsibilities of workers. The disintegration also shaped an out layer which typically seems to be substandard in various terms and conditions in connection with workers welfare. The eminence of the workers union also abridged universally with prominent declination particularly in number of membership. The review article attempts to present eminence of trade union in Indian perspective and their approach and rejoinders towards the organisational alterations along with the worker's market and worker policy in a nutshell.

3. Trade Unions in India

At present in the India Trade unionism is passing by a perilous period and encrusted with exceptional encounters due to the mainly rapid privatization and liberalization throughout the global scenario (The percentage decrement of 'Trade Unionism' of few major developed country has been presented in Fig:1.0) . Workers are pushed intentionally towards the unorganized sector from organized sector, which prominently decrease job security and also increase the numbers of casual Labour day by day. Economizing, modifications of existing managerial structure, not putting proper attention on incorporating of labour laws and the lethargic approach of the administration, are seems to be the foremost issues, responsible for this state of affairs. The deficiency may be caused due to the lack of required skill among workers and absence of proper planned training (Dhal and Srivastava, 2000). Bhattacharjee (2001) separated the era into four segments while go through the progression of industrial relations. Since the year of 1950s to 1960s were considered as a first phase, emphasised mainly with the communal sector unionism, ruled by the registered political unions which also called as 'state pluralism'. Since the year of mid 1960s to 1979, connected with immobility in manufacturing sector mainly, deteriorating employment opportunity and production which directly establish a link with exaggerated the union bustle and collective bargaining process in a production or manufacturing based organisation. The stage shown customs of remonstrance as go-slow and behave generously in case of proliferation of violence. The administration retorted with national crisis by appending the right to strike. Immediate next it comes the third stage (1980 to 1991), it was acknowledged for the limited industrialization and import liberation, monetary progression, conversion scale strategy and through send out temptations that joined by disappointment of various governments which prompted in primary changes. In the year of 1982, alteration to the "Modern Debates Act" required the businesses

utilization in excess of 100 representatives (decreased from more than 300) to take authorization from the public authority to lay off and protect the interest of labours. The fourth stage which is the time of progression (1992-2008) accomplished by depreciating the rupee, brought down tax, decreased import share prompted development of low-wage occupations, easygoing and provisional labours. Worker's Unions or trade Unions were addressed as a decreasing 'sectional vested party' while administration continuously attempted to preserve interest of buyer above the rights of an association. Since the year of 2008-2017, is considered as a fifth stage which includes the development of world-wide transnational association which blast in the assistance economical progression was defaced by this recession. Trade unions were practiced the bottom down approach as downgraded by progression of informal sector, and due to the lack of plan, in the domain of Information Technology , various manufacturing industries have seen to cultivate the beginning of novel practice which may called as an efficient unionism. During 1850s, development of various processing plant in different locations of India were represented as unhygienic conditions of working, no limit in working hours with wretched states of activities. In the year 1926, the “Trade Union Act” (1926), gave the privilege to the specialists for getting sorted out. The 2nd and 3rd phase of modern relations (examined at top) led towards the assortment of trade unionism. The politicization of the union and mixture of enlistment within a solitary section prompted the development which saw a gradual decrement. Fig: 1.1 explores the curve between the trade union density (in number only) and the membership (numerical no. only) since the year of 1952 to 2014. Prior to receiving the conversation on the system of worker's organization it is likewise basic to know the significance of the prominent Trade Unions, working in India.

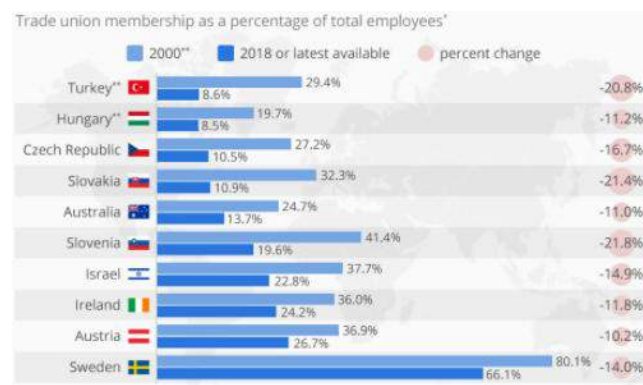


Fig:1.0 Trade Union Membership Weakened The Most

Source: OECD/Selected OECD Countries

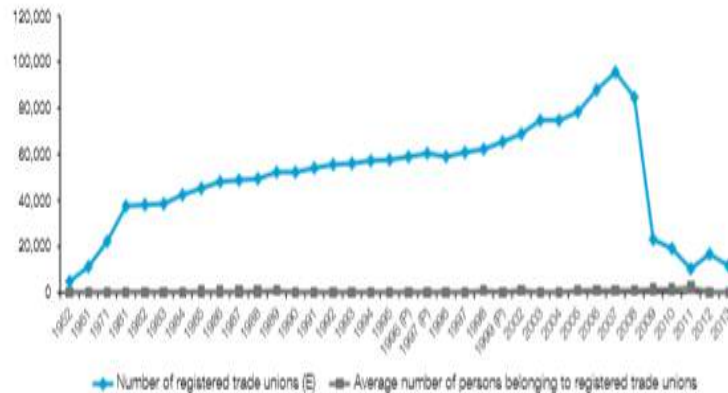


Fig:1.1: Number of Trade Unions and its Membership in India

[Source: Compiled from indiastal.com]

Table 1:

Per year data is presented of Major Central Trade Unions in India and its Membership

Name of Trade Union	Year of Establishment	Political Affiliation	Membership (in million)
All India Trade Union Congress (AITUC)	1920	Communist Party of India	14.2
Indian National Trade Union Congress (INTUC)	May 3, 1947	Indian National Congress	33.3
Hind Mazdoor Sabha (HMS)	Dec. 24, 1948	Samajwadi (Socialist)—Independent	9.1
Bharatiya Mazdoor Sangh (BMS)	July 27, 1955	Bharatiya Janata Party	17.1
All India United Trade Union Centre (AIUTUC)	April, 1958	Socialist Unity Centre of India (Communist)	4.7
Centre of Indian Trade Unions (CITU)	1970	Communist Party of India (Marxist)	5.7
Trade Union Coordination Centre (TUCC)	1970	AIFB	1.6
Self-Employed Women's Association (SEWA)	1972	Unaffiliated	1.3
All India Central Council of Trade Unions (AICCTU)	May 1989	Communist Party of India (Marxist-Leninist) Liberation	2.5
Labour Progressive Federation (LPF)		Dravida Munnetra Kazhagam (DMK)	0.6
National Front of Indian Trade Unions (NFITU)		Unaffiliated	
United Trade Union Congress (UTUC)		Revolutionary Socialist Party	0.38
Independent Labour Union (ILU)		Unaffiliated	

[Source: Compiled from various sources]

3. LEADING TRADE UNIONS IN INDIAN POLITICS

Various Trade Union organizations those are prevailing their existence into the politics of India are enrolled and recorded by yearly returns under the 'Trade Union Act' (1926). INTUC acknowledged as the supreme among seven focal trade unions operating within the periphery of Indian politics,

including participation of 33.3 million active members. The focal administration perceives that the trade union mainly dependent on their participation. According to current models, any association like to be qualified as a focal trade union that should have at least 500,000 memberships along with active presence in minimum four states and also in four enterprises as well. Table 1 addresses significant focal worker's organizations in India. The information truly exposes the strong political linkages among association and their immediate contribution in the public domain. Based on the ruling government, the membership of the different trade union also keeps fluctuating. In spite of the fact the numbers of enrolled Trade Unions are exceptionally high in India, but unexpectedly few of them are presenting yearly returns. While worker's organizations or trade unions are begun their activity from the assembling area, they couldn't discover their way into different areas, especially the areas which involved enormous number of casual labours. India has practically 92% of the workforce working in the casual and chaotic area. Aside from assembling, different areas which experience high unionization in India are transport and capacity; horticulture, ranger service and fishing; development; data and correspondence; and monetary and protection exercises. Table 2 features a point by point image of the business division-wise worker's organization and its enrolment as a member of Trade union.

4. STRATEGY OF TRADE UNION

The procedure of trade union characterized by the 'Trademark Method' of communication' in connection with its association along with its surroundings (Boxall, 2008). It is always considered that the Trade Union will utilize formal strategies for its mode of operation, though it is not true consideration in all circumstances. The associations will always utilize the scope of redesign in managerial level along with political progression to adopt and cultivate various techniques like standing advisory groups, yearly meetings, casual arrangements among power-representatives, the board orders, direct activity, etc. Like other association it also like to display its strong points and shortcomings of present hierarchy of the concern system and also fix a route which helps to execute the way of eliminating any perceived risk as far as possible along with the possible welfare of the workers as per law of the Indian Constitution—these characteristic approach of trade unionism will perform to generate true sense.

Table 2

Represent Industry wise numbers and membership of Trade Union in India, 2018

Industry	No. of Unions Submitting Returns	Membership		
		Men	Women	Total
Agriculture	273	1875018	1023493	2895511
Mining	51	87917	4185	92102
Manufacturing	1774	1151433	477587	1629020
Electricity, Gas	113	179531	9180	188771
Water Supply, Sewerage	55	21190	4059	25249
Construction	281	825515	399502	1225317
Whole sale and Retail Trade	197	188567	45012	233579
Transportation	652	985284	183441	1168725
Food Service	75	14421	3473	17894
Information and Communication	217	109522	4632	114454
Insurance Activities	216	134891	59074	193965
Technical Activities	138	82522	71867	154389
Administrative and Support Service	40	44516	10456	54972
Public Administration	181	87456	37040	124496
Education	58	7156	66226	73382

Human Health	52	58504	28825	87389
Recreation	54	25528	1963	27491
Other Service	270	218565	80774	299639
Activities of Household	12	579	852	1431
Activities of extra territorial bodies	46	371146	200582	571728
Total	4785	6470161	2712223	9182384

It can also be included that the diversified challenges are very frequently added to the system of our daily life due to the impact of globalisation and the operation of trade unionism is not completely out of this essence of continuous change. This ever changing situation can also be called as 'essential float' which may subvert and critically bargain with an association's for its financial suitability and its social authenticity (Boxall, 2008). Association of the panel is getting more prominent consideration in UK work practice and in the scholastic writing as developing quantities of associations and superiors can go into formal arrangements to construct long term connections dependent on participation as opposed to adversarialism (Haynes and Allen, 2000). Irreconcilable situation are settled through close to home relationship which depends on trust, common responsibility and specific sorts of data sharing which are secret to one another. 'Regular interest board of trustees' was additionally framed to advance association for the executive's collaboration. While considering the worker's organization methodology in Spain Köhler and Jiménez (2015) encountered that association renewal system basically requires an adjustment in the direction of aggregate activity. Worker's organizations accept a job with the intend of social change and social development as well on the interests of the average workers and also reached out to resolve different issues, related to citizenship. This is a way to deal with the existing difficulties by widening and enhancing the job opportunities for the development of work. An audit of Poland's association uncovers that post-pilgrim state also placed the clean association uninvolved of pre-increase and post-promotional discussion.

Classic Unionism - Workers Relation: Servicing plus solid organization - Employers Relation: No incorporation	Partnership Unionism - Workers Relation: Servicing plus solid organization - Employers Relation: Extensive cooperative practices
Paper Tiger Unionism - Workers Relation: Servicing only - Employers Relation: Formalistic	Consultancy Unionism - Workers Relation: Mostly Servicing - Employers Relation

Source: Boxal and Haynes 1997

Fig: 1.3 Four Pattern of Trade Union Strategic Choices

Trade Union chiefs have taken up the activity to rethink the public interest to acquire support at the EU level (Lis, 2014). Associations in neo-liberal setting can be depicted as far as the nexus between two complex measurements: unionworker relations and association business relations. The basic decisions in specialist relations are related with unique excellence between the overhauling and coordinating model. In the adjusting model, labours are viewed as purchasers of association administrations like backing in aggregate questions and individual complaints, legitimate guidance and a scope of non-modern advantages. Boxall and Haynes (1997) while contemplating the essential decisions among Trade Unions in New Zealand distinguished four wide examples of association vital decisions: exemplary, paper tiger, consultancy and organization unionism. Fig 1.3 clarifies the subtleties of these four vital decisions by matrix form. Payne (2001) found that worker's organizations are utilizing 'Deep rooted learning' as a system to keep them side by side with the changing positioning into the market. The association chiefs think that it's difficult to convince the individuals from the benefit of mastering new abilities. The worker's organizations or 'trade unions' were discovered to be the drivers of social change and playing a part of strengthening not seen as the detached survivor of worldwide powers. Associations are cooperating with preparing and enlightening organizations to grow new learning opportunities for individuals.

5. STRATEGIC CHOICE OF TRADE UNION IN INDIA

Worker's organizations in India have not changed their procedure and way to deal with the work relations and manager relations. Notwithstanding the monstrous bit of the workforce (around 92%) actually utilized in the chaotic areas, the associations have not put sufficient undertaking to sort out the easy going and provisional work. There is additionally gigantic contrast among the states in India as far as work market adaptability (Mahmood, 2016). Unionism in India as well as in Bengal is decreasing consistently. The quantities of strikes were recorded high during 1961-74 at all India level. Notwithstanding, the post-progression time has likewise encountered various cross country strikes, showing also, battles challenging neo-liberal strategies of the public authority.

Table 3: Strategy Adopted by Indian Trade Unions

Name of the Trade Union	Operation Strategy
All India Central Council of Trade Union(AICCTU)	Revolution, Conflict
All India Trade Union Congress(AITUC)	Communism
All India Trade Union Centre(AIUTUC)	Communism
BharatyaMazdoorSangha(BMS)	Nationalism, Productivity, Consumer Interest
Central Indian Trade Unions (CITU)	Communism, Protest, Strike
Hind MazdoorSabha(HMS)	Industrial Democracy, Workers Education
Indian National Trade Union Congress(INTUC)	Industry under national ownership and control

The vast majority of them were in open area units and were fruitful. The mild status of trade union is credited to the presentation of new work practices, for example, enrolment freeze, re-appropriating, expanding utilization of contractors, opportunity to recruit and terminate, freedom to close modern endeavours, delicate work assessment framework, tolerance to present work saving advances, annulment of lawful arrangements with respect to reward, deliberate retirement plans—VRS, and

privatization of non-reasonable public undertakings (Patel, 2016). Ventures began to progressively use contractors both in enduring and non-perpetual work. They use provisional work for adaptability as well as for lessening creation cost (Kumar, 2015). Worker's organization Act, 1926 gives the arrangement to association for enlistment just while the law is quiet about the interaction of association acknowledgment and aggregate haggling. Because there is no reasonable principle with respect to association acknowledgment in India as well as in Bengal also, the aggregate dealing establishments are moderately frail (Balasubramanian, 2015). Today trade union is likewise showing another pattern of relationship alongside the worldwide communication. Association for data innovation and empowered administrations (Joins together) is one of the soonest worker's organizations in India in the IT's area especially to battle against the short-term cutback by the businesses (Bisht, 2010). By investigating the accounts accessibility in the site and fundamentally assessing the method of their activity the creator attempted to set up a concentrate method of activity for the significant focal associations of India (See Table 3). Utilizing the focal points of Boxall and Haynes (1997) the above associations can likewise be ordered into bunches dependent on their essential reaction. Worker's organizations, for example, AICCTU, AITUC, AIUTUC or TUCC are generally following the paper tiger unionism as their method of activity is established in the adversarialism, dissent, strike and progressive changes. In spite of the fact that 'CITU' has a comparative methodology, it tends to be named to be a displaying exemplary unionism as an advantages that might be given to them for their getting sorted out exercises. 'INTUC', probably the most seasoned association in India, has not performed much in reviving itself however they attempt to adjust among collaboration and conflict. 'INTUC' follows a consultancy unionism. 'BMS' and 'HMS' can be sorted to receive association unionism as these associations has moved past the contest system and spotlight on market pertinent factors like mechanical majority rule government, efficiency, purchaser premium, labours' schooling and helpful development. 'HMS' is the association which follows a communist mentality and not subsidiary to any ideological group. The other significant worker's organization which has developed as a social development is independently employed Ladies' Affiliation (SEWA). SEWA has an enrolment of 1.3 million. This development is to engage the independently employed women's and sort out themselves for full business. This development has totally altered the course of unionism and can be an exercise for different associations which are battling for enrolment. 'SEWA' has a methodology of advancement with two objectives—full business, and Independence.

6. STRUCTURAL REFORMS AND RESPONSE OF TRADE UNION

According to the report of McKinsey (Woetzel et al., 2017), the development of India's economy is perceived at a rate of 6.6% each year somewhere in the range of 2011 and 2017. In spite of numerous anxieties and strains, like a rising monetary shortage, high shopper swelling, the breakdown especially in various quarrying area (e.g: Asansol, Raniganj etc. in Bengal), have created a deadlock in connection with framing the projects, this is also perceived even after significant stagnancy in macroeconomic sector from 2011 to 2013, when Gross domestic product (GDP) has shown development of 5.5% each year. Starting from 2013 to 2017, development recuperated to 6.8% in every financial year, transforming India into one of the fastest developing country on the planet. In view of the yearly studies, the all-out number of occupations in India from 2011 to 2015 developed by around 7,000,000, from 456 million to 463 million. Notwithstanding, the clear languor in work creation masks provide huge underlying change: horticultural business fell by 25 million and non-ranch business rose by 32 million, or by in excess of 8,000,000 positions every consecutive year. As per the true fact is concern, the speed of non-ranch work creation plunged during the financial stoppage stretches a long from 2011 to 2013 to as low as 11 million, and rose pointedly to 23 million during the accompanying two years. Work moved out of agribusiness into development, exchange and accommodation, and transport, the backbones of the non-ranch work market in many non-industrial nations; these three areas created a day and a half positions from 2011 to 2015. On the other hand, mining and assembling lost positions during the stoppage, despite the fact that blue collar positions appear to have developed somewhere in the range between the years of 2013 and 2015. The significant initiative has taken for carrying underlying changes towards the work transformation in public authority in India also proposed effectively the bill of 'Mechanical Relations Code' in the year of 2015. The critical changes recommended in this bill incorporate permitting organizations utilizing up to 300 specialists (expanded from 100) to lay off labours with no administration authorization. All the focal worker's organizations collectively restricting to the proposition as 85% of the organizations will be tossed out of the ambit of the demonstration (as of now covered under Modern Questions Act, 1947) empowering managers to recruit and terminate representatives voluntarily (Monetary Occasions, Oct 15, 2015). The way has proposed to perceive the worker's guild and also provide a thrust towards the existing situation give with 'arranging specialist solely' if the association gets 51% or more votes. Due to the pressing factor from the worker's organization, the bill was additionally

overhauled to incorporate every one of the associations tying down over 15% votes to have their agent on the arranging load up without two third dominant part with any worker's guild at the hour of casting a ballot.

7. CONCLUSION

Though the international trade union movement prominently has struggled to spread its essence in various geographic locations throughout the world, the leaders of the Indian trade unions still not believe in this revolution. This arrogance has forced management to focus more on the reconstruction process, to automate the manufacturing process, technological advancement, communication and adoption of union prevention strategy. This article reveals that the trade union's new approach in connection with industrial relations (IR) and in-house structural modifications could support administration to oppose labour layoffs and mass labour import strategy from cross boarder sources or from different states as well. In a nutshell of the entire discussion it can be summed up as trade union should focus on providing a living wage, enhancing efficiency of the existing workers and protecting worker's from exploitation by the rulers. They should fix their intention to fetch the larger informal sector underneath the jurisdiction of its own by combining membership along with the social cohesion.

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9. Revisiting Colonial Bengal through the Lens of Motherhood: A Vidyasagarian Perspective

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Abstract

So far as the Education Scenario of West Bengal is concerned, Pandit Iswar Chandra Vidyasagar held a significant position and as a member of the then enlightened community of the Bengal Renaissance, his ideas and enthusiasm overwhelmed his countrymen in many ways. As an administrator, he had the responsibility to design the education sector of Bengal from its core and the periphery of Women Education got a significant focus during his time. Though the conceptual pattern had been borrowed from foreign ideology, it has its root in the soil of Indian Education system also. This paper wishes to focus on the domain of Women Education and the empowerment of women as envisaged by Vidyasagar with a just appreciation of the historical scenario that has ultimately designed his stance as a deliberate strategy to attune society with the conceptual pattern of the idea of Motherhood.

Keywords: Education System, Motherhood, Ideology, Bengal Renaissance, West Bengal.

Introduction:

Some people are to be found within the periphery of society who make paths of their own to walk on. Confrontation with deception and misbehaviour from their fellow people does not bow their spirits down; rather it strengthens their self-confidence and self-sustainability. And, in course of time, their dexterity finds a way out like a great volcanic eruption. Their works levels themselves up beyond every single deceptive stance, faced previously; and the aura of their talents casts a special spotlight on them on the stage of the world. Pandit Iswar Chandra Vidyasagar is the name of such an indomitable spirit whose relentless efforts are still worthwhile to discuss in a topsy-turvy society of recent years.

V There are great names like Rousseau (1712-1778), John Dewey (1859-1952), Maria Montessori (1870-1952), Paulo Freire (1921-1997) or even Rabindranath Tagore (1861-1941) and there are several other names to increase the length of the list. Those Thinkers belong to different paths of Educational philosophies like, Idealism, Pragmatism, and Naturalism and so on. But there is no way to term Vidyasagar as an Educational thinker in the truest sense of the term. This is because; he did

not advocate, propagate or subscribe to any particular Educational philosophy. Rather, his contributions can definitely be measured by the avenue he had paved through his intellectual gifts, charismatic persona and indomitable spirit for the countrymen for the years ever after.

Colonial Bengal, Vidyasagar & Idea of Mother-hood

It can definitely be said that Pandit Iswar Chandra Vidyasagar was a philanthropist. There are a lot of phrases, coined by different eloquent writers, like, 'Traditional Moderniser' that may be applied to conceptualise the persona of Vidyasagar. But it is of no doubt that the stance that he had taken throughout his life, was guided by a strong passion that came from the inner self which tested the sufferings of the society through its close encounters. His practical outlook helped him to formulate questions against the set rules of society and it was obviously driven by that passion which may better be termed as humanistic outlook. Obviously one may term him as secular, pointing to the nature of services that he did for the sake of the society.

After one twenty nine years of Vidyasagar's demise, there are a lot of discussions on his works and most of them tried to revisit the personality through different shades of character that came out as a necessary part. The periphery of such discussion indeed encompasses different philosophies ranging from Idealism to Pragmatist outlook keeping in view the condition of the Colonial Bengal in mind. In this line of thought, this paper also tries to figure out the idea of gender justice as deciphered from the practical stances taken by Vidyasagar. But this would be done under the aegis of an ideology that poses women as 'Mother'. The burgeoning spirit of Nationalism that could be found in Colonial Bengal at that time caught hold of the image of Mother to represent the aspirations that had a flavour of nationalism. This idea, obviously, had a root in the cultural practices of rural Bengal.

The glorification of motherhood has had an ideological flavour that ultimately ensures deprivation of women in practice. The power that is associated with motherhood, is in actuality, a construct created by the patriarchal society. It is also supported by Engels according to whom capitalist patriarchy overthrew the actual right of mother. It is also supported by other scholars like Bagchi (1990) who said that the loving and nurturing mother prized with a boy is a projection created by the patriarchal society. Simon de Beauvoir opines that men with his patriarchal stance, becomes subject and paves the way to be exposed to the world outside; while women, in the same stance, gets the position of object and is doomed to inwardness. Although, scientific inventions create opportunities for women to get control over her own body and self, still women have to wait till the twentieth century for such blessings.

The idea of a mother, in Colonial Bengal, was often associated with divine entities that had exalted features. Scholars like Bagchi (1990) pointed it out as a stance taken by the so called patriarchy to reclaim power over women so far as the reproduction system was concerned. This image of mother goddess, according to Dasgupta, can definitely be found in earlier societies, SukumariBhattacharji also supported this idea and opined that Aryans considered Aditi as the mother of Sun-god, and worshipped her as well. But in later eras patriarchal notion grasped the power orientation process and used this concept of Mother to claim their superiority over women in general. But the process, in Colonial Bengal, got another additional component—a sense of alienness was associated with the English people. The spread of education system which was European in nature, created an urge amongst the Bengali people specially the educated ones, that could separate them from the alien rulers coming from England. Thus a sign was needed and in search of such a sign, the concept of ‘Bengali Mother’ was created. The all-enduring nature of mother with a specific role in the reproduction system and also with a caring and self-sacrificing entity came to the surface.

Vidyasagar’s Role in Women Education and Social Reformation

In the year 1829, Rev. W. Adam addressed to Lord William Bentinck and focussed on the necessity of ‘an educational survey’. The Governor General who was deeply interested in the education of the lower strata of Indians, appointed Rev. Adam as a special commissioner for the survey of the state of education in Bengal. In his report, Rev. Adam opined that the condition of female education in Bengal was in a most deplorable state. Girls did not get any type of educational instructions as compared to boys. This state was also related to a superstition that was associated with being widow. These beliefs were mainly cherished by the women in the families with a proper backing by the patriarchal society. But Rev. Adam also found that these beliefs prevailed extensively not universally and some people from the higher societies were free from such feelings.

It is found in a report from 1820, that there were concentrated efforts made by different persons regarding the establishment of female schools and women education. In this regard one may mention the report provided by Raja Radhakanta Deb who expressed his satisfaction on examining the reading abilities of native girls. It is noteworthy that this examination system was arranged by Female society. This society later established schools for female students in different parts of Calcutta and Raja Radhakanta Deb supported them by providing ‘StriSikshaBidhayak’, a pamphlet written by himself, whose object was to propagate the necessity of female education as beneficial to society. But after 1824, without proper and timely supports such efforts went in vain and the schools were closed.

Education for women got proper support with John Elliot Drinkwater Bethune who was the founder of Calcutta Female School. Bethune, at that time, was appointed as a member of the Supreme Council of India and subsequently became the President of the Council of Education. Pandit Iswar Chandra Vidyasagar was the Assistant Secretary of the Sanskrit College by that time. In Bethune, he found a true companion in his venture for establishing a framework for women education system. Vidyasagar was also taken the responsibility of a secretary of Bethune's school and had taken every necessary steps to institutionalise and popularise female education.

In 1855, Vidyasagar was appointed Special Inspector of Schools for the Districts of Hooghly, Midnapur, Burdwan and Nadia. From different letters and reports, it can be found that Vidyasagar had founded thirty-five schools for female students within less than six months' time (1857-1858). A tabular form may help to ponder over this matter.

District	Areas	No. of Schools
Hooghly	Potba, Daspur, Boichi, Digsui, Talandu, Hatina, Hoyra, Napara, Udayrajpur, Ramjibanpur, Akabpur, Siyakhala, Mahesh, Birsingha, Goyalsara, Dandipur, Deypur, Raujapur, Malaypur, Bisnudasapur.	20
Bardwan	Ranapara, Jamui, Srikrishnapur, Rajarampur, Jot-Serampur, Daihat, Kashipur, Sanui, Rosulpur, Bontir, Belgachi.	11
Midnapur	Vangabandha, Badanganj, Santipur	3
Nadia	-	1

Vidyasagar had a hope that government would take the financial responsibilities of these institutions but it was not granted initially. The mutiny was already taken place and the 'department was literally dying every inch out of sheer starvation. Grants to aided schools were in some cases withdrawn'. It is also to be noted that by 1862, Vidyasagar had already established forty three female schools and among them only nine got supports from people like Countess Canning, Raja Pertaub Chunder Sing and a few like-minded persons. But the relentless spirit did not give up and continued his pursuit to get financial support for British government. And within 1863 the conditions were come to a satisfactory state.

He constantly had taken into account the financial conditions of the established institutions and suggested revisions as per requirements. As an ardent educational administrator, he had shown his interest in establishing female Normal schools and later with the help of Miss Carpenter, the normal school was established as a part of Bethune school. Although Vidyasagar pointed out different problems that could pose barriers regarding female education system, the government was more interested in listening to Miss Carpenter's report. But within three years the normal school faced a closure.

This is an account of Vidyasagar's active and relentless efforts regarding the popularization of women education in Bengal. In fact, in most of the time, he was driven by a strong passion for doing something right to the neglected section of Bengali families. In this connection one may refer to his efforts in the reformation of society through legalizing widow remarriage. It is imperative to mention that if the spread of female education system was a part of Vidyasagar's societal reformation, supporting Hindu widow's remarriage was his another step on the same avenue. The very first instance of it is to be found in an essay, written by him, on 'SarvaSuvakariPatrika' on 1850. But this idea was not a new one. According to Dr. Rameshchandra Majumder, Maharaja Rajballabh had already secured the approval of the orthodox ones in favour of widow remarriage as early as 1756. Later other reformers also supported this and among them Raja Rammohan Roy, Motilal Shil, Ramjoy Sharma, Ramchandra Vidyabagish, the Dirozians are worthy to be mentioned. In fact, different newspapers had already started discussions on this matter and The Bengal Spectator, on 1842, noted a sloka from Parasara Samhita, the same sloka based on which Vidyasagar tried to establish his views on Hindu widow remarriage (Sastri,).

In fact, Vidyasagar had to fight against all societal odds to establish his ideas. He wrote the first pamphlet on 'Should Widow Remarriage Be Instituted' on January, 1855. The second pamphlet came out in October and by that time there was a wave of writings and pamphlets against widow remarriage. Even Raja Radhakanta Deb opposed him by sending petitions to government with approx. 37000 signatures. But Vidyasagar, through his authentic and well researched notions established his arguments and succeeded in convincing the British government. On 26th July, 1856 the Hindu Widow Remarriage Act was passed. But even after this, Vidyasagar had to write several books and articles in order to establish the norms in the society in general.

Vidyasagar expressed his thoughts against child marriage on the same writing that was published on 'SarvaSuvakariPatrika' on 1850. The nature of his ideas were too much modern at that time. He

pointed out the necessity of biological developmental process within female body that takes place during adolescent period. This development is important and also a preparatory step for female body before bearing child. He unmistakably opined the necessity of love as a psychological basis for a suitable match. When he was asked to present his opinion, Vidyasagar said that the child bride should be protected only up to twelve years of age and after that the 'adult husband would be legally and socially free to indulge in the sexual exploration of the child bride.' He also proposed 'that the age of consent be tied to the girl's first going into menses' (De,2020). As a result, the Age of Consent Act was passed by the British government on 1891, that raised the age of consent for sexual relationship from ten to twelve years in all jurisdiction and the violation of which would be termed as rape.

At the same he started his venture against the practice of polygamy. On 27th December, 1855 he submitted a petition signed by approximately 25000 persons, to the British government 'praying for a law against polygamy'. In the next year he submitted another petition supported and signed by 21000 persons. He wrote several letters in support of his petitions highlighting the necessity of such act. But the Bengal government did not want to interfere with the social system of the Hindus and the situation became rigid after the mutiny of 1857. Still, Vidyasagar continued his efforts to form public opinions against the practice of polygamy throughout his life.

A Review of Vidyasagar's Perspective

The question that is evident here focusses on the works done by Vidyasagar to change the deplorable conditions of Bengali women and the mindset of the traditional Hindu society of Bengal. Through his relentless efforts, it is of no doubt, he was partially succeed. But it is also evident that there were other persons who had already tried the same course of action in past years. Vidyasagar's success lies in the facts that he, in most of the cases, succeed to convince the British rulers for providing necessary legal supports and he was able to bring his ideas into practice within the framework of the society.

But his actions were not completely free from the patriarchal notions prevalent in those days. It is true that he had tried his best to popularize women education system, but no future direction was envisioned by him except better maintenance of families and moral development of women. The syllabus of Bethune's school was restricted within basic versions of reading and writing, basic arithmetic, basic knowledge of materials, basic geography, and embroidery; learning English language was not a mandatory practice. This syllabus speaks of the prevalent social conditions and customs that restricted women for obtaining higher levels of knowledge similar to men. Although one

may argue that this was only a first step towards development, but it happened, obviously, because of the prevalent mindset and even Vidyasagar was not completely free from it. Again the glorification of motherhood got a firm stand thorough his efforts and probably it was a just stance as the reformation could not be brought without the change in the mindset of the majority of people, which is a rather slow and time consuming process.

In case of his efforts for the abolition of child marriage, it is noteworthy that Vidyasagar was in favour of raising the age of consent for girls' in case of marriage from eight to twelve years. Although he had scientifically supported his ideas but even in twelve years of age the risk of a girl's life while bearing child always remained. Moreover, he supported the necessity of love as a psychological basis for a suitable match but he overlooked the issue of age-differences between brides and grooms. He was not able to be free from the societal practices even in this case.

In fact all his reformatory efforts cannot deny their relationship with the cult of motherhood that was seen as the 'ultimate identity' of Bengali women prevalent in that period. It may be termed as an overt ideological control of the male over the female counterpart of the society. 'It was an excellent ploy to keep women out of privileges like education and profession that were being wrested by their men and glorifying women-hood only through her reproductive powers.' (Bagchi, 1990). But the social reform era shifted its focus from the ideal to the living women and 'tried to bring colonial machinery upon their lives'. It was indeed a first step towards modernism of the Bengali society and thus Vidyasagar had to seek evidences from the grand narratives of Hinduism. The changing nature of the society was also of another important concern. Christianity had already taken a firm footing on the soil of Bengal and the traditional Hinduism was confused to embrace the change within the society. Vidyasagar realised this and through his movements he tried to reform the so called Hindu society that would be free from narrow prejudices. The cult of mother had also helped him in this regard.

This image of mother was later associated with nationalistic thoughts that helped to bridge the chasm between political and social lives of the colonial Hindu Bengali society. It is imperative to mention that Bankimchandra Chattopadhyay made the first attempt to establish the concept of Bharatmata in his celebrated song VandeMataram. This song found its context within the framework of his novel Anandamath.

Lastly, it is imperative to mention that the avant-garde poet of Bengal renaissance, Michael Madhusudan Dutt described Vidyasagar as a great social reformer who had 'thecourage of a lion, the

energy of an Englishman and the heart of a Bengali mother'. And it is very true, in every respect, to refer to that great spirit as an all enduring Bengali Mother who had experienced tremendous oppositions from the Hindu Society itself, specially the Bengali middleclass people. In fact, what Vidyasagar did, were mainly for the Hindu people and the Muslim counterpart of the society was somewhat overlooked at that time. Still the importance of his works cannot be undermined as these were the first great leap towards the modernisation of Bengali society in the Colonial period.

Conclusions:

The stances taken by Vidyasagar were no doubt important as these endeavors helped to pave the way towards the modernization of the then traditional society of Bengal. It is also true that his ideas were definitely shaped by the prevalent patriarchal notions but his actions, sometimes, went beyond such notions to elevate the status of women in society in particular and that was a real revolutionary step taken at that time. This step had been backed by the notion of a Bengali Mother and thus he also advocated moral upliftment of women through the actions taken by him and that, again, ensures his subscription to the patriarchal stances. The idea of moral upliftment is indeed questionable as the associated parameters vary with the demands of the society and time. But the 'Traditional Modernizer' though, constantly dealt with the ebb waves of the oppositions, paved the path for the future generations by securing a position from where women could step forward with the years to come.

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10. Digital Literacy: Need to Incorporate and Promote among Stakeholders of Education

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Abstract

Today, virtual reality has become the new reality for the educational sectors due to the pandemic crisis. Technology has connected people in a new way at the global level. People have to learn and develop different digital skills and competencies to use new technologies to interact with others. i.e. 'connecting knowledge connecting people'. Digital skills that are required to navigate the digital world have been combined under the term 'digital literacy'. The crises have realised the importance of digital literacy. The research article sheds light on what exactly is digital literacy and their importance in the current scenario. It also focuses on the existing limitations to incorporate and promote digital literacy among the individuals at the mass-level in the country. It also explored different digital competencies that are required to be more competent for the 21st century.

Keywords: Digital literacy, Digital Competency

Introduction

The Covid-19 pandemic has created unprecedented challenges in the educational system. Educational activities- classes and exams have transitioned to remote learning or e-learning. The rapid shift to remote learning has adversely affected the learner. Around 463 million students have suffered due to cut off from education (UNICEF, 2020). In countries like India, 32 million students were already out of school before pandemic due to certain reasons (NSSO, 2014). The pandemic added 320 million students more which have adversely affected due to school closures (Modi&Postaria, 2020). To alleviate the impact of sudden school closures, countries worldwide have invested rapidly in remote learning (Mascheroni et al., 2021). As a result, new educational platforms—Virtual Tutoring, Video conferencing, Google Meet, Zoom App, Google Classroom, and Skype have emerged and become new norms for the learners. To be competent on these digital platforms, there is a need for certain digital skills as well as complex skills such as cognitive, motoric, sociological and emotional skills among individuals in order to use digital environments effectively (Eshet-Alkalai, 2012). There is a need to update and develop education models for mass movement to incorporate and promote digital literacy in the country that suit the new era's demands in view of the fast-changing technology (Naidu, 2020). To amidst this pandemic transition, the Ministry of Education (MoE), India has

launched New Education Policy-2020 on 29th July, 2020 which proposes to promote digital learning and to enhance required infrastructure (NEP, 2020). However, due to regional diversity and social economic, there exist multiple bottlenecks like–digital divide and literacy gap in the country to promote widespread digital learning among the people (NEP, 2020).

Digital Divides: Missed Opportunities

Digital access has become a dividing line for the millions of children who could not benefit from the digital space. Around 346 million (29%) of youth worldwide are not online (Ayivi, 2017). Lack of access to the digital space puts the world's youth deprived of new opportunities for the 21st century workspace. Digital divide can be seen between men and women, urban and rural areas, poor and rich, educated and uneducated and between developed and developing countries (ITU, 2016a). Children who are unconnected to the digital world missing the rich global resources and information as well as unable to explore and express with new friends (Stoilovaetal., 2016). There exists huge digital deprivation in India even before the challenges bought by pandemic. The major challenges to promote remote learning are issues of equitable accessibility of digital devices along with adequate penetration of internet and technological services. As per the report of NSSO (2017-18), the availability of computers and internet in the households are shown in the table-1.

Table 1: Percentage of households with computer and internet facility

Availability of computer and internet	Rural Households	Urban Households
Households having computer	4.4%	23.4%
Households having computer with Internet	14.9%	42%

According to the data presented in table-1, there exists a huge difference between rural and urban households in the context of having computers and internet facilities. Along with this, 70 per cent of the rural population did not have an active internet facility and some of the states (Bihar, Jharkhand, Madhya Pradesh, West Bengal, and Telangana) suffered with low internet penetration (Nelson, 2019). Today Smartphones are the primary access to attend online classes. Around 500 million smartphone users were reported till 2019 which is expected to swell to 820 million by 2020 in India. But still 800 million people did not have access to smartphones (India Cellular and Electronics Association, 2020). Apart from that, lack of access to smartphones and due to mounting pressure to attend online classes, some of the worst consequences of students' suicide had been reported (Koshy, 2020&Basumatary, 2020).

Literacy Gap

Now-a-days the problem of inadequate literacy around the world remains a stunningly large and pressing issue. In the age of information, many policy makers have been raising the issue of individual standards for knowledge of Information and Communications Technologies (ICTs)—what is often called ‘technological literacy’ (Wagner, 2020). As per the NSSO (2017-18) there exist significant gap between male and female with regards to competence to use and operate computers and internet in rural and urban areas.

Table 2: Percentage of persons able to operate computer and use internet

Ability of digital device/internet	Rural		Urban	
	Male	Female	Male	Female
Able to operate a computer	12.6%	7%	37.5%	26.9%
Able to use Internet	17.1%	8.5%	43.5%	30.1%

There exists a persistent gender literacy gap at global level in context of internet use and access and it is widening day by day. There is a significant increase in global gap (11 per cent in 2013 to 12 per cent in 2016) in internet use (ITU, 2016b). This digital literacy gap is creating negative impacts in areas of educational attainment, skills development and even in online digital participation.

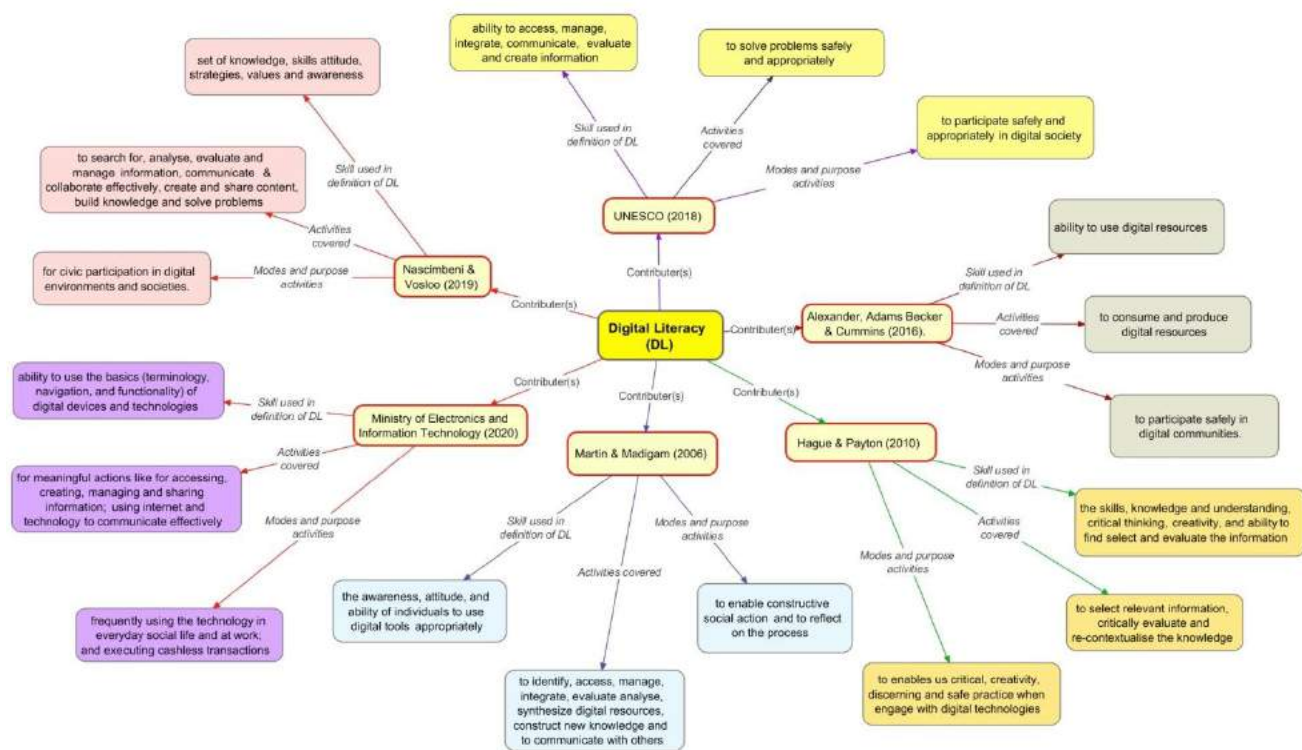
Digital Literacy in Education

Today, digital literacy has become an important entitlement for the youth worldwide. It furnishes skills to communicate, to select relevant information, critically evaluate and decontextualize the knowledge to participate in the digital society. Researcher have purposed the concept of digital literacy in different dimensions as per the advancement in technologies as the time being.

Table 3: Proposed definitions of digital literacy

Researchers& Contributors	Skillsused in the definition of digital literacy	Activities covered	Modes and purpose activities
Martin &Madigam (2006)	the awareness, attitude, and ability of individuals to use digital tools appropriately	to identify, access, manage, integrate, evaluate analyse, synthesize digital resources, construct new knowledgeand to communicate with others	to enable constructive social action and to reflect on the process

Hague & Payton(2010)	the skills, knowledge and understanding, critical thinking, creativity, and ability to find select and evaluate the information	to select relevant information, critically evaluate and re-contextualise the knowledge	to enables us critical, creativity, discerning and safe practice when engage with digital technologies
Alexander, AdamsBecker, & Cummins (2016).	ability to use digital resources	to consume and produce digital resources	toparticipate safely in digital communities.
UNESCO (2018)	ability to access, manage, integrate, communicate, evaluate and create information	tosolve problems safely and appropriately	to participate safely and appropriately in digital society
Nascimbeni&Vosloo (2019)	set of knowledge, skills attitude, strategies, values and awareness	to search for, analyse, evaluate and manage information, communicate & collaborate effectively, create and share content, build knowledge and solve problems	for civic participation in digital environments and societies.
Ministry of Electronics and Information Technology (2020)	ability to use the basics (terminology, navigation, and functionality) of digital devices and technologies	for meaningful actions like for accessing, creating, managing and sharing information; using internet and technology to communicate effectively	frequently using the technology in everyday social life and at work; and executing cashless transactions



Concept Map on Digital Literacy

In dealing with the evaluation in digital technology– concepts, levels and dimensions of digital literacy get broad day by day. Now it is not limited to the ability to use and understand information from a variety of digital sources as defined for the first time by Gilster (1997). Two major categories i.e. standard operational and conceptual definitions have emerged with regard to defining digital literacy (Lankshear&Knobel, 2008). Operational definitions are related to computers and ICT specific skills and abilities needed to complete certain tasks. In contrast, conceptual definitions are broader in contexts that emphasize mastery of ideas, careful evaluation, intelligent analysis, and critical reflection of literacies. Different kinds of skills i.e. browsing, searching and finding information quickly, critical thinking, e-safety, effective communication and collaboration with peers, solving technical problems, creating digital contents, protecting from online thefts and malware and critical thinking and evaluation and effective use of digital platforms and cashless transactions etc., get integrated in the concept of digital literacy. Further digital literacy is not confined with the ability to use digital devices or software or consuming and creating digital contents. It includes a continuum of meanings and can be seen as an umbrella term having a number of sister concepts such as computer literacy, 21st century skills, media literacy and information literacy (Brown et al., 2016).

Digital Literacies in Everyday Life

The unprecedented disturbance caused by the pandemic has highlighted the importance of digital literacy and also accelerated the digital transformation over the globe. Restrictions on face to face interaction with one another forced us to explore new ways to work collaboratively. People got sudden exposure to digital platforms like Zoom App, Google Classroom, Google Meet, Skype, Google Hangouts etc., that facilitate video conferencing, instant file sharing and messaging. Many of them do not have any experience before to use these platforms and expect to get to grips with it in a short period of time. Sudden exposure to new digital platforms for online learning and remote working does not mean they become digital literate. They still need different digital literacy competencies to work and interact more effectively with one another (Kanchev et al., 2016). Thus, day-by-day, the importance of digital literacy has increased in the context to find, use, evaluate, create and communicate information appropriately in the digital world.

To be a digitally literate one should have certain kind of digital competences that need to incorporate and promote among individuals which was purposed by Law., Woo., Torre & Wong (2018) in the ‘Global Framework of Reference on Digital Literacy skills for Indicator 4.4.2’.

Device and Software Operation

A digitally literate person should have possessed knowledge of the basic principle of computing devices such as computer hardware and software operations. To be more competence in the context of digital literacy one should have the certain abilities within them-

- able to physically operate the hardware tools and devices.
- have the basic understanding of data, information and digital content to operate the software tools and technologies.

Information and Data Literacy

Information and data literacy are complementary areas that possess the variety of skills like cognitive and technical that provide the fundamentals of managing digital environments. It enabled us to use the digital technologies effectively to search information and judge its quality and authenticity (Cordell, 2013). To be more competent in this area one should have competences like:

- able to browse and search for the data from authentic sources.
- able to organise, store, retrieve and filter data, information and digital content by critically evaluating its credibility and reliability of sources.

Communication and Collaboration

Effective communication requires knowledge, skills, understanding of digital communication tools as well as behavioural norms to share thoughts and ideas appropriately in the digital world. So, people must be empowered to communicate better in the current digital age. Similarly, in the context of collaboration, there is a need to understand how to collaborate in shared space or digital space and what are the collaborative skills required for the effective interaction and work. Collaboration helps them to develop skills of debate, compromise, flexibility, cooperation, team work and listening to others (Hague & Payton, 2010). So, there is need to inculcate the basic communication and collaboration competencies like-

- able to interact on different digital platforms through a variety of digital technologies.
- collaborate effectively in digital space through different digital tools & technologies and have the ability to share data, information and digital contents with others.
- need to follow certain behavioural norms while using and interacting digital technological environments.

Digital Content Creation

People should become a responsible effective digital contents/e-contents creator in addition to content consumers. A step to be taken towards creating digital contents instead of findings, evaluating and consuming digital contents. People who are gaining digital literacy must have certain competences like-

- able to create, edit, improve, modify, refine and integrate information and digital contents.
- able to understand the process of applying copyright and licence to the data, information, and on digital contents and using it without violating its defined norms.
- have basic knowledge and understanding about programming that are used to perform the specific task.

Digital Safety

As per the EU-KIDS research, youth of the globe were more fluent in using media but at the same time they needed intense support to strengthen in the area of safety (Pyzalski et al. 2019). They need to update time to time about new digital and hybrid threats. Teachers needed to take responsibility to develop the skills and knowledge among students that protect against digital threats (Eyal, 2012). To be safe in this digital era, one should have certain digital competence to be safe in this digital world like-

- able to protect personal devices, digital contents, data and privacy while navigating in digital space.
- able to protect physical and psychological health and avoid risk and threats associated with it while using digital technologies.
- able to use the technologies without violating environmental harmony.

Problem Solving

Problem solving is the broader idea that reflects one's competency to navigate and use multiple digital resources in order to achieve or complete personal and professional goals. Digital problem solving required a cognitive and nimble use of skills, strategies and mind-sets to navigate and respond to the task (Castek, etal. 2018). In order to achieve basic competence in area of problem solving strategies one should have certain capacity within them-

- able to solve technical problems related to operating devices.
- able to solve conceptual problems related to digital content creation, innovative processes and products.
- able to process a computable problem into sequential and logical steps as a solution that can be understood by human beings as well as by computer systems.

Careerrelated Competences

The dependence on digital technologies related to career related activities has increased rapidly. There is demand for a wide range of technical, interpersonal and conceptual skills for the career related competences. In the new digital world, there is need to think about the competences to operate as an effective career. One should have understanding about changing career contents and have ability to interact effectively across a range of digital platforms to build a relationship to assess the usefulness of career information (Moore, 2017). To be competent in the area of career competence one should have -

- able to identify digital tools and technologies which are specialized to use in a particular field.
- able to understand, analyse and evaluate specialized data, information and digital content for a particular field within a digital environment.

Conclusion

Today's digital literacy has become important life skills among youth across the globe. It enables the youth to take several advantages of digital technologies and as well as help them to keep safe and alert while navigating the digital world. But still the large portion of the population fail to get

opportunities in the digital world due to insufficient digital literacies, digital divides, digital environments, infrastructure, and awareness. There is a need to invest more in educational policies, programs and in research to facilitate digital learning environments and infrastructures to foster digital skills and knowledge among the students, teachers, administrators and other stakeholders to connect with new situations. In order to strengthen digital literacy, there is a vast need of technological devices, infrastructures, more qualified, skilled, and competent teachers as well as high awareness of the issues among the people.

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11. Need Analysis for developing a MOOC: Process and Benefits

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Abstract

Teachers' Professional Development (TPD) has remained a point of discussion for a long among policy planners and educators. In a country like India, providing TPD is a herculean task as nearly 1.5 million teachers exist in the higher education system. The study has revealed various common challenges teachers face while attending the formal PDPs and suggested that MOOC can be an effective alternative. In order to develop an effective MOOC for TPD, need analysis is an important aspect. Researchers have explained the process adopted by them for need analysis and concluded that there is enough acceptance and potential in MOOC for TPD if it is being developed with due consideration of all relevant facts and aspects, in which need analysis plays a crucial role.

Key Words: TPD MOOC, Professional development, Need analysis.

Introduction

Teachers' professional development (TPD) is an area on which much stress has been given in the recent past, not only in India but also globally. Many initiatives have been taken to enhance the opportunities for TPD in India. Under Pandit Madan Mohan Malviya National Mission for Teachers and Teaching (PMMMNMTT) schemes started in 2016, the initiatives specifically focus on TPD in education. With the enhanced technology-mediated interventions and starting India's own Massive open online course (MOOC) platform, SWAYAM, the focus has been shifted to use MOOC for TPD. In 2018, the Government of India launched a vertical on MOOC for professional development of nearly 1.5 million higher education teachers in the name of ARPIT. Nearly 75 subject/area-specific national resource centres (NRCs) have been established to offer MOOCs for professional development. The official data available with MoE (2020) shows that nearly 0.2 million teachers benefitted from the initiative. Apart from ARPIT, various other TPD-MOOCs were also developed and offered by various institutions/individual faculty/practitioners on various other platforms and LMSs along with SWAYAM. While developing a MOOC for professional development, what should be kept in mind, and how to progress for developing an effective MOOC for TPD are a few essential questions to ponder upon. Though it has not been established that MOOC is the only way for

professional development, it has been noticed that the connections amongst MOOCs and higher education are a prevailing theme in the specialized press (León et al., 2015).

It has been a widely accepted fact that it is challenging to give TPD opportunities to all staff members within face-to-face training facilities. Keeping the teachers up-to-date and providing them with continuous professional development opportunities is a big challenge. (Misra, 2018). Through MOOCs, many teachers can be benefitted from the internet; further, MOOCs provide a chance for teachers to be accessible with the updated knowledge through the web (Mohammad and Bibi, 2017).

NEED AND SIGNIFICANCE OF THE STUDY

For developing any training programme, planning and execution are the keys to its success. Worldwide, when any MOOC is developed, it requires much preparation. Several aspects need to be kept in mind while developing a MOOC. Many researchers have identified the factors contributing to the success of a professional development MOOC. Factors like 'Focus of subjects' (Nagashima, 2014; Azevedo & Marques, 2017) influence the participants' enrolment in a MOOC, 'Timing' influences the enrolment and continuance in a MOOC, both (Azevedo & Marques, 2017), readiness to learn (Bond, 2015; Davis, et. al. 2014; Kop et al. 2011) motivations for enrolment (Mackness, et. al.; Nagashima, 2014), perceived ease of use of the online system; and perceived usefulness of the online system (Yang et al., 2017) are some key identified determinants.

Nevertheless, how a MOOC developer will get an idea about all these? It requires to need analysis. Need analysis is a process that the researcher, trainer or institution undertake to identify the target group's need, identify the knowledge gaps, find out the concerns and challenges, and know the target group's attitudes. "learning needs analysis" is a method to identify the specific requirements of learners along with the profile of the target group, the goals and objectives (Kop, et.al., 2011; Humanitarian Leadership Academy, 2020). A MOOC developer also collects information about the educational background, the familiarity of learners with online learning, their prior understanding (Emanuel, 2013; Koller et al., 2013; Rayyan, et. al., 2013), etc. It is an essential step because MOOCs' successful learning is increasingly learner-driven and determined (Hood & Littlejohn, 2016). A successful MOOC is based on participatory design, and relevant activity-based learning facilitates (Hew, 2014; Istrate&Kestens, 2015). It has been observed that many contemporary MOOCs tend to have a content-driven focus of knowledge transmission, deviating from their original focus of knowledge generation (Karunanayaka, et. al., 2018), as many of these have not been

developed with a proper need analysis. Keeping this in mind, the researchers have undertaken a need analysis to find out a relevant area for developing a MOOC for TPD.

OBJECTIVES OF THE STUDY

The objectives for the study were:

- To identify the preferred areas for professional development of teacher educators.
- To find out significant concerns and challenges being faced by the teacher educators in professional development.

POPULATION AND SAMPLE

For the present study, all the teacher educators in India have been considered population, but their exact number was unknown, and no centralized data was available. Also, basic qualifications are almost similar across the nation due to regulatory bodies' intervention; it was decided to use the non-probability quota sampling technique. Total 400 respondents have filled the need analysis tool developed by the researchers; 372 tools were found filled, these 372 (241 in online mode using google form and 131 in face-to-face mode) respondents have been considered as a sample for the study.

TOOL FOR DATA COLLECTION

Researchers have developed a self-made standardized tool for "need analysis for professional development through MOOC". The content validity of the tool was established by getting validation from 11 teacher educators of the country. The tool was prepared in printed as well as in e-tool form. The data has been collected in both modes, i.e., online and face to face.

The tool for need analysis has five sections- A, B, C, D, and E. Section A comprises 'personal information'. Section B has the items related to the 'status of professional development of respondents.' Section C comprises items related to 'challenges in professional development'. Section D has the items related to 'preferred areas for professional development of teacher educators, whereas Section E is related to 'preferred mode for professional development'.

ANALYSIS AND INTERPRETATION OF DATA

Section 'A': Personal Information

Under this section, analysis of some essential demographic information like sex, 160 (43.01%) male and 212 (56.99%) female, the age group in which most of the respondents (76%) were from age group 26-45 years, followed by 23.92% of age group 46 to 65 years.

As far as the highest qualification is concerned, most of the respondents (73%) possess a Ph. D. as the highest degree in Education where they have varied teaching experience.

As shown in figure 1, 17% of respondents belong to universities, 24% to affiliated government institutions, 51% to private self-financing institutions and 8% to other institutions.

Figure 1: Nature of Institution

Analysis of data under section A reflects variability in the respondents in terms of sex, age, nature of institutions in which they are working.

Section 'B': Status of Professional Development

Items in the tool under this section were placed to ascertain the present status of professional development of teacher educators. Three hundred six respondents (82.26%) teacher educators have responded their familiarity with professional development programmes, whereas the rest 66 (17.74%) have informed that they are not aware of any professional development programmes. It reflects that even in this digital and information-led era, teacher educators have no familiarity with professional development programmes. The variety of professional development programmes is known to many as they have enlisted programmes like TLC and HRDC's faculty development programme (FDP), seminars, conferences, workshops, orientation programmes, refresher courses, etc. 278 (74.73%) have informed that they have attended at least one professional development programme and 94 (25.27%) have not attended any TPD.

As the respondents mentioned various programmes in the previous item, only the relevant ones were given in options when the respondents were asked to choose the professional development programme they attended. As shown in figure 2, 102 (27.42%) respondents have attended orientation programmes, followed by 78 (20.96%) refresher programmes. Surprisingly, 82 (22.04%) respondents have said that they have attended online programmes. The others were like 12.63% for short term courses, 14.52% for ODL mode programmes, and 15.05% have attended other types of professional development programmes.

Figure 2: Types of Professional Development Programmes

The next item in the section was 'Were their expectations regarding their needs met while attending these professional development programmes?' The responses of the respondents were found quite varied. Only 12.09% respondents endorsed 'to a great extent', but the most extensive number was with 'to a moderate extent,' i.e., 44.89%, around 21.50% respondents found the satisfaction of their expectations up to 'to some extent', only, while 19.89% were 'not at all' satisfied with it.

Figure 3: Level of satisfaction of the Expectations with PD

In an open-ended item, respondent teacher educators were asked to enlist the topics they attended the PDPs. Topics like basic ideas of higher education, university system, API system, UGC rules, leadership and institutional management, research methodology, SPSS, open educational resources, ICT tools, MOOC, MOODLE, soft skills, etc., were found in the list.

Section 'C': Concerns and Challenges in Professional Development

In this section of the tool, researchers have tried to collect information on the teacher educators' various concerns and challenges. It was a choice item in which they have to choose the statement with which they agreed. Freedom to choose more than one option was given to the respondents. The responses have been tabulated below in figure 4.

Figure 4: Challenges faced in Professional Development Programmes

The data shown in figure 4 is reflecting that the most faced challenge is "addressing the requirements and needs of teachers in PDP", 58.06% of respondents have marked it as a challenge, followed by "non-availability of a long spell of vacations/leaves for attending professional development programme", which is endorsed by 55.91% respondent teacher educators. The third in the list of challenges is "availability of quality resource persons", which was enlisted by 52.96% of teacher educators in the study. The challenges like 'lack of opportunities to attend the Professional Development Programme' by 47.04% respondents and 'quality of content in professional development programme' by 45.97% are also significant contributors to the findings. The reasons like "teaching-learning facilities at FDP centres" (24.46%) and "unawareness of available professional development programme" (19.09%) are the least endorsed challenges by the teacher educators.

On an open-ended item, some respondents have mentioned reasons like no opportunity for teachers in self-financing, no use of FDP, unfriendly attitude of college administration, no financial assistance, etc.

Section 'D': Preferred Mode for Professional Development

As the need analysis study developed a TPD-MOOC, researchers have asked the respondents to arrange the given options in their preferential order. Total 05 options were given to give the preference over the type of professional development programme they would like to attend. Their responses have been shown below in figure 5:

Figure 5: Choice for Professional Development Programme

The responses shown in a comparative chart in figure 5 indicate that most of the teacher educators have shown their interest in online programmes on the MOOC platform as for 44.89% of respondents, it is their first choice, and for 34.14% of learners, it is their second choice. The next choice is short-term courses of 7 to 10 days, which 19.89% of learners prefer as their first choice and 23.12% as their second choice. The traditional FDPs like Orientation and Refreshers programmes are not now the preferred choice of teacher educators as per data shown above. Even the FDP through ODL mode is among the least preferred programmes because most learners, 33.87%, have placed it as their last choice.

When teacher educators were asked to give their choice about the duration of blended/ online programmes for their professional development, 50.54% (188) of them have preferred the certificate programme of 4-5 weeks duration, followed by a short term awareness programme of 7 to 10 days, by 34.14 % (127). In comparison, only 15.32% (57) of respondents prefer a three to six months diploma programme.

Section 'E': Preferred Areas for PDP of Teacher Educators

As the need analysis exercise was done to develop a professional development programme for teacher educators, it was essential to know the areas in which they would prefer to complete a MOOC. In this item, the respondents were asked to choose any Seven (07) most preferred topics from the list given to them. Learners were also asked to add the topic of their choice if they wish to add. Later the preferences given to each topic were calculated, and the outcomes were analyzed; the result shows as per figure 6 that in order of preference, the following topics were in the choice of most of the teacher educators:

Figure 6: Preferred Choice of Topics/Themes for TPD-MOOC

As it is evident from figure 6, that the most preferred topic is "Research Ethics and Plagiarism", followed by "Using OERs". The third preferred topic is found as "ICT tools for classroom transaction and assessment". The topic in fourth place is "Making the classroom inclusive" followed by "languages across the Curriculum" in fifth place. The sixth preferred topic is "Constructivist pedagogy in the classroom", and the topic at the seventh place is "Indian Knowledge traditions and methods". The participants' few other topics were action research, technology integration, blended learning, understanding disciplines and subjects, curriculum development, and understanding Self.

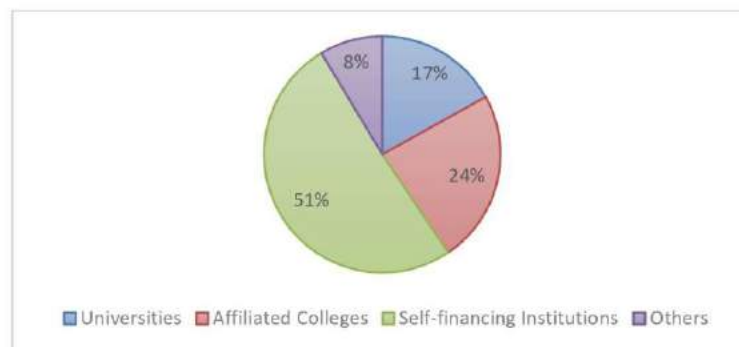


Figure 1: Nature of Institution

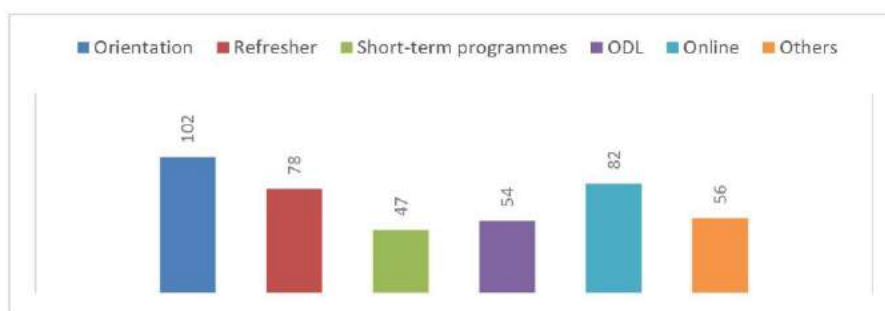


Figure 2: Types of Professional Development Programmes

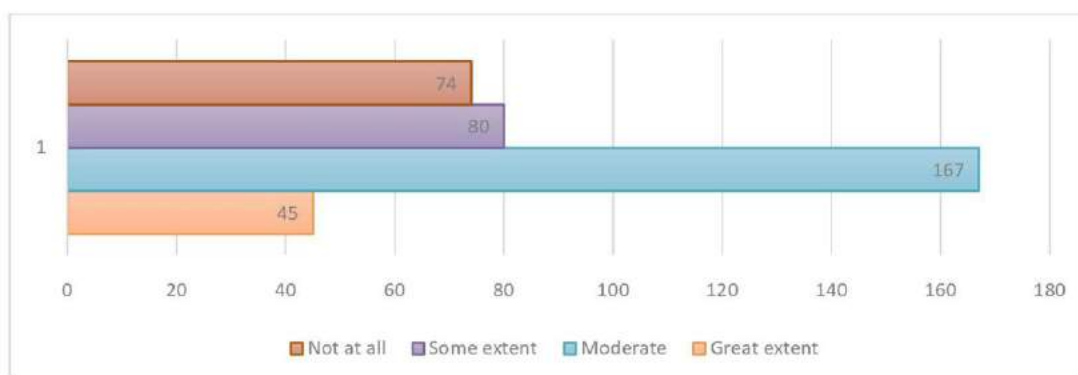


Figure 3: Level of satisfaction of the Expectations with PD



Figure 4: Challenges faced in Professional Development Programmes

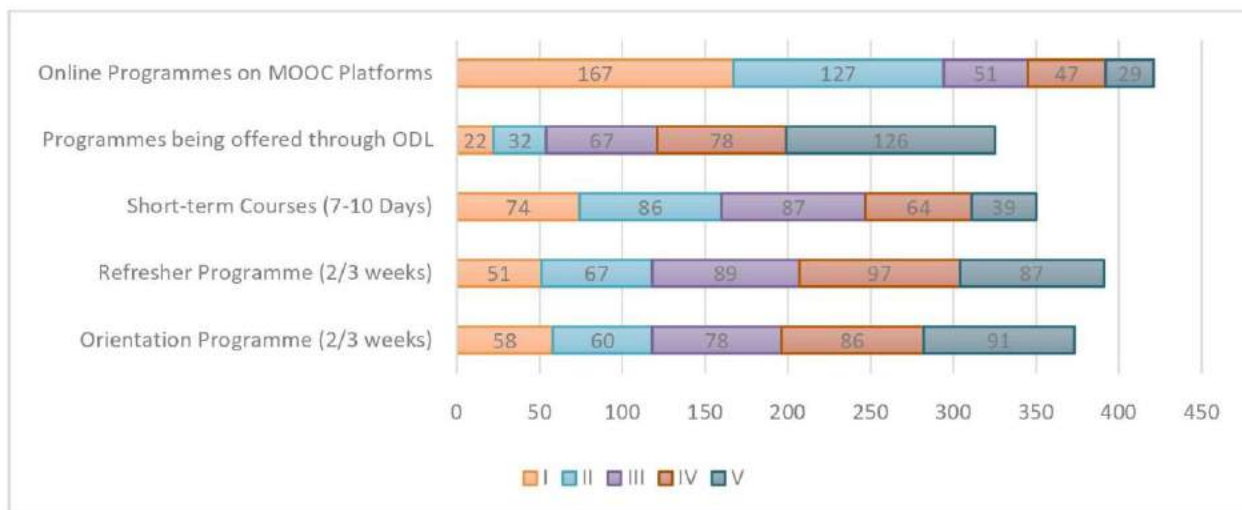


Figure 5: Choice for Professional Development Programme

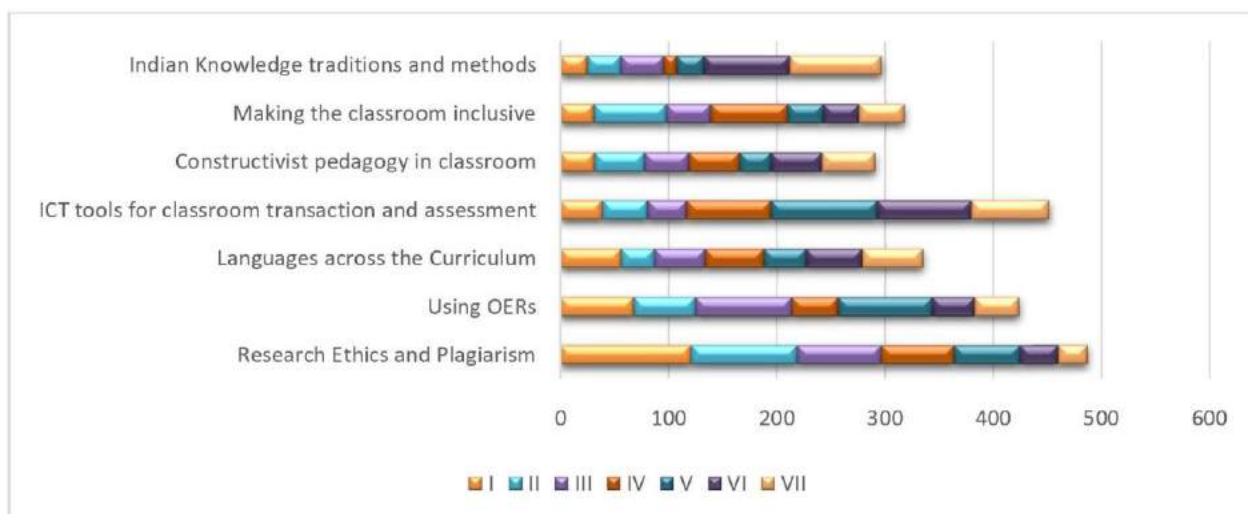


Figure 6: Preferred Choice of Topics/Themes for TPD-MOOC

Findings and Discussion

Analysis of the data reflects a positive receptivity towards MOOC for offering programmes for Teachers' Professional Development (TPD). Significant findings of the study have been summarized as follows:

- A. Most of the teacher educators, i.e., 82.26%, are aware of professional development programmes, but there are still a few (about 17.74%) who are not aware of any professional development programmes. Though the teachers, who have attended by PDP are lesser in number.

- B. Orientation (27.42%) and refresher (20.96%) are among the popular PDPs, but the proportion of online PDPs is also significant as (22.04% of respondents have attended at least one online PDP.
- C. As far as meeting PDPs' expectations, only 12.09% of respondents are satisfied "up to a great extent", and 44.89% are "to a moderate extent". There are around 21.50% respondents satisfied up to 'some extent' only, while 19.89% were 'not at all' satisfied with it. These findings reflect that the traditional PDPs are having some serious issues concerning match with the participants' expectations. If we plan a MOOC in which the content is adaptable according to the circumstances and the needs of the participants (Choy et al., 2006), this problem can be resolved. Massive open online courses can address the need for teacher professional development essentially (Larry, 2013).
- D. Participants are facing major challenge like addressing the requirements and needs of teachers in PDP, non-availability of a long spell of vacations/leaves for attending professional development programme, non-availability of quality resource persons, lack of opportunities to attend the Professional Development Programme, the question on quality of content in the professional development programme, etc. the challenges associated with teaching-learning facilities at FDP centres and unawareness of available professional development programme are the least endorsed ones. This reflects that MOOC developers need to focus on various quality aspects because the pedagogical design of CPD MOOCs needs crucial attention, a more creative and open-minded approach is desirable (Bayne & Ross, 2014; Laurillard, 2014).
- E. Online TPD programmes on the MOOC platform are the most preferred ones, followed by short-term courses. The traditional FDPs like Orientation and Refreshers programmes are not losing their charm, and so the traditional ODL programmes.
- F. Many participants have preferred the certificate programme of 4-5 weeks duration as blended/ online programmes, followed by a short-term awareness programme of 7 to 10 days.
- G. The topics like Research ethics and Plagiarism, Using OERs, ICT tools for classroom transaction and assessment and making the classroom inclusive are the most popular MOOCs for TPD. This reflects that teachers prefer MOOCs on recent topics and topics of their practical use rather than theoretical or philosophical topics. MOOCs offer the opportunities teachers need to develop a more diversified techno-pedagogical view to support their educator role (Sezgin, 2020).

Conclusion

The study has emphasized the need for a careful design and considering various aspects while planning for a TPD-MOOC. The study has highlighted the challenges and issues associated with

traditional PDP programmes and argued that MOOC could be a viable solution to overcome these challenges. MOOC is a natural solution to tackle challenges like lack of opportunities to attend PDP and non-availability of a long spell of leaves to attend PDP. Need analysis done before developing MOOC (Jung's NBISD model, 1997) helps get idea and insight about duration, type and themes to be covered in TPD-MOOC. This helps in enhancing interest and motivation and keep the participants engaged in the course. The finding indicates that the participants consider MOOCs as a potential professional development tool, which affects their pedagogical and technological knowledge in educational practice. (Goto, Batchelor, & Lautenbach, 2015; Koutsodimou&Jimoyiannis, 2015). No one can deny that MOOCs have become a natural source of TPD (Viswanathan, 2012).

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12. Achievement of Primary School Scheduled Tribe (ST) Students With Reference to Their School Environment

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Abstract

The Present study is an attempt to examine school environment and academic achievement among primary school tribal students in three district of Odisha. The sample comprised of 300 primary school Scheduled Tribe(ST) students from 15 primary schools. The objectives of the study are to study the achievement of ST students in Orissa, to compare the achievement of ST students with good, average and poor school environment and to study the relationship between achievement of ST students with their school environment. Investigator has employed survey method and School Environment Inventory by K.S. Mishra and self prepared achievement test consisting of Language, Social studies/Environmental studies and Arithmetic were used as tools for collection of data. The collected data were analysed by using percentage, t-test and accordingly interpretation was made. The study found 26% of ST students are high achiever with mean performance of 30.607, 47% of students are belong to average with mean achievement score of 18.25 and 27% of students are low achiever with 7.740 as mean achievement. There is a significant difference in achievement of ST students having good and poor, good and average, poor and average school environment. The relationships between achievement score and school environment score of whole ST students is negligible (0.125).

Keywords: Achievement, Scheduled Tribe, School Environment

Introduction

Education may not be a panacea for all our ills, but is definitely a means of change and a change for all the betterment of the society. Education has always been considered an instrument of social change. In the present day society, education has been considered a sound economic investment and that is the reason that in all the developed and the developing societies greater attention is being paid to education.

Government planners see education as indispensable for helping tribal peoples cope with national integration. Education will also determine their prosperity, success and security in life. The tribes which remain either deprived of or negligent toward education will suffer the consequence.

Compared with the literacy rates of 29.34% for the general population, literacy among tribal peoples in India is at most 6%. The Union and the state governments have spent considerable sums of money for tribal youths' education, but the results are meager. The Commissioner for Scheduled Castes and Scheduled Tribes asserts that unless exploitation among the tribes is combated and eliminated through education, no improvement in tribal welfare will occur. Within tribal areas, education can be the basis for integrated development.

Background of the Study

Article 46 of the constitution states that "The state shall promote with special care the educational and economic interests of the weaker sections of the people and in particular of the Scheduled Castes and Scheduled Tribes and shall protect them from social injustice and all forms of social exploitation". After independence, the Government of India has taken a number of steps to strengthen the educational base of Scheduled Tribes. Provision of educational institutions on priority basis in the areas predominantly inhabited by these communities, provision of incentives like scholarship, free ship, mid-day meals, uniforms, books and stationary, reservation of seats in educational institutions, relaxation of standards for admission to institutions of higher learning, provision of coaching classes for competitive examinations, remedial coaching and provision of hostels are some of the steps, which have contributed significantly in raising the educational levels of Scheduled Tribes. There are some special problems involved in the educational development of ST communities, which are mostly socio-economic in nature. Although educational facilities have been extended to almost all the habitations of these communities, however, on account of their socio-economic backwardness, they are not able to fully avail of these facilities. The average literacy rate among the tribes is 22.01 percent (10.2 per cent for tribal women), much lower than the state average of 63.1 per cent accounting to 2011 census.

Education is the key to development, but the education scenario among the Scheduled tribes of Odisha is in a very bad shape. It has been observed that the literacy level among the tribal people in general and the tribal females in particular is abysmally low. Education of tribes is one of the

cardinal basic inputs, the importance of which cannot be over emphasized. Educational development will create awareness and rapid economic development among the tribes. Besides, advancement of education will also ensure bringing a sizable section of the population i.e. the tribes to the mainstream of national consciousness.

Significance of the Study

Primary education is a corner stone of national development and a principal means of improving the welfare of individuals. It promotes economic growth thereby reducing poverty and enhances political, economic and scientific institutions. Primary education also ameliorates the health and nutritional consequences of poverty (World Bank, 1990 and 1993). The benefits of primary education for development stem largely from the improved cognitive skills, it imparts: literacy, innumeracy and problem solving ability. Our country's commitment to universal primary education dates back to 1961. The Directive Principles of the Indian Constitution stipulated that the state would endeavor to provide free and compulsory education to all children up to the age of 14 years. This commitment was reinforced when we developed National Policy of Education, 1986 at the instance of late Sri Rajiv Gandhi, former prime minister of India. The World Declaration on Education for All (EFA) adopted on Jomtien in 1990 gave further push to the national commitment to provide basic education for all. A very important step forward has been the 86th constitutional Amendment Act, passed in December 2002, making free and compulsory primary education a fundamental Right for all children. Primary education is still inadequate in India.

Various committees, commissions, working groups and study teams were constituted to formulate policies, programmes and strategies for promotion of education of ST communities e.g. Renuka Ray committee 1959, Elwin committee 1960, Education commission 1964-66. As a result, the programmes- such as establishment of Ashram schools, preparation of text books in tribals languages, free supply of school uniforms, study materials, residential facilities etc were taken up. Despite these, formal education did not attract many children.

There are so many research conducted on Scheduled Tribe students. But, there is no research evidence of studying on achievement and school environment of Scheduled Tribe students. The school environment play significant role for educational development of Scheduled Tribe students. Educational development will create awareness and rapid economic development among the tribes. So, realizing this present study has under taken.

Operational Definition of the Terms Used

Primary School

Primary school means first five classes i.e. from 1st standard to Vth standard of the formal school system.

Achievement

In the present study achievement is measured by percentage of marks secured in achievement test by ST students studying in class Vth .

School environment

School environment means the totality of external surroundings of a school, including physical conditions, location of the school, educational and professional background of the teachers and teaching learning environment of the school.

Objectives of the study

- i. To study the achievement of ST students in Odisha.
- ii. To compare the achievement of ST students with good, average and poor school environment.
- iii. To study the relationship between achievement of ST students with school environment.

Hypotheses of the Study

- i. There is no significant difference between achievement of ST students with good and poor school environment.
- ii. There is no significant difference between achievement of ST students with good and average school environment.
- iii. There is no significant difference between achievement of ST students with average and poor school environment.
- iv. There is no positive relationship between achievements of ST students and school environment.

Methodology

The present research is a descriptive survey type study, designed to obtain precise information concerning the achievement of primary school scheduled tribe students with reference to their school environment. The sample for the study consists of 300 hundred Vth class ST students from 15 primary school. The sample was selected by using purposive sampling techniques. Self developed achievement test consist of Language, Social studies/Environmental studies, Arithmetic for students and standardized School Environment Inventory , developed by K.S. Mishra were used for measuring

achievement of students and environment of school respectively. The achievement test consist of 45 multiple choice questions divided into three sections namely mother tongue (Odia), mathematics and social/environmental studies. The test item were content validated and test retest reliability was estimated (0.85). School environment inventory contains 70 items. The collected data were analyzed by using percentage and “t” test. This study is delimited to 300 primary class ST students studying in class Vth from 3 blocks of the 3 selected districts of Odisha. Further its scope in finding out the impact and relationship between and among the two things namely achievement and school environment.

Analysis and Interpretation

1. Achievement of ST Students at Primary Stage

The first objective of the study was “to study the achievement of ST students in Odisha”. For this the investigator collected data about the achievement of ST students by using achievement test. The investigator calculated different descriptive statistics such as mean, SD and percentage. The Table No.1 gives details of descriptive statistics of achievement level of ST students.

Table No. 1 : Achievement Level of ST Students

Sl. No.	Groups	Frequency and % of students	M	SD
1.	Low	81 (27%)	7.740	2.206
2.	Middle	141 (47%)	18.250	4.467
3.	High	78 (26%)	30.607	4.324
4.	Whole Sample	300 (100%)	18.666	9.249

From table No.1, it is found that 26% of students are high achiever with mean performance of 30.607. Further the table indicates that 47% of students have middle level achievement with mean achievement of 18.25. The table also find that 27% of students are low achiever with 7.740 as mean achievement.

The table also reveals that mean achievement of whole sample is 18.666 out of total score of 45. So it can be said that majority of students have less than 50% of achievement score.

2. Achievement of ST Students with Reference to School Environment (SE)

The second objective of the study was “to compare the achievement of ST students with good, average and poor school environment”. For this, the researcher collected data about the achievement of the ST students by using achievement test and school environment inventory. The schools were grouped into three such as good, average and poor on basis of score on school environment inventory. The researcher calculated ‘t’ value which is presented in the following tables.

Table – 2 : ‘t’ Value of achievement of students belonging to good and poor school environment

Groups	N	M	SD	df	‘t’ value	Result
Achievement of students from good SE	79	21.050	10.075	156	2.129	Significant at 0.05 level
Achievement of students from poor SE	79	16.886	9.292			

Table - 2 depicts that the mean score of achievement of good and poor school environment groups of student are 21.050 and 16.886 with SDs 10.075 and 9.292 respectively. The calculated ‘t’ value 2.129 is greater than the table value (1.98) at 0.05 level of significance. Therefore, null hypothesis “There is no significant difference between achievement of ST students with having good and poor school environment” is rejected at 0.05 level. Hence, the alternative hypothesis “There is a significant difference between achievement of ST students with good and poor school environment” is accepted. So it can be concluded that school environment influence students achievement.

Table – 3: ‘t’ Value of achievement of students belonging to good and average school environment

Groups	N	M	SD	df	't' value	Result
Achievement of students from good SE	79	21.050	10.075	219	3.801	Significant at 0.01 level
Achievement of students from average SE	142	18.330	8.510			

Table – 3 reveals that the mean score of achievement of good and average school environment groups of students are 21.050 and 18.330 with SDs 10.075 and 8.510 respectively. The obtained 't' value 3.801 is greater than the table value (2.60) at 0.01 level of significance. It implies that null hypothesis "There is no significant difference between achievement of ST students with having good and average school environment" is rejected at 0.01 level. Hence, the alternative hypothesis "There is a significant difference between achievement of ST students with having good and average school environment" is accepted. So it can be concluded that school environment of influence students achievement.

Table – 4 : 't' value of achievement of students belonging to poor and average school environment

Groups	N	M	SD	df	't' value	Result
Achievement of students from poor SE	79	16.886	9.292	219	4.083	Significant at 0.01 level
Achievement of students from average SE	142	18.330	8.510			

It is observed from table - 4 that the mean scores of achievement of poor and average school environment groups of students are 16.886 and 18.330 with SDs 9.292 and 8.510 respectively. The

calculated 't' value 4.083 is greater than the table value (2.60) at 0.01 level of significance. Therefore, null hypothesis "There is no significant difference between achievement of ST students with having poor and average school environment" is rejected at 0.01 level. Hence the alternative hypothesis "There is a significant difference between achievement of ST students with having poor and average school environment" is accepted. So it can be concluded that school environment influence students achievement.

3. Relationship between Achievement and School Environment of ST students.

The third objective of the study was "To study the relationship between achievement of ST students with school environment". To know the relationship between achievement and school environment of ST students, the researcher calculated coefficient of correlation which are presented in following tables.

Table – 5 : Coefficient of correlation between achievement and school environment

Groups	N	Coefficient of correlation	Interpretation
Students achievement of whole sample	300	0.125	Negligible
School environment of whole sample			

It is found from the above table -5 that the coefficient of correlation between achievement score of school environment of whole students and score of their school environment is 0.125. So, it can be interpreted that relationship between achievement score and school environment score of whole ST students is negligible.

Table – 6: Coefficient of correlation between achievement and poor school environment

Groups	N	Coefficient of correlation	Interpretation
Low level of achievement	79	-0.141	- Negligible
School environment			

Table-6 reveals that the coefficient of correlation between achievement score of poor school environment and score of their school environment is -0.141. Hence, it can be concluded that there is negative negligible correlation between achievement score and school environment score of poor school environment ST students.

Table – 7: Coefficient of correlation between achievement and average school environment

Groups	N	Coefficient of correlation	Interpretation
Middle level of achievement	142	0.078	Negligible
School environment			

It is found from the above table-7 that the coefficient correlation between achievement score of average school environment and score of their school environment is 0.078. So, it can be interpreted that relationship between achievement score and school environment score of average school environment ST students is negligible. In other words, students school environment does not substantially relate with their achievement.

Table – 8: Coefficient of correlation between achievement and good school environment

Groups	N	Coefficient of correlation	Interpretation
High level of achievement	79	-0.134	-Negligible
School environment			

Table – 8 indicates that the coefficient of correlation between achievement score of good school environment students and score of their school environment is -0.134. Hence, it can be concluded that there is a negative negligible correlation between achievement score and school environment score of good school environment ST students.

Major Findings of the Study

Findings relating to Achievement level of ST students

- 26% of ST students are high achiever with mean performance of 30.607, 47% of students are belong to average with mean achievement score of 18.25 and 27% of students are low achiever with 7.740 as mean achievement.
- The mean achievement of ST students is 18.666 out of total score of 45. In other words majority of students have less than 50% of achievement score.

Achievement of ST Students with Reference to School Environment

- There is a significant difference in achievement of ST students having good and poor school environment at 0.05 level of significance.
- There is a significant difference in achievement of ST Students with good and average school environment at 0.01 level of significance.
- There is a significant difference in achievement of ST students having poor and average school environment at 0.01 level of significance.

Relationship between Achievement and School Environment

- The relationships between achievement score and school environment score of whole ST students is negligible (0.125).
- There is a negative negligible correlation (-0.141) between achievement score and school environment score of low achiever ST students.
- The relationship between achievement score and school environment score of middle achiever ST students is negligible (0.078)
- There is a negative negligible correction (-0.134) between achievement score and school environment score of high achiever ST students.

Educational Implications

- The present study points out that majority of ST students have less than 50% of achievement score in Vth class. So, care should be taken to find out why achievement percentage is low in class Vth, which is still regarded as important stage of entire education system. Govt. of Odisha should take different positive steps through SCERT and other nodal agencies to

manage nationalized curriculum and syllabus in different classes for massive achievement as per present need. ST Development Department's educational system must be managed accordingly to cope with others. Parents and teachers meeting should be done in each month to make them well known, regarding the education of their children.

- Necessary steps should be taken for improvement of school environment in primary schools. Proper study atmosphere, recruitment of dedicated teachers may be appointed in all tribal population dominated primary schools. Schools run by ST welfare department and school and Mass Education Department of Government of Odisha need to take steps for recruitment and training /orientation of teachers.
- Adequate infrastructure facilities viz school building, hostels, electricity, portable water supply, girls toilet and road communication should be provided in primary schools in tribal concentrated districts. The non residential schools should be taken care of, in term of adequate provision of infrastructure.
- Language and text books should be written in bilingual mode at least upto primary stage. In the local school (primary), local tribal persons should be appointed as teachers. It will reduce the gap between the school and the home environment of tribal learners as it will be based on the tribal cultures. Multilingualism should be encouraged.
- The teacher working in the tribal schools should have working knowledge of tribal language to have rapport with the students. So, that thing would be easier in teaching and learning process. For this in-service training should be provided to teacher.
- In devising and selecting methods of teaching for tribal children, their learning styles viz. learning by doing, learning by memory, story telling, oral reading, recitation, repetition, group method and role playing may profitably be applied to teach tribal children. Besides the self esteem, self discovery, introspection and self identity may also be effective for ST Children.

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13.The Missing Factor of Teacher Preparedness in Low-Fee Private Schools: Exploration through Teachers' Perceptions

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Abstract

Neo-liberalism has been dominating the policies of education after liberalization of Indian economy during 1990s. It promotes privatization in and of education. 'In' in the sense that the existing institutions and services are handed over or intervened by private players. 'Of' denotes the shrinking role of the state in financing education and opening up of new schools and colleges. This policy move is rationalised in the name of efficiency and cost-effectiveness as the state schools are considered to be dysfunctional. Therefore, the lower-income parents are inclined to access low-fee private schools - a shadow form of schooling that is rapidly growing in India. Srivastava (2013) elucidated mushrooming of private schooling targeting the socially and economically disadvantaged groups in developing countries, she points a dearth of scholarship on private schooling that helps us to understand the inclination of the disadvantaged groups in private schools. She coined the term 'low-fee private school' (LFPs) to describe the private schools that target the socially and economically disadvantaged groups. The widely debated place of LFPs in the post-globalisation education scenario in India has tended to elide the role of teachers in these schools, and their role in providing education to lower-income families. If fact, public debates engendered by market reforms tend to downplay teachers' significance and suggest that teachers in low-fee private schools (LFP) perform better on around one-third of the salary of regular teachers. This paper looks at perceptions, views, dispositions and attitudes of teachers working in LFP schools. It is based on an empirical study conducted in three low-fee private schools in Ranga Reddy district of Hyderabad city. The study is aimed at accumulating the perceptions of teachers about learning of disadvantaged students. The paper argues that 'teacher' - a major actor who influences learning of the children from socio-economically disadvantaged families, lacks adequate knowledge- especially about the ways in which disadvantaged students learn. As a result, private nature of school does not improve the learning of the poor.

Key Words: Neo-liberalism, Teacher Preparedness, Teachers' Perceptions

Introduction

After the structural economic reforms of liberalisation there is a noted change in the arena of education in India¹. The neoliberal vision of the state has shrunk its role in financing education and opening up new institutions. It seeks more intervention of private players to meet the increasing demand of education from the masses. Parents' aspiration to English education for their wards have inclined the lower-income families to private schools. There is this shadow institutional framework often described low-fee private (LFP) sector emerged as a separate institutional arrangement to educate the children of lower-income families (Srivastava, 2008). The number of schools in this category is ever increasing (Kingdon, 2017). Manjrekar (2013) draws attention to changing landscape of education with neoliberal reforms and lack of research that tells us about the life and work of teachers - especially women teachers as they are predominant in these schools.

There is a heated debate about private schools in the context of meeting Education For All and Millennium Development Goals (MDGs). But this debate hardly includes teacher as a prime stakeholder in providing education to lower-income families. In fact, one end of the debate downplays teachers' significance and suggest that teachers in LFP schools perform better on around one-third of the salary of regular teachers. Jain and Saxena (2010) confront this argument stating that it is injustice with the teachers to compel them work in less salary. In addition, Nambissan & Ball, (2010) assert that the proponents of LFP schooling are transnational and some local advocacy groups and organisations who want to create market for investment in education.

In contemporary education policy sphere in India, neoliberal thinking propounds that educational institutions can run along market principles. Kumar (2011) states that the burden of outcome-oriented institutional culture which private institutions propel falls more on teachers. Neo-liberalism regimes in education have played a prominent role in taking away teachers' autonomy and dignity which lead teaching towards de-professionalisation. It is likely that historically oppressed communities are overrepresented in LFP schools. Tradition of critical pedagogy that advocates teacher's role as a transformative intellectual (Giroux, 1991) for the liberation of the oppressed communities is far from imagination in neoliberal regimes. Apple (1988) points towards proletarianisation and feminisation of teacher workforce to control teachers' work and their

¹ Excerpts of this paper were presented at the National Seminar on "Livelihoods and Skills for Progress in India", Tata Institute of Social Sciences (TISS), Mumbai.

resistance. Rationale of managerial structures are taking over autonomy of teachers. He states that in such a restructuring of teaching jobs, the work of teachers is being deskilled because of managerialism, reductive behaviourally based curricula, testing, and pre-prescribed competencies and procedures is leading towards controlling teachers' work. Intensification and technicization of teaching act by increasing technical workload of teachers is being misrecognized as a symbol of teachers' increased professionalism.

In today's growing phase of industrialisation and the so-called knowledge economy, institutionalised model of education lay a great emphasis over formal education. Indian constitution under the article 45 promises free education for children upto fourteen years of age. In order to meet this constitutional mandate state run government schools have been established for the masses. The schools are faced with many challenges such as inadequate infrastructure, underprepared teachers, lack of teaching-learning material, bureaucratic control and so on. These issues make government schools dysfunctional to the extent that the children of the poor do not even learn to read, write and do basic arithmetic (ASER, 2016). With the increased awareness and accessibility about education even marginalised parents look for good learning environment (Nawani, 2014) that empowers to lead a dignified life for their children. Even after parliamentary enactment of Right to Education Act (RTE, 2009) and guaranteeing free and compulsory education to all children even marginalised parents are compelled to access low-fee private schools² (LFP) in search of learning, since free education imparting government schools are perceived to be failing. Another important reason which makes the parents to choose low-fee private school is English education (Sarangapani, 2009). Because as Nambissan (2010) has argued that upper tier of middle classes in India have actively participated in the education system from the position of social and economic dominance that has allowed them to shape the system and define what 'good education' is, as well as desirable cultural resources for success. She stated that key route to upper middle class status in India has been the exclusive English medium private education, which followed the tradition of British public schools. These are among the schools the upper middle class elite Indians have always sent their children. This tradition, she argued, has set the trend and played down the standards of 'good' in education.

The neoliberal agenda of policy that marks the present day policies in India, promotes privatisation of education. Srivastava (2013) has elucidated that mushrooming of private schooling targeting the

² There are various nomenclatures used for LFP schools such as budget schools, low-cost schools, affordable schools etc. in the education literature. For this paper, I am using the term low-fee private (LFP) schools.

socially and economically disadvantaged groups in developing countries particularly in Africa and Asia have emerged over the last couple of decades. She has raised a concern towards academia who takes for granted the participation of disadvantaged groups in private schools. Moreover, poverty research which mainly looks at poverty as a barrier to education but do not consider its impact upon the educational experiences which actually determines knowledge, skills, attitude and dispositions one acquires from schooling (Batra, n.d.). This paper is based on an empirical study conducted in five low-fee private schools in Ranga Reddy district of Hyderabad city. The study is aimed at accumulating the perceptions of teachers about learning of the disadvantaged students. The paper argues that teacher - a major actor who influences learning, lacks adequate knowledge especially about the ways in which disadvantaged students learn, hence private nature of school does not improve the learning of the poor. Consequently, hamper their way to empowerment via education.

Rationale of the Study

Many teachers in LFP schools are women, professionally untrained and are paid two-third to three-fourth salary of the regular government school teachers. Scholars who promote free-market ideology in meeting educational needs of the poor advance that LFP sector is affordable, cost-effective and efficient as it provides parallel, if not better quality of education to the poor. The attempt of this paper is to understand the perceptions, views, attitudes and dispositions of teachers. This gains significance and urgency to probe if we consider the studies in the field of educational research that have found that classroom interactions and teacher-pupil relations increasingly crucial factors in learning. Earlier, the educational research was looking at teacher as a prime stakeholder in classroom, assuming that teacher controls the classroom environment. Therefore, teacher characteristics, beliefs, attitudes, expectations and teaching styles have been inquired in several ways. But when classroom transactions were researched from different perspective it has been seen that teachers' views and behaviour was often directed towards individual students. Henceforth, the research shift was changed from studying merely teachers to teachers' views about the pupils and classroom behaviour (Veldman & Worsham, 1983).

It has been seen in the earlier studies (Flanders, 1960 as cited in Goldberg (1964); Rosenthal & Jacobson, 1968) that teachers' perceptions and expectations have long lasting implications on students' learning experiences. If teachers' perceptions are inaccurate it would lead them to adopt incorrect teaching-learning practices (Ready & Wright, 2011). Therefore, to the large extent, teachers' perceptions of students do have an impact upon their academic achievements. Studies

(Thelen, 1963; Edwards 1974) have endorsed the fact that every teacher's perceptions about the students differ from student to student.

In this backdrop, this study endeavoured to explore the perceptions of elementary teachers' about learning of the disadvantaged students in LFP schools. Srivastava (2013) has pointed out that education of the disadvantaged groups or poor in private schools in India is an under-researched area and considerable efforts are required towards this because ever increasing numbers of disadvantaged students in private schools.

Methodology

The paper is based on empirical study conducted in three low-fee private schools selected via convenient sampling in Ranga Reddy district of Hyderabad. Fees of the schools ranges from five to seven hundred rupees per month for elementary grades. It is a qualitative study and exploratory in nature. Twelve teachers are selected as a sample via non-probability (purposive) sampling. Among them Nine are Women and the rest three are men. In order to maintain diversity in the sample teachers from different subject teaching, gender and social category are included. Data has been collected exclusively from the teachers who teach to fifth and sixth grades. This is because of two reasons; firstly, according to Piaget's theory of cognitive development, children upto 12 years come under concrete operational stage. He says that children at this stage learn to think rationally, become mature and adult like. Secondly, it was assumed that the teachers who knew the students from last five-six years, it would be possible for them to form beliefs, hold perceptions on the following aspects.

1. Judging behaviour and academic performance of the students
2. Forming concrete perceptions and views about the students
3. Making decisions on impact of factors like parental involvement, socio-economic condition of the family, language effects on the learning process of the students.

Open ended questionnaire has been used to collect data of this study. In this, respondents have given responses in written form which gave them time to reflect. Teachers were given their time to return the questionnaire. The researcher was not in the search of apparent responses. He was looking for the kind of responses which are well thought without any time constraint. This would not have possible through interviews or any other communicative data gathering techniques. This decision was also made on the basis of unavailability of time from the teachers during school hours. It was also not

possible to interview teacher outside the school. And hence the questionnaire was used as a time-efficient technique for the collection of the data.

Research Questions

1. Are the disadvantaged students capable of learning?
2. What is the role of language in learning of disadvantaged children?
3. How much part parents play in their education?
4. In what ways home environment and socio-economic condition of the family affect their learning?

Analysis and Findings

This section of the paper presents main findings of the study.

Blame Children, Parents and Conditions

The teachers prioritise behavioural aspects of the students in their notion of a 'good student'. For them, the one who is not mischievous, disciplined and attentive in class are the qualities that a student should have. In addition, a good student, for them, is the one who is good at studies and enthusiastically take part in learning and extra-curricular activities. But when it comes to students from the disadvantaged families, majority of the teachers blame the parents, socio-economic condition of the family and students for their poor academic performance. In personal attributes, teachers have pointed that they lack self-confidence and ability to be successful. The teachers who found disadvantaged children to be disengaged in learning, expressed that these children did not pay attention in the class, make minimal efforts to learn and are disinterested in studies. In contrast, Giridhar (2019) demonstrates narratives of remote rural area government teachers who joins teaching with intermediate and professional qualification of Diploma in Education (D.Ed), but keep on learning and improving as a teacher with the help of teachers learning communities, technology and workshops and seminars. Hence, contrary to the LFP school teachers, he shows rural government school teachers not only consider the socio-economic milieu of the children but also take the responsibility for learning of children. While the teachers in LFP schools certainly lack such opportunities of learning to be better at their work. They are found to be blaming the children and parents for poor learning and do not owe the responsibility of learning of the students.

Looking at the responses of the teachers it appears that disadvantaged students stand in contrast with their notion of a good student as they view a good student to be disciplined, engaged in learning activities, attentive and interested in studies. Teachers find the prohibition of corporal punishment

enshrined in the Right to Education (RTE) act, 2009 problematic. Teachers hold an entrenched belief that disadvantaged children cannot be made to learn and discipline without punitive measures. They retrospect some of their own experiences of teacher authoritarianism and punitive approach to discipline students. The teacher education programme, their own practice and intuition has not prevented them to unlearn such things and accept the change that has taken place with the neoliberal structure of schooling and in the understanding and the use of concepts of reinforcement and punishment in disadvantaged context.

Language of the Students

Language of the disadvantaged children emerged as one of the major issues in responses from the teachers with respect to classroom behavioural tendencies. Many teachers have pointing out towards non-formal language of the children. The teachers have mentioned that the students' speak non-standard language and use bad words (speak slang). This affects their interaction in the class and also leads them towards inferiority and lack of confidence. Some teachers have pointed out towards inferiority (impurity) of the language itself. They say that language of the disadvantaged students is rude as per their home. It is not fair for a teacher to assess students on the basis of standards of language because native person who is speaking the language from childhood cannot be impure. One may call it non-standard but it is not "impure" in any form (Berntsen, M., 1987). Maxine Berntsen points out towards variation in language according to educational level. The people who are educated enough speak standard form of language, and illiterate or less educated use non-standard language. The question also arises here to ask who decides the standards of language. The dominant sections of the society hold the power to decide these standards. The important thing to remember here is that teacher should not let the child sense inferior in language, in any way, because it leads the child feel inferior in the class. Rather teacher should encourage children to speak in whichever, standard or non-standard, form to develop language skills and improve vocabulary at early stage. This understanding seems to be missing among the teachers, as a result, they judge the students on the basis of standards of language. If this is the case then along with students the teachers are also disadvantaged (Lin, 1967) because they lack knowledge to understand the issues of underprivileged students.

The kind of language disadvantaged children use slows down their pace of learning and result in committing mistakes in reading and writing. Most of the teachers are agreed upon the fact that language of the disadvantaged students is somewhat different from the other students of relatively privileged families. Studies, as cited in Raph (1965) (John & Goldstain, 1964) and (Irwin 1948a, b)

found that disadvantaged children did not receive better verbal orientation from the parents during infancy, therefore they lag behind in early language development.

The language mismatch of the disadvantaged students with their well off counterparts restricts the students within group and holds them back from mingling with other students. The study conducted in Waynad district of Kerala was also found ST students being hesitant to mix with the other students (Carry & Joseph, 2008).

Few teachers found to be expecting some kind of respect from the students. They have mentioned that children of disadvantaged families lack mannerism who come from lower socio-cultural background with minimal education of the parents. Teachers stated that such children are unaware about the ways in which teachers have to be given respect, for instance, if the teachers say something they speak back. In this regard, Thelen (1963) had maintained that by judging students on the grounds of such aspects teachers exploit the purpose of classroom which is meant for learning. If teachers expect respect from students that disadvantaged children do not know, make them feel inferior which would lead them not to give any feedback or answer which will not like to the teachers. This would prevent students from expressing freely in classroom. For Thelen (1963) this kind of teacher is not an educator rather an “autocratic boss” who wants respect from his employees.

Impact of Home Environment and Parental Involvement

Teachers' views are also collected on the ways in which home environment and parental involvements in education affect the learning of the disadvantaged students. It is obvious that when we say disadvantage it means they lack some kinds of advantages at home that affects learning. However, the teachers also hold the same belief. Teachers perceive that disadvantaged students' underachievement in learning is simply because they come from the disadvantaged background. Almost all the teachers have pointed out towards lack of parental guidance, encouragement and motivation, support, care and security along with basic facilities like convenient learning environment. Carry & Joseph (2008) was found disadvantaged students to be as capable in learning as the other students from well-off families, but motivation and encouragement to study lacked in them. So the major responsibility of teachers is to motivate and encourage students towards learning. Teachers hold beliefs in relation with the ideal home environment that it should be encouraging, peaceful, disciplined, and parents should supervise daily homework. It has found that instead of overall socio-economic condition of the family and physical facilities, teachers generally tend to expect parents' values upon education, availability of books and peaceful and encouraging

home atmosphere (Edwards, 1974). This study finds a lack of knowledge among teachers regarding disadvantaged children. The teachers failed to understand the issues of children belonging to disadvantage sections especially their learning needs.

The claims of the neoliberals that private nature of governance of schooling and market principles through school choice mechanism can widen the educational opportunities for the disadvantaged sections of population seems ineffective unless the teachers are adequately equipped with professional knowledge, skills and dispositions to deal with the issues of disadvantages of unacademic socio-cultural milieu, lack of education among parents, and formal language. This seems imperative if we consider Batra (n.d) who outlines significant differences in the perceptions, views, conceptions and disposition of two groups of teachers. The first group who is not adequately educated perceived students as 'lacking in something'. On the other hand the another cohort of teachers who is educated to engage with social and individual differences, find and share ways of engaging children to participate, facilitate fearless and non-threatening learning environment and opportunity of peer learning. They question the prevalent notion of the 'educability' of children from disadvantaged contexts, and acknowledge the larger responsibility of teachers in enabling social justice.

Socio-Economic Condition of the Family

Economic backwardness of the family affects education of the children in many ways. The parents find it difficult many times to pay school fees on time. Sometimes school management keeps the students stand after assembly who have not paid fees. It must have adverse effects on their self worth and esteem. Also, parents struggle to provide necessary education material as well as they are unable to finance school trips and extracurricular activities. This is one of the major effects of poor economic condition has on their involvement with the school.

Nevertheless, many teachers are agreed upon the fact that parents of disadvantaged children do value education. However, a paradoxical situation emerges as teachers expect co-ordination from parents to educate children, on the other hand, they are unable to regularly meet teachers. The reasons teachers provided for minimal involvement of the parents lie in economic problems and lack of education among the parents. They primarily work as domestic servants, daily wage labourer or in informal sectors of employment resulting into minimal family income. Yet, teachers' expectation that parents should supervise studies at home, meet them regularly, and teach children basic mannerism manifests

teachers' social insensitiveness. The vision of NCFTE³, 2009 to have humane and socially sensitive teachers in Indian classrooms seems blurred in the context of LFP schooling.

Conclusion

Lower-income families these days want to provide their children a good quality education with the growing awareness, aspirations and accessibility. However, the impact of lack of learning affecting factors such as un-conducive socio-cultural milieu, informal language, minimal parental education, and unavailability of educational material is entrenched among teachers.

It's an inappropriate belief to hold on the part of teachers. They also found to be practising praxis for instance, judging students on the basis of standards of language, seek respect from students and form stereotypes, stigmatised identities of children based on behavioural aspects and their family background. This demonstrates that the teachers lack adequate knowledge about disadvantaged students' education - especially the ways in which they learn. So marketisation of schooling for the lower income groups through neoliberal policies appears to be ineffective until the teacher - a major actor who influences learning is not adequately prepared to deal with the issues of disadvantages, and continuous opportunities of professional development are not provided. This paper, therefore, provides a critic to neoliberal theory of education which advances LFP schools for the lower-income families and promote school choice mechanism to impart education.

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³ This National Curriculum Framework for Teacher Education (NCFTE, 2009) elaborates the context, concerns and vision underscoring teacher education to create professional and humane teachers.

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14. Corollary Strategies on Reading Comprehension - A Case Study of ESL Undergraduate Students

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Abstract

The purpose of this research paper is to explore how learners experienced in a second language classroom which is exclusively designed to increase their emotive and cognitive reading performance. Research in the areas of vocabulary acquisition, reading strategy, comprehending the text, time spent reading, engagement, and reader self-perception provided a foundation for the study. Qualitative data was analyzed and described with description regarding what the learner's action and words which shared their thoughts with one another as they actively participated in the classroom activity. Learners with special needs are in need of consistent, sequential reading instruction. Providing a framework for educators makes the reading process convenient and fascinating for both the teacher and the learner. In the recent scenarios, reading is certainly not left out. The National Council of Teachers of English and the International Reading Association has collaborated to provide standards for reading. The pattern assumes that learning begins before the child enter any official schooling system as they start learning at home even before they admit themselves to school. So, the facilitators try to develop the skills that children bring to school. Most significantly in the transit of reading acquisition, is to cultivate a passion for reading. In our own lives, when we like to do something, we do it again and again. It is the same with reading. The more a learner reads the more proficient he becomes and the more likely he is going to engage in this activity.

Keywords: Basic, Applied, Developmental Research, Classroom Implications.

Introduction

Out of the four language skills, reading is the most prominent proficiency that the language learners have to achieve, because while a person intends to learn a language he should deal with different materials such as books, magazines, newspaper and texts to get familiar with the structure and concept of different combinations of words in the target language. As there is a fact that declares for being a good writer first you have to be a good reader. However, Reading is considered to be one of the essential skills for learners as it is an important gateway for gaining and learning more knowledge. Reading contains several actions like understanding the main idea, recognizing the main and important information, comprehending and learning, evaluating the passage in the academic circumstance.

The importance of reading

Good readers can understand the individual sentences and the organizational structure of a piece of writing. They can comprehend ideas, follow arguments, and detect implications. They know most of the words in the text already, but they can also determine the meaning of many of the unfamiliar words from the context - failing this, they can use their dictionary effectively to do so. In summary, good readers can extract from the writing what is important for the particular task they are employed in. And they can do it quickly!

Educational researchers have also found a strong correlation between reading and vocabulary knowledge. In other words, students who have a large vocabulary are usually good readers. This is not very surprising, since the best way to acquire a large vocabulary is to read extensively, and if you read extensively you are likely to be or become a good reader!

So if you want your child to be successful at school encourage him or her to read. Reading non-fiction in English is probably the most important, but English fiction and any reading in the mother tongue - if done extensively - will help your child develop the reading competence that is essential for academic achievement.

Implications for Reading

All reading is an active, reflective, problem-solving process. We do not simply read words; we read ideas, thoughts that spring from the relationships of various assertions. The notion of inference equations is particularly powerful in this regard. Readers can use the notion of inference equations to test whether or not the ingredients for a given inferences are indeed present. To show lying, for instance, a text must show that someone made a statement that they knew was incorrect and that they made that assertion with the specific purpose of deception. If they did not know it was wrong at the time, it's an error, not a lie. If they did not make the statement for the specific purpose of deception, we have a misstatement, not lying.

Surface and Deep Approaches to Reading

Learning a discipline involves developing familiarity with the ways of being, thinking, writing, and seeing the world of those experts in the discipline. Reading academic texts published by those disciplinary experts permits students to immerse in the culture of the discipline and facilitates learning its conventions, discourse, skills, and knowledge but, this is only possible if students take a deep approach to reading.

A surface approach to reading is the tacit acceptance of information contained in the text. Students taking a surface approach to reading usually consider this information as isolated and unlinked facts. This leads to superficial retention of material for examinations and does not promote understanding or long-term

retention of knowledge and information. In contrast, a deep approach to reading is an approach where the reader uses higher-order cognitive skills such as the ability to analyze, synthesize, solve problems, and thinks meta-cognitively in order to negotiate meanings with the author and to construct new meaning from the text.

Involvement of Teachers of English

Phonemic awareness

Phonemes are the smallest units making up spoken language. English consists of about 41 phonemes. Phonemes combine to form syllables and words. For example, the word stop has four phonemes (s-t-o-p), while shop has three phonemes (sh-o-p). Phonemic awareness refers to the ability to identify and manipulate these phonemes in spoken words. It is also the understanding that the sounds of spoken language work together to make words.

Phonics

Phonics is the understanding that there is a predictable relationship between phonemes (the sounds of spoken language) and graphemes (the letters and spellings that represent those sounds in written language). Readers use these relationships to recognize familiar words and to decode unfamiliar ones. Phonics instruction is a way of teaching reading that stresses learning how letters correspond to sounds and how to use this knowledge in reading and spelling. The goal is to help learner understand that there is a systematic and predictable relationship between written letters and spoken sounds.

Vocabulary development

Vocabulary development refers to the knowledge of stored information about the meanings and pronunciations of words necessary for communication. Vocabulary development is important for beginning reading in that when a student comes to a word and sounds it out, he or she is also determining if the word makes sense based on his or her understanding of the word. If a student does not know the meaning of the word, there is no way to check if the word fits, or to make meaning from the sentence. Vocabulary development is also a primary determinant of reading comprehension. Readers cannot understand the content of what they are reading unless they understand the meaning of the majority of words in the text.

Reading fluency, including oral reading skills

Fluency is the ability to read words accurately and quickly. Fluent readers recognize words and comprehend them simultaneously. Reading fluency is a critical factor necessary for reading comprehension. If learner read out loud with speed, accuracy, and proper expression, they are more likely

to comprehend and remember the material than if they read with difficulty and in an inefficient way. Two instructional approaches have typically been used to teach reading fluency. One, guided repeated oral reading, encourages students to read passages out loud with systematic and explicit guidance and feedback from their teacher. The other, independent silent reading encourages students to read silently on their own, inside and outside the classroom, with little guidance or feedback from their teachers.

Reading comprehension strategies

Reading comprehension is the culmination of all of the reading skills and the ultimate goal of learning to read. The purpose of mastery of each of the four previous skills is to enable comprehension. Likewise, reading comprehension facilitates mastery of the other four skills. For example, reading comprehension is clearly related to vocabulary knowledge and development. Comprehension is an active process that requires an intentional and thoughtful interaction between the reader and the text that can be explicitly taught through text comprehension instruction.

Fluency and English Language Learners

Fluency is a tricky area when it comes to English Language Learner reading instruction. For native English speakers, fluency and reading comprehension often share a strong correlation because fluent readers recognize words and comprehend at the same time. This is not always the case for ELLs; however, many ELLs can be deceptively fast and accurate in their reading because they are good readers in their primary language and have strong decoding skills. Yet they may demonstrate little understanding of the text, and hearing the text out loud may not necessarily provide a step towards comprehension as it is likely to do for native speakers.

Classroom strategies that Teachers can use to create Readers

When it comes to young adults and reading, teacher Chris Crowe (1999) focuses on the reading preferences and attitudes of his own teenage learner. According to his view students should be helped to realize that reading books can be a refreshing and rewarding alternative to TV, movies, shopping, or hanging out with friends. Allow students to discover, or remember, the pleasures of reading. Require and encourage outside, elective reading, and steer them toward good Young Adult books. Help them to connect with what they read, and nudge them to works related to what they've just read, or, if they're in a reading rut, nudge them into something different. Read yourself and talk to your learner and their classmates about what you read. Read some of what they read. Read aloud in class, and give them time to read in class. Find out more about engaging teenagers with reading.

Free Voluntary Reading

Teachers making time for independent, free-choice reading is a powerful way to create readers. Key features of “free voluntary reading” are: Provide access to the sort of material that will engage student readers in the classroom, from the library, and during out-of-school time allow “easy reading” that is engaging, enjoyable and effortless. Avoid the idea that “if it isn’t challenging it isn’t good for you”. Ideally 95 percent of text as known words is optimal. Increased reading volume will compensate for lack of extension in individual texts, and readers tastes do gradually develop and broaden.

Provide access to a variety of reading material

If book exposure is high, students have greater opportunities to start reading. Is there a bountiful, well-displayed accessible classroom library? Does the school request books from National Library for additional classroom resources? How often and how well do the institution use library with the students? Is there a range of resources to read - fiction, comics and magazines, graphic novels, sophisticated picture books, non-fiction, poetry? Explore the potential of e-books.

Make time for independent reading every day

Reading volume plays a key role in shaping the mind. It is a powerful predictor of vocabulary, comprehension, general knowledge and cognitive structures. Make time each day for students to read, and encourage students to read at other available moments during the day. These opportunities might include: during Sustained Silent Reading (SSR), while eating lunch, when they have finished class work early, various waiting times during the day, on the bus, sometimes setting an individual or class challenge encourages reading mileage and illustrates how small amounts of reading. For example, 3 x 6 minutes per day, can add up over a year to a large amount of reading mileage. Variation in Amount of Independent Reading (Adapted from Anderson, Wilson, and Fielding)

Give students free choice about what they read

Talk to students about how we read and our right to read in different ways, including the right not to finish a book. Daniel Pennac's book the Rights of the Reader (beautifully illustrated by Quentin Blake) makes a plea to parents, teachers and librarians to instill the joy of reading by ensuring everyone has:

- The right not to read
- The right to skip
- The right not to finish a book
- The right to read it again
- The right to read anything
- The right to mistake a book for real life

- The right to read anywhere
- The right to dip in
- The right to read out loud
- The right to be quiet

Reading is not just a solitary activity. In fact the social aspect of reading such as discussion with peers can be a powerful motivator. Encourage informal discussions about reading and books helping ensure students can express opinions freely and safely. Help students understand what they read. Engaging students with what they read is one of the most effective ways of helping students think about and make sense of what they read.

Keep track of reading mileage

Open a Library account for the class with recommended books, set up a class blog for your students to post up book reviews, share relevant websites and blogs with students such as Good Reads and create Readers Blog, reading logs can become onerous if they require too many details and may even hinder rather than foster the reading habit. A simple record of a title and rating works well.

Conclusion

Researches, opinions, and suggestions regarding the teaching of the reading exist in extensive amount, and this summary of reading implications is by no means exhaustive. However, teachers can better take advantage of the most useful methodologies, strategies associated with the different approaches. What is important to bear in mind is that relying too much on either top-down or bottom-up processing may cause problems for beginning ESL/EFL readers; therefore, to develop reading abilities, both approaches should be considered, as the meta-cognitive approach suggests. Considering my own experience in teaching reading to Indian students, I have found that the students who managed to read English text effectively are those who approach texts in a painful, slow, and frustrating word-by-word manner. By improving their decoding skills, they are freed to concentrate on global meanings. So, both the psycho and the linguistic” aspects must be emphasized in EFL reading classes.

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15. An Investigation into the Entry Level Teaching Aptitude of Trainee Teachers: Gender and Locational Perspective

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Abstract

Teaching Aptitude forms the backbone of teaching profession. It is the latent potential which is mandatory for becoming a successful teacher. Trainees possessing teaching aptitude are like uncut diamonds waiting to be polished. The study desired to examine the effect of Gender (male & female) and locality (rural & urban) on the Teaching Aptitude among the entry level teacher trainees of two different districts of West Bengal. This survey research used Simple random sampling to collect a sample of 133 trainee teachers from two B.Ed. colleges of West Bengal. Researchers constructed a close ended questionnaire based on past literatures and scored it on a four point scale. Results of the study showed significant differences in the teaching aptitude based on gender and locality.

Keywords- Gender, Location, Trainee Teachers, Teaching Aptitude, Teacher Training College.

Introduction

Role of a teacher has undergone a sea change in the modern world. One of the basic fundamental right viz. Right to Education has thus come under the scanner. Now often the news of degradation of social and moral values of teachers comes to the forefront. Teacher is the main foundation stone in the process of education and national progress. Kothari Commission believes that a Nation's destiny is being shaped in their classrooms. Kothari Commission (1966) feels among all the factors affecting education, teacher and their characteristics determine quality of education and the progress of nation it brings about. NEP (2020) stresses on teacher training to develop proper aptitude in teachers as it states improvement in teachers will improve education system and develop the human resources for future generation. NCTE is given the responsibility to come up with National Professional Standards for Teachers (NPST) by 2022. Interview during appointment is stressed to examine proficiency and comfort with local language. 50 hours of continuous professional development every year is recommended to improve aptitude and keeping abreast with time. Does our education system help in the development of ideal teachers? Success in teaching profession mostly depends upon the selection of individuals with positive and proper aptitude towards teaching. A teacher trainee must possess certain positive personal qualities, as well as teaching attitude and teaching aptitude which can be refined by the teacher education system.

Aptitude is a natural ability or talent, it is also described as an inclination or tendency towards certain objects, tasks and activities (Merriam Webster Dictionary). It is the innate or acquired capacity for a task leading to increased readiness in performing the task (Dictionary.com). Aptitude can be considered a competence in doing tasks and the performance reaches a significant level. Aptitude can be both physical or mental but is an inborn potential. According to *Bingham* aptitude refers to “those qualities characterising a person’s way of behaviour which serve to indicate how well he can learn to meet and solve certain specified kinds of problem.” According to *Freeman* “aptitude is a combination of characteristics indicative of an individual’s capacity to acquire (with training) some specific knowledge, skill, or set of organized responses, such as the ability to speak a language, to become a musician, to do mechanical work.”

Sharma defines, “an individual’s aptitude, as the capacity to acquire proficiency under appropriate conditions, i.e., his potentialities at present, as revealed by his performance on selected tests that have predictive value.” *Mangal* takes a different stance and defines “aptitude may be described as a special ability or specific capacity, distinct from the general intellectual ability, which helps an individual to acquire the required degree of proficiency or achievement in a specific field. Aptitude is thus an individual’s ability to learn or to develop proficiency in an area if provided with appropriate training or education.”

Present study focusses on teaching aptitude which in the words of *Bingham* is described “as a condition symptomatic in his readiness to acquire proficiency, his potential ability, and readiness to development of an interest in developing his ability.” Teaching aptitude is a specific ability, potentiality, interest, satisfaction, and fitness in teaching profession.”

Review of Related Literature

Many researchers have studied teacher aptitude and found interesting results caused by gender and location difference. Kant (2011) examined whether teaching aptitude and responsibility feeling differs on the basis of teacher’s gender (male or female) and location (rural or urban). In this normative survey, 100 secondary school teachers (50=male=female; 25 =rural male =rural female= urban male= urban female) were studied using Teaching Aptitude Test Battery (Singh & Sharma) and The Responsibility Feeling Test (Gupta). Results revealed no significant difference in teaching aptitude based on gender, no significant difference in teaching aptitude between male rural and male urban teachers, no significant difference in teaching aptitude between female rural and female urban teachers.

Sharma & Bedi (2016) assessed the secondary school teachers' aptitude about their profession, tried to determine and appraise the aptitude using a specially developed aptitude test, and examined whether gender wise and strata wise difference in teaching aptitude occurs. Survey design was used and result revealed though no significant difference in teaching aptitude of male and female teachers exist in urban area but significant difference in the teaching aptitude of male and female teachers in rural area is prevalent.

Rao & Asthana (2018) compared teaching aptitude of male teachers working in government aided and self-financed institutions in NCR, Delhi. They also investigated difference in teaching aptitude of female teachers working in government aided and self-financed institutions in NCR. 180 teachers (male = 50, female = 130; govt. aided = self-financed = 90) were selected using random sampling and the study used TAT scale (Jaiprakash & Srivastava) having dimensions as cooperative attitude, kindness, patience, wide interest, fairness, moral character, optimism, enthusiasm, discipline, scholarly taste to measure teaching aptitude. Results showed teaching aptitude of male teachers is more intense in self-financed institutions than government aided institutions due to awareness, flair and competition for survival. Teaching aptitude of female teachers is more intense in self-financed institutions than government aided institutions due to taste, ability to learn and clarity.

Kalaivani & Pugalenth (2015) examined the effect of gender, locality and school environment upon teaching aptitude of 100 high school teachers (male = 40, female = 60; rural = 64, urban = 36; govt = 34, govt. aided = 28, self-financed = 38) using survey design. Teaching Aptitude Battery (Karim & Dixit, 1986) was used which comprised of 80 questions measuring 10 areas- co-operative nature, wide interest, scholarly taste, fair mindedness, moral character, discipline, attitude, motivation, dynamic personality. Results found no significant difference in teaching aptitude with respect to gender (male or female), locality (rural or urban) and institution type (govt, govt. aided, self-financed).

Islahi & Nasreen (2013) undertook a descriptive research using purposive sample to collect 482 secondary school teachers (male = 245, female = 237) from 54 secondary school. Correlation, t -test and ANOVA result showed gender does not influence the effectiveness of secondary school teachers while marital status and training does. Urban secondary school teachers were comparatively more effective than rural secondary school teachers.

Appadurai & Saraladevi (2015) examined Teaching Efficiency in relation to Teaching Aptitude and Teacher Attitude using random sampling method to select 400 high school teachers (100 = govt =

govt. aided = corporation = matriculation; 50 =male=female from each institution).This survey used Teacher Efficacy Questionnaire (Gibson& Dembo, 1984), Teaching Aptitude Battery (Karim& Dixit, 1986), Teacher Attitude Questionnaire (Cook, Leeds&Callis, 1951). Results found significant relation between teacher aptitude of males and females in govt. and matriculation schools.

Khan & Khan (2016) examined relationship between teaching aptitude and teacher adjustmentsecondary & highersecondaryschool teachers; compare teaching aptitude male & female secondary & highersecondaryschool teachers. Survey used 440 teachers of which 240 secondaryschool teachers (male = 130, female = 110) & 200 highersecondaryschool teachers (male = 110, female = 90). Teaching Aptitude Test (Jaiprakash & Srivastava) & Teacher Adjustment Inventory (Mangal) were applied to measure the variables. Relationship between teaching aptitude and teacher adjustment score was low, positive and negligible. No significant difference in teaching aptitudewith respect to gender (male or female) or level (secondary & highersecondaryschool) was reported. Teaching aptitude was overall low for teachers of Jaipur.

Parh&Benjongkumba (2019) investigated level of teaching aptitude of B.Ed. student-teachers, and investigated difference of teaching aptitude between male & female B.Ed. student-teachers. Analytic survey method was used and random sampling technique was implemented to collect a sample of 180 student-teachers from 6 Teacher Education Institutions (IASE or CTE). Teaching Aptitude Test Battery(Singh & Sharma) was used and results show significant difference in teaching aptitude of male & female B.Ed. student-teachers, female having higher teaching aptitude.

Tasleema& Hamid (2012)examined the aptitude of elementary and secondary teacher educators separately and also compared the teaching aptitude of elementary and secondary teacher educators. Sample consisted of 300 teacher educators (150 = elementary = secondary) and the researchers used teaching aptitude test battery (Karim & Dixit). Results obtained from t-test shows elementaryteacher educators and secondary teacher educators differ on certain characteristics as co-operative nature,considerativeness, fair mindedness and impartiality, moral character anddiscipline, motivational aspect and dynamic personality aspects measured with teaching aptitude test battery while they were similar in optimistic attitude, wide interest and scholarly taste.

Objectives of the study

To find out the effect of Gender (male & female) and locality (rural & urban) on the Teaching Aptitude among the entry level teacher trainees of two different districts of West Bengal.

Objective 1- To find out whether there exists any significant difference in the teaching aptitude of the male and the female teachers of rural B.Ed. colleges of West Bengal

Objective 2- To find out whether there exists any significant difference in the teaching aptitude of the male and the female teachers of urban B.Ed. colleges of West Bengal.

Objective 3- To find out whether there exists any significant difference in the teaching aptitude of the male teachers of rural B.Ed. colleges and the male teachers of urban B.Ed. colleges of West Bengal.

Objective 4- To find out whether there exists any significant difference in the teaching aptitude of the female teachers of rural B.Ed. colleges and female teachers of the urban B.Ed. colleges of West Bengal.

Hypotheses of the study

H01: There exists no significant difference in the teaching aptitude of the male and the female teachers of rural B.Ed. colleges of West Bengal.

H02: There exists no significant difference in the teaching aptitude of the male and the female teachers of urban B.Ed. colleges of West Bengal.

H03: There exists no significant difference in the teaching aptitude of the male teachers of rural B.Ed. colleges and the male teachers of urban B.Ed. colleges of West Bengal.

H04: There exists no significant difference in the teaching aptitude of the female teachers of rural B.Ed. colleges and female teachers of the urban B.Ed. colleges of West Bengal.

Methodology

Sample

The trainee teachers from two B.Ed. Colleges were selected one from rural area and one from urban area in West Bengal. These two colleges belong to two different districts of West Bengal and are geographically far away from each other (300 km) of Kalyani and Midnapore in the beginning of the first semester formed the sample. Simple random sampling was conducted and a sample of 133 trainee teachers were collected, of which 45 were rural trainee teachers (rural male=23, rural female=22) and 88 urban trainee teachers (urban male=41, urban female=47) who have just started the B.Ed. programme.

Tools

The research tool has been designed by researchers to test the teaching aptitude of trainee teachers at entry level. Researchers themselves constructed a questionnaire consisting of 25 items scored on a 4-point Likert scale ranging from strongly agree to strongly disagree. Items are classified into areas as

mental ability, attitude towards children, professional knowledge and interest in teaching. Some of the items of the questionnaire were as “what is required for good teaching?” “If you are unable to get a job of teacher, then you will” “You like teaching profession because...” The Cronbach Alpha value of the entire scale is 0.650 and the inter-item -total validity range is -0.128 to 0.590.

Scoring

The scale had a total of 25 items all of which is scored on a 4-point scale ranging from strongly agree to strongly disagree where strongly agree is given a score of 4 and strongly disagree is scored as 1. The undecided option is omitted to prompt the trainee teachers to provide an opinion. The final score of each trainee is obtained by summing the scores of each individual item. The maximum score that can be obtained by any trainee is 100 and minimum score is 25.

Design

The researchers employed a Descriptive Survey Method of research in the present study for investigating the teaching aptitude of the teacher trainees from two different districts of West Bengal. Survey method was conducted through use of close ended questionnaire which was constructed by the researchers and filled by teacher trainees from two different districts of West Bengal.

Procedure

The data was collected using online Google forms which students completed at their convenience. Thirty minutes time was allotted for the questionnaire. Permission was obtained from Principals of both the B.Ed. colleges and each student was asked to provide their informed consent. Only those who participated voluntarily were asked to fill the questionnaire. Students were requested to provide honest responses and confidentiality of data was ensured. All ethical considerations were considered and maintained.

Result

Four *t*-test were conducted to test all the four hypotheses. For testing H_{01} , Levene's test of equality of means is not *accepted* (5.362, $p = .025$), so we reject the null of Levene's test and equal variances not assumed row is taken while interpreting *t*-test results. The *t*-test ($t = -16.451$, $p < .001$, $MD = -21.2$) shows that teaching aptitude of the rural male is significantly different from teaching aptitude of the rural female, i.e., rural male has better teaching aptitude.

Table 1

Independent Sample t-test between Gender (rural male & female) and teaching aptitude

		Levene's Test for Equality of Variances				t-test for Equality of Means			
		<i>F</i>	Sig.	<i>t</i>	df	Sig. (2-tailed)	Mean Difference	95% CI of the Difference	
								Lower	Upper
TA	Equal Variances Not Assumed	5.36	.025	-16.45	31.70	.001	-21.19	-23.82	-18.57

Note.TA = teaching aptitude

For testing H_{02} , Levene's test of equality of means is not *accepted* (.000, $p = .992$), so we accept the null of Levene's test and equal variances assumed row is taken while interpreting t-test results. The t-test ($t = 17.307$, $p < .001$, $MD = 5.64$) shows that teaching aptitude of the urban male is significantly different from teaching aptitude of the urban female, i.e., urban female has better teaching aptitude.

Table 2

Independent Sample t-test between Gender (urban male & female) and teaching aptitude

		Levene's Test for Equality of Variances				t-test for Equality of Means			
		<i>F</i>	Sig.	<i>t</i>	df	Sig. (2-tailed)	Mean Difference	95% CI of the Difference	
								Lower	Upper
TA	Equal Variances Not Assumed	.000	.992	17.30	86	.001	5.64	4.99	6.28

Note.TA = teaching aptitude

For testing H_{03} , Levene's test of equality of means is not *accepted* (11.371, $p < .001$), so we reject the null of Levene's test and equal variances not assumed row is taken while interpreting t-test results. The t-test ($t = -18.216$, $p < .001$, $MD = -11.67$) shows that teaching aptitude of the rural male is significantly

different from teaching aptitude of the urbanmale, i.e., rural male has better teaching aptitude than urban male.

Table 3

<i>Independent Sample t-test between Gender(rural male &urban male) and teaching aptitude</i>									
		Levene's Test for Equality of Variances				t-test for Equality of Means			
		<i>F</i>	Sig.	<i>t</i>	<i>df</i>	Sig. (2-tailed)	Mean Difference	95% CI of the Difference	
								Lower	Upper
TA	Equal Variances Not Assumed	11.371	.001	-18.21	29.19	.000	-11.67	-12.98	-10.36

*Note.*TA = teaching aptitude

For testing H_{04} , Levene's test of equality of means is not *accepted* (30.108, $p < .001$), so we reject the null of Levene's test andequal variances not assumed row is takenwhile interpreting t-test results. The t-test ($t = 13.022$, $p < .001$, $MD = 15.16$)showsthat teaching aptitude of the rural female is significantly different from teaching aptitude of the urbanfemale, i.e.,urbanfemalehas better teaching aptitude than rural female.

Table 4

<i>Independent Sample t-test between Gender(rural female & urban female) and teaching aptitude</i>									
		Levene's Test for Equality of Variances				t-test for Equality of Means			
		<i>F</i>	Sig.	<i>t</i>	<i>df</i>	Sig. (2-tailed)	Mean Difference	95% CI of the Difference	
								Lower	Upper
TA	Equal Variances Not	30.108	.000	13.02	22.60	.000	15.16	12.75	17.57

Discussion

Present research shows that there is significant difference in teaching aptitude between male and female trainee teachers both in rural and urban B.Ed. colleges. While rural male trainees showed better aptitude than rural female trainees, urban female trainees had better aptitude than urban male trainees. This result is partly supported by a survey research conducted with secondary school teachers' which revealed though no significant difference in teaching aptitude of males and female teachers exist in urban area but significant difference in teaching aptitude of males and female teachers in rural area is prevalent(Sharma & Bedi, 2016). Using Analytic survey method on a sample of 180 student-teachers from 6 Teacher Education Institutions (IASE or CTE), Parh & Benjungkumba (2019) showed significant difference in teaching aptitude of male & female B.Ed. student-teachers, female having higher teaching aptitude. Another study investigating the effect of gender, locality and school environment upon teaching aptitude of 100 high school teachers found no significant difference in teaching aptitude with respect to gender

(Kalaivani & Pugalenth, 2015). Earlier researchers found no significant difference in teaching aptitude based on gender (Kant, 2011). In another study, Menka (2016) found trainee teachers from different B.Ed. training colleges mostly possess a positive teaching aptitude and there exists no gender related difference in teaching aptitude. Among secondary school teachers also gender was not a predictor of the teachers' effectiveness but marital status and training were significant predictor (Islahi & Nasreen, 2013). In a study conducted with Junior Basic trained teachers of Himachal Pradesh using Teaching Aptitude Test Battery (Singh & Sharma), Pany (2013) pointed out no difference in teaching aptitude based on gender and between graduate and undergraduate level as well as graduate and postgraduate level. Though difference in teaching aptitude exists among undergraduate level and postgraduate level. In another study investigating the association between teaching aptitude and teacher adjustment secondary & higher secondary school teachers, the researchers concluded no significant difference exists in teaching aptitude with respect to gender (male or female) or level (secondary & higher secondary school) was reported (Khan & Khan, 2016). Results also revealed that there is significant difference in teaching aptitude based on location of B.Ed. colleges (rural and urban). While rural male has better teaching aptitude than urban male, urban

female has better teaching aptitude than rural female. Previously in a normative survey, with 100 secondary school teachers (50=male=female; 25 =rural male =rural female= urban male= urban female), Kant (2011) found no significant difference in teaching aptitude between rural male and urban male teachers, no significant difference in teaching aptitude between rural female and urban female teachers. A survey study using high school teachers revealed no significant difference in teaching aptitude with respect to locality (rural or urban)

(Kalaivani & Pugalenth, 2015). Similarly, a descriptive research using purposive sample of 482 secondary school teachers Urban secondary school teachers were comparatively more effective than rural secondary school teachers (Islahi & Nasreen, 2013).

Our investigation shows a fair difference between the male and female trainee teachers. Urban Female trainee teachers are having better teaching aptitude towards teaching profession in relation to urban male trainee teachers. In contrast rural male teachers have better teaching aptitude than the rural females. This study is an indicator of the growing gender difference towards teaching profession based on locality, i.e., rural or urban. Steps should be undertaken to minimize these differences. Rural female trainee teachers may be influenced by a number of factors like early marriage, male dominance, unequal opportunities etc. and urban male may be attracted towards other professions. Quality should never be compromised in this profession and entry of many misfits coming into the teaching profession should be prohibited at entry level. We need people who have the right type of aptitude for teaching. So, the entry level Teaching Aptitude tests are an essential tool to choose the right talent for teaching profession for admission to teacher education programmes like B.Ed., D.El.Ed. Teaching profession should attract the brightest minds, finest personality and most committed young people.

Limitations and Future Research

1. The study was confined only to two training colleges of two different districts. Researchers are working without any grants so larger sample was difficult to access.
2. A small sample of 133 trainee teachers were used. This study was conducted by online survey and response rate was about 50%.
3. Self-Report instrument was used to measure teacher aptitude. Researchers felt aptitude is an inherent quality which is best assessed by individual themselves.
4. Teacher trainees with disabilities in an inclusive classroom setting were not studied. This was intentionally done as researchers desired to study only effect of gender and locality.

5. Survey design was used to collect data as the scope of experimental or longitudinal research was limited in Covid situation.
6. Further research should be undertaken to upgrade the teacher training programmes and build better aptitude.
7. Further research needs to be done as to why the teaching aptitude of the urban male is different from urban female and also why it is different among the rural male and rural female.
8. Other factors like intelligence, creativity, personality, etc. were not studied. Future research on these variables along with those studied can open up new dimensions.

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16. Life Skills in Classroom: Strategies and Activities for the Teachers

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Abstract

Adolescents of today will survive and flourish in the 21st century only if they are skillful and psychosocially competent. They are full of energy and potential but due to lack of proper guidance they are unable to channelize their energy in the proper direction. Nowadays students have abundance of theoretical knowledge but they are found to fail in practical life. Their bookish knowledge does not suffice them to face the adversities of life. Life Skills increases their psychosocial competency so they respond to real life problems in positive and responsible way. Holistic development of the children can be achieved only if they are exposed to various Life Skills. Students those who have sufficient life skills recognize their potential and grow well in the society. Life Skills Education would play a crucial role in growing years of children. So our education system must be blended with the Life Skills Education. School provides opportunities to foster and improve life skills. But school also faces some challenges in the implementation of Life Skills Education. Teacher can overcome these challenges if they are competent enough. Life Skills should be practical based and requires experiential learning hence its transaction needs trained teachers. Teacher should make effective strategies and activities and integrate these skills during class teaching without any extra burden on teacher and students both. With organized strategy and planning teacher can inculcate life skills among young aspirants to succeed in all spheres of life and represent themselves as intuitive citizens of a bright tomorrow. This paper aims to explore the Life skills education, core life skills, problems faced by teachers in Life Skills Education implementation and suggested strategies, activities and teachers' efforts with which they can successfully implement life skills during class teaching especially at upper primary level.

Keywords: Life Skills Education, Life Skills, Psychosocial competency, Teaching, Strategies, Activities, Teacher, Classroom, CBSE

Introduction

Education is important for overall development of the child. Today knowledge is growing at an escalating rate, but knowledge with no practical consequences is fruitless for students. Education has transformed from rote learning to application and skills based now. Practical knowledge can be applied in real life situations and makes individuals in handling realities of life. Life Skills Education

has tremendous capacity to fulfill this objective. Life skills enable individuals to live happy balanced and meaningful life. It will help the young people to empower in challenging situations.

1. Life Skills Education

Learning Life Skills Education is a basic need for all the individuals. It aims at providing individual with strategies to make healthy choices that contribute to a meaningful life. In this context, a meaningful life is achieved through self-awareness, physical, mental, emotional wellbeing of individuals. The Mental Health Promotion and Policy (MHP) team in World Health Organizations (WHO) Department of Mental Health defined as “Life Skills Education is designed to facilitate the practice and reinforcement of psycho social skills in a culturally and developmentally appropriate way; it contributes to the promotion of personal and social development, the prevention of health and social problems, and the protection of human rights”(WHO,1999). Life Skills Education is a value addition programme for the youth to understand self and able to assess their skill, abilities and areas of developments, which also enable them to analyse their capacity to enhance the function in a most productive way (Nanaware & Palanaethra, 2017).

2. Life Skills

There are certain important skills that young people will acquire as they grow up such as reading, numbers, technical, livelihood skills etc. Life Skills are distinctly different from these skills. Thinking skills, social skills, and emotional skills are also needed for survival of individuals that enhances their psychosocial competency together constitute the Life Skills.

World Health Organization (1997) defines Life Skills as "the abilities for adaptive and positive behaviour that enable the individuals to deal effectively with the demands and challenges of everyday life". Here 'adaptive' means that a person is flexible in approach and is able to adjust in different circumstances and 'positive behavior' implies that a person is forward looking and even in challenging situations can find a ray of hope.

United Nations International children's Emergency Fund (UNICEF) defines Life Skills as "a behaviour change or behaviour development approach designed to address a balance of three areas: Knowledge, Attitudes and Skill. (as cited in CBSE Life Skills Teacher's Manual,2013).

Yarham (1919) defined Life Skills as “the personal competence that enables a person to deal effectively with the demands and challenges confronted in everyday life (as cited in CBSE Life Skills Teacher's Manual, 2013).

World Health Organization (WHO) in 1997 has identified ten core Life Skills which help the person to develop into healthy responsible and productive citizen. The ten core Life Skills are- Self Awareness, Empathy, Critical Thinking, Creative Thinking, Decision Making, Problem Solving, Effective Communication, Interpersonal Relationships, Coping with Stress and Managing Emotions. These Life Skills fall into three basic categories: **Social skills, Emotional skills, and Thinking skills.**, which compliment, supplement and reinforce each other. All these skills are interrelated and together, they are responsible for enhancing psychosocial competence of individuals.

3. Role of Schools in Transacting Life Skills Education

The present generation needs life skills which can be acquired formally and non formally both, and in formal mode school plays an important role. Schools are the appropriate place to develop and strengthen life skills. The schools provide Life Skills Education by creating a friendly, supportive, teaching- learning environment. Schools are catering to the needs of all the students and promoting individual empowerment. In school teachers and peers play a crucial role in developing life skills. National Curriculum Framework (NCF)-2005 clearly outlines that rather than a stand – alone programme the Adolescent Education Programmes (AEP) should become an integral part of school education and the responsibility of transacting Life Skill based education to secondary school students are fixed up on secondary teachers. National Curriculum framework: 2000(NCF: 2000, published by NCERT) and National Curriculum Framework for Teacher Education: 2009 (NCFTE: 2009) has rightly focused on life skills education among the prospective teachers (Chauhan, 2016 p 91). Central Board of Secondary Education (CBSE) has introduced Life Skills Education as integral part of its curricula for class VI to VIII consecutively in 2003-05 and published Teacher's Manual for Life Skills for transaction of the ten core life skills in 2013. These manuals provide guidelines for each of the ten core Life Skills, methods of its transaction and mode of assessment.

4. Major problems in the implementation of Life Skills Education

Schools are failing to implement life skills due to certain challenges faced by the teachers. Extra workload on teachers, load of course completion, lack of fund, time and resources, proper teaching methods for life skills's transaction, lack of trained teachers, shortage of effective life skills materials, unawareness regarding Life Skills Education, parents and students focus more on academics than life skills activities, students are not taking interest in life skills activities are some of the problems identified from the studies of Abobo (2012) Adhiambo (2013), Mugambi & Muthui

(2013,) Abobo & Orodho (2014), Wairimu (2015) Behrani (2016), Tiwari, P. (2018) on life skills implementation in school.

5. Teachers' effort for successful implementation of Life Skills Education in Classroom

Teachers can make some efforts to overcome above identified challenges to some extent and strengthen life skills among students.

- First of all, teachers should familiarize them with life skill education. They should make themselves competent enough by receiving training on teaching life skills or joining any course of developing life skills.
- To avoid extra workload on the teachers and students both, teachers should mingle life skills activities with curriculum in inseparable way. Teacher should plan academic contents delivery in such a way that it can develop life skills simultaneously. In this way teachers can avoid problems like load of syllabus completion, lack of time and resources to some extent.
- If teachers give importance to life skills activities only then students and parents will value these just like other academics activities.
- Peer educators trained by expert teachers can present an effective resource for mentoring students as peer interaction is the most successful form of transaction of Life Skills.
- Teacher should make powerful strategy, select proper methodology and approaches of teaching Life Skills. Teacher should design interesting interactive activities to create interest among students about life skills.
- Preparation and planning in different phases of teaching should be done.

Different **Phases of Teaching** requires some preparedness by the teacher for Life skills transaction in the following ways:

In Pre-active Phase: Teacher plans about the content and filters out the topics related to life skills from the syllabus.

- Focus on subjects that enhance Life skills Education.
- Review textbooks to identify relevant elements of Life skills.
- Find out content that can enhance Life skills .
- Correlate topics of syllabus with daily life examples to enhance Life skills.

In Interactive Phase: Actual interaction occurs between teachers and students where by creating reflections of real life situations life skills can be developed. When the teacher exposes the content to the learners they should go beyond book and connect with real life. Teachers should

- Ask Life Skills related questions during class teaching.
- Help students to solve Life Skill based question given in the book.
- Share real life experiences, give real life examples and ask students to share their experiences also, this will enhance students' understanding about Life skills.
- Incorporate interactive Life Skills activities during classroom teaching to internalize these skills among students (List of such activities as Life Skills Enhancers are given in the section 6(B) of paper)

In Post-active Phase: Assessment is done by teachers to find out the behavioural changes among students about Life skills along with other objectives. Both group and individual assessment can be done. This will evaluate their performance also, and then the teacher makes necessary modification in their teaching accordingly to achieve the educational objectives.

If life skills are taught consistently then gradually results will become visible. Life skills develops in combination. Activity suggested for a skill can develop a number of skills. Life skills can be implemented as a module or as individual exercises. Some examples of these exercises are suggested here to familiarize teachers with different life skills activities and plan their strategy accordingly during class teaching.

6. Suggested Strategies and Activities for Life Skills integration in Classroom Teaching

For proper development of life skills in classroom there should be collaboration among disciplines, teachers and students. Teachers are advised to adapt an **inter-disciplinary approach** for transacting Life Skills Education. The transaction should be both through curricular as well as co-curricular activities. Teachers incorporate life skills activities during classroom teaching, some activities can// be planned in advance and some/ while others are formulated as per need. Life Skills Education cannot be learned in an abstract and theoretical way rather it can be learned in concrete and practical way as through **experiential learning and participatory learning**. Two types of approaches- Teacher centered approach (Expository approach) and Learner centered approach (Heuristic approach) are suggested to the teachers for empowering Life skills among students (CBSE Life Skills Teacher's Manual, 2013).

With proper selection of strategies a teacher can transact life skills. **Strategies** such as classroom discussion, brainstorming, role plays, small group/ buzz group, educational games and simulations, storytelling, debates, situation analysis and case studies can be used to inculcate life skills in classroom setting. (Nivedita & Singh, 2016; Kumar P., 2017; Prajapati, R., Sharma, B. & Sharma, D,

2017; Tiwari & Bajpai, 2018; Saravanakumar, & Devi, K.R.P, 2020). These strategies are suggested in the teacher manual for Life Skills Education & CCE of IX & X (pp10-15).

6(A). List of Activities as Life Skills Enhancers

Every day, teachers organize number of activities as part of the curriculum. With skillful selection of theme of these activities can enhance life skill of students. Following enhancements are expected from teachers to develop Life skills in class teaching:

Type	Examples
Individual activity	Innovative assignment, projects based on different life skills
Group activity	Group discussion, Cultural task, Science club, Compassionate Children's Club to offer a helping hand to the needy students.
Daily activity	Maintaining notice board, News board, Word and Thought of the day, etc.
Weekend activity	Maintaining bulletin board, soft board, art and craft etc.
Monthly activity	Preparation of birthday chart, star chart, calendar making activity etc.
Annual activity	Organization of annual functions, sports events, science exhibition, book fair etc.
Dialogue based activity	Assemble talk to develop communication skill and reduce public speaking phobia, poetry recitation, debate, quiz, TED talks etc.
Writing based activity	Poetry writing, story writing, essay writing, article writing, diary writing etc.
Performing arts tasks	Dancing, singing, role play of famous personality, creative play, drama related to life skills etc.
Problem solving activity	Puzzle solving, riddles, provide different problematic situations to solve
Organizes lecture/ visual programmes	Motivational lectures, stories of famous personalities, Show visual clips from films based on life skills etc.
Games	Various indoor and outdoor game -Chinese whisper game, dumb charades, musical chair, chess, etc.

Above mentioned areas are just suggestions teachers can find more situations where they can integrate such life skills activities along with content and develop life skills simultaneously without any extra effort.

6(B.) Examples of activities related to different Life Skills

Examples of activities related to ten core Life Skills are suggested here for the teachers so that they can transact life skills without the need of extra resources and extra burden on them.

Activity 1. Self Awareness: My Flower Fair

Life skills to be enhanced: SA, E, CrT, CT, DM

Ask students to make a drawing of flower and write their strength/ likes on each petals and weakness/ dislike on the leaves. Ask them to not write their name on the sheet. Now in next class display drawings of students. And ask to guess the name of the student on the basis of character written on the petals and leaves.

Activity 2. Empathy: Today's Show

Life skills to be enhanced: E, CT, PS, DM

Teachers arrange a movie show based on any skill suppose Empathy. Ask students to analyze the situation of the movie and ask question as 'What would you do if...' it is followed by discussion and finally conclusion done by the teacher.

Activity 3. Critical Thinking: Newspaper collage

Life skills to be enhanced: CT, CrT, PS, EC

Ask students to explore different media as newspapers and magazines for any current topic for several days /weeks and collect relevant content, pictures and make beautiful collage showing their creativity. Display all the collage in the class and arrange group discussion on the topic.

Activity 4. Creative Thinking: My own story

Life skills to be enhanced: Cr T, CT, DM, EC

Students are asked to narrate creative story from given words/pictures, expanding their mind in new ways, share it with the whole class followed by discussion.

Activity 5. Decision Making: Our Representatives

Life skills to be enhanced: DM, CT, EC, SA, IR

Engage students in the selection of class representative, soft board in charge, cultural head, discipline head etc. of the school. Organize a discussion on student's decisions also so that they can evaluate their decision.

Activity 6. Problem Solving: My problem-My Solution

Life skills to be enhanced: PS, SA, CT, Cr T, DM, EC

Identify and make list of some common problems faced by students in their daily life or problematic situations in the surroundings. Divide class into group of 5-6 students. Assign one problem to each group and ask to prepare a role play on this. Performance of each group is followed by discussion with the whole class for getting final positive solution.

Activity 7. Effective Communication: News Time Now

Life skills to be enhanced: EC, SA, CT, Cr T, DM

Students are asked to report news of class/school and present it before the class, they can assign role of different media person among them. Debate and discussion can also be done on any burning issue.

Activity 8. Interpersonal Relationships: The Right Fight

Life skills to be enhanced: IR,E, SA, CrT, CT,EC

In the class teacher often faces many small fights amongst the students. Identify such problems and make a list. Now ask students for solution of each problems. Students respond to a particular problem differently, so variety of solutions will come out, compile the solutions and discuss it with students laying stress upon its pros and cons. Solutions suggested by students will be helpful in decreasing fights among them and maintaining good relationships.

Activity 9. Coping with Stress: My Stress Diary

Life skills to be enhanced: Co S, SA, EC, ME, DM

Ask students to write down their feelings in a diary when feel stressed. Observe common themes and coping strategies used by students. Check whether their coping strategies are positive or not and arrange group discussion on the positive strategies.

Activity 10. Managing Emotions: How to Handle Anger

Life skills to be enhanced: ME, SA,CrT, PS

Ask students to write their ideas on how to manage anger or any other emotion. Make list of innovative positive ideas shared by different students and discuss it with the whole class.

These are some suggested activities, a teacher can plan various other interactive activities according to their skill, situation, and as per the need of students to integrate life skills during class teaching.

Abbreviation used: Self Awareness(SA), Empathy(E), Critical Thinking(CT), Creative Thinking(CrT), Decision Making(DM), Problem Solving(PS), Effective Communication(EC), Interpersonal Relationships(IR), Coping with Stress(Co S) and Managing Emotions(ME).

Conclusion

Life skills education prepares learners to adapt to new situations and enables them to adjust in the ever-changing future by making them strong, competent, innovative and balanced to make pace with the changes, accepting and adjusting with realities of life. If they obtain quality education, acquire more practical knowledge i.e. knowledge of Life skills then they will develop self confidence, positive behavior and will be able to take wise decisions and solve problems effectively. It will promote social interaction, reduce stress and evolve them into productive and responsible future citizens. Hence, life skills must be introduced in our education system to provide quality education to the students. It can improve academic performance of the students along with enhancing their psychosocial competencies. School education has an abundant potential to develop life skills amongst young aspirants. There is need to bring Life skills Education into curriculum compulsorily. It has been suggested that whether life skills education is a part of the curriculum or not, teachers must transact these skills among students by using their practical experiences and skills during class teaching. By creating healthy and supportive teaching- learning environment teachers can get better outcome. Students will more actively get involved in a dynamic teaching- learning process, if teachers focus more on practical aspects than theoretical knowledge, students will learn life skills better. Depth of acquired life skills can be deepened further by powerful and effective teaching strategies and activities adapted by the teachers. So teachers should think out of the box to innovate interactive and effective activities to internalize these skills among students.

It depends on teachers' skill how they can design classroom activities to get positive results. They must acquaint themselves with life skills, its importance methods of its transaction. School management should be think about teachers' need and should arrange training and capacity building programmes, orientation programmes for them. Government should take this into consideration and make necessary arrangement for increasing awareness, development of life skills materials, teachers guide book for life skills, training of teachers for life skills transaction and make mandatory policy regarding Life Skills Education. In this way teacher will be proficient and will be able to handle the challenges in empowering young generation. By integrating life skills during classroom teaching a teacher would be able to contribute to the welfare of society.

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17. Attributes and Pedagogical Practices of Successful Teachers

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Abstract

The term 'successful teacher' does not reveal the underlying attributes of such a teacher. Factors like pedagogical satisfaction of students, the extent of learning outcome achievement and expected competencies, ability to develop higher order thinking skills may be some of the measuring aspects of a teacher's success. However, the term 'successful teacher' is not yet explicitly defined. This study is an attempt to identify the attributes and pedagogical practices of successful teachers through interviews and classroom observations. 20 principals, 162 teachers, 270 students and 142 parents of students of middle schools were interviewed and classroom transactions of 30 identified successful teachers were observed. The thematic analyses of the responses revealed the attributes and significant pedagogical practices of successful teachers.

Keywords: Attributes, Middle School, Pedagogical Practices, Successful Teacher

Introduction

It is not easy to determine the success of teachers as there is no set definition of a successful teacher and there are no definite parameters or criteria for this. Therefore, it is important to know which tasks or set of behaviour or practices declare teachers as successful or what makes an ordinary teacher a successful teacher. Many factors are responsible for the teacher's success or on which the success of the teacher depends. There are some aspects of the personality of the teacher which is beyond teaching that makes him/her an extraordinary teacher and leaves a lasting impact on the lives of students. Those effective teachers who have a positive impact on students' minds are generally known as successful teachers in the students' community. Some researchers mention student achievement in the classroom as a teacher's effectiveness and focus on the high-performance ratings of many teachers, and many times rely on the observations of supervisors, students, and administrators. Teacher quality is the most important factor in determining student achievement (Lovat, 2007).

Most teachers have the basic qualities of teaching, but some teachers use unique techniques to make their teaching more effective, which makes them successful among their students, peers, and

administrators. Teachers who influence their students' activities and have cognizance of how students learn, what they learn, how much they learn, and how students share their experiences may be considered as successful. Teachers who have a positive impact on students' minds may be known as successful teachers in the students' community. Different researchers have different opinions on this subject. The purpose of the present study is to identify attributes and pedagogical practices of successful teachers which can be used to lead to significant student learning in the present scenario.

Literature Review

With the changing times, the objectives of education, needs of learners, and society are changing. To match this pace of change, policymakers and researchers are incessantly exploring required pedagogical strategies (Goodyear & Zenious, 2007; Toom, 2012). Pedagogy is a set of teaching practices and the primary goal of pedagogy is to build on student's prior learning and to stimulate students towards learning (Westbrook, Durrani, and Salvi, 2013). Pedagogy refers to synergistic relation between teachers and learners (Loughran, 2013). The quality of the relationship between students and teachers is considerably important factor in teaching (Opdenakker & Dammea, 2006; Lovat, 2007). A successful teacher identifies the students' reluctance to work, helps them think through their choices and brings academic momentum (Strahan, 2008). Teachers' personality, teaching ability, contextual attitude and professional knowledge are important aspects of quality teaching (Beizhuizen, Hof, Putten, Bouwmeester & Asscher, 2001; Arnon & Reichel, 2007). Some researchers report student achievement in the classroom as a measure of teacher's effectiveness and focus on the high-performance ratings of teachers, and rely on the observations of supervisors, students, and administrators. Certainly, subject mastery is a component of teacher professionalism, not only knowledge of professional skills, but also skills, attitudes, and motivational variables that contribute to students' learning (Guerriero, 2015). Good teachers are ideal, analytical, dutiful, competent, expert, reflective, satisfying, diversity-responsive, and respected (Cruickshank and Haefele, 2001). Several factors are attributed to good or effective teaching. Effective teachers must take into account the unique needs and characteristics of their students (VanTassel-Baska & Hubbard, 2016). Trilling and Fadel (2009) opined that to be an effective teacher requires a shift from teacher-directed to student-centered learning, direct instruction to interactive exchange with students, content knowledge to application of it, content to problem-solving activities, basic to top, facts and principles to search the questions and problems, theory to practices and working with real-life projects.

Arnon, Sara, Reichel, & Nittir (2007) examined different types of teachers according to students' perceptions. Two groups were formed one was pre-service teachers and the second was in-service teachers. The sample consisted of 89 students from two colleges. The analysis was qualitative and based on the open-ended questionnaire. Pre-service and in-service teachers have given equal importance to the personal qualities of the ideal teachers. In-service teachers attributed less importance to knowledge than the personal qualities of the ideal teacher. The findings of the research reveal two main categories, first was personal quality and the second was instructive knowledge of the subject. Both groups described the characteristics of the ideal teacher as empathetic, attentive, and knowledgeable.

Bullock (2015) conducted a study to identify the characteristics of good teachers and how good teaching affects the students. Students of fourth grade and teachers were included as samples. There were two groups one was of students ($n = 21$) and the other of teachers ($n = 15$). In this research, supportive, compassionate, flexible, and caring teachers for students were identified as good teachers. Kolesnikova, (2016) conducted an experimental study to compare the effect of traditional teaching methods and active learning with the traditional teaching method. It was concluded that the active learning plus the traditional teaching method is a feasible alternative to the traditional teaching method only. Thus, active learning can be one of the important factors to successful teaching-learning process.

Most of the researches reviewed have discussed the characteristics of effective and good teachers and teaching (Arnon & Reichel, 2007; Bullock, 2015; Cruikshank & Hafele, 2001; Devine Arnon, Sara; Reichel & Nirit, 2007; Gargani, Strong & Hacifazlioglu, 2011; Gargani & Strong, 2014; Hardiman, 2012). In the present study, the attributes of successful teachers as opined by students, their parents, peer-teachers and principals have been explored.

Method

The qualitative research method was to identify the attributes and pedagogical practices of successful teachers. The related literature was reviewed and suggestions were sought from experts. The words related to successful teachers and successful teaching like the effective teacher, ideal teacher, good teacher, professional teacher, great teacher, expert teacher, quality teaching, effective teaching, good teaching, and creative teaching practice were included in the search. The sources and the measures used to collect data as in the study of Berk (2005) are mentioned in Table 1 and Table 2.

The experts interviewed to identify the attributes and pedagogical practices of successful teachers, included pedagogues, best teacher award winners and members of the Advisory Committee on Education - a think tank.

The questions asked to the experts were - “Who are successful teachers according to you?” and “What are the attributes of your ideal teacher that you still follow in your life?”. “What are the pedagogical practices of successful teachers”? Interviewing with more experts was stopped when the responses reached saturation i.e. a similar pattern of responses was recognized. Total twenty-six experts were interviewed. The responses to these questions from experts are enlisted in Table 1 and Table 2.

The data was collected from other significant sources such as students, parents, peer-teachers, principals, and teachers. To identify the pedagogical practices of successful teachers, the classroom transactions of teachers were observed. The design of the study is outlined in figure 1. The prior permission was sought from the Dean and Head of the Institute, School Principal and parents to collect the data. It was assured in writing to all concerned that the identity of the participants will not be revealed and data will be used for the said purpose only.

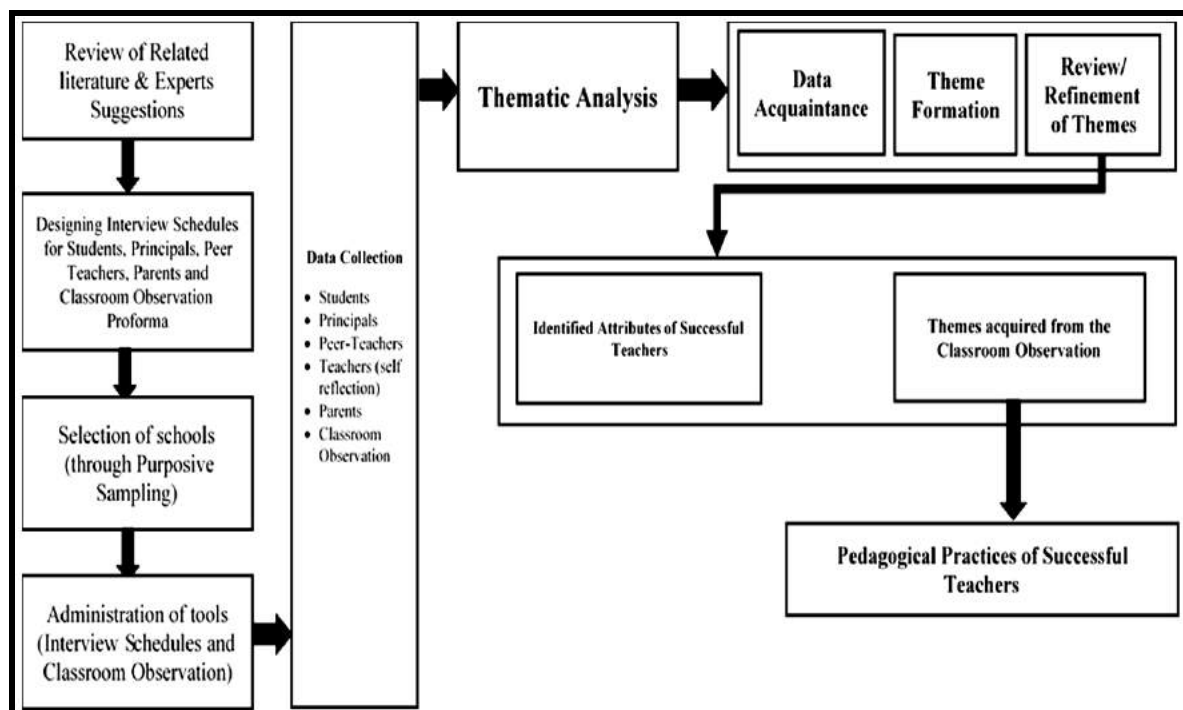


Figure 1: Design of the Study

Sample

The data was collected from eighteen schools selected using purposive sampling. The consent was sought from the school principals to interview/observe teachers, students and peer-teachers of middle school. In the study, middle school referred to grades 6, 7 and 8. The teachers, peer- teachers, students, and their parents were selected using purposive sampling. A total of 20 principals, 162 teachers, 270 students of middle schools, and 142 of their parents were interviewed. One common question was asked in the interview to the principals, peer-teachers, students, and their parents i.e. ‘Who is your favorite/successful teacher in school?’. The list of such teachers was prepared. The separate list was prepared from each source (students, parents, principals and peer-teachers). The final list of successful teachers was prepared with the condition that the name of teacher appeared in at least two more sources other than the students’ list. The final list had names of 30 teachers. A teacher's success can be identified by students' feedback. The opinion of students is considered to be the most influential and relevant source to identify the attributes of successful teachers. Therefore, the opinion of the students was given prime importance in this study. The identified successful teachers were interviewed to further explore the attributes of successful teachers. The classroom transaction of these teachers was observed to identify the pedagogical practices used by them.

Data Acquaintance

A structured interview using open-ended questions was conducted with students, peer-teachers, parents, principals, and teacher’s (self-reflection) to gather the data. The interview schedules were prepared to seek the opinion of experts, students, peer-teachers, parents, and principals. The observation proforma was prepared to study the pedagogical practices of identified successful teachers. In the first draft, items were based on the review of related literature. The first draft was distributed to experts for content validity. Based on suggestions of experts, some items were modified/restructured/deleted to (i) reduce the length of the sentence while retaining its meaning (ii) avoid the duplication of items (iii) avoid items conveying the same meaning. The questions included are as follow.

The Students

“What is the name of your favorite teacher?”, “Why is he/she your favorite teacher?”, “Any special quality of the teacher, which makes him/her different from others?”, “According to you, what are the qualities of successful teachers?”

The Peer-Teachers

“Who is a successful teacher in this school?”, “According to you, why is he/she a successful teacher?”, “Which practices can be included in teaching to make teachers successful?”, “How is the behavior of the teacher with the students in the class?”, “How much is he/she passionate about teaching?” “What efforts does the teacher make to solve the personal or academic problems of the students?” “If the teacher is not able to solve the problems of the students, does he/she make an effort to solve it by discussing with a peer teacher or others?” and “Any special quality of the teacher which makes him/her different from others?”

The Parents

According to you, “Who is a successful teacher according to you?” “Who is the favorite teacher of your ward?”, “Which type of information your ward passes to you about that teacher?”, “What are the extra efforts by the teacher for teaching to your ward?” and “Any special quality of this teacher, which makes him/her different from others?”.

The School Principal

“Who is a successful teacher according to you?”, “Who is a successful teacher in this school?”, and “Any special quality of this teacher, which makes him/her different from others?”

The Teacher (Self - reflection)

The thirty teachers identified from the above-mentioned process were also interviewed. The questions asked from them were - “What is the most challenging factor in classroom teaching?” “How do you deal with slow learners”? “How do you prepare your lesson plan”?, “Which method do you use for teaching a topic which is found to be difficult by students?”, “Which teaching methods do you usually use in teaching?”. “How often do you use teaching aids in the classroom?”, “What strategies do you adopt to make teaching effective?”, “How do you solve the problems of the students in the classroom?”, “What are the attributes of a successful teacher according to you?” ‘Which pedagogical practices do you adopt in the class for optimum student learning?’

Classroom observation

The classroom transaction of teachers was observed using an observation proforma. The aspects included in the observation proforma were - Introduction of the lesson; Development of the lesson (preparation of lesson, use of the developmental questions, variation in questions, use of different techniques and methods); Use of teaching-learning aids (Audio, video, any other media/technology); Blackboard work (writing, font size, sketching, organization); Explanation of teaching points; Non-

verbal communication of teachers; Classroom management (Sitting arrangement of the impaired students, and rotation process, discipline); Student involvement in classroom activities; Summarization of the lesson/recapitulation; Relating topic to daily life/experiences/practical utility of the lesson; Quality of questions given for the assignment, the personality of the teacher and any other.

Thematic analysis of Attributes of Successful teachers and Their Pedagogical Practices

In order to prepare the exhaustive list of identified attributes of a successful teacher, the thematic analysis of the qualitative data collected through the interview was done. The objective of thematic analysis was to recognize patterns and relationships through the dataset of attributes collected after interviewing teachers, parents, experts, students, peer teachers, and principals. The main themes identified from the literature review were teaching practices, attributes, and conclusion statements of the participants. The patterns are described through a standardized process - data acquaintance, theme formation, and review as mentioned in figure 1.

Themes formation

The themes were formed after aggregating the collected data. All attributes were entered into the MS Excel database for analysis. The data was systematically arranged for categorization of unique attributes and were put together under a theme for possible similar facts. The attributes of successful teachers identified from various sources are mentioned in Table 1 and Table 2.

Review

This phase involved the refinement of various attributes identified. Some attributes collapsed into other attributes. The data was read and re-read till meanings and patterns were identified such as ‘Students can easily share their problems with him/her’ is finally converted into the attribute ‘Approachable’, motivates students as ‘Motivational’, Uses soft language to talk as ‘polite’ and so on. The attributes identified from the review of responses are mentioned in Table 1 and Table 2.

Findings

The interview with various stakeholders and administrators like students, parents, principals, peers, and teachers themselves opened up several attributes and pedagogical practices that were associated with the term ‘successful teachers’. These attributes and practices were summarized using thematic analysis. The twenty-eight attributes and five aspects of pedagogical practices of successful teachers were found from the analysis.

Attributes of Successful Teachers

Twenty-eight attributes of successful teachers were identified. The attributes of successful teachers that emerged through this process are depicted in Figure 2.

Table 1

Responses Collected from Experts/Principal/vice-Principal and Identified Attributes of Successful Teachers

Source	Responses Collected	Identified Attributes
Experts' /Principal/Vice principal's Opinion	Understands students' problems	Understanding
	Tries to solve problems related to students	Helpful
	Guides students as and when required	
	Identifies their weaknesses & helps accordingly	Compassionate
	Has impartial behavior with students	Impartial
	Friendly behavior with students	Friendly
	Builds good relationship with students	
	Students can ask questions without hesitation from such a teacher	Approachable
	Works with dedication	Dedicated
	Works efficiently	Efficient
	Has strong work ethics	Ethical
	Works with dedication	Punctual
	Works efficiently	Righteous
	Has strong work ethics	Innovative
	Completes the work on time	Exemplary
	Demonstrates moral values	
	Implements innovative teaching methods	Ardent
	Has exemplary personality as a teacher	
	Identifies and nurtures the innate ability of students	
	Creates a congenial learning environment	
	Takes interest in solving problems of students	

Further, the attributes conveying similar meaning were clubbed together. For example, friendly was used for attributes helpful, understanding, empathetic, approachable and interactive; Ardent was used for ardent and dedicated; Motivational was used for cheerful, motivational; outcome- oriented was used for constructive and outcome- oriented; Ethical was used for ethical and punctual; Righteous was used for impartial and righteous. These clubbed attributes are presented in Figure 3. Thus, 28 attributes were reduced to 12.

Table 2

Responses Collected from Teachers, Students and parents, Identified Attributes of Successful Teachers

Source	Responses Collected	Identified Attributes
Teachers	Supports academically weak students	Cooperative
	Understands the psychology of students and treats them accordingly	Understanding
	Satisfies students' learning needs	
	Strong presenter of ideas	Eloquent
	Provides new information to students	Updated
	Uses new patterns of teaching	Creative
	Focused on learning outcomes	Outcome-oriented
	Develops and assesses critical thinking among learners	Interactive
	Promotes interaction with students	Approachable
	Inspires the intelligent students to support the weak student and works collaboratively with students	Collaborative

Students	Creates a fearless atmosphere in the classroom and students can easily share their problems with him/her	Approachable Co-operative
	Gives chance to everyone for answers and is impartial	Impartial
	Can create interest in learning and motivate students	Motivational Empathetic
	Does not humiliate students in class and listens to them patiently	Understanding
	Does not get irritated if students ask questions many times	Patient Cheerful
	Wears smile and listens patiently to the problems of students	Polite Constructive
	Uses soft language to talk	
	Shares mistakes of students without being rude and Gives constructive feedback	

Parents	Well-wisher of students	Generous
	Helps academically weak and needy students	Helpful
	Motivates students to learn and creates interest in learning	Motivational
	Is not partial to anyone	Impartial

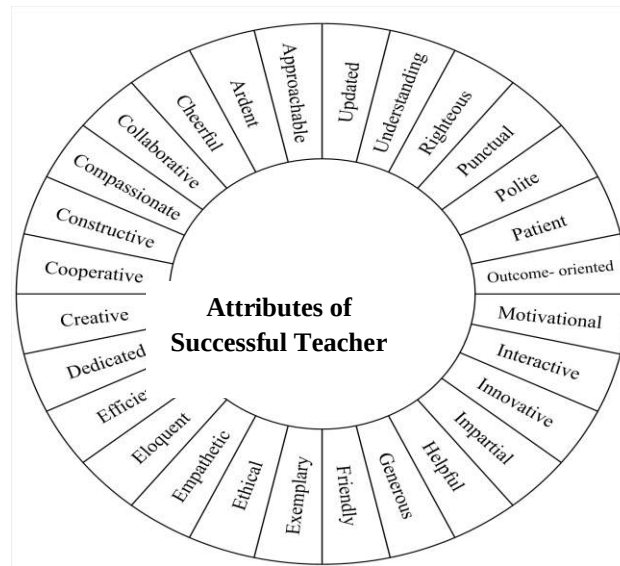


Figure 2: Identified Attributes of Successful Teachers

Attributes of Successful Teachers		
Efficient	Friendly	Ardent
Innovative	Motivational	Eloquent
Outcome-oriented	Polite	Ethical
Updated	Righteous	Exemplary

Figure 3: Identified Attributes of Successful Teachers (After clubbing)

Table 3

Pedagogical Practices of Successful Teachers

Pedagogical Aspects	Pedagogical Practices of Successful Teachers
	Orchestral immersion: Develops an educational environment of real experiences for students to learn from themselves
	Uses updated content
	Uses scientific/technical terminology
Classroom	Opportunity provider (Keeps students engaged through open-ended questions for sharing their ideas)

Pedagogy (Student-Centric pedagogy)	Encourages critical thinking
	Encourages deep thinking, discussion with peers and shares new ideas in the class
	Uses approaches like Inductive-Deductive, Analysis-Synthesis, Heuristic, etc.
	Encourages Inquiry-based classroom activities
	Adopts learning by doing approaches
	Follows maxims of teaching like concrete to abstract, simple to complex, parts to whole or vice-versa
	Caters to needs of differently-abled/special children in the class
	Develops positive relationships with students within or outside the classroom
	Teaches according to learning levels of students
	Effectual process (Ensures active participation of all students in the class)
	Behaves unbiased and maintains equality in the classroom
Use of TLM	Uses visuals (like charts/models/specimen/labelled diagrams on blackboard/audio-visual aids for explanation)
	Uses teaching-learning material for demonstration and practical learning.
Creation of a learning environment	Uses contemplative activities to regulate students' attention
	Creates a fearless environment (motivates to students to share their ideas)
Closure of the lesson	Summarises the lesson
	Provides guideline to students for preparing the assignment
Assessment Strategies	Uses application-based questions and questions that foster critical and deep thinking

Conclusion

The overall analysis reveals that successful teachers have passion for their profession. On interviewing successful teachers, the most common observation was that they had a positive and caring attitude towards students. They did not complain about students rather had an understanding attitude for their students. Another significant attribute that surfaced from interviews was the enthusiasm and dedication towards the teaching profession. Teachers who positively influence their students' activities are conscious of how students learn, what they learn, how much they learn, and how students share their experiences. The optimal student-teacher interaction was found to be the key pedagogical practice of a successful teacher.

The effectiveness in teaching of teachers becomes vital to face the emerging challenges of globalization and liberalization on the one side and mushrooming growth of the educational institutions on the other side. The schools should give recognition to such teachers and motivate them for their teaching endeavours. This research was initially aimed at identifying the pedagogical practices of successful science teachers; however, the results did not reveal any subject-specific pedagogy or the attributes of the successful teacher. The identified attributes are applicable to all teachers irrespective of their teaching subject. Future research can focus on differentiating subject-specific pedagogical practices of successful teachers.

To improve the quality of education, there is a great need to spend considerable resources on teacher training, which can lead to quality education in the future. This research will be helpful for research administrators, academicians, and teacher education institutions. This study is also an effort to highlight the factors which affect the quality of education imparted to students. It can be of help to teachers, parents, educationists, and administrators to execute the plans and work towards improvement by enhancing the positive factors revealed in the study. Through the teacher, students learn the right way to gain knowledge, but it also requires teachers to continue their professional development and learning.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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18. The Effect of Critical Thinking on Academic Performance among School Students

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Abstract

Critical thinking is a thinking style involved in problem-solving activities, and it is also an essential constituent in the decision-making process. Whereas, a person's achievement or performance is measured by the level of expertise he has gained due to training. The objective of this study was to investigate the effect of Critical thinking on Academic performance among school students. The sample involved 200 School students age ranging from 12 to 15 years from Bihar and Kerala, 100 from each state. The study used a correlational research design. The study results indicated that two dimensions of Academic performance- Academic success and Academic productivity were found positively correlated with some dimensions of Critical thinking. The stepwise regression analysis showed that the Cost value dimension of Critical thinking emerged as a significant predictor of Academic productivity ($\beta = .44, p < .01$), and Academic success ($\beta = -.17, p < .05$). Overall, Critical thinking emerged as an important predictor of academic performance. The findings of this research may be used as a reference point for incorporating a critical thinking pedagogy into the course curriculum, which would aid students in developing their skills and dispositions, hence enhancing their academic performance.

Key Words: Critical Thinking, Impulse control, expectancy-value, Academic Performance.

INTRODUCTION

Critical thinking has been found to be associated with many outcome variables such as efficiency, effectiveness and success. Critical thinking has its foundations in a variety of areas, including psychology, education etc. and in each discipline, it has its definition. It is an intellectually disciplined process "of actively and skillfully conceptualizing, applying, analyzing, synthesizing or evaluating information" (Paul, 1993).

Critical thinking-Critical thinking can be defined as the objective analysis of facts to form a judgment and is self-directed, self-disciplined, self-monitored, and self-corrective thinking. In learning how to analyze the logical structure of texts, critical thinking also improves comprehension abilities. In learning how to analyze the logical structure of texts, critical thinking also improves comprehension abilities. Critical thinking elements are dispositions, criteria, argument, reasoning, point of view, procedures for applying standards, and judging. Critical thinking is reasoned, reflective thinking focused on deciding what to believe or what to do (Ennis, 1996). Halpern (1998) believes that critical thinking is "purposeful, reasoned, and goal-directed".

One of the main characteristics of critical thinking is that it is a thinking style involved in problem-solving, and it also has a great deal in decision-making. Education experts have also mentioned the importance of critical thinking in the course curriculum. It is a vital talent for kids to acquire critical thinking to perform better, and also employers favour people who are good at critical thinking. A study conducted among employees and their supervisors revealed that employees' critical thinking positively affected employees' voices (Jiang, J., Gao, A., & Yang, B, 2018). However, many educational institutions are not exposed to critical thinking because of less competent teachers and a lack of understanding of its relevance. Various studies have shown that critical thinking can be taught through various ways, such as critical thinking-based education training, with students participating actively in various activities. It has been suggested that it should be taught not just by describing them but also by discussing their significance and applicability (Gelder, 2005).

Academic achievement or performance- As defined by Crow & Crow (1969), it is the extent to which the learner is profiting from instruction in the given area of learning. A person's achievement is measured by the level of expertise he has gained due to his training. In addition, it refers to how far a pupil, teacher, or organization has progressed toward their short or long-term educational objectives. Academic achievement is generally assessed by exams or continuous tests, although there is no consensus about how it could get judged or which factors are more important—procedural knowledge such as abilities or declarative knowledge such as facts.

Researchers have discovered that Institutional factors, such as “method of lecture delivery” (Santos et al., 2020), parent-related factors, such as parent's socio-economic status, educational background, poor parent-child interaction (Dhakal, 2020), peer group, age (Yusuf, 2020), etc. are shown to influence students' academic performance.

Even today, most educational institutions do rely on rote learning, which is based on repetition. However, it is preferable to use critical thinking like higher-order thinking for better academic outcomes and tremendous student success, as it is based on expert analysis and interpretation. Pintrich et al. (1991) identified five cognitive strategies that might influence learning: rehearsal, elaboration, organization, critical thinking, and metacognitive self-regulation. Aripin et al. (2003) found that paying attention to students' learning styles and matching them to a learning framework enhanced academic performance considerably, but a misfit between learning styles and curricula decreased performance levels. Critical thinking is linked to academic success, particularly in the cognitive processes of analyzing, evaluating, and synthesizing (Safitri et al. 2018). Several previous studies have found that employing critical thinking modules as learning material might increase students' academic progress (The, Isa, & Omar, 2018).

Rationale of the Study- Previous research has shown that Critical Thinking aids in better understanding, evaluating, and synthesizing knowledge, which may aid in a student's overall academic and life achievement. Thus, the present study has high implication value, enabling students to learn through practical and intelligent ways to improve their academic performance. Moreover, factors affecting academic performance are very much relevant in this current society.

Objectives of the Study

RO.1–The objective of the present study was to investigate the effect of Critical Thinking on academic performance among school students.

Hypotheses

H1.1 . There would be a positive association between critical thinking and academic performance.

H1.1(a)- There would be a positive association between all dimensions of critical thinking (expectancy, attainment, utility, interest value, cost) and academic success.

H1.1(b)- There would be a positive association between all dimensions of critical thinking (expectancy, attainment, utility, interest value, cost) and academic productivity.

H1.1(c)- There would be a positive association between all dimensions of critical thinking (expectancy, attainment, utility, interest value, cost) and impulse control.

H1.2 - Critical thinking would predict all the dimensions of academic performance- academic productivity, academic success, and impulse control.

METHODOLOGY

Research Design - The present study followed a correlational research design to determine the answers to the research questions regarding Critical Thinking and Academic Performance.

Sample- Sample size of 200 high school students were collected from Bihar and Kerala, 100 from each state. The age limit range of the students is between 12-15 years (High School students). The purposive sampling method was used to collect the sample.

Tools & Techniques

Questionnaires-

Critical Thinking Motivation Scale -This scale was developed by Saiz and Valenzuela (2017), consisting of 21 items with two main dimensions- expectancy and the value assigned. The second dimension comprises four sub-components dimensions-including, Attainment, Utility, Interest value, and Cost. The level of reliability was well qualified, and the reliability with an alpha value above 0.6. **Academic Performance Rating Scale.** This scale was created to acquire teacher's views of certain characteristics of a student's academic accomplishment (completion and correctness of work in several subject areas) in the context of a multimodal evaluation paradigm that would incorporate more direct assessment techniques. This scale was developed by DuPaul and Rapport (1991), consisting of 19 items that have to be filled by the student's teacher. The main three dimensions of the academic performance rating scale are- Academic Success, Academic Productivity, and Impulse Control. Inter-observer reliability of this scale is consistently above 80%, with a mean of 90% for all children; Interrater reliability for these measures was consistently above 96%, with mean reliability of 99%. Therefore, internal consistencies for the Total APRS are to be (.96).

Statistical Techniques Used - Descriptive as well as Inferential statistics have been used to test the framed hypotheses. All the analyses were conducted using the Statistical Package for Social Sciences- version 25(SPSS). In addition, correlation analyses and Regression analysis were to test the hypotheses.

RESULTS

Results analyses were carried out to assess the effect of Critical Thinking on Academic Performance among school students.

Correlation Analyses: This is a correlational study based on empirical data from the students of an educational institution. The correlation between each of the dimensions of variables has been analyzed.

Table 1.1a -Correlation between Critical Thinking (Expectancy, Attainment, Utility, Interest value, Cost) and Academic Performance (Academic Success, Academic Productivity, and Impulse Control)

	CTMS- Exp	CTMS- Att	CTMS- Ut	CTMS- In	CTMS- Ct	APRS- As	APRS- Ap
CTMS- Att	.685**						
CTMS- Ut	.415**	.594**					
CTMS- In	.472**	.466**	.402**				
CTMS- Ct	.392**	.538**	.468**	0.105			
APRS- As	-0.018	0.044	0.077	-.145*	.247**		
APRS- Ap	.175*	.302**	.255**	-0.036	.442**	.860**	
APRS- Ic	0.071	0.025	-0.104	-0.055	0.089	0.009	0.113

** . Correlation is significant at the 0.01 level (2-tailed)

* . Correlation is significant at the 0.05 level (2-tailed)

#CTMS-Exp(Expectancy), CTMS-Att(Attainment), CTMS-Ut(Utility), CTMS-In(interest value), CTMS-Ct(Cost), APRS-As(Academic Success), APRS-Ap(Academic Productivity), APRS-Ic(Impulse Control)

Table 1.1 a) shows that the dimension of Academic success was positively correlated with the cost ($r=.25$, $p<0.01$) and interest ($r=-.14$, $p<0.05$) dimension of critical thinking. The expectancy dimension is negatively correlated with critical thinking. Also, the Attainment and utility dimension of critical thinking is not related to academic success.

Academic productivity is significantly correlated with the dimensions of critical thinking like expectancy ($r=.175$, $p<0.05$), attainment ($r=.302$, $p<0.01$), utility ($r=0.25$, $p<0.01$) and cost ($r=.44$, $p<0.01$) but interest ($r=-0.04$, $p>0.01$) dimension of critical thinking is negatively correlated with academic productivity.

Impulse control is negatively correlated with both the dimension utility ($r=-.104$) & interest ($r=-.055$), and there was no correlation between the expectancy ($r=0.071$), attainment ($r=.025$) and cost ($r=0.089$) dimension of Critical thinking.

Step-wiseRegression Analyses: Regression analyses were carried out to investigate whether independent variables determine the study's dependent variables. Critical thinking is taken as independent variables, and the three dimensions of academic performance (Academic Productivity, Academic Success, and Impulse control) are taken as dependent variables.

Table 1.1b - Stepwise linear regression: Critical thinking (5; Expectancy, Attainment, Utility, Interest value, and Cost) as IVs and Academic productivity as DV

Model	R	R2	F (Eqn)	Sig. of F	β	Sign of β	Variable
1	.44	.195	48.060 df (1,198)	.01	.442	.01	Cost Value

Table 1.1b shows the regression analysis results, which have been carried out between Critical thinking and Academic Productivity. This regression analysis shows that the cost value dimension of critical thinking predicted the academic productivity dimension of Academic Performance ($\beta = .44$, $p<.01$).

Table 1.1 c) - Stepwise linear regression: Critical thinking (5; Expectancy, Attainment, Utility, Interest value, and Cost)as IVs and Academic success as DV

Model	R	R2	F (Eqn)	Sig. of F	β	Sign of β	Variables
1	.25	.061	12.865 df (1,198)	.01	.25	.01	Cost Value
2	.30	.091	6.392 df (1,197)	.05	.27 -.17	.01 .05	Cost Value Interest Value

Table 1.1c shows the regression analysis results, which have been carried out between Critical thinking and Academic Productivity. This regression analysis shows that the cost value dimension and Interest value dimension of Critical thinking predicted the academic success dimension of Academic Performance ($\beta = .27$, $p < .01$), and ($\beta = -.17$, $p < .05$) respectively. Therefore, hypothesis H1.2 was partially accepted.

This regression analysis shows that there are none of the dimensions of critical thinking that predict the impulse control dimension of Academic Performance. Overall, the findings of the study showed that the Cost value dimension of Critical thinking emerged as a better predictor of academic performance.

DISCUSSION

The present study also confirms previous studies on the notion that critical thinking impacts a student's academic performance. It has found that the two dimensions of academic performance, including academic success and academic productivity, are correlated with the dimensions of critical thinking. However, the impulse control dimension of academic performance shows no correlation with critical thinking dimensions.

Critical thinking abilities are essential for people to succeed in numerous aspects of their lives, including personal concerns, since they aid in improved decision making, creativity, problem-solving, and learning. In addition, evidence supporting the impact of critical thinking suggests that critical thinking positively impacts the average academic performance of students (D'Alessio et al., 2019). Critical thinking is the objective analysis of facts to form a judgment and is self-directed, self-disciplined, self-monitored, and self-corrective thinking. It may also entail critical thinking and logical reasoning and assisting individuals in asking questions, solving problems, and challenging ideas.

Academic achievement is a critical component for anyone planning to work because it is necessary for many jobs. It also aids in the development of self-esteem and self-confidence. In addition, it is critical to improving your ability to apply the information you have learned for better academic performance. Different cognitive skills, such as critical thinking, can help to improve this. The cognitive determinants of academic achievement have been examined in several types of research (Morosanova, 2015; Furnham et al., 2003). This research also shows that critical thinking is

a predictor of academic performance except for one of its dimensions, impulse control which means overall critical thinking is a good predictor of academic productivity.

The development of each person's creative potential is emphasized in the National Education Policy, 2020. Furthermore, it is built on the notion that education must promote cognitive skills, including "foundational abilities" like literature and numeracy, and "higher-order cognitive skills" like critical thinking, problem-solving, and social, ethical, and emotional talents. So, this study, which emphasizes the importance of critical thinking on students' academic performance, provides much applicability in the education field.

Conclusion

The dimension, Academic success, is positively correlated with the cost and interest dimension of critical thinking. Likewise, the attainment and utility dimension of critical thinking is not related to academic success. Academic productivity is significantly correlated with expectancy, attainment, utility, and cost dimensions of critical thinking, but the interest/interest value dimension of critical thinking is negatively correlated with academic productivity. The cost dimension of critical thinking predicts academic productivity, and also the cost and interest dimension of critical thinking predict the academic success dimension of Academic Performance. Thus, overall critical thinking is a good predictor of academic productivity.

Implications

The results obtained can be used for policymaking by the government in the education sector to design and implement the policies to improve the students' performance. This research may also be used as a reference point for incorporating a critical thinking pedagogy into the course curriculum, which will aid students in developing their skills and dispositions, hence enhancing their academic performance. The findings of this study might be used as a guide for researchers in teacher education, administrators, and education officials and a training strategy to improve the quality of teacher education. It may also be used to identify and study learning processes to better understand how people acquire and retain new knowledge. Finally, the findings will help academically challenged youngsters reconsider and change their behavioural habits and strategies. Using critical thinking tactics in the classroom can help students get the skills they need to deal with current societal affairs. Teachers are the moderators of society, and as such, they play a vital role in the development of a child's skills and dispositions. Thus, including critical thinking in teacher training will aid in enhancing students' overall performance.

Limitations

The students were assessed only on the variables under study. There may be differences in the quality of education and facilities in the school type, which were not considered. No interventions were administered. Socio-economic status and other such variables were not taken into account in the present study.

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19. Scope of Education for Sustainable Development [ESD] in India: A Systematic Review of CBSE Curricular Provisions

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Abstract

United Nations through UNESCO has in the past decade put sustainable development in the foreground of all its development initiatives, policy and plans. Continued efforts in SD have accentuated the need for building a future workforce that is prepared to take up environmental, social and economic challenges in more sustainable ways than their present trainers can envisage. All the 40 chapters of Agenda 21(UNESCO) describe school education as the most promising way of influencing human attitude, action and commitment toward a sustainable way of living. Volumes of UNESCO literature scientifically establishes how the global challenges that apparently look unlinked; are constituent parts of same crises. Their remedies therefore need to be in accord too. Economic slump, perpetually growing cultural intolerance, social disproportions, environmental degradation, gender and human rights insensitivity are pieces of a mammoth jig saw puzzle. Logically, academia world over upheld that a competency based learning is needed to address this pressing crisis. This is where Goal 4 of 17 SDGs (sustainable development goals) talks about ESD. UNESCO implores all the member states of UN to work on SDGs particularly ESD as this is the chief 'preventive measure' to the global crisis.

This paper builds on these understandings and involved a systematic inquiry into CBSE documents and circulars pertaining to Class IX. Data from this extended text analysis was synthesised to present a comprehensive and coherent approach of CBSE towards implementation of ESD related aspects. These were then used as the basis for analysing the scope and presence of three essential elements of ESD: Locally Relevant, School-Wide Approach, and Competency Building and related implications in the national context. The study shows that through its policies CBSE seeks to empower the affiliated schools to develop ESD related competencies [knowledge, issues, perspectives, skills, and values] in students and integrate them into the curriculum. It also conducts training and capacity building programmes to address contemporary ESD challenges and to create sustainable school communities and future society. Considering the fact that it has immense futuristic relevance, it corresponds to CBSE's aim of building future-ready citizens. Findings show that there is a need for a

forum for public participation and networking among stakeholders regarding ESD which can provide exchange and consultations to develop relevant material for ESD. This will fulfil the needs for capacity- building through requisite training, development of guiding literature and mutual sharing of experience and resources. Achieving this feat can showcase ESD initiatives of India, regionally and internationally.

Key Words: Sustainable development, Global challenges, CBSE, Curricular Provisions

Introduction

United Nations through UNESCO has in the past decade put sustainable development in the foreground of all its development initiatives, policy and plans. Continued efforts in SD have accentuated the need for building a future workforce who is prepared to take up environmental, social and economic challenges in more sustainable ways than their present trainers can envisage. All the 40 chapters of Agenda 21 describe diverse sustainable strategies ranging from reclamation of lost fertile lands to on-site waste management. Each of these chapters explicitly mention education as an essential instrument for a more sustainable future. It draws attention towards the fact that it is through Education for Sustainable Development (ESD) that the coming generations of global citizens-statesmen, entrepreneurs, judiciary, law makers, teachers and health workers are prepared for life-long learning.

To reorient existing curriculum in various member states; UNESCO guides them to identify the ESD competencies [knowledge, issues, perspectives, skills, and values] and substantiate these into the curriculum. The teaching community also needs to address sustainability and analyze which sustainability issues are locally most pertinent and ought to be part of the curriculum. Ideally, national or local sustainability goals should define the ESD curriculum. A thoughtfully aligned ESD curriculum (to the educational goals) will address local environmental, social, and economic agendas. The reorienting process is to be scaled up then to state levels or national levels where the mandated curriculums are actually prepared.

The Central Board of Secondary Education (CBSE) is a national level board of education controlled and managed by Union Government of India. There are approximately 21,271 public and private schools in India and 220 schools in 28 foreign countries affiliated to the CBSE. It has therefore a wide reach and impact; not only these schools but its policies also influence the state-level boards in India.

Research Questions

1. What are the ESD- based educational initiatives of CBSE?
2. Do these initiatives address locally relevant issues?
3. Are these initiatives advocated to be executed in the school-wide approach?
4. How do these initiatives propose to build ESD competencies?

Methodology

This study utilizes content analysis of key documents, circulars, notifications and reports to examine patterns in communication from CBSE to the affiliated schools regarding ESD in a replicable and systematic manner. Comprehensive summary of circulars (available in the archives of the CBSE website) relevant to a [research questions](#) is created to elucidate the scope of ESD at secondary school level in the current education scenario. The research findings include analysis of more than 200 circulars issued by CBSE in last 5 years at the time of writing, promising more reliable and better evidence. To minimizing bias and make valid inferences from the content; intra-coder and Inter-coder reliability has been established. The study is delimited to three elements of ESD: being locally relevant, being executed in the school-wide approach and building of ESD competencies.

Analyses and Findings

Competency Based Education (CBE)

CBSE issued Circular No. Acad-05/2019 in January 2019, which asserts for Teaching-Learning process methodically based on Learning Outcomes in all its affiliated schools. The circular laid emphasis on Competency Based Education (CBE) and asked schools to align their pedagogy to the LOs prepared by NCERT.

The Board clarified that the underlying focus of CBE is to attain the subject specific concepts applicable in real-life. It is detailed that CBE is best implemented through joyful participatory pedagogies. For this multi-disciplinary and trans-disciplinary approaches; infusing art and sports with academic subjects was suitably proposed and adequate training was conducted through various online portals. These pedagogies are considered most appropriate to develop skills of reflection, critical thinking, creativity, self-direction and key competencies that they require to face real life challenges of the present day and future world.

It further invokes the Draft National Policy of Education 2019; where Art-Integration is a suggested as transdisciplinary approach that utilizes diverse aspects and forms of art and culture as pedagogy for school subjects.

To popularise ESD in India; CBSE collaborated with UNESCO Executive Board in October 2014 for “UNESCO-Japan Prize on Education for Sustainable Development (ESD)” instituted by the Government of Japan to reward individuals, institutions, organizations or other entities for exceptional projects and programmes in the field of ESD, [Circular No. Acad-23/2019]

The SEWA [Social Empowerment through Work Education and Action] project of CBSE

It is well known dogma in education that along with cognitive learning, affective and psychomotor domains are equally pertinent. The SEWA project was initiated in the academic session 2018-19 by CBSE. This educational practice is aimed to orient children’s psyche in collaborative activities. This will also facilitate creative and critical thinking with positive outcomes. This is also essential because a harmonious coexistence with peers, community and resources; students need to inculcate values and social skills.

SEWA project is valid implementation of the integrated vision of education embellished in the four pillars of learning, to know, to do, to be and to live together (Delors, J. et al, 1996). The SEWA project is seen to build self-confidence and self-esteem in the students and to instil respect for co-habitants of natural and social ecosystems.

Through this educational initiative students are implored to recognize their responsibility towards the marginalised, the CWSN (Divyang), the less privileged of the society, and the much neglected natural ecosystems. The manner in which the board brings out the SEWA Philosophy; it presents an ideal ESD implementation where the ecological, social responsibilities, citizenship, mental and physical health and global issues are interconnected to enable the tender mind of the child recognise that all these are; deeply intertwined. The Social Empowerment component of SEWA philosophy, refers to the concept of service to the community that provide opportunities of participatory and active learning to develop key competencies such as self- awareness, empathy, creative and critical thinking and meaningful communication.

Another remarkable tenet of the project is that the teachers are asked to be non- judgemental towards mistakes of learners while working on a task. However, mindfulness and insights by students widen the scope for the learner. The project aims to train the adolescents to resolve their moral dilemmas and arrive at logical conclusions in ambiguous life situations. The Board highlights the responsibility of the school as a community which needs to be developed and established as a caring community that boosts collaboration, tolerance and skills of citizenship in the fabric of school education.

The CBSE expression series

The expression series is literature and arts competition organised for classes 1 to 12 by the Central Board of Secondary Education (CBSE) in India. It was initiated in academic session 2014. Students can submit entries in English or any of the 22 scheduled Indian languages. When the educational practice was conceived; its purpose was to create awareness for the glorious history of India, the great leaders, freedom fighters and their contributions. However, it broadened its horizons over the years and the first Expression Series for the session 2020-21 is about 'Thinking Innovatively in Unprecedented Times like Covid-19'.

Ek Bharat Shrestha Bharat (EBSB) Programme

'Ek Bharat Shrestha Bharat' (EBSB) programme initiated by the CBSE aims to create cultural awareness and celebrate diverse ethnic cultures thriving in States and UTs of India. It also aims to promote greater mutual understanding and bonding among them. In the session 2019-20 it has been dovetailed with art- integration programme of the Board.

Creative and Critical Thinking (CCT) Weekly Practice Programme

This educational programme is targeted to enhance critical and creative thinking and to generate curiosity in students by connecting learning with real life situations.

Handbooks and Manuals

In continuation with its efforts to reach the value-based global standards of education, Central Board of Secondary Education has had initiated the preparation and release of Handbooks on recent qualitative measures taken by the Board. Several Handbooks and manuals such as manual on 21st Century Skills, Handbook for Students and Handbook on Cyber Safety have been developed for self-help of students. In this context, vide circular no. 17/2019 dated October 1, 2019 and circular no. 18/2019 dated November 4, 2019, CBSE has released a total of 14 such Handbooks /Manuals.

Locally Relevant Issues

CBSE has initiated programmes based on socio-economic and environmental issues which are currently relevant. The selected themes are age appropriate and suitable pedagogy is detailed with each of such programme or event:

- Circular Nos: Acad-79/2019, Acad-31/2018, Acad-28/2018, Acad-15/2018, Acad-54/2019, Acad-44/2019, Acad-42/2019, Acad-33/2019, Acad-40/2019, Acad-43/2017, Acad-20/2016 and Acad-18/2016 focus on enabling students for conservation of natural resources.

- Circular Nos: Acad-45/2018, Acad-44/2018, Acad-73/2019, Acad-72/2019, Acad-76/2019, Acad-63/2019, Acad-59/2019, Acad-58/2019, Acad-32/2019, Acad-14/2018, Acad-44/2017, Acad-40/2017, Acad-46/2016 and Acad-03/2016 aim to provide integrated value education, foster cultural integrity and develop citizenship skills and values.
- Circular Nos: Acad-28/2019, Acad-23/2017 and Acad-19/2016 are issued to the affiliated schools briefing about educational activities and practices on gender sensitization and child rights.
- Circular No Acad-28/2016 elaborates on examination reforms for inclusive education
- Circular No Acad-14/2016 guides the affiliated schools to observe ‘no tobacco day’. Targeting adolescent issues such as substance abuse and addictions is an area that can no longer be brushed under the carpet and has to be tackled proactively by the schools.
- Circular Nos: Acad-32/2018 and Acad-32/2017 reiterate Guidelines for Safe and Effective Use of Internet and Digital Technologies in Schools and directs schools to refer to the manual issued from the board.
- As our country fought the Covid-19 pandemic; CBSE Vide its Circular No: Acad 45/2020 (dated July 1, 2020) appraises schools that in association with Tata Trust; the board has developed an interactive online course on ‘Experiential Learning’ for the Teachers and School leaders which aims to share multi-sensory pedagogic processes involved in experiential learning, sensitise Educators to implement it in their classrooms and make learning more joyful, reflective and multidisciplinary. This is aimed to create opportunities for children to learn the skills and competencies for the 21st Century and to promote experiential learning pedagogy in schools, CBSE
- Circular No.: Acad 33/2020 reiterates its policy to promote an art- integrated academic curriculum in the schools. This initiative is multidimensional and targets several locally relevant issues such as national integration, respect for culture, tolerance, and brotherhood simultaneously.

School-Wide Approach

In all its policies, circulars and manuals CBSE reiterates that it is mandatory for all schools affiliated to CBSE to follow the School-Wide Approach regarding Learning Outcomes in teaching and learning as it is central to the learning process in CBE.

- Circular No. Acad-34/2020 provides a bird’s eye view of the entire scheme of implementation of Competency Based Education (CBE)
- Circular No: Acad 33/2020, Acad- 27/2020, Acad-24/2020, Acad-83/2019, Acad-82/2019, Acad-78/2019 and Acad-68/2019 focus on holistic well- being of the students.

- Circular No: Acad 32/2020 and Circular No: Acad 31/2020 focus on the capacity building programs of the teachers to facilitate School-Wide Approach of its programmes.
- In Circular No. Acad-24/2020 CBSE implores the triad of children, parents and teachers to work on problems that they see in their community, environment and geography.

Competency Building

CBSE Vide its Circular No: Acad-44 /2020; dated 02/07/2020, apprised schools that the Board in collaboration with Central Square Foundation, New Delhi has brought out two series of cartoon books namely ‘Cogito’ and ‘The Question book’ to promote ‘Competency Based Learning’(CBE). It further emphasised that cognitive skills related to Problem Solving and Communication are indispensable for successful living in the future times. These Comic Books will focus on the importance of these competencies and to promote them through joyful reading and CBSE has also indicated that this will substantiate students’preparation towards PISA 2022. In ‘Cogito’, the animated conversation between Ankit and Ankitabrings out epistemological underpinnings of learning through logic in a lucid manner. Learning the process of problem solving and being able to apply it to the life situations is of utmost importance. In the other series “The Question Book”; Ankita and Ankit have discussed the process of problem solving.

Circulars no: Acad-40/2020 and Acad-37/ 2020 also mentions about enhancing students’ imagination, critical thinking, language skills, focus, memory and concentration through reading exercises.

Conclusion

The study brings to fore that through its various circulars and documents CBSE seeks to empower the affiliated schools to develop ESD competencies [knowledge, issues, perspectives, skills, and values] and integrate them into the curriculum. It also conducts training and capacity building programmes to address contemporary ESD challenges and to create sustainable school communities and future society. The pandemic times have proved beyond doubt that we cannot be prepared with solutions to future problems but we canprepare the future-ready generations who can deal with problems effectively and find their solutions.

Educational implications for future:

A national or state level process can be initiated that would include coming together of stakeholders to gather input regarding needs, rationale and opinion related to the implementation of ESD Initiatives. In this way a forum for public participation and networking among stakeholders will be created. Multi-stakeholder consultations can provide exchange and consultations; events, seminars,

public events can develop literature for ESD—lesson plans, assessment modules, manuals for educational practices etc.

This will fulfil the needs for capacity- building through requisite training, development of guiding literature and mutual sharing of experience and resources. Achieving this feat can showcase ESD initiatives of India, regionally and internationally.

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20. Technostress among School Teachers: A Survey

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Abstract

The information communication and technology(ICT) has played a significant role in all professions especially in educational institutions. Teachers play a pivotal role in integrating technology into the teaching-learning process. Due to increasing pressure exerted by the institutions related to the use of technology, the teachers experience technostress i.e. psychological state as a result of inability to cope with new technologies. The present study was taken up to study technostress faced by school teachers. A sample of 127 school teachers was selected from Amritsar city, Punjab, India. A scale developed by COKLAR, EFİLTİ and ŞAHİN (2017) having 28 items and five factors was used to collect data. The five Factors were Learning-Teaching Process Oriented , Profession Oriented , Technical Issue Oriented, Personal Oriented and Social Oriented. The scale was sent through google form to the selected sample. Percentage analysis was used to analyse the data. The analysis revealed that technology use makes the teachers anxious and further revealed that it requires more effort and made them dependent on the internet. The teachers also agreed that the necessity to keep up with constantly developing technology worries them and the dependence on the individuals who are better at using technology makes them uncomfortable. The study has implications that providing technical supports and opportunities for professional development with hands on experience in mastering ICT will help teachers embrace technology to make the learning rich and effective.

Key Words: ICT, School Teachers,

Introduction-

The information communication and technology(ICT) has played a significant role in all professions especially in educational institutions. Educational institutions worldwide have been advancing their agendas for education enhanced by ICT, such as promoting mobile learning, blended learning, and virtual reality-based instruction (Markowitz et al., 2018; Qi, 2019). Admittedly, these efforts are laudable and could potentially benefit learners. Teachers play an important role in the integration of technology to the learning-teaching process (Ilie, Courtney & Slyke, 2007; Miller & Sim, 2004). However, this could also exert increased pressure on teachers who are often less technology-savvy than their students (Jena, 2015; Hatlevik and Hatlevik, 2018), but have to

constantly adapt themselves to the ever-demanding institutional requirements related to the use of technologies. As a result, the incongruence or misfit between institutions and teachers may lead to the latter being subjected to technostress, “which is defined as a modern maladaptation resulting from the failure to cope with ICT and changing requirements related to the use of ICT” (Brod, 1984).

Technostress was first conceptualized in the early 1980s as “a modern disease of adaptation caused by the inability to cope with new technologies in a healthy manner” (Brod, 1984), which can result into non-acceptance of ICTs or excessive identification with the new technologies, resulting in both anxiety and stress. Weil and Rosen (1997) defined “technostress as any negative impact on attitudes, thoughts, behaviors, or body physiology that is caused either directly or indirectly by technology”. Technostress is considered to be multidimensional, and it is defined as “a negative psychological state associated with the use or the “threat” to use new technologies,” which leads to “anxiety, mental fatigue, skepticism, and sense of ineffectiveness” (Salanova et al., 2007).

Techno-Stressors

A widely accepted scientific classification of the creators of technostress is proposed by Tarafdar et al. (2007) who used a transactional approach to describe five techno-stressors.

- **Techno-Overload:** The techno-overload is that situation when the forces work and work faster with technology.
- **Techno-Invasion:** They describes person are always exposed and potentially reached anywhere and anytime
- **Techno-Complexity:** They describe a situation where they are forced to work for spending time for learning and struggle to update their internet skills. They feel stressed when feel variety of application of computer
- **Techno-Insecurity:** They are connected with situation of people feel threatened about new gadgets and other computing devices
- **Techno-Uncertainty:** The techno-uncertainty refers to perfection of individual is unable to accurately predict to understand

ICTs have taken its place as a pedagogical tool in education as of 2000s (Peeraer & Van Petegem, 2015) and have been integrated to education systems rapidly (Trucano, 2005) because of the facilities it provides. It has become inevitable for teachers to integrate ICT in the teaching learning process. Teachers especially experience techno-stress in the process of integrating new technologies, which is a common condition encountered with digital technologies (Lei, 2010; Lei & Zhao, 2007; Zhao &

Frank, 2003). Moreover, the continuation of the pressure for technology integration in education both from the institutions and the society, and the lack of knowledge and support result in techno-stress among teachers (Longman, 2013). The present study has been taken with the purpose of studying technostress among school teachers.

Objective- To study the technostress faced by teachers while integrating technology in the teaching learning process.

Methodology

Method- This research falls under descriptive research. The data was collected online through the google form using a scale developed by COKLAR, EFİLTİ and ŞAHİN (2017) .

Research Design- A self-designed online survey was used to find the level of technostress faced by teachers. Percentage analysis was used to analyse the data.

Sampling –Purposive sampling technique was used to select sample as those teachers were required who were integrating technology in the teaching learning process. A sample of 127 school teachers was selected from Amritsar city, Punjab, India.

Tool used-A scale developed by COKLAR, EFİLTİ and ŞAHİN (2017) having 28 items and five factors was used to collect data. The five Factors were Learning-Teaching Process Oriented , Profession Oriented , Technical Issue Oriented, Personal Oriented and Social Oriented. The scale was sent through google form to the selected sample.

Analysis and Interpretation of Data

Table 1.

Demographic Profile of sampled teachers			
Variables	Sub-Variables	N =127	%
Gender	Male	27	21.3
	Female	97	76.4
	Other	3	2.3
Type of School	Government	60	47.2
	Private	69	54.3

School Level	Elementary	17	13.4
	Secondary	113	89.0
Stream	Science	47	37.0
	Humanities	88	69.3
Length of Service (in years)	0-5	57	44.9
	5-10	22	17.3
	10-15	27	21.3
	15 +	22	17.3

The table 1 reveals the demographic profile of the selected sample. It can be seen that 21.3% were males and 76.4% were females. 47.2% of the teachers were from government schools and 54.3% were from private schools. Further, it can be seen that 89% teachers were from secondary level and 13.4% from elementary level. As regards stream is concerned, 37.0% teachers were from science stream and 69.3% were from humanities. 44.9% teachers had upto 5 years of experience, 17.3% were with 5-10 years experience, 21.3% had 10-15 years of experience and 17.3% had more than 15 years of experience.

Table 2

Factor 1-Issues of Learning-Teaching Process with "Technology and Education"	Likert Scale (in %)				
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
The idea that I won't be able to teach the whole course content, because technology use takes time makes me anxious.	6.3	53.2	15.1	18.3	7.1
I think that technology use require more effort in the classroom.	18.9	53.5	13.4	13.4	14.2
I feel forced to become more dependent on the Internet in the educational process.	22.8	42.5	16.5	14.2	4

I am worried because digital-technology oriented materials are becoming more common in the educational process.	12.6	47.2	20.5	15.7	4
I feel uncomfortable that technological devices are used for extra-curricular purposes during the lessons by the students.	15.1	33.3	15.9	31.7	4
I feel uncomfortable that technology leads everyone in the educational environments to laziness and freeriding.	22	33.9	15	23.6	5.5
I am worried that technology use blunts students research skills	13.5	42.9	17.5	22.2	3.9

As observed from table 2, the percentage analysis of the issues of learning- teaching process with “Technology and Education” reveals that the highest %age of teachers agree that technology use makes them anxious, requires more effort and have become dependent on the internet . The teachers have also shown their concern that the digital technology has overshadowed the educational process thus leading everyone in the educational environment to laziness and freeriding. The teachers also agreed that the technological devices are used for extra-curricular purposes during the lessons and were worried that technologies blunt students’ skills.

Table 3.

Factor 2 -Professional Oriented Issues with "Technology and Education"	Likert Scale (in %)				
	Strongl y Agree	Agree	Neutral	Disagree	Strongly Disagree
I think technology use makes the teaching profession more difficult.	4	24.8	16	47.2	8
I think the teaching profession is losing its value, because information sources have become technology Oriented.	11.2	42.4	12	28	6.4
I am worried that educational understanding might change because of technological	11.2	42.4	24	20	2.4

devices.					
I am worried that I might get unemployed in the future due to technology use.	16	24	19.2	36.8	4
I am worried that I might lose prestige, because newer teachers can use technology better.	10.4	28.8	16.8	36	8
I think technology use has increased teachers' workload	17.6	42.4	16	18.4	5.6

The analysis of professional oriented issues with technology and education revealed that the maximum percentage of teachers disagree that technology use makes the teaching profession more difficult, might get unemployed in the future due to technology use or lose prestige because newer teachers can use technology better. On the contrary, the teachers have agreed that the teaching profession is losing its value because information sources have become technology oriented, educational understanding might change because of technological devices and technology use has increased teachers' workload. Vladut and Kallay (2010) have also underlined that the increase of teachers' workload creates technostress among teachers.

Table 4.

Factor 3-Technology Oriented Issues with "Technology and Education"	Likert Scale (in %)				
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I feel uncomfortable, as I am constantly worried about infecting technologies with viruses.	14.5	42.7	29	12.9	0.9
I am worried that data I store in digital environments (memory sticks, Internet, etc.) can be lost, or change hands.	19.4	45.2	16.9	16.1	2.4
I am worried because there is too many information (password, account name, etc.) to remember for technological environments, and I might forget these.	22.8	39.8	17.9	17.1	2.4

I feel uncomfortable because technology costs a lot (purchase, repair and maintenance, paid websites, etc.).	20.2	44.4	15.3	17.7	2.4
I am worried about the negative effects of technological devices within the classroom (noise, heating, etc.).	22	45.5	11.4	17.9	3.2
I am worried about the security of technological devices (storing, keeping, etc.) at the school.	22.8	46.3	13	16.3	1.6

The analysis of technology oriented issues with technology and education reveals that maximum percentage of teachers agrees with all the issues. This clearly shows that teachers are constantly worried about infecting technologies with viruses, loss of data, too much information to remember for technological environment, cost of technology, negative effects of technological devices within the classroom and security of technological devices

Table 5

Factor 4-Personal Oriented Issues with "Technology and Education"	Likert Scale (in %)				
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I am worried that I might not be able to learn using technology, even if I want to.	11.5	20.5	26.2	32	9.8
I am worried about technology use, due to the necessity to keep up with constantly developing technology.	12.2	38.2	22.8	25.2	1.6
I feel uncomfortable with being dependent on the individuals who are better at using technology.	7.3	41.5	14.6	32.5	4.1
I might give up on using technology, as I cannot find sufficient opportunities for technology education.	9	32	21.3	33	4.7
I am uncomfortable because I am not familiar with the terminology used in	9	30.3	12.3	39.3	9

defining new technologies.					
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Table 5 reveals the personal oriented issues leading to techno stress among the teachers. It can be observed from the table that the maximum teachers have agreed that the necessity to keep up with constantly developing technology worries them and the dependence on the individuals who are better at using technology makes them uncomfortable. Whereas, it can also be observed that maximum teachers have disagreed that they might not be able to learn using technology, even if I want to and they feel uncomfortable because they are not familiar with the terminology used in defining new technologies. Further, the data also shows that almost equal percentage of teachers agree and disagree that they might give up on using technology, as I cannot find sufficient opportunities for technology education.

Table 6.

Factor 5-Social Oriented Issues with "Technology and Education"	Likert Scale (in %)				
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I feel uncomfortable that, digital technology use takes too much time.	13.6	33.6	12	33.6	7.2
I think social interaction between everyone in the educational processes is damaged due to technology use.	16.8	52.8	12	14.4	4
I am worried that I can have problems with my colleagues about technology use.	8	9.6	23.2	34.4	24.8
I am worried that technology use can cause health problems (sight, hearing, pain, etc.).	34.4	40.8	11.2	12.8	0.8

As far as social oriented issues are concerned, the maximum percentage agreed that social interaction between everyone in the educational processes is damaged due to technology use and at the same time are worried that technology use can cause health problems. The data also reveals that the teachers disagreed that technology use can cause problems with colleagues. It may also be seen from

the analysis that equal percentage of teachers agree and disagree that digital technology use takes too much time.

Conclusion

Integration of ICT has become essential in the teaching learning process to augment its quality and also a necessity in the present times of pandemic. The present study revealed that teachers do experience stress due to technology. Longman (2013) suggested to provide more support, training, and guidance from the educational leaders as important methods in overcoming technostress among the teachers in classrooms. More importantly, providing technical supports and opportunities for professional development with hands on experience in mastering ICT will help teachers embrace technology to make the learning rich and effective. Consequently, it will reduce the level of technostress without having the sense of heavy workload and job insecurity resulted from teachers' inability to deal with ICT.

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21. Effectiveness of Seminar and Webinar in Learning Experience: An Empirical Analysis

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Abstract

The purpose of this study is to analyse the effectiveness of online mode of instruction (webinar) vis-à-vis offline mode of instruction (seminar) and the possible policy implications therein. This Study has been conducted by using the primary data collected from 253 scholars ranging from students to professors through questionnaire. Various factors were taken into consideration for investigation based on review of literature. The results throw light on why scholars favour one mode of learning over the other. This study has repercussions on policy implications concerning effectiveness of distance education involving intergenerational age differences on learning outcome and experience.

Keywords: Online and Offline mode of learning, Seminar, Webinar, Intergenerational age differences, learning outcome

Introduction:

In the backdrop of Information Communication Technology (ICT) revolution the entire world is witnessing a shift in the learning mechanism (Buchanan, (1999), Peters, O. (2000)) Traditional methods are being supplemented with ICT platforms to increase the efficiency of learning outcomes (Internet Society, 2016, Kumar, B. A., Goundar, M. S., & Chand, S. S. 2020). There are studies (Wan Faezah, A., & Nor Aini, A. R., 2012) which support combining traditional methods with ICT platforms results in better learning outcomes. To be more specific active supplementation of traditional methods with ICT is found to enhance critical thinking skills (Wang, Q., Güzer & Caner, 2014) as well as play a role in enhancing communication skills. However, given the increasing trend of exposure to online learning, there is indeed a requirement to develop new techniques to make the same efficient (Bott & Edwards, 2014). In this context, this paper tries to analyse the effectiveness of Online learning (webinar) vis-à-vis offline learning (seminar) Comparative analysis of these two modes of learning does have practical relevance in the backdrop of many Indian Universities offering Bachelor and Master Programmes in online mode. The study also gives an opportunity to evaluate if differences in age do affect learning outcomes of individuals who undertake these online courses

Literature Review:

There are several studies which have concluded that use of technology in class does ensure more efficient learning outcomes (Gomleksiz,2004). The use of technology also gives ample scope for having reflection about the content among the scholars (Susman,1998). Marcinkiewicz (1994) has identified the prominent factors which determine the use of Computers by teachers in their class room. However, learning is a two way process, it not only depends upon the familiarity of teachers in use of technology, but it also depends upon the ease with which the students are able to access and utilise the same to enhance their learning experience (ACSDE, 1999). According to some studies, it was found that, current generation learners (Harris & Rea, 2019) were in more ease with regards to using technology than teaching facilitators (Bousbahi & Alrazgan, 2015). Since even in India, as online courses are gathering more leverage, it becomes pertinent to analyse as to whether the results of the above study holds true in context of Indian learners and to analyse whether it is conducive in the backdrop of Indian Policy framework. Moreover, in a multilingual country like India, it would be interesting to analyse what would be chosen as lingua franca for dispersion of knowledge via online mode, given the plethora of languages that India has. In the plain sight, English would seem to be the first choice. However, there are several studies (Xiong (2008) Sabti and Chaichan (2014)) which have identified English as a prominent barrier in using ICT for E-learning. In this backdrop, this study tries to analyse the acceptability of English among the scholars in the context of online learning. There are studies which have analysed the factors affecting online studies (Buchanan, 1999; Draves, 2000; Liu & Ginther, 1999).

Though teachers do play an important role in promoting use of technology in teaching pedagogy (Alharbi & Drew, 2014, Siyam, 2019) the perceived usefulness (Davis, 1989; Lai & Savage, 2013) of the learning technology plays an important role in determining the usefulness of the same, which is in turn is affected by external variables(Shih-Chih et al., 2011). Thus, this study tries to analyse the perceived usefulness among the learners with respect to online and offline platforms, along with external variables in Indian context. It is to be noted that, in context of India there are very few studies which are based on empirical and inferential approach which have been undertaken for making a comparative analysis between online and offline mode of learning. The present study tries to address the same issue.

Specific objectives of the study:

- To analyse the impact of intergenerational differences in learning experience.
- To analyse whether the area and infrastructure of residence among the scholars does play an important role in influencing their chosen mode of learning.
- To analyse whether any particular stream, of students are more adept in choosing one form of learning over the other.
- To analyse whether the choice of platform has an impact on the medium of instruction.

Hypothesis:

- Socio-Academic Variables like residence, designation, stream of learning & medium of instruction are having influence on preference for the mode of learning.
- Intrinsic factors like ease of access, registration process, hands on experience, academic networking along with clarity of understanding are having influence on selection for mode of learning.

Methodology of the Study:

The study is based on analysis and interpretation of primary data. The investigator has made use of online survey method to collect the data via google forms. The questionnaire was prepared and sent via electronic mail to 1643 scholars belonging to Science and Humanities streams who had recently attended E-Workshops conducted by the Departments of Davangere University. Out of 1643, 253 scholars (15.39%) have responded. The questionnaire was prepared to collect information about 13 variables. The reliability of the questionnaire was established via Cronbach's alpha which reflected a scale reliability coefficient of 0.76 which does reflect acceptable reliability. 'Chi- Square' test was used test the significance of association between the socioeconomic attributes of the respondents and their preference for seminar or webinar.

Results and Discussion

Preference for one mode of learning over the other is expected to be influenced by various socio-academic and intrinsic factors. Socio-academic and intrinsic factors which are likely to influence the choice of mode of learning have been selected based on the experience gained from the review of literature.

Table-1: Socio-academic attributes of respondents and their preference for mode of learning

Socio-Academic Background	Sub-Divisions	Seminar	Webinar	Total	Chi-Square Value
Residence	Metro	15 (50.0)	15 (50.0)	30(100.0)	10.083**
	Urban	72 (63.0)	41(36.0)	113(100.0)	
	Semi-Urban	8(32.0)	17(68.0)	25(100.0)	
	Rural	42(49.0)	43(51.0)	85(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Designation	Students	28 (40.0)	42(60.0)	70(100.0)	10.577**
	Research Scholars	40(58.0)	28(42.0)	68(100.0)	
	Asst. Professors	51(56.0)	40(44.0)	91(100.0)	
	Associate Professors & Professors	18(75.0)	06(25.0)	24(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Stream of Respondents	Social Sciences & Literature	62(58.0)	45(42.0)	107(100.0)	2.36
	Commerce and Management	07(39.0)	11(61.0)	18(100.0)	
	Science and Technology	68(53.0%)	60(47.0)	128(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Medium of Instruction	English	73(48.0)	79(52.0)	152(100.0)	5.75**
	English & Regional Language	64(63.0)	37(37.0)	101(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	

Note: Figures in parenthesis are percentage to respective row total

** indicates significance at 5 percent probability level

Primary data collected from the sample respondents pertaining to the socio-academic attributes and their preference for mode of learning have been consolidated in table-1.

Residential status of the respondents is expected to have greater influence on the choice of mode of learning as there is significant difference between metropolitan cities and rural area with respect to network connectivity, tele-density and other online mode learning environment. In the survey results,

it was interesting to observe that the percentage of respondents having preference for webinar were found to be relatively more among semi-urban (68%) and rural (51%) residents compared to the respondents from Metro and urban places. Preference for webinar over seminar was found to be lowest among respondents from urban area (36%) followed by metropolitan cities (50%). To arrive at a conclusive result for this anomaly, a separate and detailed investigation about such preferences needs to be undertaken. In order to test the statistical significance of association between the preference for mode of learning and residential status of the respondents, chi-square value has been calculated. The calculated chi-square value found to be statistically significant at 5 percent probability level. Thus, the null hypothesis could be rejected at 5 percent probability level. Hence, it could be inferred that the residential status does have significant influence on the preference for the mode of learning.

Designation of the respondents is another important variable which could influence the preference for mode of learning. Designation not only indicates the position in the profession, but it is closely associated with the age. Students are representing the youngest age group and Associate Professors and Professors are representing the most senior age group. It was observed that the percentage of respondents having preference for webinar were found to be relatively more among students (60%), followed by Assistant Professors (44%) & Research Scholars (42%) compared to Associate Professors & Professors (25%). Preference for seminar over webinar was found to be highest among Associate Professors & Professors (75%), who belonged to the oldest age group. This reflects that younger generation of learners are more adept in using technology for supplementing their learning. The association between the designation and preference for mode of learning was found to be statistically significant as per the chi-square test results. Hence, it could be inferred that the intergenerational differences does have significant influence on the preference for the mode of learning.

Stream of learning could be considered as another important variable which is expected to influence the mode of learning as the academic environment and learning process is different across various streams of learning. It was observed that the percentage of respondents having preference for webinar were found to be relatively more among scholars belonging to Commerce & Management (61%), followed by Science & Technology (47%) and Social Sciences & Literature (42%). These results were not along expected lines, as we had hypothesized that scholars belonging to science stream would be more inclined towards online mode of learning. Even Chi Square results found the

association between stream of scholars and preference for mode of learning to be statistically insignificant at 5 percent probability level.

In programmes where medium of instruction was English, 52% of respondents preferred webinar over seminar & in the programmes where medium of instruction was a combination of English and regional language, preference for seminar was relatively more to that of webinar. This was reaffirmed by calculated chi-square test, which reflected that, the association between the medium of instruction in the programme and the preference for mode of learning to be statistically significant at 5 percent probability level. Thus, we could infer that medium of instruction in the programme has its influence on preference for mode of learning.

There are many intrinsic factors of learning which could influence on mode of learning. Based on the review of existing literature, some of the most important intrinsic factors which could influence on the preference for mode of learning has been selected and they have been cross classified with mode of learning and consolidated in Table 2. With regards to Ease of access, out of 167 respondents who opined that there was ease of access to learning platform, majority favoured webinar. Thus, we could infer that, ease of access is better in webinar as compared to seminar. Moreover, among 86 respondents who opined that, they did not have ease of access to learning platform, majority were from seminar. It means that ease of access is better in webinar as compared to seminar. Since calculated chi-square value is significant at 1% probability level, we could infer that, presence of ease of access has greater influence on choosing the mode of learning.

Out of 253 respondents, 163 respondents found the registration process to be simple and 90 respondents opined that, they faced hurdles in the same. Out of 163 respondents who found simplicity in registration, maximum were from seminar. However, Chi-Square value calculated to test the significance of association between the relationship of registration process and mode of learning was not found to be statistically significant. With regards to Registration fee, among 253 respondents 137 opined that the fee was high. Among them, majority of them were from seminar. Interestingly enough, even among the respondents, who opined that, the fee was on the lower side, majority of the respondents were from seminar. Now, on the surface it may seem contradictory, but the association between registration fee and preference for mode of learning is not statistically significant.

Table 2: Intrinsic Factors of Learning Platforms Associated with Respondents' Preference for mode of learning

Intrinsic Factors	Sub-Divisions	Seminar	Webinar	Total	Chi Square Value
Ease of Access	Present	80(47.0)	87(53.0)	167(100.0)	7.719*
	Not Present	57(66.0)	29(34.0)	86(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Registration Process	Simple	87 (53.0)	76(47.0)	163(100.0)	0.11
	Faced Hurdles	50 (55.0)	40 (45.0)	90(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Registration Fees	High fees	76(55.0)	55(45.0)	137(100.0)	1.63
	Less fees	61(52.0)	61(48.0)	116(100.0)	
	Total	137(54.0)	122(46.0)	253(100.0)	
Hands on Experience	Present	100(64.0)	54(36.0)	154(100.0)	18.430*
	Absent	37(37.0)	62(63.0)	99(100.0)	
	Total	137 (54.0)	116(46.0)	253(100.0)	
Academic Networking	High Possibility	100(58.0)	71(42.0)	171(100.0)	3.98**
	Low possibility	37(45.0)	45(54.0)	82(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	
Clarity over Subject	High	102(59.0)	72(41.0)	174(100.0)	4.48**
	Low	35(44.0)	44(56.0)	79(100.0)	
	Total	137(54.0)	116(46.0)	253(100.0)	

Note: Figures in parenthesis are percentage to respective row total

* and ** indicates significance at 1 and 5 percent probability level respectively

When we analysed hands on experience, among 253 respondents, 154 respondents reported that, they did get hands on experience. Most of these respondents favoured seminar over webinar. Among the 99 respondents who opined that they could not get hands on experience, majority of them were critical of webinar. This was on expected lines, as seminar does provide greater chance to get hands on experience for the participants. Chi Square value to test the association between exposure to hands on experience and preference for mode of learning was found to be statistically significant at 1% probability. Thus, we can infer that, one of the major drawbacks of online learning mode is that it lacks personal hands on experience. Out of 253 respondents, 171 respondents stated that, there was high possibility of academic networking by participating in academic events. Majority of the

respondents acknowledged that, seminar was more conducive for academic networking compared to webinar. Among 82 respondents who felt that there was low possibility of academic networking, majority of them acknowledged that webinar was not conducive for the same. Chi-Square value to test the significance between the relationship concerning possibility of academic networking and preference for the mode of learning was found to be statistically significant at 5% probability.

When respondents' perception towards clarity over subject was analysed, 174 respondents acknowledged that there was clarity over subject, of whom majority favoured seminar. Among the other 79 respondents, who stated that, they could not get clarity over subject, majority of them were critical towards webinar. The test result clearly indicates the significant association between clarity over subject and mode of learning. Thus, we could infer that, clarity over subject would be better in seminar as compared to webinar.

Policy Implications:

The policy implications from the study are as follows:

- From the study we observe that, younger generation of learners prefer online mode of learning (Webinars) over offline mode of learning (Seminars) to enhance their knowledge. This, indeed, supports the Government's initiatives like Swayam Portal and other online learning Platforms established by universities with the aim of conducting distance learning programmes via internet. The study also found out that webinars are more popular among semi-urban and rural scholars. A strong reason for the same may be attributed to greater digital penetration due to Digital India Programme.
- According to the study, it was observed that, offline mode of learning (seminars), is popular among scholars, mainly due to personal hands on experience, it helps in getting more clarity over the subject along with provides ample opportunity for scholars to build academic network with fellow peers. Thus, this further reinforces the need for conducting National and International Seminars which are actively being funded by UGC, ICSSR and the like.

Limitations:

The limitations of the following study are as under:

- Generally speaking, online programmes would be more popular among urban scholars due to better availability of network; However the study found it otherwise. This may also be due to the reason that, rural scholars studying in urban settings may have identified themselves as belonging to rural areas.

- The data was collected in 2021 amidst pandemic, so it might have had influence among the opinion of the scholars.

Conclusion

Learning is a lifelong process. In the competitive world that we are living in, upgrading our skills and knowledge is very much relevant. With advent of modern inventions and innovations, Information communication technology has revolutionised academic pedagogy. The growing importance of online learning platforms is reaffirmed from the rising trend of offering online courses for Masters and Bachelors programs by reputed Indian Universities. In this context, it becomes pertinent to analyse, as to what extent, online mode of learning can substitute offline mode of learning. In this study, seminar was to be representative of offline mode of learning & webinar was to be representative of online mode of learning. The study was conducted with an objective of analyzing the effectiveness of offline vis-à-vis online mode of learning by considering socio academic and intrinsic factors affecting the same.

At a superficial level, webinar was preferred over seminar for its ease of access among the respondents. However, an interesting trend which was revealed in the study with regards to webinar was, online mode of learning was highly favored by younger generation of learners relative to older generation of learners, which does have significant policy implications. At the same time, seminar was preferred among the respondents for gaining personal hands on experience which enhanced their possibility of building academic network along with helping them to get a greater clarity over the subject. From this we could infer that, traditional method of learning has its own significance and relevance which cannot be completely substituted by online mode of learning. All in all, in a nation, wherein there is a need for optimum utilization of scarce resources to foster human capital formation, both offline and online mode of learning have their own significance and role.

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22. Transformational Framework for Innovative Assessment in Higher Education

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Abstract

The competitiveness and sustainability focus has been continuously increasing in the education sector. The gap between the skill sets of students and the needs of industry has necessitated innovation in the teaching-learning process. The assessment constitutes a significant aspect of the teaching-learning process of higher education since it helps to identify the extent of learning of students and also facilitates further learning. Further, it facilitates refinement of the curriculum and establishment of learning goals. The methods of assessment in higher education have evolved over the last few decades. For example, continuous assessments are being used in addition to summative assessments in higher education. Though different assessment methods are being used, there is a constant need to introduce innovation in the assessment process for enhanced effectiveness. The innovative approaches that have been practiced in assessment have not been adequate. Students need to enhance their skills and become “industry-ready” as quickly as possible, and this aspect assumes greater significance in a dynamically changing world. This has necessitated a shift towards the assessment of real-life scenarios. An innovative assessment approach based on Authentic assessment techniques is needed to enhance understanding of the learning level and skill enhancement of students. Authentic assessment enables assessing the learners from a knowledge application perspective. This paper highlights the significance of innovation related to assessment. The methodology for introducing innovation in the assessment process encompasses vision creation for the assessment approach and establishment of assessment goals, standards, tasks, as well as measures. It suggests an innovative assessment framework known as Form-Fact assessment by considering factual assessment design principles and incorporating these in a formative assessment approach. The paper indicates the various stages from assessment-related idea generation until implementation, and also maps the proposed framework with higher education assessment goals.

Keywords: Assessment goals, Assessment framework, Factual assessment design, Form-Fact assessment, Higher education, Innovation, Transformation,
Introduction

Assessment in higher education is mainly intended to periodically measure the learnings of a student in a specific course. This also helps to identify the strengths and weaknesses of an individual student

and the impact of assessment practices. It is a cyclic activity and needs constant improvisation based on students' performance and feedback. The assessment also helps to fine-tune the deliverables of the course and further map the course and program outcomes.

Effective assessment practices are always required to validate the learning of students in a particular course. There has been a continuous transformation in assessment patterns triggered by various requirements. Currently, different types of assessment methods exist in practice, and constant refinement and innovation to these practices are required to achieve better outcomes. Research reveals that higher education teachers use both authentic activities and examinations as assessment practice (Rawluysk, 2013). The objective of Authentic assessment is to evaluate the extent of new learning used by a student instead of the extent to which he or she remembers. It facilitates the learner to exhibit problem-solving skills, application of knowledge, and communication of the new information. While paper and pencil tests could be utilized, performance-oriented tasks such as portfolios, presentations and demonstrations could be utilized (Hoffman, 1996-2008). The practices related to assessment reflect and influence the way in which learning and teaching are envisaged and organized.

Background

Assessment methods in higher education keep changing frequently due to the growing number of students, change in curriculum, recruiter's demands, the learning potential of students, etc. The major influencing factor of assessment are psychometrics, theories of cognition, the nature of the curriculum, and the socio-political context of education—from 1957 through the present (Pellegrino, 2004). Assessment activities were meant to identify the areas of improvement in the curriculum and to establish accountability for learning goals so that overall education quality could be improved. Every assessment approach is unique and has a different end-objective. The evolution of assessment approaches indicates the growing aspect of gathering data about student learning and its analysis (Ewell & Cumming, 2017), rather than directly assessing learning outcomes using a test, the assessment of student engagement records the extent to which students have participated in defined learning activities (Kuh, 2009). To measure the outcomes appropriately it is required to adopt innovative assessment practices and these practices would enable the students to act strategically if they want to be competent in facing professional problems.

Nowadays authentic teaching and assessment practices are becoming popular. It means that the purpose and resolution conditions, as well as the demand, are accurate and very similar to those

found in real contexts of action (Monereo et al., 2009). Authentic assessments help the students to be active and reflect on their learnings immediately as and when the topic gets discussed. Authentic assessments focus on the learning process, sound instructional practices, and high-level thinking skills and proficiencies needed for success in the real world (Darling-Hammond & Snyder, 2000). Generally, authentic assessment procedures involve performance of tasks by the student and generation of the scores by the teacher using rubrics considering the extent of correlation between the final demonstrations and the criterion. Authentic assessment helps to transform remembering/recall of knowledge to the construction/application of knowledge. In Authentic Assessments, the teachers are encouraged to teach to the test. It is important for students to learn the method of performing well on meaningful tasks (Mueller, 2018).

Innovation in assessment need not be inventing assessment activities that no one has ever tried earlier. According to the study conducted by Bryan and Clegg (2006), these activities need to be innovative in the context of the course and the experience of students so that students respond to the task at hand and not to their preconceptions of what a particular assessment method does. Assessments are expected to be in line with “Know-how” phrases and this sets a new regime, (Eraut, 1994). There is an urgent need for higher education to give place to skills and values-centric education. There is a belief that when there is a change in curriculum it will have implications for assessment and vice versa, though in reality research has proved it to be minimal. Appropriate ICT-based tools could be selected to facilitate the implementation of innovative assessments that could include cognitive skills. Working towards meeting up with the requirements of Global competencies needs the following 21st-century skills to be focused namely creative thinking, problem-solving, communication, cooperation, and digital literacy (Care & Vista, 2017). The current assessment models are not in line with sustainable goals and need a reformation. We are in critical need of innovative assessment approaches that would bring out the hidden skills of the student and not just the scores.

Gap and Problem statement

A lot of innovation has taken place in various aspects of education including curriculum design. However, the extent of innovation in pedagogy and assessment aspects has not been substantial. Traditional assessment approaches constitute a post-mortem activity and do not meet the skill development needs. There is no effective way to measure the outcome of continuous learning based on continuous assessment. Another challenge is the “industry-readiness” of students for

meeting the current needs of industry in a dynamic and disruptive environment. Hence, there is a need to address this gap and arrive at innovative assessment approaches

Objectives

The key objectives are outlined below:

1. To illustrate the need for innovation in the assessment process
2. To propose an innovative assessment framework
3. To establish a mapping between the higher education assessment goals and the proposed assessment framework

Methodology

The methodology for introducing innovation in the assessment process includes the following:

- Creating the vision for the assessment approach
- Identifying the assessment goals
- Formulating the assessment standards
- Specifying the assessment tasks and measures
- Measuring the assessment outcome

An innovative hybrid assessment approach known as Form-Fact has been proposed in this paper. This involves the incorporation of the key aspects of a Factual assignment (Fact-based assignment) into the Formative assessment structure.

Formative assessments involve the deployment of various methods and techniques that could be utilized by teachers during a lesson, chapter, or unit to assess the understanding of concepts by students and also their learning needs. Continuous assessment is a key aspect of formative assessment design.

The factual assessment considers assessing student responses to real-life/real-world scenarios or problems. Examples of Factual assessment include making observations and inferences from a field study, interviewing industry practitioners to obtain perspectives, etc. This is integrated into the Formative assessment approach by applying the principles of formative assessment.

The Factual Assessment Design approach

The factual assessment design approach requires students to perform tasks that occur in real life or the real world. It also requires "knowledge application" or "knowledge construction". Further, it provides direct evidence of the skill acquired or developed by the student, unlike a traditional assessment approach.

The Form-Fact Assessment Design approach

The proposed Form-Fact assessment design approach is a hybrid approach involving integration of the Formative approach and Factual Assessment Design approach. It aggregates the benefits of both these approaches. This approach involves continuously assessing the learning of students using real-world scenarios throughout a chapter or unit to enhance students' learning and practical knowledge to solve real-life problems.

The Form-Fact assessment framework

The Form-Fact assessment framework is provided below:

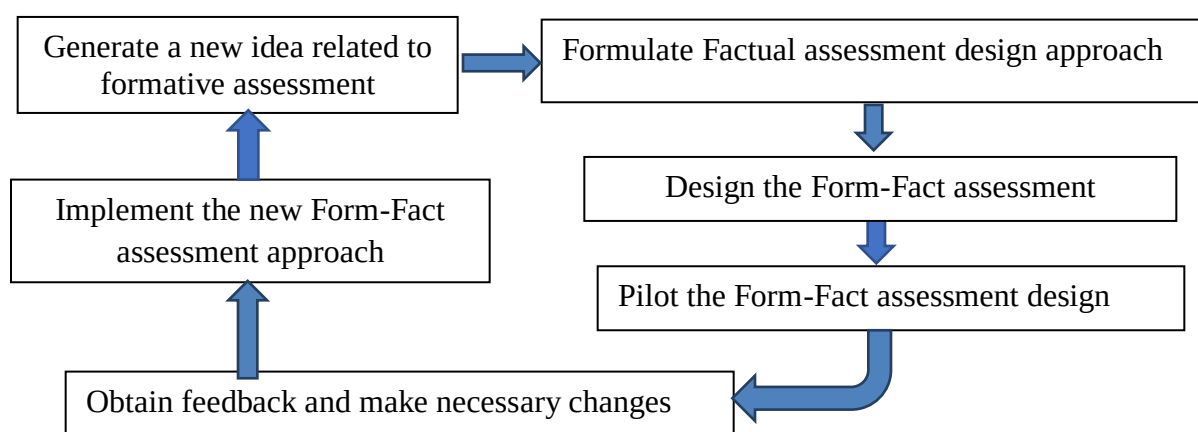


Fig.1. Innovative assessment framework

The initial stage of the innovative assessment framework (Fig. 1) involves consideration of the most appropriate formative assessment approach/technique that could be effective in a specific scenario. The second stage relates to the formulation of a factual assessment design approach in which students are provided with a real-life/real-world problem for which they have to find a solution by performing the required tasks. The third stage pertains to Form-Fact assessment design wherein the requirements and rubrics are specified by integrating principles of formative and factual assessments.

In the fourth stage, pilot testing is done for the Form-Fact assessment design before going ahead with full-fledged implementation. This would facilitate the identification of any issues, challenges, or constraints in the assessment approach so that these could be fine-tuned before the overall implementation. The fifth stage pertains to seeking feedback on the new assessment process/technique and incorporating the suggested changes. The sixth and final stage of the

innovative assessment framework relates to the implementation and utilization of the "pilot-tested" Form-Fact assessment design.

Mapping higher education assessment goals to the proposed assessment framework

The goals of assessment in a higher education scenario could be mapped to the proposed assessment framework as indicated in the following table (Table 1).

Assessment goal	Stage in the innovative assessment framework
Revamping the assessment process	Generate a new idea related to formative assessment, Incorporate the Factual assessment principles, Design the Form-Fact assessment
Move from Summative assessment to Formative assessment	Design the Form-Fact assessment
Shift the patterns in assessment towards meeting 21st-century skill requirements	Design the Form-Fact assessment
Shift to a grading system based on criteria to assess achievement of students	Design the Form-Fact assessment
Utilize technology in the assessment process	Design the Form-Fact assessment, Pilot the assessment approach, Implement the new assessment approach
Migrate from examinations mode towards a more continuous and comprehensive system of assessment	Pilot the assessment approach, Obtain feedback and making necessary changes, Implement the new assessment approach
Continuous review of progress based on assessment	Implement the new assessment approach

Table 1: Mapping of higher education goals to the proposed assessment framework

Conclusion

The changes in the higher education landscape have made innovative approaches necessary in the assessment process apart from curriculum design and course delivery. The Formative assessments provided innovation to the traditional summative assessments. The Factual Assessment design which incorporates assessment considering a real-world environment would facilitate enhancing the learning and skill-set of students to meet the challenges of a real-life scenario in a corporate world.

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23. Gamification: An Aid to Facilitate Formative Assessment

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Abstract

To measure 'learning' of an individual, assessment is needed. Assessment can be truly meaningful only when it is formative in nature. The researcher here, attempts to delineate the need for assessment, and develops a framework to carry out a gamification strategy, applying the mechanics, dynamics, and aesthetics of gaming in education to evoke 'joyful' learning. It can also be carried out in normal classroom setting in every school, without the need of constant internet and computer lab. Moreover, how gamification can act as a potential tool for formative assessment is also dealt with in detail.

Keywords: *Formative assessment, Gamification, Framework, tool.*

Introduction

From childhood onwards, it has been often heard that "A tree is known by its fruit" the similar analogy goes in the case of assessment and evaluation in the field of education. Evaluation is the understanding of the final learning outcome that has happened over a while or at the end of a semester or a year. On the other hand, the assessment itself connotes a shorter period, referring to its various kinds that are formative and summative.

Summative assessment tells in a nutshell what a pupil has achieved at the end of a time interval, in terms of learning, whose aims and objectives are on par with national standards. The period depends on the objective of the teacher and can be at the end of a topic, a term or half-term, at the end of a year or, at some key stage. Summative assessment can be a conversation, recordings, written, or tasks.

Formative assessment on the other hand takes place regularly during teaching and learning, where teachers, as well as learners, can assess their learning and progress frequently. It starts with a diagnostic assessment, indicating the known and the gaps. Fulfilling the gaps is the focus of the teacher for which the teacher facilitates everyday learning followed by feedback. The feedback leads to further modification in terms of strategies on the part of the teacher and improvement on the part of the learner. Formative assessments may be questions, tasks, quizzes, or more formal assessments.

Formative assessment is referred to as 'assessment for learning' and summative as 'assessment of learning'. Therefore in the case of formative the learners are laid open to ump ting number of teaching strategies and instructions to see which pedagogy suits them most that they improve and learn more with every single assessment followed by feedback. Teacher too reflects on their day to day teaching strategy. Summative assessment assesses 'what' has been learned till then. Therefore, formative assessment is assumed to have more effect on the learning of the learners than summative.

Though the assessment has been categorized effectively, effective and innovative assessment seems to remain quite limited according to the OECD review collected by diagnosing the assessment patterns of the OECD countries. It says that primarily classroom assessments or national assessments focus on the reproduction of knowledge and application of basic skills. The other complex competencies are neglected. For this reason, an acute approach towards teaching and learning can be adopted where the assessment system becomes a 'hidden curriculum.'

The complex and competencies that are mentioned here refer to collaboration, peer interaction, problem-solving skills, critical thinking, and intrinsic motivation for learning. If the learners of today once gain these competencies none can stop them from gaining further skills and knowledge throughout their life.

Several studies have shown that there has been a remarkable improvement in motivation, vocabulary acquisition, competence, the cognitive and behavioral learning outcomes of the students or learners when they were taught through technology and gamification. Gamification as defined by Kapp in 2012 is “using game-based mechanics, aesthetics and game thinking to engage people, motivate action, promote learning, and solve problems.”

In other words, gamification is the use of mechanics, dynamics, and the frameworks of games in a non-game context; such as education. Therefore, the focus is on how to turn the classroom learning experiences into gaming experiences, so that the children can learn as well as enjoy. Scot Osterweil, creative director of the Massachusetts Institute of Technology's Education Arcade, says that the main reason why games attract children are the "four freedoms of play": Freedom to fail (allow mistakes), Freedom to experiment (explore new strategies and knowledge), Freedom to assume different identities (have a different stance) and Freedom of effort (pause and reflect on their accomplished activities to find the solution to any problem).

The term gamification is often interchangeably used with serious games or game-based learning. Game-based learning is the use of games to learn but gamification is the use of dynamics

and mechanics of learning to learn. Game-based learning uses learning games to achieve an instructional goal. For example- A platform like knowledge guru pertains to game-based learning.

Werbach and Hunter (2012) laid down the dynamics and mechanics of learning where the dynamics represented constraints, emotions, and progression. The mechanics included aesthetics, levels, time, resources, challenge, reward, competition, co-operation and feedback.

The rationale of the study

English being a global language and the second official language for communication in India holds a vital position in academics. Children often find learning the English language when it comes to the basics like grammar, reading, writing, speaking. They are often driven by literature since it is theme and content based which they might relate to or transform the content into imagination. But in the case of grammar, they are often bored and try to avoid it.

The literature thoroughly studied by the researcher made her of the view that gamification can be a solution to this boredom. Gamification was found to have the potential to motivate, make learning fun, be immersive, competitive, collaborative, and promote critical thinking and problem-solving skills. Therefore, the researcher aimed at incorporating gamification in the classroom to teach English and assess the progress of the learners on a day-to-day basis that is adhering to formative assessment.

Researches have shown that in most cases gamification was done through a gaming platform on the web, which requires a sufficient number of computers, high connectivity of internet, computer operating knowledge by the children as well as technical knowledge by the teachers to frame and design the games. India being a developing nation still goes through an infrastructural crisis and therefore pertaining to technology and gamification on the technical platform seemed to be quite inappropriate and impossible regarding the condition of the schools apart from the elite private ones that are often witnessed. Moreover, most of the teachers are not even trained in operating and handling computers, so the dream of designing games and having technical knowledge acquired by all seems unachievable at least for recent years.

The researcher, therefore, decided to incorporate gamification in the classroom but without the technical aspect, that is turning the classroom into a gaming platform by using the elements of gamification, the dynamics, and mechanics but in a feasible form.

The objective of the study

- To design the gaming framework and experience to be provided to the learners

- To develop a design for gamification elements to work in a classroom environment
- To assess the academic achievement in English by the learners by means of Gamification.

Methodology of the study

The researcher will divide the course content to be taught into small units to be taught within a week. The units of the week will be further divided on the basis of days and each day learning will be followed by some task. The learners will be divided into groups of an equal number of participants for the task and they will be acquiring badges on the basis of their performance as individuals as well as a group member. The group members will be decided for the entire week which could not be changed. Hence, the learners will learn to work as a team irrespective of individual differences and trivial issues.

From the first day of the week till the last working day of the week, each day the performance will let learners remain competitive, motivated, and active since they will be acquiring badges which gathered together at the end of the week will fetch them the achievement or points. A leader-board will be maintained that will show the current position of the learners as a group and also as an individual.

The classroom learning will follow the design for gamification framed by the researcher where the elements of gaming, mechanics, and dynamics are defined in the context of classroom learning.

At the end of every week, the winner of that week as an individual as well as the winner group will be declared based on the position that they hold on the leader-board. Rewards can be given in the form of points or XPs which they will keep on accumulating till the end of the semester. These accumulated points can be converted into marks or prizes that can be rewarded to the learners. So the learners won't have the moment where they can quit the gamification module as it will become a continuous process of assessment and achievement.

Categorization of Gamification Elements for Classroom Learning

Dynamics	Mechanics	Elements
Constraints	Challenge	● Learners need to perform well individually as well as in group to earn more points and save their group from getting eliminated

Emotions	Reward Competition Co-operation Resources Acquisition Feedback	● Badges or small tokens can be provided ● Healthy competition is facilitated among the learners as they work in groups to achieve a united goal ● Working with peers as a team increases co-operation, as well as learners, co-operate with teachers or instructional clarity ● Points/Stars are to be collected instead of marks on the basis of intervals of marks ● Leader-board maintained shows the level of the learner and the teacher serves as the feedback machine as well as suggestions are provided for further improvement in the next level.
Progression	Reward Resource Acquisition Feedback	● Badges and tokens defining achievements ● Points/ Stars collected ● Leader-board, level followed by suggestions

At the end of each task every day, the feedback will be provided by the teacher to the learners so that they can rectify their mistakes and also perform well in the next day. At the end of each week, anonymous peer feedback can be utilized for analyzing the capability of social interaction and handling of social relations by the learners. It can also include reflection on the performance that they witness among themselves.

It has been reported by a study in Malaysia that meaningful gamification should be more use of quality-based examples; such as instead of points and rewards or any quantitative element, students can give ratings and feedback among themselves. The rewards also need to be achievable so that the learners stay intrinsically motivated, and at the same time rewarding enough to feel a sense of accomplishment. Therefore, the researcher planned the tasks that would lead to gamification as well as assessment of the learners in such a way that feedback from peers and as well as teachers are incorporated for further improvement.

Hollz in 2018 even mentioned clearly "Why gamification works?" The reasons mentioned were related to the intrinsic motivation of the learners. She said that in order to be intrinsically motivated three human needs should be fulfilled by every task and they are,

- **Autonomy:** When learners have agency over their learning and when learning is self-initiated it is more likely to elicit intrinsic motivation.

- **Competence/Mastery:** When learners feel confident with multiple communication and feedback from teachers which clarifies as well as boosts their confidence level, a feeling of mastery and competence emerges which is more likely to enhance intrinsic motivation for that activity.
- **Connectedness:** Collaboration and cooperation make learners grow life skills. When they exchange ideas and positive energy among themselves they are expected to get internally motivated to do an activity as it generates connection and relationships with others.

Expected Learning Outcomes

The expected learning outcome will be measured by means of the badges gathered every day at the end of the task which will give the number of accumulated points at the end of the week. These points will finally be converted into grades or marks for a wholesome depiction leading to summative evaluation.

The learning outcome is expected to be of positive nature affecting the cognitive, affective as well as psychomotor domains of the learners. They are expected to have embraced gamification more than the usual approach towards learning, as this innovative pedagogy provides them autonomy, mastery, and connectedness as mentioned by Hollz. Since gamification results in problem-solving skills, joyful learning, intrinsic motivation the academic achievement of the learners in the English language is also expected to be satisfactory. They will be positively motivated to learn, actively engage, decide and plan how to reach the goal and therefore accomplish their targets. They will also know the act of collaboration and co-operation while working as a team. The need for meaningful gamification would also be catered to as it will not just be a competitive environment and task of achieving rewards but also sharing, interacting, and providing and accepting feedback and suggestions both from peers as well as teachers. The leader-board therefore, won't showcase just the level of learners in academics but also would serve as a reflection of their position at the curriculum level, as a curriculum talks about not just academic achievement but also total development of the learners, thereby acquiring the 21st-century skills.

Conclusion

It is said that assessment is mainly about evaluating the effect of instructions and how it has impacted student learning. But assessment won't be truly meaningful unless it has been followed up by a couple of feedback and suggestions in order to rectify the mistakes, change the pattern of thinking, and thereby exhibiting improvement in the next assessment. Therefore formative assessment takes the lead role in the case of student learning and gamification being a potential tool to aid

formative assessment makes it innovative and appropriate enough to be implemented within the classroom environment.

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24. Cancer Stigma among Non-Patient College Students

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Abstract

In spite of advances in early detection and treatment, a cancer diagnosis is often considered a death sentence, and negative images and myths regarding cancer are continued. Stigma has a negative influence on individuals' health, creating social distance, which affects their interactions with people in the society. Cancer is connected with stigma, due to misunderstanding and myths about cancer. Cancer stigma may not just affect cancer patients but also non-patient population. College students should first be aware of cancer and cancer related false myth in society. The aim of this study was to assess the cancer stigma among non-patient college students. The cross sectional online survey was conducted using a questionnaire. Cancer stigma was measured using items from the Cancer Stigma Scale (CASS), a validated scale with 25 items. 500 responses from India participated in this study. Non-probability snowball sampling technique was used in the present study. The findings of the present study indicates that there is a significant difference in cancer stigma among non-patient college students with respect to locality, educational qualification and branch of study.

Keywords: Cancer, Stigma, Non-Patient college students

Introduction

Cancer is a leading cause of death and burden of disease worldwide and are estimated to affect about 20 million individuals worldwide by 2030. In India, approximately one million new cancer cases are detected every year with an annual toll of 0.63 million people are died from cancer disease (WHO, 2008; Globocan, 2012). In spite of advances in early detection and treatment, a cancer diagnosis is often considered a death sentence, and negative images and myths regarding cancer are continued. A large proportion of cancer deaths are preventable through early detection but there are a range of social, emotional, and financial reasons that are obstruct the effectiveness of cancer prevention and treatment efforts. One such barrier is cancer stigma.

Stigma has a negative influence on individuals' health, creating social distance, which affects their interactions with people in the society (Lebel and Devins, 2008; Ahmad and Dardas, 2016). The extent

and impact of stigma on the cancer prevention, detection and care continuum are poorly understood in India. Stigma is typically a social process, experienced or anticipated, characterized by exclusion, rejection, blame, or devaluation that result from experience, perception, or reasonable anticipation of an adverse social judgment regarding a person or group. Stigma may occur in two forms – perceived and actual. Perceived stigma refers to the shame associated with having a condition and to the fear of being discriminated against and denied fundamental welfare rights leading to social exclusion. On the other hand, actual stigma refers to overt discrimination, which may lead to feelings of guilt, shame, and threatens one's own identity. It threatens the psychological and social aspects of life.

Cancer stigma not only affects cancer care and treatment but also puts a strain on prevention efforts. It is particularly worrisome in a low and middle-income country like India. There have been attempts by the government and positive changes toward mitigating stigma, but a lot needs to be done to address this issue. The present study made an attempt to assess cancer stigma among non-patient college students.

Need and Significance of the study

India is going through a great epidemiological transition with the rising of chronic diseases such as cancer. In India, the concept of multimodal treatment of cancer emerged three decades ago; this increased the overall cancer survival rates. But not decreasing social stigma and negative perceptions regarding cancer yet. Cancer is associated with stigma, due to misunderstanding and myths around cancer. Cancer is viewed as a contagious illness; these type of myths and misunderstanding have increase the risk of stigmatization and social isolation against cancer population. Cancer stigma leading people to uncomfortable, suffering, feelings of awkwardness and fear. So cancer patients have waited more days before consulting friends or family members about signs and symptoms. Cancer stigma may not just affected cancer patients but also non-patient population. College students should first be aware of cancer and cancer related false myth in society. They are the representatives of future generation. So getting the information from this study will help the government and non-government cancer organisations and positive changes toward mitigating stigma among non-patient college students. The findings of the study will also help the educationalist to promote health education programmes related to prevention of cancer stigma.

Objectives of the study

- To study the level of cancer stigma among non-patient college students

- To study the significant difference in the cancer stigma among non-patient college students with respect to their gender, educational qualification, age group, locality and branch of study.

Hypotheses

- 1) There is a significant difference in the cancer stigma among non-patient college students with respect to a) Gender and b) Educational qualification.
- 2) There is a significant difference in the cancer stigma among non-patient college students with respect to a) Age group and b) Locality
- 3) There is a significant difference in the cancer stigma among non-patient college students with respect to Branch of study.

Methodology

Population of the study: The college students are the subject of the study.

Sample and Sampling Technique: The cross sectional survey was conducted between 2nd September 2020 and 26th October 2020. Since the data were collected during the lockdown period in India, the investigators used online survey. Snow-ball sampling technique which is a type of non-probability sampling technique was used to collect the response from the participants. A sample of 500 college students seen as representation for the overall population.

Data Collection Method: The cross sectional online survey contains a questionnaire regarding the cancer stigma. The online survey questionnaire was developed by using Google forms. Participants voluntarily participated in the study. The questionnaire contains two parts. In the first part the demographic variables such as gender, age group, educational qualification, locality and branch of study were included. The second part contains 25 questions to measure the cancer stigma. Cancer stigma was measured using items from the Cancer Stigma Scale (CASS) developed by Marlow LA; Wardle J (2014). A validated scale has 25 items. Responses were recorded on a 7-point Likert scale from 'disagree strongly' (agree slightly, agree moderately, and agree strongly) to 'agree strongly' (disagree slightly, disagree moderately, disagree strongly) and Not sure, so that all scores were from 1 to 7 and reversed score for 5 items. The total stigma score ranged from 25 to 175, with a higher score denoting more cancer stigma.

Statistical Analysis: The responses were coded and entered using SPSS and statistical analyses were done with the help of SPSS (Statistical Package for Social Sciences) 16.0 Version. Descriptive statistics, such as percentages, means, and standard deviations, were calculated.

t-test and ANOVA were used to analyse the relationship between the dependent (stigma), and independent variables (demographic characteristics of the participants). Statistical significance was considered at P value of less than 0.05 for all analyses.

Analysis of Data

A total of 500 respondents participated in the online survey and their mean score is 71.62 shows low cancer stigma among non-patient college students.

Testing Of Hypotheses

H₀- 1: There is no significant difference in the cancer stigma among non-patient college students with respect to a) Gender and b) Educational qualification

Table-1

Significant difference in cancer stigma among non-patient college students with respect to a) Gender and b) Educational qualification

Variable		N	Mean	S.D	't' value	df	Sig.	Result
Gender	Male	277	71.85	14.150	0.377	498	0.707	NS
	Female	223	71.33	16.547				
Educational qualifications	UG	357	70.77	13.913	1.96	498	0.050*	S
	PG&above	143	73.73	18.053				

(*Significant at 0.05 level) S- Significant NS- Not Significant

The table-1 indicates that the calculated 't' value (0.377) is less than the table value (1.96) at 0.05 significant level with respect to gender and thus there is no significant difference in the cancer stigma among non-patient college students with respect to gender, and hence it is failed to reject the H₀- 1(a).

The table further reveals that the calculated "t" value (1.96) is same as the table value (1.96) at 0.05 significant level with respect to educational qualification and thus there is a significant difference in the cancer stigma among non-patient college students with respect to educational qualification and hence the formulated H₀ -1(b) is rejected. The mean score of students who have PG&above qualification (73.73) is greater than that of UG students (70.77) which indicates that the student is who have PG&above qualification have more cancer stigma than UG students.

H₀- 2:There is no significant difference in the cancer stigma among non-patient college students with respect to a)Age group and b)Locality

Table-2

Significant difference in the cancer stigma among non-patient college students with respect to a)Age group and b)Locality

Variables	Source And Variance	Sum of Squares	Degrees of Freedom	Mean Square	F	Sig.	Result
Age group	Between	1023.352	2	511.676	2.20	0.111	NS
	Within	115054.686	497	231.498			
	Total	116078.038	499				
Locality	Between	7856.822	2	3928.411	18.041	0.000*	S
	Within	108221.216	497	217.749			
	Total	116078.038	499				

(*Significant at 0.05 level) S- Significant NS- Not Significant

The table-2 shows that the calculated 'F' value (2.20) is lesser than the table value (2.30) at 0.05 level with respect to age group, hence it is failed to reject H₀- 2 (a).

The table further reveals that the calculated 'F' value (18.04) is greater than the table value (2.30) at 0.05 level with respect to locality, hence the formulated H₀- 2 (b) is rejected and thus there is a significant difference in the cancer stigma among non-patient college students with respect to locality.

Table-2.1

Scheffe Test- Significant difference in the cancer stigma among non-patient college students with respect to Locality

Locality			MD	Sig.	Result
Urban	Semi urban	Rural			

70.08	69.02	---	1.066	0.774	NS
70.08	---	79.30	9.215	0.00*	S
---	69.02	79.30	10.281	0.00*	S

(*Significant at 0.05 level) S- Significant NS- Not Significant

From the analysis of table 2.1 it is exposed that the student from urban and semi urban are not significantly differ at 0.05 level. However students from urban and rural& semi urban and rural significantly differ at 0.05level.It is revealed that the mean score of students from rural (79.30)is greater than from urban(70.08) and also semi urban (69.02) students. This indicates that students from rural are more cancer stigma than the urban and semi urban.

H₀- 3: There is no significant difference in the cancer stigma among non-patient college students with respect to branch of study

Table-3

Significant difference in the cancer stigma among non-patient college students with respect to branch of study

Variables	Source And Variance	Sum of Squares	Degrees of Freedom	Mean Square	F	Sig.	Result
Studied Group	Between	8795.598	3	2931.866	13.55	0.000*	S
	Within	107282.440	496	216.295			
	Total	116078.038	499				

(*Significant at 0.05 level) S- Significant NS- Not Significant

The table-3 shows that the calculated 'F' value (13.55) is greater than the table value (2.30) with respect to studied group, hence the formulated H₀-3 is rejected. Thus there is a significant difference in the cancer stigma among non-patient college students with respect to branch of study.

Table-3.1

Scheffe Test- Significant difference in the cancer stigma among non-patient college students with respect to branch of study

Studied Group	MD	Sig.	Result
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Arts	Science	Engineering	Medicine			
71.13	76.21	---	---	5.079	0.014*	S
71.13	---	67.30	---	3.833	0.219	NS
71.13	---	---	62.75	8.385	0.022*	S
---	76.21	67.30	---	8.912	0.000*	S
---	76.21	---	62.75	13.464	0.000*	S
---	---	67.30	62.75	4.522	0.463	NS

(*Significant at 0.05 level) S- Significant NS- Not Significant

The table 3.1 shows that arts and engineering students, engineering and medicine students do not significantly differ at 0.05 levels. It is also learnt that arts and science, arts and medicine, science and engineering, science and medicine studied students are significantly differ at 0.05 levels.

However, it is revealed that the mean score of science students (76.21) is greater than that of the arts(71.13), engineering(67.30) and medicine (62.75) students, which indicates that the science students have more cancer stigma than arts, engineering and medicine students.

It is also exposed that the mean score of arts students (71.13) is greater than that of the medicine students(62.75), which shows that the arts students more cancer stigma than the medicine students.

Conclusion

The findings of the present study indicate that the level of cancer stigma among non-patient college students is low. It shows that the students who have PG & above qualification have more cancer stigma than UG students. It also shows that students from rural area are more cancer stigma than the urban and semi urban area students. It further indicates that the science students have more cancer stigma than arts, engineering and medicine students. It also exposes that the arts students have more cancer stigma than the medicine students. Thus the survey generally exposed that the UG students who are living in urban and semi urban area studying engineering and medicine group are low stigma towards cancer which means that they are positive attitude towards cancer. This may be due to area, life style and high exposure of mass media and mostly social media among college students. Life style of rural society might be the reason for high stigma towards cancer. So, PG& above studied

college students who are from rural area studying arts and science group are indicating high stigma towards cancer. Hence, the government and non-government cancer organizations and health educators should give special attention towards above mentioned category of non-patient college students who belong to high cancer stigma. Cancer related non-governmental organizations (i.e., cancer advocacy groups) should carry out anti-stigma campaigns, governments should televise public service announcements, and mass media should actively provide powerful messages to students society to prevent stigmatization. Health educators should implement cancer awareness and health education programmes related to prevent the misconception and negative thought about cancer.

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25. Perception of Under Graduate Students towards Online Assessment

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Abstract

A study was conducted to find out the perception of students about Online Assessment. The main objectives of this study were to study the perception of the under graduate students towards use of online assessment at University Level. To study the difficulties faced while using online assessment and to study the student's satisfaction with the quality of feedback received in online assessment learning activities. A descriptive survey method was used to collect the data. 200 undergraduate Students were selected randomly belonging to different programs from the Universities of Delhi NCR. A self-designed online survey was prepared for collecting the data. The questionnaire contained 25 items related to the perception of the students towards online assessment, impact on learning and student's satisfaction with the quality of feedback received in online assessment learning activities. Each item was rated on 4-point scale of Strongly Agree, Agree, Disagree, Strongly Disagree. The results show that, in general, the students had a favourable perception towards online assessment and did not face much difficulty while using online assessment. They were satisfied with the quality of their engagement and the quality of feedback across all the formative e-assessment activities offered. The results also showed that the student were satisfied with the quality of feedback received in online assessment learning activities.

Keywords: Online assessment, student perception, feedback, learner satisfaction

Introduction

The rise of COVID-19 has forced educational institutions to close and to shift all strategies to online platforms from physical face to face learning. This has been done for the welfare and safety of the students, instructors, and all the other staff members. This has affected the process of assessment also to large extent due to the sudden adoption of online education than physical education at campus. Many researchers have proved that online education has many advantages as compared to physical education. For example, there is more flexibility with time in online education. In online education the maintenance cost of classrooms is decreased since in online education there is a virtual classroom instead of a physical one.

However online learning also has its drawbacks. It extremely depends on various online tools and Wi-Fi connectivity. If servers are down or Wi-Fi connections are weak, online classes can be inaccessible

and therefore not as reliable. The server may not be able to support the number of students in the class and therefore crash. Online assessments are done by students on their own devices at home therefore they cannot be checked upon and so they are more likely to cheat risks to cybersecurity when it comes to online education.

Literature Review

Liaqat Ali and Nidal Abidal Hamid Hmoud Al Dmour (2021) conducted a research to discuss issues relevant to online assessments and the strategies used by the universities due to COVID-19 in the region of UAE. The findings of the research proved that students had benefited from the online assessment.

N. N. Omar and Y. J. Phung (2019) conducted a study to develop a prototype of an electronic assessment tool to assist lecturers in evaluating students' assignments via technology application platform. The prototype included features such as an easy-to-use template to generate assignment rubrics, operationalized the metric calculation of students' marks, an automatic generated feedback function and video capture or tapping features in assessing students' presentation

P. I. Wuisan and B. Wibawa, (2019) conducted an investigation to provide an example of e-assessment system application, describe challenges in the use of e-assessment systems in higher education based on literature, journals and research results and provide recommendations on how to deal with all of these challenges so that the assessment objectives can be achieved. The findings showed that there are still many questions for lecturers, students and institutions that pose challenges in the application of e-assessments related to validity and reliability, accuracy, fairness, security, infrastructure and human resources.

Alsadoon.H (2017) in his investigation about students' perceptions of E-assessment at Saudi Electronic University found that the Students had positive perceptions of e-assessment and valued its features such as immediate feedback and unbiased grading.

Alabi, A. T.; Issa, A. O. & Oyekunle, R. A. (2012) conducted a research on the application of Information and Communication Technology in the conduct of post-UTME and other examinations at the University of Ilorin. The findings of the research showed the widespread adoption of CBT method for the conduct of not just the post-UTME examinations, but also other University-based examinations with large population of students in Nigerian Universities owing to the success story of the University of Ilorin.

Dermo, J. (2009) conducted a research at the University of Bradford on student perceptions of e-assessment. The findings of the investigation were that students had greatest concern about the fairness of item banking. It was also found that the most positive aspect of e-assessment in the eyes of students concerned the benefits that it can bring to teaching and learning. In addition, the paper concluded that age and gender did not significantly affect student responses in any of the areas studied.

N. A. Buzzetto-More and A. J. Alade(2006) conducted a study to provide insight into the origins, evolution, and key concepts of assessment as well as illustrate some of the ways in which technology and e-Learning serve as the most promising mechanisms for satisfying assessment goals and objectives. The findings of the study were that technology and e-learning strategies can play a critical role in an overall assessment program. It also plays an important role in our efforts to evaluate instruction and learning outcomes, as well as drive the decision making that seeks to enhance educational effectiveness. The findings also showed that assessment programs are time consuming efforts that require planning and foresight. Effectiveness is dependent on institutional and administrative support as well as a long range plan for sustainability that includes technological preparedness.

Objectives

The main objectives of this study are:

1. To study the perception of the students towards use of online assessment at University Level.
2. To study the impact of online assessment on learning.
3. To study the student's satisfaction with the quality of feedback received in online assessment learning activities.

Research Questions

1. What are the student's perceptions towards online Assessment strategies?
2. What is the impact online assessment on learning?
3. To study the level of satisfaction of students with the quality of feedback received in online assessment learning activities?

Research Methodology

A descriptive survey method was used to collect the data.

Sample: 200 undergraduate Students were selected randomly belonging to different programs from the Universities of Delhi NCR.

Tools Used For Data Collection

- A self-designed online survey was prepared for collecting the data. The questionnaire contained 30 items related to the perception of the students towards online assessment. Each item was rated on 4 point scale of Strongly Agree, Agree, Disagree, Strongly Disagree.

Validity and Reliability

The feedback was taken by the experts to ensure the Validity of the questionnaire. Their suggestions were taken to modify the instrument. A pilot testing was done on 30 students to ensure its clarity and reliability.

Findings and Analysis

Students perceptions towards the use of online Assessment strategies in the universities level

To answer the question related to the student's perception towards the use of online Assessment the data collected from 10 items of the questionnaire was used. The assumption of Bandele et al. (2015) was used for analysis: means 1.00 – 1.49 (very unfavorable), 1.50-2.49 (unfavorable), 2.50-3.49 (favorable) and 3.50- 4.00 (very favorable). Almost all items ranged from favorable to very favorable for the participants.

The only item that fell into the unfavorable category was "I prefer Online assessment for all subjects". 56.7% students have the opinion that online assessment is fairer than physical assessment and only 6.9% disagree. Majority of students agree that they get immediate feedback about their performance through online assessment. More than half of the participants agree that online assessment improves the quality of learning. 70.2% of the participants agree that cheating is difficult in online assessment. Majority of students agree that Online assessment helps in improving the quality of assessment in Higher education. Online assessment improves self-learning was strongly agreed by 61.1% of students. Students also agreed that the online assessment reduces exam stress. Students also agreed that the online assessment is more effective than physical exam at the campus. Majority of the students preferred online assessment as compared to the traditional assessment. The students were not in favour of online assessment for every subject

Impact online assessment on learning

To answer the question related to the impact of online assessment on learning the data collected from 10 items of the questionnaire. According to Table 4.2 41% students strongly agreed that it was easy to follow the online instructions in online assessment but 27% found it difficult to follow the online instructions. 44% students found it difficult to take exam online. Majority of Students were of the opinion that they could manage their time well in online mode but 18% still found difficulty. Online

Assessment helps to keep a track of the past result was agreed by all the students. Online Assessment is easy to browse among web pages was the opinion of 41% students, 39% also agreed to this. 87% students were of the opinion that Online Assessment helps to correct work faster. Majority of the students agreed that Online Assessment was consistent with the teaching style. Students also believed that Cheating is difficult in online assessment and it helps to better understand the growth. 96% students also agreed that online Assessment is faster than paper pencil.

Level of satisfaction of students with the quality of feedback received in online assessment learning activities?

To answer the question related to the level of satisfaction of students with the quality of feedback received in online assessment learning activities the data collected from 10 items of the questionnaire. In general, the students were satisfied with the quality of every feedback criteria across all the formative e-assessment strategies.

If taken separately, there are variations in students' satisfaction with formative e-assessment strategies regarding feedback. The results show that the respondents were, in most cases, predominantly satisfied with the quality of feedback of formative e-assessment activities they completed in knowledge survey. 73% students agreed that feedback was appropriate to the purpose of the assignment, 85% agreed Sufficient feedback was provided, 82% agreed The feedback was provided in enough details. 100% agreed The feedback was timely. 80% students satisfied with the feedback as they felt that the feedback was describing to them about what they were supposed to be doing for improvement.

Discussion

The study investigated the students' perception about the online assessment at the university level in Delhi/NCR. Descriptive analysis of the questionnaire showed that the students of different programs at the University level in Delhi NCR were in favour of the online assessment. They were of the opinion that the online assessment is a very effective and fair means and helped them to improve their learning. Immediate feedback helped them and immediate score motivated them to improve effectively in their examination. The online assessment was far better than the paper pencil examination. It helped in enhanced self-learning. Participants appreciated the unbiased grading in the online assessment.

Conclusion

This study aims to find out the perception and attitude of the learners at the university level towards the use of online assessment. Descriptive analysis of the questionnaire showed that the participants

had a positive attitude and perception about the online assessment. They appreciated the immediate feedback received which helped in enhancing their learning. They valued that the online assessment reduced bias in grading during examination. According to the views of the participants there was no need of ant advanced skills required by the learners. The investigator recommends that while developing the online assessment in any course the developer should ensure the fairness for all the students. Continuous feedback should be taken from the students so that their problems can be addressed. Students should be given proper training to develop their skills of using online tools. Third party software can be used to avoid cheating during online assessment. Technical help should be provided to the learners during the examination. While preparing the online assessment quality and standard should be maintained.

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26. Ambedkar, Education and the Dalit Emancipatory Trajectory

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Abstract

The paper points out that though Dalits representation in education has increased over the years but they still lag behind the 'general' category in many ways. The paper dwells on Ambedkar's initiatives for educational mobilisation of Dalits. The paper is based on secondary sources and interviews of Dalit scholars in education. The paper argues that in the context of Dalit exclusion and discrimination in educational institutions the paper emphasises that there is need to re-invoke Ambedkar's approach for democratisation of education and social transformation in society.

Key Words: : Ambedkar, Education, The Dalit

Introduction

Dalits were excluded from access to education and were forced to do menial occupations. Excluding them from knowledge deprived them of their self-esteem and dignity. Such kind of culture of exclusivity and exclusion has diffused into the well-known folk consciousness in Maharashtra, as rightly pointed out by Gopal Guru (2002) and many others. As a matter of fact, the educational attainments of Dalits are low due to such historicity of exclusion and continuation of it in new forms. Though due to reservation policy there is decadal increase in educational status of Dalits but when compared with open category representation the figures for Scheduled Castes (SC)/ Scheduled Tribes (ST) are awful. The data compiled by Yadav and Deshpande (2006) from various official sources shows overrepresentation of upper caste Hindus and underrepresentation of SCs/STs in higher education in both rural and urban India. In elite educational institutions i.e. the Indian Institute of Technology half of the seats reserved for SC category and one sixth of ST category remain unfilled. The reason for low representation in elite professional educational institutions is attributed to the lack of cultural capital and discrimination based on social origin i.e. caste and tribe. The studies of Aikara (1980), Velaskar (1986), Paramji (1985) and Patwardhan & Palshikar (1992) suggest that generally students belonging to SC/ST come from relatively less well-off background and the reason for low performance is attributed to the lack of cultural capital (Weisskopf, 2004).

On the other hand, the educational institutions are not free of caste based discrimination. The SC and ST students have to face the [symbolic] violence in some of the educational institutions. They are

stereotyped and looked with prejudice. Some face insults and humiliations as they belong to low caste origin. The SC/ST students are mistreated by the upper caste teachers. Stereotyping and humiliation of these students goes on in some form or other in educational institutions.⁴Therefore, it is important to understand the Dalit emancipatory trajectory as enunciated under the leadership of Ambedkar to re-invoke its learnings for mobility of oppressed groups and social transformation in society.

Ambedkar's Educational Movement

Dr. Ambedkar's struggle for inclusion in education dates back to colonial era. It is during the colonial rule Dr. Ambedkar became critical of British education policy and exclusion of Dalits in education. In the year 1924 he started the Bahiskrit Hitakarini Sabha and Depressed Class Education Society for ameliorating the conditions of the depressed class. The Bahiskrit Hitakarini Sabha played significant role in mobilizing and putting the issues of Depressed Classes to especially Simon Commission. In order to remove social and cultural handicaps of depressed class he asked the students to "Educate, Organise and Agitate" so that they become important for the emancipation of their own community. Dr. Ambedkar accorded education not as a means of material gains but to overcome their inferiority complex that had rather made them slaves. This he made it categorical on occasion of laying the foundation stone of the permanent building of Milind Mahavidyalaya (which was then laid by Dr. Rajendra Prasad, the first President of India on 1st September 1951).

He took grants in the form of public donations, from municipalities, District Local Boards and Provincial Government and opened hostels in Solapur, Panvel and Thana for untouchable class students. Later hostels were started in Poona, Nasik, Dharwar and other places. As per Chitnis, self-respect and self-reliance was the watchwords of Dr. Ambedkar and he therefore thought of having his own enterprise for promoting collegiate education amongst depressed classes (Chitnis, 1980). Dr. Ambedkar persuaded Lord Linlithgow, the Viceroy and his Cabinet members to give grant of three lakh of rupees and an equal amount as loan without interest to People's Education Society (PES). He founded the People's Education Society in Bombay on July 8, 1945 to promote higher education for the lower middle class and Scheduled Castes in particular and started the Siddharth College of Arts

⁴The Thorat Committee Report has pointed out that how the premier medical institution of the country excludes and discriminates the oppressed groups. See, Thorat Committee Report, 2007, Caste Discrimination in AIIMS, *Economic and Political Weekly*, June 2, 2007. There have been reporting from the Indian Institute of Technology in various parts of the country of the cases of discrimination against the scheduled castes and scheduled tribe students.

and Science.⁵ This was the turning point in the social life of the untouchables and the society too. The object of PES was the education of working class in general and the Scheduled Castes in particular.

People's Education Society (PES)

Siddharth College of Arts and Science was the first college established under PES. This event was the landmark in the history of education in the city of (then) Bombay. The main objective of the PES was to promote higher education among the lower middle and working classes, especially Scheduled Cates. The college was the gift of the PES to the working people (the Arts College), taking into consideration the expensive university education which cannot be borne by the clever students coming from poor background.⁶ Besides this PES runs many colleges, schools and hostels in Maharashtra. Periodically the Siddharth college had to its credit high academic honours as well as extra-curricular activities. The college showed dynamism and vitality. Also the significant feature of the college was the Library known for its rare collection of books. Dr. Ambedkar was himself a bibliophile who was attached more with books. This Library stands even today as rich source of books and research references (Jadhav, 1965). The college had high academic credits and in extracurricular activities and sports. The college had the test cricketers on its roll of which more famous were Ramakant Desai, Dilip Sardesai, Tamhane, etc. ⁷

The Milind Mahavidyalaya was established in 1950 in Aurangabad as there was no facility of higher education in entire Marathwada except the Government Intermediate College at Aurangabad. Milind Mahavidyalay had made pioneering work in Marathwada for education and especially for the Scheduled Castes. Milind Mahavidyalaya caters to the needs of the poor in education in Marathwada. In one study by Indrajit Alte of the Dalit students those come from Marathwada to Aurangabad to study in Milind Mahavidyalaya he took the sample of 500 Dalit students to understand how they have come and in what conditions they take education here. Indeed, he pointed out that those who come from interiors, backward and remote areas from Marathwada. The interviews with the teachers and

⁵See, Chitnis, M.B. Dr. Babasaheb Ambedkar as an Educationist, 1980, Milind College of Arts, College Annual, 1980-81 and D.G.Jadhav Chairman, People's Education Society, on the occasion of the opening ceremony of The Siddharth College Hostel "Siddharth Vihar" on 12th July 1964, in Siddharth College of Arts and Science Magazine, 1965

⁶See Memorandum of Association of the People's Education Society, Bombay as prepared by Dr. Ambedkar Founder Chairman, in Milind Kala Mahavidyalaya, Milind Varshikank, 1989-90

⁷See, Siddharth College of Arts and Science, Silver Jubilee Number, 1973-74

students of Siddharth College and Milind Mahavidyalaya reveal the fact that they get not only admission here but also a livelihood source which is crucial for overcoming pauper conditions. They pointed out that the Milind College do not take fees from the backward class and gives admission first and recovers the fees from the scholarship amount.⁸

The alumna of the PES shows that the PES has created and produced a substantial amount of scholars, bureaucrats and officers of good rank like IAS, IPS, IFS and Judges (revealed in discussion with R.K. Kshirsagar ex-Principal of Dr. Ambedkar College of Commerce, Aurangabad). Kshirsagar pointed out that most of the Dalits who are in the academics have come from PES especially Milind Mahavidyalaya. He gave the example of the batch of Mandavdhare and Waware when 10-15 students had got admission in Jawaharlal Nehru University, New Delhi. The ex-Vice Chancellor of University of Mumbai Prof. Bhalachandra Mungekar is product of PES, University Grants Commission Chairman Prof. Sukhdeo Thorat is a product of Milind Mahavidyalaya and so as the Vice Chancellor of Gujarat Central University Prof. R.K. Kale. All of them have to say that if PES would not have been there they could not have achieved such top positions in academics (though they have to struggle).

The Decline of Cultural Movement

Gokhale (1993) has pointed out that there were two institutions for educational and cultural movement i.e. People's Education Society and the Bauddha Jana Panchayat Samiti. The Bauddha Jana Panchayat Samiti has played a role in socio-cultural transformation and spread of education amongst the Dalits (Gokhale, 1993). The post-Ambedkar Buddhist movement (dalit movement in general) though had made significant efforts for the socio-cultural transformation but largely the cultural movement started declining. The PES colleges and schools were very significant for the educational development of the Dalits but post 1970s its effectiveness had dwindled. In discussion with Wankhede.G.G he pointed out that this was because of the internal politics in PES, vested interests of its members, corruption and regionalism. Also post 1970s the PES created only degree holders from their colleges largely and most of the students were unemployed or in low paid occupations. This is elaborated in the study of Wankhede (1978). He noted his disappointment as

⁸See, Milind Magazine, 2008-09, Milind College of Arts, Nagsenvana, Aurangabad

there was limited degree of social mobility seen for what he says that ‘the career trajectories of dalits moves from traditionally poor background to disadvantaged educated classes. Dr. Kshirsagar was of the opinion that one could see that educational development of dalits is better in South Indian states especially Andhra as he says that in Maharashtra the students of the PES are becoming volunteers and activists in the fractured RPI’s movement.

Concluding Remarks

Dr. Ambedkar’s initiatives in the field of education are significant in many ways. Besides affirmative action for ‘depressed classes’ the movement has brought cultural revolution to overcome their social and cultural handicaps. Ambedkar believed that a democratic pedagogy was needed especially for the educational development of depressed class (Nagsen-Milando: teacher student relations). This was liberating and empowering particularly for the depressed class. Also this was important for making social and natural science disciplines egalitarian as far as pedagogic practices is concerned. He also emphasized on educative politics for bringing about social transformation in society. The Dalit movement has developed especially the culture of politics of emancipation in educational institutions established by him. Further the Dalit movement has worked (and continues to do so) for emancipation of Dalits in rural areas, fighting for their rights and fighting against caste discrimination. But after 1970s the Dalit cultural movement has declined due to its institutional problems as well as diversion of the agents of cultural transformation (as Guru, 1997 suggests). It is important that institutions should be developed to teach Dr. Ambedkar’s approach and train the students in constitutional rights and how to fight for it to achieve it. The committed Dalit ‘organic intellectuals’ should take lead to develop cultural capital amongst the Dalit. This should be done as majority of Dalits still remain poor, excluded and discriminated sections of the Indian society.

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Interview with Prof. R.K. Kshirsagar, Ex-Principal, Dr. Ambedkar College of Commerce, Aurangabad

Interview with Prof. G.G. Wankhede, Tata Institute of Social Sciences, Mumbai

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27. Technology Driven Assessment and Accreditation Framework for Institutions of India

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Abstract

In India, the assessment and accreditation of institutions offering general programmes is undertaken by National Assessment and Accreditation Council (NAAC). Recently the agency has introduced the paradigm shift in the philosophy of external Quality assurance by automating the entire process driven by quantitative indicators. This model is seen as bringing enhanced objectivity in the process of Assessment and Accreditation. This attempt also forms the base for entire community of Quality assurance agencies world-wide specifically after outbreak of COVID-19. The paper is an attempt to highlight this revised model of NAAC and also to analyze the effects and emerging directions originating out of it.

Keywords: External Quality Assurance (EQA), Assessment and Accreditation (A&A), NAAC, CGPA, Criteria, Weightage, Grade

Introduction

Indian education system is the third largest in the world and is known for catering the pupil of diverse backgrounds and culture. Beginning in late 80's various measures and reforms for improving access to higher education took place such as privatization, massive expansion, increased autonomy and introduction of programmes in new and emerging areas. This has resulted in raising concerns to quality and relevance of the higher education with global trends those days. With the expansion of education, there was a need felt for an agency to address these concerns.

The National Policy on Education (NPE, 1986 (National Policy on Education(NPE 1986 as modified in 1992)) and the Programme of Action (PoA, 1992) spelt out strategic plans for the policies, advocated the establishment of an independent National accreditation agency. Consequently, the National Assessment and Accreditation Council (NAAC) was established in 1994 as an autonomous body of the University Grants Commission (UGC) (GOI, UGC, 1956) with its Head Quarter in Bengaluru. The mandate of NAAC as reflected in its vision statement is in making quality a defining element of Higher Education Institutions (HEIs).

As a general philosophy, assessment expects institutions to introspect themselves for their functioning thereby evaluate and promote accountability in higher education. Thus it takes on a gamut of

challenges in the quality assurance in higher education. To accomplish this task appropriate policy measures needs to be taken at the institutional level, State government level, Central government level and at regulator level to improve the quality of higher education.

Evolution of EQA in higher Education and NAAC (Prasad, 2004)

The movement for quality in higher education in India began in late eighty's. In 1986 UGC constituted a 15-member committee on Accreditation and Assessment Council under the chairmanship of Dr. Vasant Gowariker (Sharma & Sharma, 2005). During period 1987-1990, nine regional seminars and a national seminar organized to debate Gowariker Committee report. The outcomes of these seminars mostly were in support of the report. In 1990 Dr Sukumaran Nair's (Stella & Gnanam, 2001) project report submitted to UGC that reflected a consensus to have an accreditation agency accountable to UGC. In 1992, the revised New Education Policy reiterated all round improvement of educational institutions. In 1994, Prof. G. Ram Reddy committee appointed to finalize the memorandum of association and rules and regulation of the accreditation board (July 1994). Finally 1994, National Assessment and Accreditation Council established at Bangalore (September 1994).

The assessing and Accrediting Institutions of Higher learning is the main responsibility of NAAC. In order to do so it sets up the framework focusing the higher education diversity of India. NAAC undertakes consultations from academia, organizes awareness programmes, and takes feedback from various stakeholders before building up various norms and standards of the framework. The framework is reviewed periodically in tune with dynamic changes happening in Higher Education. The Outcome of these reviews had facilitated of decision making and ensured representation of diverse sectors of higher education. Some of the major highlights with regard to NAAC process are given as follows:

1. NAAC would concentrate on institutions as the unit of assessment, with a provision for assessing individual departments as well, and for
2. Accreditation made to be a voluntary process, initially but in 2013 it was made mandatory with the mandatory regulation of UGC.
3. The outcome of which would be valid for a period of five years. But later with the advancement in the journey of quality of HEI's policy for extending accreditation validity upto

seven years was brought in for best performing institutions. Therefore if institution secures best in class grade for consequent three cycles, validity will be extended to seven years.

4. Unlike many accreditation agencies the Assessment performance is measured in terms of differential grade instead for simply accredited or not accredited. This is provisioned in order to cater huge diversity of Indian Higher Education system. Grading Pattern revised over the period in past, accordingly the definition of best performance is modified in context to the period when accreditation is granted.
5. The process of assessment has always been the combination of self evaluation and external evaluation especially in case of NAAC, i.e. Institution will write their self study report (SSR) and submit it to NAAC for external assessment.
6. The outcome report of accreditation and SSR is mandated to be disclosed to public by uploading these on institutional website for stakeholders.

Quality Assurance model of NAAC

NAAC has been doing Quality assessment in the Institutions of Higher learning since past 25 years. The model for assessment is relooked and revised at frequent intervals to accommodate all institutions of the country and to be in tune with the various changes taking place in higher education sector of country. Recent revision of the assessment model took place in July 2017. The assessment pattern usually rests on principle where the future course of action is dependent on the previous outcome of assessment; it is not wrong if an analogy can be established with the control loop used in feedback control system shown in Figure 1. A control loop is widely used terminology defined in feedback control system (Tannenbaum, Francis, & Doyle, 1992) where the objective is to minimize the system error or to drive output towards reference value. The parameters measure the performance of the system in attaining the same. Similar such case exists with higher education, wherein reference means the set of standards defined for educational elements. Objective of Quality assurance and accreditation is to bring the institutions to attain this reference or level of standards and move continuously towards excellence and quality improvement.

In case of any Higher Education Institution (HEI) as far as Quality is concerned the objective can identified such as to maintain at least minimum standard and enhance quality by planning long term and short term goals. With this aim, goals are set by HEI's and accordingly they work towards achieving goal in which the report of the previous accreditation cycle accreditation forms indicators

for bringing corrective action of the feedback loop. Using the analogy of control system engineering reference input is set at the beginning for a period and output is measured continuously by implementing systemic changes in the various components of the educational process. The Data and information (D/I) according to goal set forms the input to the Assessment and Accreditation. The input is converted to score by coder. In the process of NAAC's revised framework of accreditation, there are two parts of coder i.e. the first part is conversion of data to score done by application of benchmarking technique. The other is conversion of Qualitative information and peer judgement into score. Once scores are finalized from these two, then weighted score is calculated which when divided by the total weight gives CGPA score and categorizes it into one of the seven grades. Apart from this the recommendation from the peer team forms the basis for forming the data along with information for the next cycle. Thus the output for NAAC process is grade and recommendations for further enhancement.

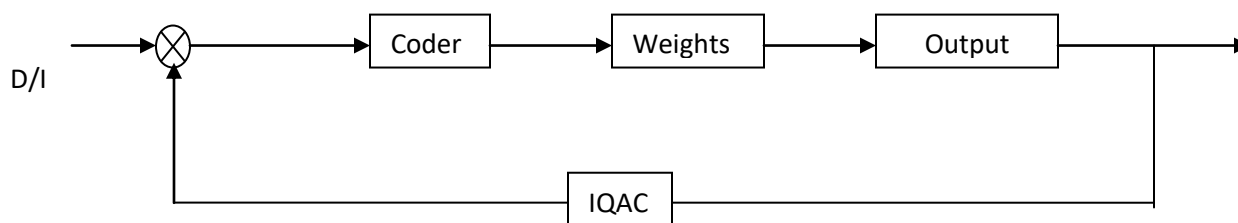


Figure 1 Control loop for Assessment and Accreditation

Let P be the measurement of performance of individual metric then

$$P = \frac{\sum_{i=1}^n \text{Benchmark}_i}{n}$$

or

$$P = \frac{\sum_{i=1}^n \text{Benchmark}_i}{n}$$

Let there are n number of metrics in the evaluation framework of NAAC and each of the metric has got a predefined weight, then Grade, $G = \frac{\sum_{i=1}^n w_i P_i}{\sum_{i=1}^n w_i}$

PARAMETERS OF QUALITY ASSURANCE IN NAAC

NAAC model for assessment and accreditation considers the Institutional accreditation where it measures the performance in 7 broader heads called criteria. The list of seven broader head or criteria is as follows:

1. Curricular Aspects
2. Teaching-Learning and Evaluation

3. Research, Innovations and Extension
4. Infrastructure and Learning Resources
5. Student Support and Progression
6. Governance, Leadership and Management
7. Institutional Values and Best Practices

Each criterion (Cr) is made up of key indicator (KIs) and each key indicator is further delineated to metrics. The internal weight distribution of individual criterion, key indicator within framework and individual metric within criterion is driven by expert opinion and accordingly the scoring is done. The CGPA and the grade for the institution will be calculated as follows:

$$\text{Institutional CGPA} = \frac{\sum_{i=1}^7 \text{CrGPA}_i \times w_i}{\sum_{i=1}^7 w_i}$$

Let for criterion j , where $j=1, 2 \dots 7$, KI represent the Key indicator then

$$\text{CrGPA}_j = \frac{\sum_{i=1}^m \text{KI}_i \times w_i}{\sum_{i=1}^m w_i}, \text{ where } m \text{ depends on the criterion number.}$$

Key indicator weighted grade point is the function of the individual quantitative metric (Q_{NM}) score, which in turn dependent on the data of the institution and evaluation benchmark set for the metric. The framework is characterised by predefined fixed benchmark ranges.

THE VALIDATION OF DATA

The methodology of assessment and accreditation rests on the validation and extensive interaction with stakeholders such as students, teachers, staff, alumni, parents including community and society. The focus of the revised framework of NAAC is the information and data in a pre-designed format. The data is driving the performance of institution pertaining to criterion set by NAAC. Therefore the validation of data (Zio, et al., 2016) is very important for getting the accuracy in the result. The data is evaluated by third party agency in the current methodology who gives the recommendation about the correctness of data. These agencies are advised to look into various external sources apart from institutional sources such as AISHE (All India Survey on Higher Education(AISHE), 2011), UGC (GOI, UGC, 1956), AICTE, other Statutory Regulatory Authorities (SRA), funding Agencies and Universities. After recommendation by third party agency on final data the clearance for the peer team visit is scheduled for those which attain pre-qualification score.

SUPPORT AND PUBLIC DISCLOSURE

Once SSR is submitted to NAAC, it is necessary for the Institution to upload the reports to their own website for the stakeholders to get informed about the quality status submitted to external agency. The NAAC also displays the outcome of Assessment of the Institution, the peer team report and the other relevant documents on the website after results get declared. With regard to support a dedicated incidence management system is integrated in portal and centralized complaint monitoring system as well as feedback monitoring cell is functional for NAAC processes.

AUTOMATION FRAMEWORK MODEL

NAAC has come up with the complete automation mode for Assessment and Accreditation where the entire process starting from registration till the declaration of result is automated through use of technology and ICT. Manual intervention is kept minimal throughout the process including selection of peer team members. The SSR, IIQA and DVV processes are completely automated and there is no other mode of communication except the online. The process flow is represented in figure 2.



Figure 2 Work flow of Assessment and Accreditation

PROCESS FLOW FOR ASSESSMENT AND ACCREDITATION

Registration: After introduction of Revised Accreditation Framework, all the processes of NAAC are handled only through online portal. All the communication and correspondences are handled using portal by different users. Therefore all the institutions interested in NAAC have to compulsorily register themselves in the online portal. The sequence wise process of registration is available on NAAC website apply online link (NAAC, 2017).

The users of NAAC portal are 1. Institution 2. Peer Team Assessor 3. Third party Agencies for data validation and Logistics arrangement 4. NAAC staff and Government.

IIQA (Institutional Information for Quality Assurance): This is first online form to be submitted by the institutions applying for assessment and Accreditation. It is to be noted that the process of

Revised Accreditation framework is a time bound process and there are timelines associated. The registration is the only process in which there is no timeline for an Institution to come forward to apply for IIQA submission.

IIQA is a form which talks about the profile of the Institution about the number of students, about the numbers of teachers about the number of Programmes and their affiliations and approval with regulatory authorities of higher education. While the documents are being checked manually the process is handled through online system. The automation can be extended further with automated reading of the documents submitted and decision making.

SSR (Self Study Report): After the approval IIQA the institution are required to submit online SSR through Portal. The data of last five years along with the supporting document upload forms the basis for requirement of SSR. The data and document is used by the third party agency to give the recommendation on data. Looking at the diversity of the institutions select option for exclusion of non-essential metrics is given to the colleges. There are some process based qualitative questions around 30%. These qualitative questions to be evaluated by team of peer assessors empanelled with NAAC. The assessors will go through the reply response to the questions and based on the evidences found onsite they are going to give scores for all Qualitative questions.

Third party data validation:

The data which is submitted by Institution is raw data which is subject to interpretation of individual; therefore, NAAC has made a provision of authentication of data submitted to NAAC by third party agency. The third party assignment is objectively system driven for an Institution.

The Guidelines for evaluation is commonly shared among the empanelled third party agencies to minimize the deviation in the evaluation done by different agencies. These guidelines are available on NAAC website in the form of Standard Operating Procedure (SOP). Also there are set of templates associated with each Quantitative Metric in which Institutions are supposed to compile the data and upload in SSR. SOP's and data template are available on website of NAAC for prior preparation of SSR. (NAAC, 2017)

Stakeholders's Role in the Process:

Being an accreditation agency, NAAC has been always accountable to various stakeholders of higher education in India. The framework development process involves stakeholders from Academics and before applying over the institutions it is ensured to have huge consultations from policy makers,

Vice chancellors, Government and Statuary Regulatory Authorities and Institutions themselves. External experts visiting Institutions are assessors who are been oriented with NAAC processes during various Orientation and Training sessions. Feedback is taken from them and also by Institution after each visit. Not only this, assessment criteria itself is based on the feedback from students, parents teachers, employers etc. The basic philosophy of NAAC stresses towards participatory management, therefore the emphasis is given on the participation of all the staff of the Institution for the preparation of NAAC assessment. The channelization of this process is managed by Internal Quality Assurance cell of the Institution.

Future directions from QA framework

NAAC is housing data of institutions who had come for assessment over last two decades and credit of performing more than 13000 accreditations (category wise numbers given in Table[1-3]. However not all the data is usable and available in machine readable form. The present revision w.e.f. July'17 is opening the direction for implementing machine learning models for future revision. There is a database getting developed with major factors indicating quality of higher education. As we know that data is the fuel for implementing machine learning algorithms, hopefully NAAC in future will be able to develop this data set for category wise institutions, which may be used for training machine learning algorithms. At present the data of accreditation is maintained on NAAC website in valid accreditation list (March 2020 before pandemic). The numbers of Institutions accredited by NAAC category-wise is given in table-1. The implementation scheme proposed to utilize standard model shown in Fig. 3.

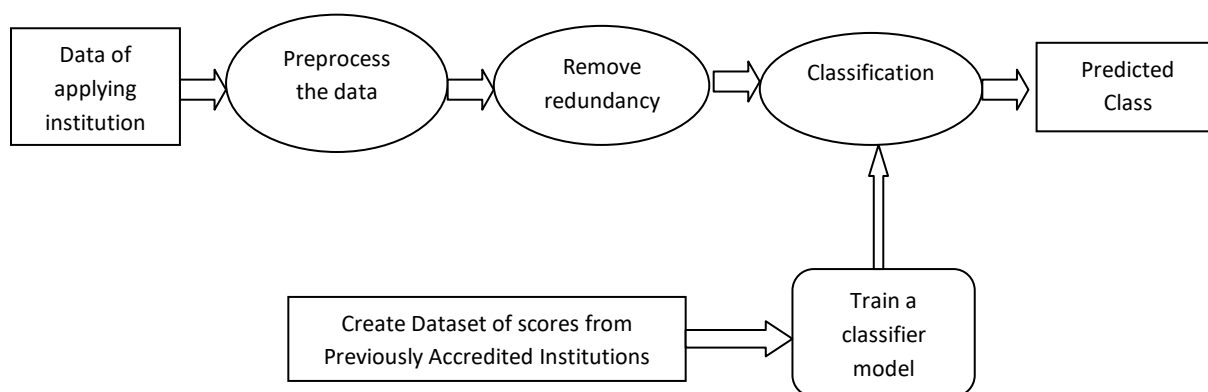


Figure 3 Proposed Methodology

Table 1 Total Number of Valid Accreditations (Status as on 09/09/2019) (Accreditation Status)

Grade / Category	Colleges	University
A++	16	8
A+	134	24
A	1155	155
B++	512	24
B+	691	23
B	2321	75
C	494	11
Total	5323	320

Table 2 Total Number of Accreditations (Status as on 09/09/2019) (Accreditation Status)

Category/Grade	First Cycle	Second Cycle	Third Cycle	Fourth Cycle	Number of Accreditations
Universities	358	166	74	2	600
Colleges	8038	3485	984	35	12542
Total	8396	3651	1058	42	13142

Table3 Grade Break Up of Institutions accredited (As on 09/09/2019) (Accreditation Status)

Category/Grade	A	B	C	Total
Universities	208	139	11	358
Colleges	1685	5401	952	8038
Total	1893	5540	963	8396

The Machine Learning Techniques are already being used in many areas of our daily life ranging from prediction in stock market to the estimation of width of a nerve in the medical science by many researchers. In common language Machine learning is “a field of computer science in which statistical techniques are used to make computer systems to “learn” and make performance improvement progressively with data, without being explicitly programmed for a particular task”. These trained NN models, now are used for classification purpose. This model can be used for future

revisions in the framework. This will also help Government and other policy makers to draw conclusions or to predict the parameters of quality.

The present NAAC's framework grading is based on predefined weights of components; however there are redundancies observed in complete set of indicator's. The removal of redundant information may be attempted using feature selection and feature extraction algorithm of machine learning. This data is also useful to develop regional benchmarking.

As explained in section 2 of this paper the scores are arrived at by encoded benchmark for each question. The present encoding is based on defining discrete scores for range of values. The ranges are arrived at using expert opinion and pilot survey conducted over sample of institutions.

With large data, the expert opinion can be replaced to data driven artificial intelligence based methodology. This AI based algorithm can facilitate in bringing out solution for

1. Obtaining ideal benchmark ranges for region specific, country specific demographically distributed HEI's in order to facilitate regional benchmarking as spelt out in NEP-2020 (Committee, 2019) and EQUIP report of MHRD
2. NAAC grading is always seen as reference to provide objective data to stakeholder for quality of education, building reliable public dataset will facilitate more researchers to undertake policy research related to education.
3. NAAC process of Assessment and accreditation may emerge as a field specific accreditation models extracting field specific indicators or through outputs being normalized to specific fields.

Conclusion and future Scope

After launching of Revised Accreditation Framework, NAAC has opened up an opportunity to build the data warehouse of Institution. This will give lot of opportunity for the researchers who wish to contribute in Higher education by doing research or for the policy makers for decision making.

There may exists a possibility that this warehouse will become the main source of validated data since it contains a verified data by third party agency which is not done by any of the agencies such as NIRF (NIRF, 2015) or AISHE (All India Survey on Higher Education(AISHE), 2011). After implementation of the proposed method, the Outcome of this model is expected to provide

1. The indication towards the policy reforms of NAAC for future course of action.
2. Identification of redundancy in parameters of assessment methodology for future, if any

3. Analysis of Higher education system of the country to understand the input out model for the higher education system.
4. Acting as a driving force for various connected policy decision by facilitating data driven decision making.

The future outcome of this data driven revised framework is that it is opening an interesting resource for educational data which is expected to be largest in size. The data of the institution may form the basis of further revision of the framework using the emerging techniques of Artificial Intelligence. The techniques of Pattern Recognition may help in building big data set of the institutions for the refinement of weightage and removing redundancy in the framework for future revisions.

However some of the major challenges emerging out of automation in process are highlighted as follows:

1. Cope up challenges of associated with Change Management for stakeholders such as staff, Institutions and peers.
2. While the validation of the data submitted by NAAC keep a good level of checks and balances in the process but 100% reliability can't be ensured about the data since higher education system lack centralization of records. Often in order facilitate decentralization of the governance in tune with the constitution of India, where education being listed in concurrent list of state and central government, a centralization of records and lack of strong communication between the two was the key component for the basis of validation.
3. Readiness of the data of last five years. The framework is launched in July 2017 with retrospective effect on the process. Since the earlier framework was not based mainly on data, it took time for the people to get the old data in new format for few metrics; however this change resulted contribution for much bigger notion of bringing objectivity in the process.
4. While the process is end to end solution for A&A, a component of which is based on peer assessment. The future may also demand for removing this component as we move forward with various technology interventions such as Geo tagging as well as Machine learning and augmented and virtual reality.
5. With centralized data housing all the records of higher education the validation would also become light and probably most of the processes will move towards self disclosure and good governance practices requiring no validation.

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28. Industry-Academia Collaboration in Global World: Indian Perspective

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Abstract

Globalisation and modernisation are posing challenges before education for developing global talent. The individuals who are capable of surviving in multicultural society and working in a fast-changing economic scenario can only be able to face the challenge. In this situation educational institutions need to cross its traditional responsibility and move toward working directly for the economic growth and development of the country. For fulfilling this objective Industry and Academia should work in collaboration. In other parts of the world different initiatives have already been taken in this context by India is still in a nascent stage. There is a lot which needs to be done for effective and productive collaboration of Industry and Academia. The best strategy is based on the financial, technical, material and human resources of the country. As both industry and academia are supposed to collaborate so both the partners should zero in on few common objectives and motives. After deciding on that, a beneficial strategy should be planned on the basis of different resources. A collaboration structure with few of the collaboration strategies has been suggested after analysing various successful practices already running in other countries. The strategies if used effectively would be beneficial not only for both the partners but also for the country at large.

Key Words: Industry-Academia Collaboration, Best Practices, Strategies

Introduction

The National Policy on Education of India (1986) characterizes higher education as a ‘Crucial Factor for Survival’, which provides the Indian people with an ‘Opportunity to reflect on the critical, economic, cultural, moral and spiritual issues.’ Education has always been meant to fulfil the needs of the society. Since time immemorial it is doing its job efficiently but due to globalization and modernisation education system is facing complex issues and is finding itself in a critical situation. Global society is posing challenges before educational institutions to nurture global talent by fostering technological skills, reflective and creative thinking, economic expertise and human values. The time has come when the complex concepts of economics, science and technology like demonetisation, e-wallets, digital payments, satellites, global warming, digital classrooms, social

networking sites etc have become a part and parcel of common man. These concepts are not confined only to class economists and scientists.

The education system is producing quite a huge number of unskilled, frustrated, depressed and anxious degree holders who, due to lack of experience of real work culture, are only increasing the unemployed population of the country. To reduce the gap between theoretical and practical knowledge government, educationists, policy makers, industrialists, and economists have realised that the collaboration between academia and industry is highly required.

Dr APJ Abdul Kalam in his book titled '**Ignited Minds (Unleashing the power within India)**' rightly says that our intellectual forums, political platforms, academic institutions and Chambers of commerce should all work together to take up the full responsibility of the development of the country.

Thrust of different Policy Reports

The UNESCO report (1997) 'Learning to Be' states as -“The correlation and collaboration between society and education are so complex that simplified explanations cannot possibly give an adequate account of them. This remark is valid for the tasks which education assigns to society and society to education reciprocally and also for statements concerning the objectives to which such tasks are designed to lead”. Describing education as an image of society, this report has added that education being a sub-system of society, necessarily reflects the main features of that society. It would be vain to hope for a rational human education in an unjust society.

National Policy on Education adopted by the Parliament in **1986** emphasized the need for the collaboration by stating that-“The curricula of technical and management programmes will be targeted on current as well as the projected needs of the industry or user systems. Active collaboration between technical or management institutions and industry will be promoted in programme planning and implementation, exchange of personnel, training facilities and resources, research and consultancy and other areas of mutual interest... Networking will have to be established between technical education and industry, R and D organisations, programmes of rural and community development, and with other sectors of education with complementary characteristics”.

The Technology Policy statement, (DST 1993), has emphasized the collaboration between engineering educational institutions and industry through statements such as:- “Attention will be directed to further enlarge the base of polytechnics, technical and vocational institutes, and engineering institutions and launch programmes for training and retraining industrial and technical

personnel in numbers significantly more than what has been attempted hitherto. Industries will be involved in this process of upgrading the human skills...Pursuit of R and D as a career prospect will be deliberately encouraged through further concrete measures so as to attract scientists and technologists to the challenges of creative science and innovative development with a target of doubling their number in R and D by 2000 AD.”

The DST Policy Statement 1993, further states in the context of Policy Implementation that “RandD institutions including academic institutions would be encouraged to interact with industry and other agencies for contract projects. Income arising out of such initiatives will become additional resources for furthering their infrastructure and other RandD activities... All measures will be taken to accelerate the process of transformation of a laboratory technology to an industrial operation through strengthening and involvement of design, consultancy, and project implementation groups”.

The most of the funding at present for Research and Development in the country is coming from the Government side. There is an urgent need for more contribution from industry in research, besides greater private sector involvement. The industry should engage itself more in terms of not only funding but also in skill development, innovation and entrepreneurship. The gap between the academia and industry has to be bridged to enhance employability of our people. While China spends more than one and a half per cent of its Gross Domestic Product on research and development, India spends less than one per cent, and thus needs a closer partnership between industry and academia (Pallam Raju, 2013).

The review of above mentioned policy reports reflects that collaboration of Industry and Academia is the need of the hour. Resources of both the partners should be used effectively to achieve the objective of nurturing global talent to make them capable of leading the country in global world. For the productive and constructive collaboration it is required that all the policy makers and implementers from all fields of interest like economists, scientists, educationists and government should come together to plan strategies. It is said that a single flower makes no garland so to transform India into a developed country a garland made up of united minds of industrial and academic worlds is required.

Need and Benefits of Industry-Academia Collaboration

Ideally Educational Institutions and Industrial organizations should both work towards the development of individual in particular and society in general. The incongruence in achieving this goal occurs when they both work alone and in certain situations they work in different directions. To make their working styles congruent they should both work in the same direction by keeping common outcomes in mind.

A number of committees have made several recommendations on this issue, significant among which are the **National Knowledge Commission (2008)**, the **Kakodkar Committee Report (2011)** and the **Narayan Murthy Report (2012)**. All these reports emphasize the significance of enhancing industry-academia collaboration for augmenting research, innovation, employability and greater productivity through various measures.

The following reasons clearly mention the need for industry-academia collaboration:

- Due to globalization the industrial world has become more complex.
- Industrialization and modernization has developed new challenging needs.
- Demand of highly skilled, competent and cooperative personnel.
- Huge gap between demand and supply of educated individuals.
- Industry requires experienced and productive human resources which is only possible if they are trained beforehand about the working demands and culture of the industry.
- Need of the fresh recruits equipped with knowledge, skill and attitude.

Bisoux (2003) has explored the relationship between academics and industry. He says that corporations are placing growing emphasis on finding the ‘right person’. It forces the business schools to think more carefully on whom they hire and therefore the role of industry in the entire business school model becomes important.

Benefits

Whenever any collaboration is made it needs to be kept in mind that both the parties should be at winning end as no one wants to make an initiative which is at loss. Therefore, both the parties should be made clear that for them collaboration is a win-win initiation. In the following table few of the areas have been chosen to show that how collaboration is useful for both Industry and Academia.

Table No.- 01

Benefits of Industry-Academia Collaboration

Areas of Concern	Industry	Academia
Knowledge Creation	Database of Knowledge would help in using more effective ways of production	Applied knowledge would certainly enhance and strengthen the theoretical concepts
Consultancy	The ideas and views from the experts of concerned area would add new perspectives for better planning and implementation	The first-hand experiences shared by the industry experts would broaden the horizons of creative thinking of the students and faculty
Research	Cost-effective and innovative researches can be done as per requirement. Flexibility in research projects	Facility to get better equipment's for researches. Challenging research projects. Financial support for research projects Publication of research papers. Filing of Patents
Human Resource Development	Recruiting efficient and skilled human resources More experienced, innovative and creative personnel Globally talented and technically modern individuals	Faculty Development Programs Internship Facility for students More exposure of Industry of interest Development of values like collaboration and cooperation Getting a glimpse of work culture before entering into it

Keeping in mind the various issues along with the benefits of collaboration there is a dire need of collaboration strategies in the era of globalization and modernization. If India has to play an important role in global market then it should give priority to collaboration strategies.

Best Practices of Collaboration in different parts of world

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In India the collaboration between industry and academia might be at budding stage but in other parts of world a lot of work has been done in this regard. Many collaboration models are running successfully. In the following table the level and outcome of partnerships between industry and academia can be understood properly. There are few initiatives which have been taken in India also but there is lot more to be done.

Table No. – 02

Outcome of Partnerships Between Industry and Academia

Partnerships	Parties Involved	Outcomes
Caltech Boeing Strategic Agreement 2004-Present	Caltech is a world-renowned educational institution with a long history of fulfilling the needs of industries, governments and academia. Boeing is the world's leading aerospace company, providing products and services to over 150 countries	Caltech faculty played a key role in certification of Boeing's new 787 Dreamliner. Boeing has hired researchers in key areas Students are encouraged internship Patent applications have been filed Boeing provides Scholarships
Scintillator Development for Medical Imaging 2005-Present	University of Tennessee and Scintillation Materials Research Center, Siemens Medical Solutions.	Various researches are being done due to partnership Several patents have been filed Articles are published
Improving Research and Education By	Syracuse University and JPMorgan Chase	The Center of Excellence in Information Security by JPMC Tech Center at SU

Bringing the World of Practice On To Campus 2007-Present		The domain of Global Enterprise Technology (GET) was developed. This domain focuses on large complex information systems that are typically not covered via traditional university courses. All internships provide value to the student (work experience), to the company (GET curriculum enables more productive students), and to the university (enhanced student learning). The Institute for Veterans and Military Families (IVMF) Supporting at-risk Youth through Mentoring Programs etc.
The International Food Safety Training Laboratory: A Partnership that Improves the Safety of Food Globally 2011-Present	University of Maryland's Joint Institute for Food Safety and Applied Nutrition (JIFSAN) and the Waters Corporation (Waters	Trained professionals from various countries Valuable feedback from trainees has been taken for improvement Students have been hired for key roles
MoU of NASSCOM with UGC and AICTE 2012-Present	National Association of Software and Service Companies (NASSCOM) and the All India Council for Technical Education	Curriculum Review Preparing Training Modules Workforce Development

	(AICTE)	Mentorship Programs Organising Workshops and Seminars
IIM Ahmedabad- IDEA Cellular	Indian Institute of Management Ahmedabad (IIMA) and IDEA Cellular Company	Establishment of Telecom Centre Exploring business issues together
Anna University- TCS	Anna University and Tata Consultancy Services (TCS)	Organizing Workshops for students Faculty development programs Awards for best performances Internship opportunities

The analysis of the above mentioned initiatives clearly indicates that other countries are doing well in this field when it comes to outcomes. They are achieving their goals successfully, whereas in India there are only a handful of cases in which such type of collaboration is done. A thorough and outcome based planning is required. This should be followed by rigorous implementation. Strict supervision and evaluation plan should also be prepared so that funds are utilised properly.

Realities and Expectations

Hon'ble Minister Dr. Pallam Raju pointed out that the current stage of Industry-academia linkages in India today is still quite nascent. He further pointed out that a more sustaining and long term structural plan for continuous development would serve to be mutually beneficial for the industrial sector as well as the academic research driven sector of India. Dr. Pallam Raju, speaking in Workshop on Academia- Industry Collaborations, New Delhi (March 5, 2013) organised by Ministry of Human Resource Development, Government of India, pointed out the following challenges that are faced by the industry and academia today:

- Even though India prides itself on significantly enhancing the service sector, the capacity and competency of the manufacturing sector in India requires serious attention
- Despite excellent work being carried out in academic research labs, there is a considerable lag in converting into a marketable commodity
- There are weak linkages between the research and development taking place in academic laboratories and the various manufacturing companies that could use that research to advance

In the afore-said workshop the ways and means of achieving the goal of collaboration between industry and academia and the need to transform our institutions from centres of knowledge dissemination to hubs of knowledge creation were also discussed. The following areas were stressed upon to achieve this goal:

- Recognizing each other's strengths
- Documenting success stories of industry-academia collaboration for dissemination
- Efficient use of government funds
- Research, Innovation and Entrepreneurship (focusing on Manufacturing, Defence and Sustainable Development and key national goals such as Social Development, Health and Water)
- Skill Building and Employability (focusing on Manufacturing and Services)

While there are many priorities before the MHRD, the three areas requiring top-most attention are:

[a] Improving learning outcomes at all levels;

[b] Meeting the shortage of faculty/ teachers at all levels;

[c] Industry-academia linkages to ensure better employability of our graduates and also skilling of our youth.

It is expected from both the industry and academia that collaboration should be strengthened to make the country an efficient competitor among global leaders.

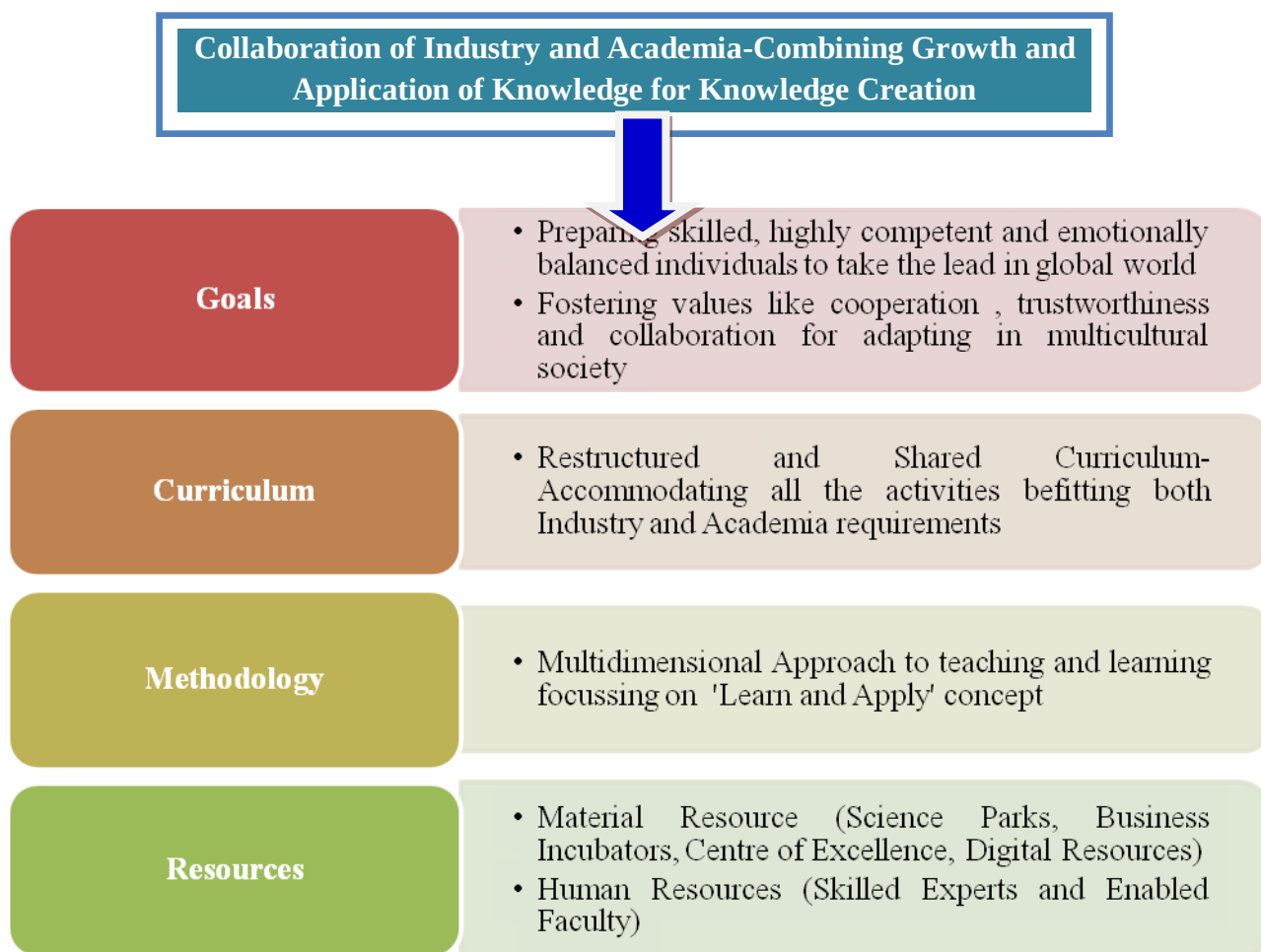
Industry-Academia Collaboration Structure

The following is the analysis of various collaboration models, which are running successfully in other parts of the world and certain points which need to be kept in mind while using them in

Indian scenario. Indian realities are different than other parts of the world. On the basis of realities and expectations a brief outline can be prepared to fulfil the goal of Industry-Academia Collaboration.

Figure No. – 01

Realities and Expectations



Curriculum of academic institutions needs to be restructured so as to enable the industry to integrate its activities. Well-equipped laboratories are required to complete the projects based on industry problems. Both the industry and academia should work in tandem to create an environment of co-learning, co-teaching, co-working and co-researching.

Collaboration Strategies

There are various strategies which can foster the collaboration structure in a better way. If used properly and effectively then it would certainly help in a strong and successful collaboration.

Promote

- Projects that deliver Innovation, Collaboration and Successful Implementation
- Filing of Patents
- Awards to motivate concerned individuals
- Research publications

Figure No. – 01

Collaboration Strategies



Identify

- Problems of Industry for providing Innovative and Creative plans for better production

Facilitate

- Selection and Evaluation of projects that provide sustainable benefits
- Long-term Sustainable projects
- Recruitment of skilled and competent individuals

Enhance

- Experiential Learning and Entrepreneurship Education
- Performance focus

- Global Talents

Create

- Online Portals and other Communication fostering processes
- Conducive Environment for self discipline and self motivation
- Funds from various organizations

Connect

- Research Teams and Industry Teams by allowing access to resources and knowledge database
- Experts from Academia and Industry through Workshops, Seminars, Webinars etc for discussion and sharing stimulating and innovative ideas

Evaluate

- Students should be Evaluated mutually by Academia and Industry

Absorb

- Certificate-holders should be absorbed by the Industry

The discussed strategies would be helpful in making the collaboration effective and productive. The strategies are briefed after reviewing the existing practices of India and other countries. Each pair (Industry and Academia) of partners should make the plan more specific depending upon their common goals, financial, technical, material and human resources to make the collaboration more effective.

Conclusion -Threats and Hopes

Collaboration seems beneficial for Industries and Academic Institutions if planned and implemented properly. On one hand industries are finding that knowledge and research base provided by the academia would help them in getting more production and benefit. On the other hand universities are moving beyond their traditional role of teaching and research toward addressing the demands of industry and contributing directly to economic growth of the country. For successful collaboration both industry and academia need to support each other's missions and motives. Both the partners are looking forward to the effective collaboration as it is beneficial for the socio-economic growth of the country but there are some threatening factors also, which should be kept in mind

before making any collaboration strategy. The research orientations of industry and academia are primarily different, where academia is more interested in developing and creating knowledge base and industry is more focussed on commercial use of that knowledge. It might be possible that academic researches start working in business lines than on fulfilment on academic goals. There is another issue of new patents and products. When it comes to publication, industry is more interested in patenting the research outcomes than on sharing them. There is incongruence in the underlying goals of the industries and academic institutions. The major thrust of academic institutions is on welfare of the society whereas for industries it is on commercial benefits only.

In view of these obstacles to university-industry collaboration, a report of the Joint Project of the U.S. National Council of University Research Administrators and the Industrial Research Institute recommends the following guiding principles for university-industry endeavours (NCURA 2006):

- Successful university-industry collaboration should support the mission of each partner. Any effort in conflict with the mission of either partner will ultimately fail.
- Institutional practices and national resources should focus on fostering appropriate long- term partnerships between universities and industry.
- Universities and industry should focus on the benefits to each party that will result from collaborations by streamlining negotiations to ensure timely conduct of the research and the development of the research findings.

The above mentioned recommendations can be applied to India also keeping in mind the financial and human resources of the country on one axis and capability of producing applicable and useful research outcomes for better production on the other. The ultimate aim of both industry and academia should be the socio-economic development of the country. We can be hopeful for the productive collaboration if we keep the common goal in mind and are able to remove the threats in the larger interest of the country.

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29. Social Science and Democratic Citizenship: Reflections on developing democratic skills among students at secondary school level.

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Abstract

The present article tries to explore the practices, principles, processes which foster democratic engagement among students in the school. What are the various aspects in the school, which can facilitate the democratic functioning in the school? What is the school ethos and how does it function? How schools can be a 'democratic' place(s) for fostering active political participation, and democratic citizenship among students so as to prepare them for future roles in a democratic country, like India. The present paper tries to examine the functioning of democratic principles, and processes in an educational institution-the school, and understands how it aids in democratic citizenship. The paper describes varied aspects of social life to contextualise how democratic principles are inculcated among students. The analysis helps us to understand how democracy is a lived ideal, as it aids in developing the democratic process as an individual and collective action and prepares one to actively engage in the political process of the country.

Key words: Democratic ethos, social science, democratic citizenship, Indian Constitution.

Background: Understanding Democracy in the Indian context-

Rousseau said "Man is born free, but everywhere he is chained" in his famous quote, in the theory of the social contract which explains how a state and a nation must function for society to function properly. Democracy is again one of the concepts that govern and operate in society, but we must also

understand that democracy is not only an activity of society, but is reflected first and foremost at the individual level, then community, social and nation. Though discussing the discourse on democracy, a popular phrase given by Abraham Lincoln is usually deliberated particularly in social science classes. Defining it in very simple terms “it is a form of government where the head of the state is elected by the people”. However, it is not to say that every country which holds elections can be called democratic. This just reinforces the fact, that people are indispensable, valuable and the most important part of democracy. Similarly in our education system ‘students’ are the most important part, the **core**. The school system is built around them and can’t be fulfilled without them. Imagining an education system without students would be like running a vehicle without wheels which has everything- a driver, seats, AC, seat belt etc. but can’t move.

On the other hand, it is the government which is elected by the people, it is obvious to think that people who are in majority in society tend to overpower, and dictate, so at times majoritarianism and democracy are used interchangeably. The question which arises “is majoritarianism actually democracy?” Some may say ‘yes’ however it is not so. There is a high possibility that the majority may be more visible in the forefront, since they are large in number but the interests of the minority and socially or educational backward cannot be ignored.

Fundamentals of Democracy and Citizenship-

Democracy is a way of life which starts from the individual and their nature which is reflected in the society. No other system relies on citizens as much as democracy does. So, it becomes very important for the citizens to be skillful, aware, informed, evaluate and participate in the democratic process on a regular basis, not just during the election time. For a successful democracy, it becomes very important for the citizens to have willingness and ability to face up their responsibility

and fulfill their roles in accomplishing the vision, ideas and goals of democracy. For the reason that is, defined in the earlier part, citizens are the wheels of democracy without them it would be very difficult and almost impossible for the vehicle to run. In this section, the basic edifice of democracy is reconnoitered.

SCHOOL- as a Democratic ‘space’--

Social Science is an integral part of our life and various aspects of an individual’s life is surrounded by it. The rights and duties one performs and recognises as part of being a citizen, helps one to take active part in the democratic functioning of the country. “India being the largest and vibrant democracy has values of justice, liberty, equality, fraternity and the unity and integrity of the nation enshrined in the Constitution, it is essential to appreciate and value them”. Thus, in order to develop these skills, abilities and attitudes among students the school life incorporates varied aspects of the democratic processes, and helps them to be an empowered, reflective citizen in the future. The school is the place where all the democratic aspects come to life, and it is a way to get trained in the democratic process.

The educational aims and objectives are quite democratic, which are essential to develop our nation, and let the citizens participate in the democratic process of the country. The educational curriculum at all levels, have varied aspects, and emphasis on the democratic ethos, the textbooks, and the various activities aid to empower the students. The various co-curricular activities at school, help the students, develop respect for each other and participate in the celebrations, competitions cultural events at school, like our national days Republic Day, Independence Day, or various festivals like Diwali, Christmas, Id, etc. where all respect diversity and give an opportunity for students to participate in various events which are held to mark these days, or events. This helps to foster the

spirit of inclusion and develop basic democratic principles. India despite it's multiple challenges, has always been very positive and aspire to become a very functional and empowering democracy, which promotes values and democratic ethos. It is essential to develop a broad outlook, analyze critically, appreciate, have a balanced judgment, foster cooperation among each other, work together for bringing prosperity, peace and promoting international cooperation.

The following section puts forth how democracy can be meaningful and functional in the school-

First is the *time table*- Time table is the lifeline of the school, an aspect which everyone needs to follow in school whether they are students or teachers. Without the timetable, there will be chaos in school. Time table acts as a road map of the school, and helps to regulate the functioning of the school. Within the timetable, schedules and periods are mentioned, like subject classes, break, library period, games period where the students learn to cooperate and are free to practice the sports they like, the books they like to read and talk about the things they like with their peers under minimum regulation.

Next aspect where the school can enhance the democratic values among their students is the *school assembly*. An assembly of a school informs about the way a school functions. The prayer which is being sung to how the students are standing or to the things which are being presented in the school in the assembly by the students whether be it the news of the day, to thought(s), speeches or some poem, song etc. The student Council constituted by Head Boy, Head Girl and prefects etc. is a way to understand the democratic function at school, as students will in future participate in the democratic process in the country, vote and stand in the elections of the country at various levels- central, state, local etc.

During the social science classroom various debates, discussion help to develop, engage and analyse the democratic values enshrined in the Constitution and understand the democratic ethos. The UN Model assembly, working of Panchayat are all activities carried out in school to understand the basic distinctions and nuances of democratic process, and community living. The role of the teacher is to initiate the students to democratic living, the important aspect to note here whether the assembly is coordinated by students themselves or is being done under the guidance of the teacher, it aids students learn the nuances of democratic living and citizenship. For the students, to blossom to their full potential it is important to have faith and trust on them and motivate them so that they are fully capable of taking responsibility and become capable of executing an event efficiently and independently.

Next aspect which the school provides for harnessing the democratic values in students are the ‘*varied aspects*’ of school life, which help to promote the democratic principles and values among students. The next section briefly deals with these aspects-

Most of the schools have student councils and electing monitors, prefects, or heads of various cultural groups or clubs in school. The heads of these may be sometimes elected in a democratic way by voting by the students and letting them participate in the election process. However sometimes in some schools in India they may be nominated by the teachers directly and voting process may not take place. Annual Functions and various cultural, social events etc. are also a way to expose students to democratic principles, as community living and working as a team is crucial, to understanding democratic ethos. The participation of students in various school events and competitions, helps develop a feeling of team spirit and solidarity. Further, being able to participate at different events in school helps in developing feeling of competition and apart from developing skills of negotiation, and diplomacy.

Besides, there are parent teacher meetings held in order to ensure the proper growth and prosperity of the students. So, for the school to function properly, productively and harmoniously it is very important for teachers', students' and parents' to work together in a coordinated manner. Thus, for the school, all need to work in close coordination as they are all interdependent on each other, which is another way to train and evolve in the democratic principles.

Firstly, classroom seating arrangement may also play an essential role in increasing and decreasing students participation in the class. The traditional way in which students sit in majority of classrooms in India, all facing the front tends to ignore some students at the back and may leave them out of classroom discussions. However, given the large strength and limited space in classrooms it may seem challenging for teachers to experiment with any other form of arrangement. However, the teachers may incorporate group work in the form of projects, activities to involve students which helps students learn analytical and critical skills and work in a team spirit which are essential for democratic living.

From the above, it is very clear that various activities which take place in schools broadly imbibe the democratic values in students.

Role of Social Science in DemocraticCitizenship Education-

The disciplines that are included in the social sciences namely history, geography, political science, economics and others have distinct methodologies and boundaries. The boundaries of the discipline needs to be unlocked so that plurality of approaches may be adopted to understand a given phenomenon, as interdisciplinary approach or multi-disciplinary approach helps enhance students contextual understanding.

Social Science enable us to understand citizenship education as it helps us live in the nation and take care of the needs of the community. A reflective citizen must be fully aware of the socio- cultural issues and its working and contexts. A well informed citizen should have the understanding of the society and act in a responsible manner. He should possess moral values and proper knowledge through education. Besides values, education helps to acquire and empower citizens.

India is a country with unity in diversity. The diverse society has a multitude of languages, religions, tribes, castes, sects, associations and political parties. It is very much important note the contribution of social science education in teaching a citizen to empower themselves. Infusing the knowledge of Citizenship among school students, it is utmost important to develop a clear thinking among them regarding social and natural world, for that the education should encourage to analyse things systematically and objectively. The school investiture system, or student council, provide opportunity to take decisions, work with students, coordinate with teachers and Head of Department, Vice- Principal, Principal, which empowers them and learn the basic skills of democratic living and functioning.

Democracy is based on the tents of freedom, and its success rests on its citizens, as they have to stand and take up the challenge of participatory democracy and make it functional and empowering for the society. Democracy is one of the popular form of participation, on such a wide scale, so much freedom to express views, opinions freely even if they are critical of those in power or authority, a from which shows such high levels of tolerance to diversity and inclusion. The principles on which democracy is based on “equality, freedom, participation, protecting the rights of minority to its best gives every individual a sense of dignity and respect”. As stated by Jermeý Bentham that “democracy work (as based on the utilitarian principle) to promote the greatest happiness for the

greatest number". While J. S. Mill advocated for the harmonious development of human capacities. It is also essential that community needs are taken care of, group consensus and solidarity is key to achieving democratic participation.

The democracy we see today is the result of struggle against unequal distribution and concentration of power and wealth in the hands of the very few. This struggle proves that our socialisation is a part of the historical process which validated the inequalities and hierarchies. The present socialization and the cultural capital which the children and individual accumulate is very much affected by these old historical identities which their family or environment carries. Therefore, to establish democracy and to practice it at an individual level, is a difficult struggle which one can't accomplish alone. That is why it needs to be given its appropriate space in classrooms to debate, discuss, deliberate, engage within the school and make it part of the educational experiences of children.

Besides, if we try to look at it this way, out of the 24 hours available to the student one spends 5-7 hours in school and out of the remaining time 3-4 hours doing things related to school, some homework, preparing for next day arranging bag according to timetable and other things and the remaining time maybe with family, playing with friends accompanied by minimum 8 hours of sleep. Clearly one can see that a major part of every child's life and practically is spent studying. These have immense influence in shaping one's views, opinions and the person one will become in the future apart from family socialisation and values. It becomes the responsibility of school, teachers and the way our education system functions as a whole, to make the child aware, informed, skilful, and a well equipped citizen for the future.

Even within the subjects which are taught in schools social science holds utmost importance when it comes to democracy and citizenship. This subject is all about human behaviour and one's context, surroundings, no other subject gives place to experiences of day to day life of students, their family, the community they come from, the country they live in as the way social science does. It makes students' analyse-- why and how of things happening around them, in their family, in society about all the relationships, contexts, interdependence and hierarchies which exist in our society.

Teacher's Role in the Classroom- Developing Democratic Citizenship

This section attempts to examine the “banking model” proposed by Paulo Friere and the role of a teacher in the classroom as facilitator and co-builder of knowledge. This model that we can see in most schools across the country is what Paulo Friere calls the banking model. However, Friere also talks about changing the banking model with a novel one: the “problematic” model. Unlike “banking”, the “problematic” pedagogy is grounded on communication and dialogue, and it encourages human freedom. It transforms student-teacher relationships, uniting them into teacher-student and student-teacher. Everyone in the class teaches and learns from each other. While the “bank” model includes active teachers and passive students, the “problem” model makes the two groups work together. As this helps to liberate the mind and emphasize human freedom.

The teachers can develop methods in which students are actively involved in the process of search, defining problems, setting hypothesis, taking position to develop critical thinking. In this the newspaper clippings, collages, e-resources etc. may be used as a resource for enabling students to understand the different concepts of the subjects of social sciences which are related to everyday life. The motive is to connect the classroom to the outside world and raise the awareness of students, with

what is going on around them that is the current affairs, so that they feel connected and can relate to it.

The aim is to develop democratic, critical citizens who actively and responsively participate in the society. The general assumption is that classrooms discussions facilitate the development of important skills and attitudes, so that students can participate in the democratic process as citizens. However, not every kind of classroom discussion may be equally effective for the development of skills and attitudes that are important for citizenship education. The quality and depth of classroom discussion are important for learning process. The social science classes provide ample scope for discussion, debate, dialogue etc. on various aspects of the country and assisting students to understand the governance in the country.

The role of teacher in guiding classroom discussions in such a way that they provide learning opportunities for citizenship education is a very challenging and multifaceted task for teachers and it is different from the usual educational practices followed at school. Engaging students in classroom discussion means that teachers must change their daily practices including shift in 'power relations' to be a facilitator. Teachers are used to be perceived as authorities on the subject matter at hand and purpose is to control the content of classroom interaction. Teachers need to become less directive and make more opportunities for students to express their views and the discussions should be more collaborative rather than authoritative.

Teachers initiate the interaction followed by response from students and provide feedback. In most classroom interactions the teacher is the expert and directs the discussion to a predetermined conclusion. In addition, students may also be accustomed to a form of classroom interaction in which teachers expect certain answers, which depends on the school culture, and how teacher uses the classroom 'space', in a collegial environment using appropriate pedagogical tools. The role of the

teacher is quite crucial to develop an atmosphere for free and open discussion, and prepare the students to take up the challenges and participate effectively in the democratic process of the country. The varied opinions, perspectives, attitudes need to be discussed which aids in democratic living and gives the students a true flavour and push towards developing 'enlightened citizens' for the future.

Conclusion-

The thrust of democracy and its principles affect our everyday life, it is also basic that the civil and political rights are sufficiently guaranteed so that people can express and share freely without fear, as it's based on collective process, further, as democracy by nature is inclusive and thrives on participation by all. Though, it may have some lacunas but the very beauty of it, is that we can dialogue about varied issues, contexts, opinions etc. that exist in our society and work on the solutions. Further, as suggested by John Dewey, "...it is participative and part of our everyday interactions'. Thus, the school's role is crucial in developing values to promote democratic citizenship, the varied activities promoting tolerance, cohesion, solidarity, resilience and inclusion which help in achieving the wider context besides promoting the democratic ethos for the country.

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30. Anand Bazar: The Fair of Joy and its Educational Implications

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Abstract

In 1901, Rabindranath Tagore established “Bhrambacharya Ashram” in Santiniketan. From the inception of its journey a unique educational system was experimented and practiced in this “Ashram”, where Tagore adopted a multidimensional curriculum along with a range of co-curricular activities. Various events were also planned and practiced apart from regular school activities and most importantly students, teachers, other staff, guardian and other members of the nearby community participate spontaneously in such events. “Anand-Bazar” is one of such important mega event that is practiced in Santiniketan annually on the day of Mahalaya or better to say, a day before the autumn vacation in Visva-Bharati. It is an integral part of the life of the students of Visva-Bharati. It inculcates various values and boosts up the overall development of the students, specially of the school section. In this study, an attempt has been made to find out and explain the educational implications of Ananda-Bazar, in the life of the students and in development of their social traits.

Keyword: Ananda-Bazar.

Introduction

Functions and festivals are the integral part of Visva-Bharati as well as the Santiniketan community life. Rabindranath Tagore developed a unique educational system which has multidimensional objectives and practices to ensure the fullest development of the students as well as the overall development of the society, nation and the world. On one hand, this educational system is based on the principle of environmentalism while on the other social reconstruction through education has been set as a prime objective of Visva-Bharati educational system. Tagore defined education as “*The highest education is that which does not merely gives us information, but it makes our life in harmony with all existence*”. (Tagore. R, 1917, p.116).

A different kind of curriculum was also designed and implemented by Tagore in his educational institution. A wide range of subjects like dance, singing, clay modeling, woodwork, painting, weaving, nature study etc. were also incorporated in the curriculum with the equal importance with the traditional school subjects. Along with all these various events are also organized on regular basis in form of functions and festivals around the year to ensure the fullest development of the student. Some of these programs are based on the nature and help the students to develop

affection, appreciation, awareness towards the natural environment while some other festivals are also being organized with the aim to develop some personal and social characteristics of their life. In short, it can be said that all the functions and events of Tagore's Visva-Bharati are intimately related with the educational system of the institution, integral part of the curriculum and indispensable of Santiniketan community life, which has the ultimate aim of joy of mind.

Ananda- Bazar or Ananda -Mela: A Fair Of Joy and For Joy:-

Ananda-Bazar also referred as Ananda-Mela, is generally organized in Visva-Bharati on the day of "*Mahalaya*", which is the last working day for the school section before the autumn vacation. At the same time after the fortnight long "*Sarodutsava*", it is a type of grand finale, where the main aim is to get in touch with the limitless joy. Almost all the students and teachers starting from the nursery level to the university level actively participate in this occasion and dip themselves in the ocean of joy. Since the time of Tagore, this was there in practice. "*Karmimandali*" or the Staff Association of the university takes the initiative to organize this event every year with immense sincerity. Students mainly from the school section with their full energy and enthusiasm built up their own shop, decorate it, make food, craft items, painting and other materials by their own, sale those items, keep an account of the shop and submit the profit to the Seva-Vibhag of Karmimandoli for charity. This event formally starts at three o'clock in the afternoon and continued up to eight o'clock in the evening. But these five hours give the student of the university many important lessons from the very beginning of their childhood. "*Gourprangana*", which is very much located at the ashram established by Tagore, is the ground or the venue where generally this event is being organized annually. The concept of Ananda-Bazar is creation of the creative mind of Tagore. In 1915, Ananda-Bazar was organized for the first time in Santiniketan at "*Amra- Kunja*"; very much located in the ashram. This year it was organized on 15th of April which was the first day of the Bengali New Year. After that this fair had been shifted to various places within the campus as its size continued to increase and finally it was placed in the ground of *Gaurprangana*. Similarly there was change in its date also and ultimately it was fixed on the day of *Mahalaya*.

Early in the morning at around 04:30 a.m. or 05.00 a.m. students of the school section and the university level gathered in the ground and the mark their plots for their shops. Then they collect the iron railing supplied from the office and drag those to their shops. With the help of their teachers,

seniors even with the help of their guardians they construct the temporary structure with bamboo and with the railing. All the students are divided in their small groups or according to their department and make their shops and decorate the shop by their own. Various funny names are given to make their shops unique and attractive. During this time the traditional drums of rural Bengal popularly known as “*Dhaak*” are being played all around the ground. Singing, dancing with joy, playing, constructing the shop etc. make the entire place not less than heaven and someone can easily assume the limitless amount of bliss that is being created here and everyone dips their mind in joy in such great event. While a group is busy in constructing the shop, other members of the group or the team are busy in making various vegetarian food items and various craft items like mask, painting, cards, clay modeling, wooden items, painted pot and many more. The students start making all this craft items around 15 to 20 days before this event. At three o'clock in the afternoon the fair or Ananda-Bazar is declared to be open and slowly it reaches to its peak. Small children who are the shop owner convince the visitor to visit their shop and buy something from there. As the evening rolls all the student of the university, teachers, staff, guardians, dwellers of the nearby areas assemble here, meet each other and spend the evening with great joy. Teachers take care of the students and the staff of the university always ready to solve any sort of problem during these hours. At eight o'clock in the evening the fair is closed. At the end, the students make the account of their selling and calculated the profit. After that all the students are back to their hostels and homes with their utensils and other things that they brought from their hostels and homes. The very next day the team leader of each shop deposit the profit amount to the “*Seva-Sakha*” of Karmimandali and that entire amount is used for charity.

This one day of Ananda-Bazar, no doubt brings flood of joy to the life of Visva-Bharati and Santiniketan community. At the same time, this event also helps the students to get many experiences and important lessons of life by participating in such a great as well as creative event and it must be said that along with all the festivals and functions of Santiniketan, this particular event Ananda-Bazar has multidimensional educational implications which ultimately strengthen the overall development of the personality of the students who are the future citizens of the nation.



PLATE: 1- ANANDA- BAZAR IN PATHA-BHAVANA, VISVABHARATI

Educational implications of Anand-Bazar:

- **Creativity:** One of the major aims of the education is to development of creativity within the students, for which various activities are arranged and incorporated in the curriculum. Ananda-Bazar is such a kind of event where at every step the students get ample opportunities to show and enhance their creativity. Making of shops, decorating the shops, for preparing craft items etc. at every stage the students get exposure to the creative world.
- **Cooperation:** All of us are social being and cooperation is one of the major principles to live a peaceful social life. In school level it is one of the major aims of education that is to develop a positive co-operative attitude within the students which will ultimately let us towards a sustainable social life. Ananda-Bazar is such an event which needs teamwork and cooperation at every step starting from its very beginning. The success of this purely depends upon helping each other in putting out their best. So it can be said that, by participating in an event like Ananda-Bazar, positive

attitude towards work and cooperation automatically build up within the students which has the ultimate aim to strengthen the social base.

- **Leadership Quality:** Development of leadership quality within the future generation of the nation is considered as a main objective of education. It is expected that this particular quality will help in getting good leadership in the development of the nation. In Ananda-Bazar the students in each group take all the decisions by their own to run their own shop. At the same time they also develop strategies to earn maximum profit. So through participating in this event student get a direct experience to develop their leadership quality within themselves.

- **Management:** It is another important lesson that the students get by participating in Ananda-Bazar. The students not only have to manage their own shop but at the same time they have to play a vital role in managing the entire Ananda-Bazar. Taking permission for installing the shop, distribution of duties among the team members etc. for the entire process has to manage by the students themselves.

- **Esthetic Sense:** Esthetic Sense is a sense that comes with the creativity within the students. Various arts like painting, clay modeling, wood work, weaving etc. are the integral part of the Visva-Bharati curriculum specially in the school level. In Ananda-Bazar, the students make shop and prepare various colorful craft item for selling. It is needless to say that the variety of items available during the Anand-Bazaar, shows the excellent esthetic sense that the students develop and exhibit over the years. The junior students also learn all these from their seniors which also ensure the flow of culture from one to another generation freely.

- **Vocational Education:** Almost every educational commission in India, after independence had emphasized on the development of Vocational Education in the nation. Vocational Education is also an indispensable part of Visva-Bharati educational system. Specially the curriculum of “*Sriniketan*” gives much importance on vocational education. Apart, in school level also various art and craft subject are incorporated in the curriculum in a view to develop a positive attitude of the students towards the vocational education. Ananda-Bazar is an event where the students specially the school students of class nursery section, primary section or school section probably for the first time earned by selling products made by their own. It not only give them pleasure and satisfaction but at the same time a practical experience and a positive attitude towards vocational education and professional life.

- **Accounting:** Dealing with currency and keeping an account of that during the Ananda-Bazar gives the students an important practical lesson of their day to day life. In Anand -Bazar the students keep an account of their own shop by their own. Collection of fund or contribution for making the shop from the group members, making plans for expenditure, taking prices from of the product after its selling, returning the balance to the customer and at the end accounting the total sale, repaying the initial contribution made by the members if they want and finally calculating the profit of the day, indeed gives them a hand on experience about an important lesson of life. It is intimately related with the vocational learning of the students also.

- **Attitude towards Healthy Competition:** In a fair where many shops with similar kinds of product and with different rates are available, one can imagine the level of competition to maximize the profit of the own business. About a week or ten days before the event, the students advertised for their shops and its uniqueness by posting in the school campus with decent decorative posters, poems and by painting to attract customers as much as possible. During the fair the student continuously try to convince the customers to purchase their products, even many a time the small shopkeepers for say the students of class one, two or three, force their teachers, seniors or other customers to visit their shop and purchase something from there. But all these things happened in a healthy manner. The student of one shop often found to help others to install their shop or decorate it. By this way, an environment of perfect and healthy competition of Anand-Bazar helps the students to develop an important trait of their personality.

- **Respect to Labour:** Success of Ananda-Bazar is a result of unconditional team efforts. Every single member of every shop put beyond their limits to make the entire event a grand success and make a memorable day for lifetime. This is a special day in the life of every student of Visva-Bharati. All the students in group executed their responsibilities and at the same time learn to admire and respect the effort and labour put by the other members of the team also. Probably these gestures of respecting each other, multiply the joy of the students many a times. This gives them a vital social lesson. We the human being have a social life and in our social life each and every member has a defined responsibility to move our society and nation forward. Due to this, every member of the society is equally important and each effort made by each member must be equally appreciated and respect by other also. Ananda-Bazar gives an orientation about this important social responsibility to the future citizen of the nation.

- **Practical Knowledge:** A good educational system always organized such event as a part of its curriculum or co-curricular activities which will be helpful to bridge the gap between the theoretical as well as the practical knowledge. Ananda-Bazar provides the students, specially those from junior classes a lot of practical knowledge which are important for lifetime. Apart the opportunities of social interaction and exchange also help the students to get and exposure to the outer world and develop their personality..

- **Self-dependency:** Ananda-Bazar provides the student a good opportunity and exposure to the decision-making process related to each and every activity of them on that particular day. Planning, formation of group, establishing shop, selling of product, calculating profit etc. in each phase students even the junior students also take decision by their own and this practice no doubt boost their self-confidence and help them to be self dependent.

- **Expression of Self:** Expression beyond any hesitation is an important trait of personality. Balance development of personality is one of the major aims of education. Students in Ananda-Bazar express themselves in the best possible way with their friends and to greater world, in a fearless, burden free environment. Convincing the customers, bargaining with the customers, advertising for the product even to the unknown person help the students to express themselves energetically in a joyful environment.

- **Source of Inspiration:** Tagore educational system always advocate for such a natural environment where the student could learn according to their own interest in a barrier free atmosphere. Imposing unnecessary restrictions in learning was against his principle. An in-depth study of different kind of functions, festivals, events, activities, co-curricular activities clearly indicates Tagore's unique effort to motivate and inspire the students to learn various lessons about the natural environment and about the society. Ananda-Bazar is also a kind of event where students inspire by just staying in the environment. Different types of craft, different innovative decorations, organizing the events, playing responsible role, getting priceless lessons of life, active participation even in the smallest task, facing and overcoming different types of problems and most importantly doing everything by sharing joy, no doubt motivate and inspire the students to perform in the best possible way and grow up physically mentally and socially.

- **Discipline:** Rabindranath Tagore in his educational system practice for self-discipline, which according to him will automatically grow within the students in a free and joyful environment. He was totally against any kind of strict discipline which is imposed on the students in the educational

system or in learning. “*Ashram-Sammilani*” was another unique creative creation of Tagore, which for the first time introduced in his Vidyalaya in 1912 for the student self-governance. An array of activities were set for the students. Students are divided in different groups to ensure their participation in every activity. Students by their own organize all the activities of “*Ashram-Sammilani*”. Responsibilities are given to the student to organize all these activities and that automatically help the students to develop the discipline within them. Ananda-Bazar is a unique tradition of Santiniketan initiated in 1915 where maximum responsibilities of organizing the fair are there on the shoulder of the students. This sense of responsibility plays a vital role in developing discipline in them not only for the day but also for lifetime.

- **Personality:** A balance personality is the desired goal of education. Tagore in most of his educational writing emphasized on the fullest development of an individual keeping harmony with all existence. Each individual with a bright personality will play an active role in the possible development of the society and the entire human civilization. Ananda-Bazar by nature is a fair of five hours, which experiences huge gathering with a very small span of time. Students in this crowd established their identity with their confidence and interact. Apart creativity, self discipline, corporation, leadership quality, respect for other, self-dependency, skill, adjustment power, self-control etc. which are supposed to be important characteristics of one's personality grow effectively by participating in Ananda-Bazar.

- **Time Management:** One of the important traits of a perfect personality is management of time. Learning of organizing time and performing responsibility on and according to time make the future generation dependable and admirable to all. Accomplishment of the event Ananda-Bazar largely depends upon the management of time which keeps the students on their toes to perform their responsibility in time and which ultimately make the students sensible towards time management.

- **A Place of Meeting and Exchange:** Visva-Bharati in its name, spirit and practices carries the meaning and essence of the Vedic Hymn “*Yatra visvam bhavati ekanidam*” “where the entire world rests in one nest”. Tagore always believed that the meeting with others and outer world, exchanging ideas, mixing of thinking and emergence of new ideas and concepts, are the main driving force of human civilization. It was one of the reasons for him to organize such a large number of functions and festival as a part of an academic institution. Ananda-Bazar may be a small event, organized by the school children but during their school life or students life in Visva-Bharati,

they get opportunities of mixing with mass from different discipline, culture, religion, region and even from different nations also, irrespective of caste, creed nationality or any bias which gives everyone the feeling of unity, harmony, national integration, universal brotherhood and ultimately being a world citizen.

- **Blend to Social Responsibility:** Ananda-Bazar has a great objective of orienting the students with the activities of social work. The profit earned by the students is been spend for a noble cause by visiting and distributing essential materials to the nearby villages. The student themselves organized another event known as "*Gramparidarshana*". During their visit they distribute things like books, clothes etc., many a time construct toilet in the rural libraries, purchase books for the rural libraries, install water pump, financial aid for medical or academic purposes, etc. to the distress people in the nearby villages. It is needless to say that, through these activities not only a blend towards social work developed within the students but many other social traits also grow effectively within them. So it can be concluded that Ananda-Bazar not only gives joy to its participants but at the same time it also brings smile to few more children or people who are not directly related with this event. Probably here lies the success of the thinking of the creator.

- **Development of Value:** Development of values among the students through proper education is one of the prime objectives of any educational system. In Tagore's educational practice at Visva- Bharati, along with the formal curriculum and co-curricular activities various functions, festivals and different kinds of events are also included which make it a unique educational system to the learned world. Among the various aims this curriculum, value development among the students is a prime objective. Ananda-Bazar is such a kind of activity which ensures the development of various kind of moral and social values like honesty, hard work, respect another, esthetic value, environmental values, cooperation, team effort, healthy competition, creativity etc. within the students.

- **Mental excellence:** Mental wellbeing and excellence is the most desirable aim in individual life and equally it is needed for a healthy social life also. Anand-Bazar is such an event that ultimately helps the students in their mental excellence along with all those persons who are directly or indirectly involved with this event. Along with all other sides of Ananda-Bazar, it ensures a wonderful time to the students and some unforgettable experiences and memories for lifetime.



PLATE: 2: ANANDA-BAZAR AT PATHA-BHAVANA, VISVA-BHARATI

Conclusion:

Basant.K..Raj; a student of tenth standard of Patha Bhavana, in his essay entitle “Ananda-Bazar” very interestingly mentioned that “*Mela x Bazar = Ananda*” or “*Fair x Market = Joy*”. This was published in a magazine named “Anand-Bazar” by the students of Patha-Bhavana, on 23rd of September 2014. This clearly indicates that, the boundless joy which touches the soul of the students which is the main aim of this event. All the events of Visva-Bharati are equally important and special but the student, specially the school students eagerly wait for Ananda-Bazar which is being organized in the yard their school. This event may be a small, local, ordinary in nature but its effect on the student are very valuable. This event has a silent positive psychological impact on the upbringing of the students. Important lessons of life are hidden in this event which brings a socially desirable behavioral change in the students.

Educational system of Visva-Bharati is unique in many senses. Closeness to the nature, ancient traditional principles of education, teaching-learning under the roof of blue sky, selection of subject according to students own choice, along with a range of craft subjects like singing, dancing, painting, clay modeling, weaving, designing, playing music instruments, wood work, leather work etc. are also included by Tagore in his educational practices and curriculum with equal importance with the regular school subjects. Apart functions and festivals are indispensable part of Visva-Bharati life. Around twenty festivals are organized in Visva-Bharati round the year. Each and every event has some educational values which unknowingly enrich the knowledge and experiences of the students by ensuring individual development. Social reform is an important objective of Tagore Educational Philosophy. Ananda-Bazar is such a kind of event which ensures the development of the characteristics of personality as well as the positive attitude towards social responsibilities. Entire day, all the students are busy in making their shop, preparing different items and selling those to the customers at the end of the day reconciliation the profit for the day. Distribution of essential goods to the distress with the profit earned by the students in Ananda-Bazar is another very important lesson to the life. So, in concluding remarks it can be said that Ananda-Bazar is one of such event of Visva-Bharati Educational system where the students learn various important lessons and gather real life experiences for lifetime. It is the most unique experiment of Tagore and an example of learning with joy and education for fullness that ensures the fullest development of the individual as well as the society.

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31.Impact of Cooperative Learning (Jigsaw Methods) on Biology Achievement of Rural and Urban Students at the Secondary Level

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Abstract

The present research study was conducted to determine the impact of informal cooperative learning strategies (Jigsaw I and Jigsaw II methods) on the academic achievement of secondary students in Biology. To investigate the research question an experiment had been carried out by using randomized group Pre-test Post-test design. A total of 62 students of class IX have comprised the sample of the study. The BAT (Biology Achievement Test) and layout plans based on informal cooperative learning strategies were used as instruments for the present investigation. The BAT comprised of 100 items and its reliability coefficient calculated by the K20 method and Cronbach reliability formula, which were found 0.67 and 0.83 respectively. Two-way ANCOVA was used to test the hypotheses at 0.05 levels of significance. The finding shows that the Informal cooperative learning instructional strategies significantly enhance students' achievement in biology and the mean gain of rural students is slightly more than their urban counterparts.

Keywords: Rural and urban; achievement; cooperative learning; Science; Biology; Jigsaw methods

Introduction

Science education has given due emphasis in the contemporary educational scenario because it educates the young minds about the science subjects and, also develops the scientific temper, ability of reasoning, curiosity, creativity, positive attitude, and problem-solving approach. Not only availability and accessibility of science as a body of knowledge is necessary for all but the understanding of science as a process and way of thinking is also important. Learning science at the secondary level is very important for students of the 21st century to develop their scientific interest and participate in the economic and social development of the nation. Science learning can contribute to develop critical thinking and, prepare the learner to struggle with prejudice and superstitions. It also empowers students to question the social beliefs, notions, and practices that perpetuate social inequality. Therefore for a developing country like India science education is an indispensable need of an hour for sustainable development in a present globalised world. Many efforts and reforms have been made over the last quarter-century to improve science education in the country but still, we are

far miles and our country is lagging in science and technology compared to developed countries (status of science education). In the fast-growing globalised age, we cannot imagine a part of the country enjoying high speed 4G and 5G accessibility of internet speed and other part is deprived of accessibility of basic facility of electricity. For true development and progress of nation quality science education, must accessible for all and steps should be taken to fill the gaps in science education in terms of gender, socioeconomic class, caste and region.

Equity in science education is a major issue for a democratic nation in India. In most of the states in India science is a compulsory subject up to secondary school. But, even after completing their secondary school many pass outs are 'scientific illiterates' or would soon lapse into this state (NCF-2005). Research studies from several countries in the contexts of science education report the differential participation and low achievement of students belonging to different socio-economic regional, cultural and gender backgrounds (Baker, 1998). The provision of education in rural areas and schools has been associated with inferior opportunities for student achievement (Arnold, et al., 2005). The difference in urban and rural achievement is particularly seen in science and mathematics (Graham, S. E., and Provost, L. E. 2012). This rural-urban disparity is mainly because of lack of infrastructural facilities, trained teacher and inefficient resources in the school of rural areas (Young et al., 1997; Steele et. al., 2007; Monk et al., 2007; Elam et al., 2012; Goodpaster et al., 2012). Although many of the good secondary schools in most of country are located in the suburban area which is accessible to both urban and rural students. But the students come from rural background face many difficulties in these schools and especially in the learning of science subjects because in most of these school science has taught in English. Therefore, students coming from rural locality feel alienated from this language and unable to digest the concepts of science in English. Since most of them are a first-generation learner and their rural family background altogether different gradually they lost their interest in a science subject. On the other hands, overcrowded classes are also a big challenge in front of science teacher due to this they are not able to give individual attention to each student. To achieve creativity and overall national development, a teaching strategy that captures the interest of secondary school students in science concepts is crucial (Nnorom, 2015). In this perspective, some effective pedagogies are required that can helps students to develop the feeling of belongingness inside the classroom and science subject too. So, we must quest for some innovative practices that can enhance students' achievement and also inculcate self-esteem and confidence among themselves. Implementation of cooperative learning in science may prove fruitful in this

direction that can enhance the achievement of every student that provide opportunities to interact with each other harmoniously to promotes the learning of every member. An experimental study by Lau, Kwong, Chong and Wong (2014) found CLS improve the students` teamwork skills. The study also, indicated that students were willing to help out other team members to achieve a common goal.

Researchers have been carried out to compare the students from rural and urban areas on a variety of social, psychological, and educational outcome variables in South Africa (Liddell, 1994; Mwamwenda, 1992), Nigeria (Akande, 1990), Australia (NTDE, 1992), India (Singh & Varma, 1995), and Peru (Stevenson, Chen, & Booth, 1990). Studies suggested that the rural-urban differences in cultural, economic, and political conditions can differ drastically from one country to another therefore it appears to be a global issue. The academic improvement of rural student provides one significant barometer for monitoring the progress of a country because not only participation but the success of all students in science is remarkable instead of their locality.

Cooperative learning is theoretically grounded and extensively studied pedagogy. After a meta-analysis of several types of research Slavin (1991) and Johnson (1999) found that enhance the achievement in the different subject area. The positive effect of cooperative learning was also found in all grade levels, in urban, rural, and suburban schools, and for high, average, and low achievers.

There are very few studies carried out on the implementation of cooperative learning in rural schools to improve the academic performances of students in science subjects. A study conducted by Hossain. A. and Tarmizi, R. A. (2013) revealed that implementation of cooperative learning enhances students` performance in mathematics and attitudes towards mathematics. Similarly, in an experimental study of students` academic performance in senior secondary schools in Sokoto State, Nigeria, Haliru, U. (2015) revealed that there is a significant difference between the academic performance of students taught Geography using CLS and those taught using traditional lecture method and also reports a that urban student scored more than their rural counterpart when exposed to CLS. A study conducted by Molla, E. and Muche, M. (2018) to investigate the impact of cooperative learning methods on students` academic achievement and laboratory proficiency in biology subject. Multivariate analysis (two-way ANOVA) was used to analyze the test scores exposed by teaching methods the results show that there were significant differences in the pretest and posttest results of the students` laboratory proficiency by teaching methods. It has been found that cooperative learning methods significantly improve students` performance. Yaduvanshi, S. and Singh, S. (2018) studied the effect of cooperative learning (STAD method) on the Biology achievement of rural and urban

students at the secondary level. In this study, the researcher found that the structured cooperative learning strategy (STAD method) enhanced students' achievement in biology. By motivating this study researcher also inspired to investigate the effect of informal cooperative learning on the achievement of rural-urban students since these two methods of cooperative learning study is quietly distinguished from each other.

More so, there is a dearth of study in the Indian context on cooperative learning that examines the influence of locality of students' on their achievement in biology. The study, therefore, aimed to investigate the effect of Jigsaw I and Jigsaw II methods of CLS on students' achievement in Biology concerning the locality of the students'.

2. Materials and Methods

For the present study, the investigator adopted the 'Randomized group Pretest Post test' design where the students were randomly placed into two groups.

R	O1	X	O2
R	O3	C	O4

where R – random selection; O1, O3 – pretest scores; O2, O4 – post-test scores; X – experimental group; and C – control group.

2.1 Population

All IX standard students of Varanasi city constituted the population of the present study.

2.2 Sample and Sampling Technique

For the present study purposive method of sampling was used by the investigator for the selection of school i.e school from the suburban area where rural as well urban types of students were enrolled. Students were randomly assigned into control and experimental groups.

2.3 Instrumentation

To fulfil the objectives of the present study the following four instruments were constructed by the researcher and used to collect the relevant data:

2.3.1 Biology Achievement Test (BAT)

The data collecting instrument was named BAT developed by the researcher herself. BAT is consisting of 100 items and validated by experts of test and measurement and three experienced biology teacher for face and content validity. The reliability coefficient of the test was calculated by using Kuder-Richardson formula 20 and Cronbach coefficient (split-half method of reliability) method, the values were found 0.67 and 0.838 respectively.

2.3.2 Layout Plans on Jigsaw I and Jigsaw II methods of cooperative learning

The layout plans on Jigsaw I and Jigsaw II methods deals with the theme of organization in the living world and cover four units of Class IX NCERT (National Council of Education Research and Training) science textbook include units; Cell: The fundamental unit of life, Tissue, Diversity in Living Organisms, Why do we fall ill? The plans included instructional objectives, a list of materials needed, group size, assignment to roles, and arrangement of the room.

2.3.3 Opinionnaire

To assess the perception of students' for cooperative learning: A questionnaire of 15 items was prepared by the researcher to assess the perception and feedback of students towards cooperative learning.

2.3.4 Experimentation

The experiments consisted of three phases, the pre-treatment phase, treatment phase and post-treatment phase (Table - 1).

2.4 Participants of the study

A total of 62 students of class ninth have participated in the study. They were randomly divided into two groups, out of which 30 students comprised the experimental group and 32 students in the control group. One group called an experimental group which is taught by the Informal cooperative learning and the other group called as a control group is taught by the traditional lecture cum demonstration method.

Table -1. The various phases, duration and activities of experiments.

S.No.	Phases	Duration	Activities
1.	Pre treatment	2 hours	Pre test was administered on all students. After pretest students were randomly assigned into control and experimental group.
2.			Experimental group Control group
	Pre	3 days	Orientation on the cooperative
	Treatment Phase	(2 hours)	learning process. -----

	Treatment Phase	45 instructional classes	Taught according to lay out plans based on Jigsaw I and Jigsaw II methods of cooperative learning strategy	Taught by the lecture cum demonstration method covering the same units of biology
3.	Post treatment	2 hours	After treatment both group were subjected for post test.	

2.5 Treatment

The experimental group was given an orientation for cooperative learning for 3 days at the same in the previous experiment, and then treatment was given. For the implementation of Informal CLS, investigators used both types Jigsaw i.e., Jigsaw I and Jigsaw II according to the suitability of the contents. In this method, the topics to be studied were segmented in sub-topics and member of each group was assigned a particular subtopic of the lesson. All members sharing the same sub-topic were formed into expert groups where they discussed their content to master and become experts. They finally reconvened where each member explained his unit to other members of his/her group. While in Jigsaw II individual prepare his/her topic and taught within their respective group. The scores of members of the group were summed together to form the group's score which was used to award prizes to the best group.

3. Results

The results of the present study were investigated by two-way analysis of covariance (ANCOVA) to assess the interaction effect of independent variables (instructional methods) and moderating (locality) on the dependent variable (achievement in BAT). All values were expressed as mean ($\pm$ SE). P-value < 0.05 was considered significant in the present study.

Two way ANCOVA table to analyze the interaction effect of locality with methods of instruction has been represented in the following tables (Table 2-4).

Table – 2. Descriptive statistics for effect of Jigsaw methods in relation to locality

Group	Area of residence	Mean	Std. Deviation	N
Experimental	Rural	69.77	10.248	13
	Urban	66.41	7.229	17
	Total	67.87	8.669	30
Control	Rural	49.83	12.059	12
	Urban	54.65	9.438	20
	Total	52.84	10.574	32
Total	Rural	60.20	14.916	25
	Urban	60.05	10.274	37
	Total	60.11	12.241	62

Table – 3. ANCOVA for effect of Jigsaw method in relation to locality

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	8169.883 ^a	4	2042.471	119.981	.000
Intercept	801.563	1	801.563	47.086	.000
Pre_Total	4418.316	1	4418.316	259.546	.000
Group	2050.221	1	2050.221	120.436	.000
Locality	1.104	1	1.104	.065	.800
Group * Locality	25.920	1	25.920	1.523	.222
Error	970.326	57	17.023		
Total	233181.000	62			
Corrected Total	9140.210	61			

a. R Squared = .894 (Adjusted R Squared = .886)

Table – 4 Estimated marginal means - 4a. Group, 4b. Area of residence, 4c. Group* Area of residence**4a. Group**

Group	Mean	Std. Error	95% Confidence Interval
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			Lower Bound	Upper Bound
Experimental	66.252 ^a	.769	64.713	67.791
Control	54.223 ^a	.763	52.695	55.751

a. Covariates appearing in the model are evaluated at the following values: Pre Total Score Test = 22.05.

4b. Area of residence (Locality- adjusted scores)

Area of Residence	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
Rural	60.101 ^a	.826	58.447	61.755
Urban	60.374 ^a	.681	59.011	61.737

a. Covariates appearing in the model are evaluated at the following values: Pre Total Score Test = 22.05.

4c. Group * Area of residence

Group	Area of Residence	Mean	Std. Error	95% Confidence Interval	
				Lower Bound	Upper Bound
Experimental	Rural	66.784 ^a	1.159	64.463	69.105
	Urban	65.720 ^a	1.002	63.714	67.725
Control	Rural	53.418 ^a	1.212	50.992	55.844
	Urban	55.028 ^a	.923	53.180	56.876

a. Covariates appearing in the model are evaluated at the following values: Pre Total Score Test = 22.05

The ANCOVA analysis to find out the effect of Jigsaw methods about locality shows $F(1,61) = 0.065$, $p = 0.800$, here, $p > 0.05$ for the locality of students. It indicates that there exists no significant difference between the achievement of the rural and urban student. The interaction effect of group on locality shows $F(1,61) = 1.523$, $p = 0.222$; $p > 0.05$ indicates that the interaction effect of Jigsaw with locality had no significant effect on the achievement of students in Biology.

Table 4a showing the interaction effect of group and locality shows that the adjusted post total BAT score of rural students in the experimental group is 66.252 which is significantly higher than the scores of rural students in the control group (54.223). Table – 4b showing the adjusted total post-test scores of rural is 60.101 and urban is 60.374 there, is a very slight difference in the adjusted mean scores of rural and urban students; indicates that there is no significant difference in the achievement of rural and urban students. The scores of adjusted mean scores of rural and urban students in the

experimental and control group are represented in table 4c shows that the adjusted means of rural students (66.784) is slightly higher than the score of urban students (65.720) in the experimental group. Also, in the control group, this difference is insignificant as the adjusted means of the rural student is 53.418 and urban student is 55.028. These scores clearly show that there is no significant difference between the achievement of rural and urban student, but the achievement of both types of students i.e. rural as well as urban enhances through Jigsaw methods.

4. Discussion

This study reveals that both Jigsaw methods significantly enhance the achievement of rural as well as urban students in comparison to the control group. The mean gain was a little in favour of rural students. This indicates achievement of rural students is considerably enhances within a cooperative learning group. It was found that the Jigsaw method had no significant effect concerning the locality of the students. In a cooperative learning situation, the students learn in small heterogeneous groups, all group members have their definite role and they all participate fully in cooperative learning activities; interact with each other and the boring science lessons become interesting to them. In an experimental study Lau, Kwong, Chong and Wong (2014) found CLS improve the students' teamwork skills. Implementation of Jigsaw methods in the science classroom can enhance achievement every student also provide opportunities to interact with each other harmoniously to promotes the learning of every member. The group discussions and group presentations during Jigsaw methods provide opportunities for all types of learners to express their views without hesitation. Since the medium of instruction in school was the English language and students from rural background feel diffident and hesitated to express their views during the traditional method of teaching due to language inefficiency. But the students had many opportunities for self-expression during group works. The views and voices of all members were respected because the group members were positively interdependent on one another to achieve common learning goals. The group discussions and group presentation provided opportunities for all types of learners to express their views without hesitation. These components of cooperative learning strongly promote cultural parity to fill the gap between rural and urban students. The results of the present study are aligned with the findings of Slavin and Oickle's (1981) and Hossain & Tarmizi (2012). The findings of the present study are in contrast with the result of Haliru, (2015) who reported a significant difference in the academic performance of urban and rural students when exposed to CLS in favour of urban students. The feedback of the students on the opinionnaire regarding their perception of

cooperative learning also reveals that learning in a cooperative learning situation is very interesting and joyful to them. They all feel that these methods are helpful to them to develop more friendly relation to other and improve their way of communication a lot.

5. Conclusions

It has been concluded that Jigsaw-I and Jigsaw-II methods of informal cooperative learning strategy are significant instructional methods for learning biology as compared to the traditional method of lecture-cum demo method. The study also confirms that Jigsaw-I as well as Jigsaw-II both methods of cooperative learning create a harmonious and pleasant classroom environment where everyone got an opportunity for joyful learning and develop the good skill of communication among them.

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32. New Education Policy 2020 and Teacher Education

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Abstract

The present paper reviews the New Education Policy 2020 with a specific focus on the teacher education. The policy seeks to bring reform in the education system by bringing structural changes in the system with minimal financial input and other resource allocation. It is argued that the policy carries forward the neoliberal agenda of market and promotes privatisation of education at all the levels. Teacher education is one of the main areas to improve the quality of education in the country. Only qualified educated and trained teachers can help towards the building the Nation and contribute towards economic development. This paper also examines and discusses how the new education policy attempts to bring drastic changes in the teacher education and attempts to improve teaching profession using the incentives such as performance based promotions and tenure. The policy presents a minimalist vision of education and teachers.

Keywords: Teacher Education, Teacher, Policy, Performance, Learning

Introduction

The New Education Policy 2020 brought by Central government is going to define the vision of education in many ways. A number of structural alternations are proposed in the new education policy 2020 to uplift the face of Indian education system. The policy promises to allocate 6 percent of GDP towards education budget. It is one of the most welcome steps as financial support can bring substantial quality improvements in the provision of educational services.

From pre-primary to higher education level new the policy attempts to bring drastic change in terms of structure and function of the entire system. The main highlights of the NEP 2020 are changing the 10+2 school education system to 5+3+3+4 and introducing four year bachelor programmes. The proposed four year bachelor course will make the Indian students fare better and get admission in foreign universities.

The policy recognises the fact that the increasing demand for education cannot be met only state provision but private sector also needs to be encouraged to provide educational services. To bring changes in the Higher education system many reforms are proposed. Teacher education is no exception to this re-structuring and the policy proposes drastic measures for teacher training intuitions as well as the

teaching profession. Teachers are one of the important pillars of the education system. However, teacher in government schools are at the centre of criticism and quite often blamed for the poor performance of the students at elementary level (Atherton &Kingdon, 2010). The policy also sets the precedence how learners and teaching leaning process would be perceived. The overemphasis on the acquisition of cognitive skills of reading and writing undermines the child centred approach. The new policy tends to reform teaching-learning process as well as the teacher is going have serious implications particularly for the future of elementary education in the country.

Performativity and Teachers

Ball argues (1999) that the education policies across the globe despite socio-cultural and economic diversity are being organised around two main thrust or principles; first is entry of ‘market form’ and second is ‘performativity’. It is argued that introduction of market in education will provide choice and incentive to compete with different service providers hence improving the quality of education. Within this context performativity entails setting of targets and performance indicators to “drive, evaluate and compare educational products” (Ball, 1999:1). Both these principles form the core of New Education Policy, 2020 in India. At all the levels of education provision privatisation is encouraged and regulation and performance standards are set exclusively for government funded educational institutions while privately owned intuitions have the freedom to self-report.

The policy also reduces the notion of education using the narrow lens of learning and mere tutoring. The policy curtails teacher’s autonomy by defining learning achievements for children, where the primary emphasis is on acquisition of fundamental numeracy and literacy skills. This also reduces the notion of education to a ‘minimalist’ level. The policy says,

“The highest priority of the education system will be to achieve universal foundational literacy and numeracy in primary school by 2025. The rest of this Policy will become relevant for our students only if this most basic learning requirement (i.e., reading, writing, and arithmetic at the foundational level) is first achieved” (GoI, 2020:8).

It also tends to underline the principle of homogeneity and such statements render teaching learning activity devoid of the context and unidirectional. However it is important to keep in mind that teaching work is bidirectional and it not only includes helping children to acquire cognitive skills of reading and writing but also requires the teacher to have a set of cultural, social and emotional competencies,

sensitivity, and communication skills besides subject expertise to make the teaching learning process effective.

Improvement in quality of education in terms of students learning can only be achieved by bringing substantial changes in how we prepare our teachers and what kind of support in-service teachers require. The teacher appointment criterion for teachers at elementary level of schooling is not in sync with the provisions of Right to Education Act 2009. The Act highlights the importance of availability of professional trained teachers to bring reform in the school education (RtE, 2009, section 23). Ball and Forzani (2009) have distinguished between the work of teacher and other mundane 'teaching' by parents and other individuals in everyday life. They argue that,

"The work of a teacher is instead specialized and professional in form and nature. Decisions about what to do are not appropriately rooted in personal preferences or experiences but are instead based on professionally justified knowledge and on the moral imperatives of the role. Intuition and everyday experience are poor guides for the specialised work and judgement entailed by teaching" (ibid: 500).

To support the teachers to meet the changing expectations of the curriculum, pedagogy and technological advancement a robust system is required. But in the policy no attempt towards institutional capacity building is taken. No effort is made to develop professional competencies among the teachers through per-service and in-service programmes. For instance, in year 2015-16 only 14 percent of elementary school teachers received any kind of in-service training (DISE, 2017) it shows that the provisions for in-service support for teachers is inadequate.

The NPE 2020 clause to hire 'eminent' local person to teach at school further damages the teaching profession as it sets the precedence that subjects such arts, crafts, music and physical education are not worthy to be taught by trained teachers in these field. It is also important to note that in a society like India which is based on the principle of graded inequality who will decide this 'eminent' person worthy to be invited to teach in the schools. It is also against the constitutional spirit of equal opportunity. Such provisions fail to take cognisance of failure of outsourcing services done in past to deliver education services effectively through NGOs. A classic example in this regard can be cited from policy documents such as SSA (SarvaShikshaAbhiyan, 2000) which talks about participation of civil society to spread awareness about the education for all programme and encourage participation of local people in the delivery of primary education. This led many NGOs modifying their working to suits the requirements of the government funded projects. The problem with such an approach is that such NGOs tends to focus on short terms gains as they exclusively depend upon the funds provided by

the donor agency. This output oriented attitude leaves much to be done on the ground. As many of the NGOs in their attempt to scale up their models fail to release the complex social structure of different regions and diverse needs of disadvantaged social groups which might prove counterproductive.

The new policy attempts to bring competitiveness among the teacher by introducing performance based tenure and other benefits. Provision are made to hold teacher accountable for the learning achievement of the children which also reflects the larger market ideology of commodification of education, performativity, instead of making provision of career advancement based on incentives, the teacher are de-incentivised at various levels in terms of job tenure and pension benefits. The policy states,

“Teachers doing outstanding work must be recognized and promoted, and given salary raises, to incentivize all teachers to do their best work. Therefore, a robust merit-based structure of tenure, promotion, and salary structure will be developed, with multiple levels within each teacher stage that incentivizes and recognizes outstanding teachers. A system of multiple parameters for proper assessment of performance will be developed for the same by State/UT Governments that is based on peer reviews, attendance, commitment, hours of CPD, and other forms of service to the school and the community or based on NPST given in Para 5.20 (GoI, 2020:22).”

The bias towards detailed monitoring and setting performance parameters for teachers in educational institutions often pervades the view that teaching is not a creative endeavour and only thorough extrinsic motivation of reward and punishment better ‘quality’ education/set goals can be achieved. It is also important to note that at elementary level of school majority of the teaching workforce in India is female. Such regressive measures will have long term implication for the education of women in the country. Despite many contestations teaching at elementary level is feminised and in popular culture it is seen as the ‘most viable’ ‘career’ path for girls. According to DISE data in year 2015-16 the percentage of female teachers was 48 at all India level. For the same year, the percentage share of teachers belonging to SC and ST communities was 12 and 8.5 respectively (DISE, 2017). For optimal functioning of the education system and effective delivery of educational services to the diverse population requires incentives to retain and motivate the teachers. The policy tends to prefer ‘merit’ over diversity. The merit based structure for career advancement without giving any heed to the socio-cultural, gender and economic background of the teachers tends to marginalise those coming from already disadvantaged background. There is no commitment towards increasing the recruitment and

retention of the teacher to cater to the demands of the ever growing student population. On contrary the state started hiring teachers on contractual basis from 1990s and the teachers were called ‘para’ teachers. The new policy also carries forwards this trend and the teaching profession is disincentivised by linking promotion to the performance. The policy curtails the avenues of salary increment and promotions it mentions,

“Promotions and salary increases will not occur based on the length of tenure or seniority, but only on the basis of such appraisal. The professional standards will be reviewed and revised in 2030, and thereafter every ten years, on the basis of rigorous empirical analysis of the efficacy of the system” (GoI, 2020: 22-23).

There were 543,671 contract teachers in India in 2008-09 (Atherton & Kingdon, 2010). No attempt is made to bridge the gap between regular and ‘para’ teacher at school level and casualization of teaching profession is encouraged. The new policy also fails to recognise the huge number of teachers employed in private schools especially the low fee private schools. Most of the low fee private schools function on ‘hire and fire teachers at will policy’ (Verma, 2017) and most of the time they are underqualified and paid very less as compared to teachers of elite private and government schools. There is no mention of any kind of service regulations and incentives for private school teachers and regulation private schools itself. The progressive step to end the inspection regime and allow schools to self-report is going to complicate the situation given the fact that there is a lot of ambiguity in the functioning of private schools and it is well known that quite often is under and over reporting of certain facts and the data. Also, this will create problems for data collection for the purpose of research is going to be very challenging.

Teacher Training Programmes

The policy acknowledges acute shortage of teachers in public elementary education system. To supply the demand for trained teachers the policy emphasises the need to increase the number of teacher training institutions and establish multi-disciplinary universities. At present there are a variety of teacher training institutions existing in the country under government as well as private management offering different degree and diploma courses. Approximately 90 percent of the teacher training institutions are under private management (Srinivasan, 2015) and many of these are of sub-standard quality. Much remains to be done in terms of regulating the privatisation in the field of teaching education as the

policy is silent on the question of private B.Ed colleges. It can be said provision and regulation of teacher education institutions was never on the top priority of the government. New Education Policy 2020 also carry forwards this legacy and instead of regulating the operations of standalone teacher education institutions the policy recommends developing multidisciplinary universities where teacher education can be located. The teacher training programmes are under scrutiny and many committees have evaluated and recommend reforms in these programmes. The much debated issue whether teacher education institutions should be brought within the ambit of higher education are put on hold by NPE 2020. The standalone teacher education college will cease to exist by year 2030. The policy talk about ‘light’ but ‘tight’ regulation in disguise to curtail whatever autonomy is left with public funded educational institutions and teachers,

“A ‘light but tight’ regulatory framework to ensure integrity, transparency, and resource efficiency of the educational system through audit and public disclosure while encouraging innovation and out-of-the-box ideas through autonomy, good governance, and empowerment” (GoI, 2020:5).

It is pertinent to mention that the policy integrates the pre-primary classes with grade 1 and 2, but there is no road map to prepare teachers for the same. Nothing is said about appointment of regular teachers and how this would affect the work of Anganwadi workers. The policy absolves the government from making appropriate provisions for the training of pre-primary teachers by suggesting the use of technology and distance learning. However the research in the field of early childhood care and education and the National Policy on Early Childhood and Education 2013 recommends that teachers should be trained to develop theoretical understanding of the child development and pedagogy. The private schools are given the freedom to continue with nursery classes as it is. However nothing is said about the appointment and training of teachers in private schools.

In the name of innovation and out of box thinking privatisation is promoted. Instead of allocating funds and resources to uplift the face of teacher education institutions, the most apparent reform is to increase the duration of teacher training programmes. It suggested that the teacher training courses will be of four year duration to improve the quality of teachers. However, the policy contradicts this provision by making provision for ‘special shorter’ local teacher education programmes to be available in BITEs and DIETs. No time period is given to define this ‘shorter’ local teacher education programmes. Such disparities will exacerbate the already existing quality issues in terms of teaching learning activities. The ‘special shorter’ courses would fail to bridge gap between theory and practices. Sarohe (2018) based on a study of Bachelor of Elementary Education (BELED) programme and

Diploma in Education (DEd) programme which are of four year duration in Delhi attempts to understand the notions of citizenship from teacher-students and concludes that most of the candidates hardly get any space and scope to engage at an in-depth level with stereotypes around citizenship and notions they have about many social issues. Hence it can be said the teacher education programmes find it hard to challenge the social and cultural biases and teacher-students are socialised in the dominant curriculum.

Deliberate Gaps

The Indian education system is marked by hierarchies and social inequalities in terms of caste, class, gender and religion. To maintain the social cohesion and harmony it is essential to give equitable representation to diverse social groups in the education system especially at higher education level. However very slow progress is made in this regard and majority of the marginalised social groups remain outside the purview of higher education. Batra (2005) argues that education system in the country is dominated by upper caste which tends to maintain the 'status quo on the questions of caste, community and gender asymmetry'. The policy tends to set the course whether education is going to be an agency for social change and strengthen democratic principles or its going to exacerbate the existing social hierarchies. Sarohe (2018:32) highlights the importance of Teacher Education programme in shaping the worldviews of teachers as well as students; she argues that, "teacher education programme should underline an active engagement in public life, and dispositions and sensibilities for living in plural societies with a commitment to social cohesion and solidarity." The new policy deliberately avoids using the constitutional mandate for positive discrimination at higher education level for disadvantaged, marginalised, and religious minorities. This is going to have discernible effect on the future of teaching profession as there would limit diversity in terms of representation of individuals from SC, ST, OBC and other minorities. The teacher education training intuitions have failed to cater to diverse classroom as majority of the teachers are urban based. The teacher education programmes do not cater to the needs of diverse population groups across the country. The policy attempts to wish away with these issues by bringing structural and curricular reform but it fails take any substantial efforts to make the educational institutions more democratic spaces. It can be said that teachers are public intellectuals who shape the future of the country.

To a great extent the quality of any education system at intrinsic level depends on the quality of its teachers. In the name of performance and improving standards the policy promotes neoliberal agenda of privatisation and managerial regime in the field of education. The policy document uses

authoritative language and there is no scope for dialogic process and equitable representation of different social and cultural groups. To conclude it can be said that the policy promises to re-establish the lost credibility of the teachers but in spirit it tends to further degrade the value of teachers.

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33.School Choice through Educational Voucher System: A Move towards Decentralization and Heterarchy

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Abstract

Education is the backbone of every individual in this world. Education is necessary not only for self-development but also for the success of the whole nation. Every nation has its unique education system and each country tries to sort out its educational problems in its own way. The Government of India has attempted to universalize education, yet the goal seems to be far away. It has been reasoned that children are out of school because lack of infrastructure and poor quality of education offered in them (Weidrich, 2003). However, the lack of facilities and infrastructure in schools is attributed to mismanagement and not shortage of funds. The problem is therefore with the inefficient and corrupt utilization of the budget.

Time is ripe for the Indian education system to reorganize itself for better to enhance its quality. One potential step towards the enhancement of quality education could be the adoption of the voucher scheme. The educational voucher scheme is not only enhances the quality of education but also allows parents the volition of the choice of schools, be it government, private or non-government schools. It increases the competition among the schools to enhance their quality to be able to attract greater number of vouchers.

Keywords: Education voucher, quality education, universal education, educational choice, schools

Introduction

The importance of education for the growth of individual and society is indisputable. In the attempt to universalize education, the Government of India has time and again initiated various programs. One such program was the SarvaShikshaAbhiyan (SSA) that attempted to universalize elementary education through community ownership of schools. Similarly, the 'Right to Education' bill was initiated to ensure 100 per cent accessibility to education. Report claims that 90 percent of the country's population have access to a primary school within one kilometer radius of their house(Singh, 2010).Evidentially in their paper 'Caste and Labor Mobility', economists Hnatkovsha, Lahiri, and Paul (2012)observed significant convergence in occupational distribution, wages, and

consumption patterns of SC/STs and non-SC/STs. Similarly, between 1983 and 2005, the percentage share of SCs/STs in blue and white collar jobs have improved. Further, the wage gap between the SC/STs and the mainstream population have declined across all the income group except for the highest income quintile (Sharma and Chandrashekar, Report of Deccan Herald, 2019). The primary reason for this convergence can be attributed to the spread of education among the underprivileged section.

However, it is observed that majority of the underprivileged section attend government schools, simply because it is affordable to attend such school. In the process, they are deprived of the agency to make choices to avail educational services. Research reveals that education provided in government schools is below quality and students often face discrimination in schools. Many of the government schools suffer from lack of adequate teachers, high student-teacher ratio, infrastructural shortages like shortage of classrooms, laboratories, washrooms, playgrounds, drinking water facilities, etc. In fact, DISE data between 2011 and 2016 reveal that total enrollment in governmental schools declined by 13 million and enrollment in private schools increased by 17.5 million (Kingdon, Report of the Times of India, 2017). Such decline in enrollment has rendered many of the governmental schools dysfunctional and 40 per cent of the governmental schools have less than 50 students (Kingdon, Report of the Times of India, 2017). The poor quality education coupled with the discrimination they face in schools, makes the underprivileged children to eventually drop out of the system. Few students who are able to sustain themselves through such a system are too weak to academically compete in the mainstream society. The poor comprehension level of such students was reflected from the result of the PISA test where India ranked only ahead of Kyrgyzstan (Bhasin, Report of OpIndia, 2019). It is thus evident that such education system renders the students inept of availing full advantage of reservations prevailing in higher educational institutes in India.

As around the globe, the Indian education system is heavily hegemonized by the rich and educational success is correlated with strongly associated with greater wealth, status, and privilege (Tesse, 2000; Feinstein et al., 2008; Thomson, 2009). Echoing Bourdieu, we contend that education system largely preserves the status quo through the reproduction of social and economic privileges (Bourdieu and Passeron, 1977). Bourdieu argues that children coming from families having ‘cultural capital’ that matter in schools are more likely to be successful in schools. These differences are exacerbated by the differences in school pedagogies (e.g. in the government vis-à-vis elite private schools), curriculum, books referred, method of examination adopted, etc. As Velaskar (1990: 136) puts it, “*The school*

system in India-its organization, the internal structure of individual schools, content and processes of schooling, indeed its total ethos-is structured towards bringing about the unequal outcomes that it eventually does.”The dominance of the rich/powerful in the private schools is maintained by the ‘sorting’ and ‘selecting’ mechanisms adopted by these schools in the form of admission test/requirement at the entry level. In sum, educational diffusion in the country has thus parochially benefitted the “upper, middle class Hindu men” and the underprivileged are forced to remain content with constraints arising out of structural inequality in society (Velaskar, 1990).

Not only discrimination is class-based, it exists at the level of caste and gender as well. A study by Jeffrey et al (2005) reveal how the upper castes in Uttar Pradesh have successfully defended their privileges over the years and have successfully used their disposable income to offer private schooling to their children and thereby preserve their hegemonic standing in society. Similarly, owing to the patrilineal nature of society, the Indian parents prioritize educational expenditure for their son’s rather than their daughters (Dreze and Sen, 2002). Evidentially speaking, studies by Kingdon (2005) lends credence to the pro-male enrollment bias existing in low-fee private schools in Uttar Pradesh. Therefore, in India there exist a correlation between access to good quality education and various ascribed variables (like class, caste, and gender) (Harma, 2009). Though the access to quality education is hegemonized by the rich, the underprivileged remain desirous of offering quality education to their children in private schools as revealed by the study of Tooley and Dixon in 2006. In their study of schools in Hyderabad they found that the low fee private schools, which otherwise were inaccessible to the underprivileged, were made accessible to them by means of numerous concessions and scholarships. Echoing Sen (1999) we argue that by denying the underprivileged the access to quality, we are denying them the freedom to act and develop their capabilities (Harma, 2009). Development to Sen (1999:74) is to allow “substantive freedoms-the capabilities-to choose a life one has reason to value.”

The Educational Voucher Scheme

Empowering people to have a say about the education of the children is an essential facet of the democratic society (Levin, 1991).One way of lending more voice/involvement of the stakeholders in the system is to make schools accountable to both educational authorities and parents (Shah, 2009). This can be done by providing access of governmental funds to parents or children instead of schools. Thus, they will have the agency to choose schools, be it government, private, or non-governmental schools. This precisely is the conception of the educational vouchers, whereby the needy students are

able to choose schools. Stated differently, in the current education system ‘money follows schools’ (figure: 1) but in the voucher system ‘money follows students’ (figure: 2). **Basic Difference between Present Education System and Voucher Education System**

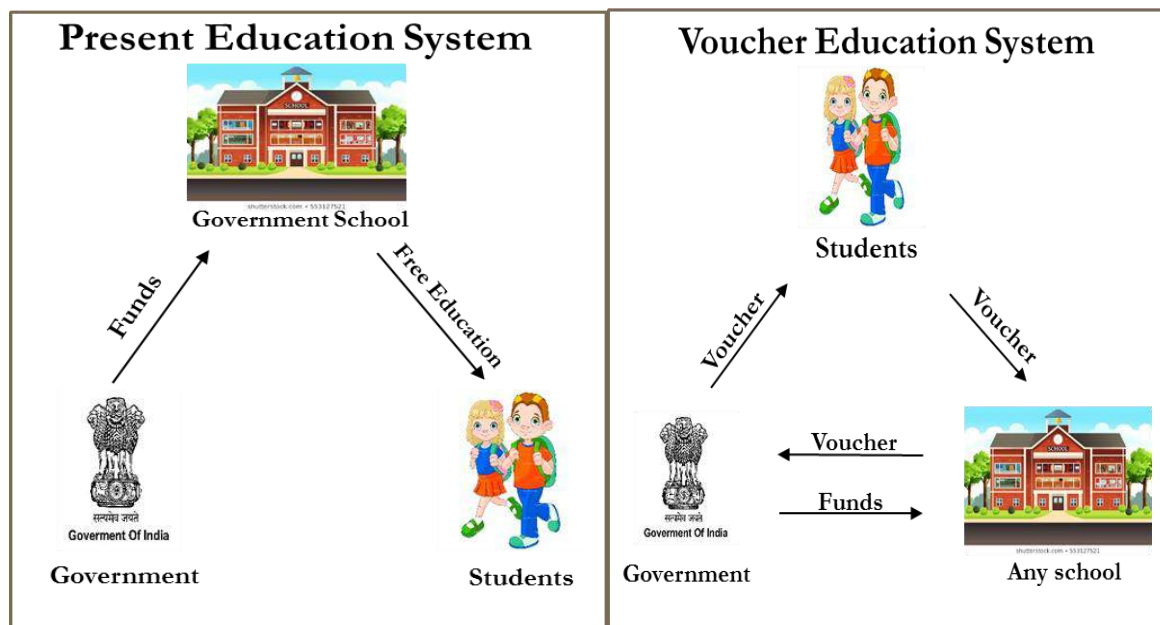


Figure: 1 Figure: 2

Therefore, schools compete with each other to attract and retain greater number of vouchers by the way of attracting greater number of children and thus strive to improve their quality (Singh, 2010). Voucher scheme or ‘direct benefit transfer’ is thus expected to improve the quality of education, enable agency of the parents and students to choose schools, and provide equal opportunities for education to the underprivileged (Gomathi and Sudhakar, 2014). Borrowing from Hirschman (1978) we thus contend that voucher scheme lends ‘voice’ to the parents to bring about change in the system, and lends them the option to ‘exit’ a particular school if they are not satisfied with the quality of schooling.

The brainchild of economist Milton Friedman, the ‘voucher scheme’ aimed at improving educational outcomes (Friedman, 1955 and Weidrich, 2003). The idea was to end the discrimination meted out to the poor by providing them access to good private schools of their choice. The scheme did not entail the government spending more money on education but transferring the money to educational vouchers instead of schools. Under the scheme, the schools that did not perform well had to be shut down (Friedman, 1962 and Weidrich, 2003).

However, there seems to be some contradictions remaining in determining the cost of vouchers. Experts opine that voucher amount should be based on the cost of schooling in a particular area. In case of high school fee, parents may use additional funds to complement the vouchers. One way of making the voucher system more equitable is to determine voucher value based on family income. Therefore, instead of offering equal value vouchers to all, poorer children may be provided with higher value vouchers (Kingdon, Report of the Times of India, 2017).

Vouchers may be of two forms: public vouchers and private vouchers. When the government issues public vouchers, there is a set of regulations that shape the performance in education. The access to voucher may be dependent upon school recognition, regulations pertaining to governing of the schools, hiring of government-licensed authorized teachers, abiding by governmental regulations pertaining to syllabi, text books, and examinations, teacher's qualifications and salaries, determination of teacher-student ratio, availability of necessary infrastructure at schools, etc. (Weidrich, 2003).

On the other hand, in private voucher programs, the schools are relatively free to decide where to allocate the money. Thus, in an attempt to attract more students, schools may offer free meals, transport, uniform, etc. Arguing in favor of private voucher scheme, it may be said that gives the parents the liberty to choose schools of their choice for their children (Weidrich, 2003).

Voucher Scheme-The Success Stories

(a) Bangladesh

Bangladesh's Female Secondary School Assistance Project (FSSAP), though not a voucher program, provides testimony to the benefits that can be accrued out of the incentive system. It exemplifies a successful initiative of providing monetary incentive to girls in order to encourage their participation in schools. The project entailed providing stipends ranging from US\$ 12 in standard sixth to US\$ 36 in standard tenth, to girl students who have enrolled themselves in secondary schools and meet the eligibility criteria (Liang, 1996 and Weidrich, 2003). The stipend meets the direct cost of schooling involving personal expenses (such as the school fees, tuition support, stationary items, books, transport, etc.) and examination fee allowance in higher grades/standards. Besides, the stipend is inclusive of the occupational skill training, maintenance of school infrastructure, expenses relating to promoting public awareness, etc. (Weidrich, 2003).

It has been estimated that the stipend cover 30-54 percent of the schooling expenses and amount is directly transferred to the bank account of the girls. Besides, FSSAP offers tuition assistance but the

financial assistance is given to the school in which a girl is enrolled. Report reveals several positive effects of the FSSAP as it led to positive surge in enrollment, attendance, lesser number of drop-outs, and better performance in schools. As a matter of fact, it was found that in July 1997, the difference between the enrollment of boys and girls were considerably reduced and girl students receiving Secondary School Certificates surged from 42 percent in 1996 to 52 percent in 1997 (Weidrich, 2003). Despite such achievements, it is imperative to mention that there a considerable portion of girls who do not attend schools. The support system needs to be tailor-made for them, bigger stipend needs to be offered to the poorer section of the society.

(b) United States of America

Cleveland Scholarship Program, was the first publically funded American voucher program. The voucher provides 90 per cent of the student's tuition and goes to a maximum of the US\$ 2250. The award of the voucher is based on examination of the students and interviews of the parents. The program was popular among the parents because it allowed them to offer their children quality education and to be convinced about the safety of the school environment (Weidrich, 2003).

Similarly, in Milwaukee, Milwaukee Parental Choice Program (MPCP) was privately sponsored in response to the high drop-out rates, underperformance of children and increasing educational disparity across social classes. The scheme substantially increased the involvement and satisfaction level of the parents and the scheme has been highly successful in increasing the socio-economic diversity in schools. Thus the Milwaukee Parental Choice Program (MPCP) exemplifies one of the most successful implementation of the voucher scheme and represents the creation of an inclusive model of education (Weidrich, 2003; Singh, 2010).

(c) Success Stories from India

ENABLE

In 2011, a London-based charity organization, Absolute Return for Kids (ARK) collaborated with the Centre for Civil Society (CCS), a Delhi-based think tank, to implement a voucher scheme namely, "Ensure Access to Better Learning Experiences (ENABLE)". The focus area of the program was 20 square kilometer, the Shahdara slum area of Delhi. The target group was underprivileged children, aged five to seven years, belonging to families earning less than Rs. 8,000 per month (i.e. the families that are below the poverty line). In order to enroll for the voucher scheme, verification of the children's details, income levels, birth certificate, BPL card, etc. was done. Moreover, it was conditioned that children enrolling in the voucher scheme have not attended school before or were not

currently enrolled in government schools. The selection was done through the lottery system and out of a total of 1618 children who applied for the scheme, 835 children were randomly selected. The lottery winners received four vouchers namely, for covering tuition cost, uniform, books, and meal. An annual voucher worth Rs. 4800 covered the tuition cost, vouchers worth Rs. 600, Rs. 900, Rs. 1000, covered the costs of school uniforms, books, and lunch respectively. The total cost of the voucher amounted to Rs. 7,300, which was provided annually for five years. In an attempt to save extra expenditure of the parents, this scheme did not allow parents to add money to the vouchers or schools to charge extra money. Though the scheme did not provide for transportation expenses, some private schools did provide school bus services. Further, the scheme allowed students to transfer schools, provided that the student was having a valid reason for transferring and he/she was transferred to a school covered under the ENABLE program. Parents were also offered counselling so that they can take informed decision about school choice. The Centre for Civil Society (CCS) reveal the satisfaction of the parents about the voucher scheme in the following words, *“A high majority (94 percent) of the voucher parents mentioned that their children were happy with their present school. 61 percent of the voucher parents felt that their children had also become more regular with school work. More than 50 percent of the voucher parents noticed that their children had become more disciplined and studied more in their new school.”* It was also found that voucher students exhibited better performance in English, Mathematics, and Hindi across all grades, in comparison to those studying in government schools and was comparable to general students studying in private schools (Dixon et al., 2019).

Moreover, there was an agreement that the participating schools would admit maximum possible ENABLE students and a minimum of five students. The schools were also prohibited from offering any form of financial inducements to parents to induce them in sending their wards to a particular school. It was stipulated that the schools would not charge any additional fees from parents for a period of five years and not discriminate against students enrolled through the ENABLE scheme. Finally, in order to assist parents in choosing schools, a handbook was designed. It entailed information about facilities that are being offered by the participating schools of the neighborhood and provided guidance to parents about availing the voucher scheme. The publicity of the scheme was done by various promotion techniques like banners, posters, use of public address system, distributing leaflets, puppet shows, etc. (Dixon et al., 2019). Significantly, the voucher-awarding ceremony took place at a venue that is frequented by the elites of the city and were distributed by the

then chief minister of Delhi (Ball, 2009). Additionally, in an attempt to further boost the program, CCS has put the following appeal on its website for prospective donors:

“Each voucher worth up to INR 6000 will fund one child’s education in the school of their choice for a year. The voucher will be given until they complete their primary education from their preferred school...You can support this pioneering effort by sponsoring one or more vouchers. You will thus brighten a child’s future by giving her the power to choose her school. You had a choice, give her a choice.”

The Rumi Education Foundation

The programme was run by the Rumi education foundation also known as Rumi Bright Futures (RBF) Voucher Programme. The organization based in Hyderabad and has been offering education vouchers to school dropouts since 2009. It aims to target students who were dropped out of schooling system for more than a year. The organization however does not encourage students leaving existing schools to avail the scheme. The basis of selection is an eligibility test and those who perform well in the eligibility test are permitted to enroll in the voucher scheme. Moreover the organization keeps the parents in the loop by explaining them the process and functioning of the voucher scheme (Gomathi and Sudhakar, 2014).

RBF have till now covered 151 students from 110 families in the scheme. The students have been trained in 18 schools of RBF and given education voucher till the standard 10th. The RBF created awareness about the voucher scheme among the community through various sources like door-to-door canvassing, through print media (leaflet), word of mouth through schools, teachers and parents. The role of school manager was crucial in spreading the awareness of the voucher scheme (Gomathi and Sudhakar, 2014).

Discussion

While these success stories of the voucher scheme are very encouraging towards large scale adoption of the scheme in India, there are certain issues that needs to be addressed. For instance, when the major beneficiaries of the voucher scheme are the underprivileged sections of the society, the voucher amount should be big enough to cover the costs of high-quality education. Additionally, echoing Weidrich (2003), we contend that vouchers need to be ‘academically and economically impressive’ and sufficient number of schools need to be covered under the scheme. The scheme also needs to be responsive to the needs of the differently abled children, migrant and refugee children, out-of-school children, street children, girl students, orphans, SC/ST/OBC children, etc. As a matter of fact,

voucher scheme may be distributed more widely among the girl students to improve upon the female enrollment in school. The scheme needs to cover both public and private schools and each municipality should decide upon how many vouchers to fund, depending upon the funds allocated to the area by the central government.

Not only does the educational voucher scheme strive to educational level of the underprivileged, it also puts pressure on the government schools to improve the quality of education imparted in them. However, extensive groundwork is needed to implement the scheme—for instance, the government needs to decide the voucher amount per student and the number of students to be enrolled under the scheme. This may require involvement and co-operation of some locally active NGOs and municipalities (Makwana, 2011). This apart, the success of the voucher scheme program like the ENABLE and the RUMI foundation exemplified the importance of private actors in the process. This was precisely evidenced when Akshara, an NGO formed by the CEO's wife of Infosys, supported schools for the underprivileged in Bangalore or charitable foundations established by corporate houses (namely the Azim Premji Foundation by WIPRO and Pratham by ICICI) made substantial contribution towards quality improvement in government schools. Similarly, multinational banks like HSBC, Citicop, Standard Chartered offer micro-finance loans encouraging private school 'start-ups' and HSBC ran a programme, entitled EQUIP (Enabling Quality Improvement Programmes in Schools). Precisely speaking, the support from the corporate sector to the trailing government schools can be in the form of infrastructural upliftment, development of curriculum, provisioning free meals, etc. In short, there are a plenty of options open for public-private partnership in the domain of the voucher scheme (Ball, 2009).

However, critics question as to whether the voucher scheme will really improve the quality of education in government schools. They contend that government schools will be undermined at least for some initial years, if the scheme is introduced. It will result increased migration of students from government to private schools. Consequently, this will result in the growth of private schools at the cost of government schools (Kalra, Report of the Mint, 2007). The government may tackle the issue by consolidation i.e. merging the dysfunctional, low-enrollment schools with the successful ones (Kingdon, Report of the Times of India, 2017).

With the implementation of the voucher scheme, the country would be able to offer quality education to each and every student. Parents would also have the agency to choose the best school for their children and it will particularly be beneficial for students who are facing financial problems and drop

out of school. Thus voucher system is a footstep towards mainstreaming the underprivileged children in terms of education. It will potentially be helpful in promoting educational opportunities among the female and minority students. Stated differently, voucher scheme have the potential to make the schools culturally diverse and inclusive places. Moreover, it makes the parents more involved and confident about the education of their children. In short, by successfully implementing educational voucher, the country would be able to achieve the objective of access to education for all, with accountability, cost-effectiveness, and sustainability. Such drift towards ‘decentralized’ and ‘heterarchical’ education system is indeed a progress towards post-modern India!

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34. Problem Solving Ability of Undergraduate and Post Graduate Students of Tezpur University: A Case Study

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Abstract

The present study investigates the Problem Solving Ability of Undergraduate and Postgraduate Students in Tezpur University. The objective of this study were- to compare the problem solving ability with respect to its gender, area of residence, course of study and discipline. Descriptive survey method was adopted in the study. The researcher used L.N Dubey Problem Solving Ability test tool. 120 students of Tezpur University were selected as a sample and random sampling technique was used. Statistical technique such as Mean, Median, Standard Deviation, t test and ANOVA were used for data analysis. The findings revealed that no significant difference between male and female, rural and Urban undergraduate and postgraduate students in Problem solving ability. Moreover, significant difference was found between Engineering, Humanities, Management and Science students towards problem solving ability. Further, it was found that Science and Engineering students are superior in problem solving ability.

Keywords: Problem solving Ability, undergraduate & postgraduate students.

Introduction

In the present study Researcher studied the Problem solving ability of the University students. It is very much important for each and every individual success in life. Each and every individual faces different problems. But the process to find out the solution may not be same for all. It mainly depends upon the cognitive ability, thinking capacity of the individual. In order to be a success in life the individual should be able to solve problems as well as able to think in a scientific way. Problem is a state of difficulty that needs to be resolved. Individuals face problems in their day to day life in every fields. Some problems may be big, some may be small, some may be complicated and some problems may be easily solved. New strategies as well as different methods should be enlarged in order to solve problems rapidly and effectively. 'Individuals' problem solving skill is connected to their concentration of the problem and their evaluation of the self. Problem solving skills will help to resolve the problems in our academic as well as in our day to day life. Problem solving is a process

which requires different strategies and techniques to solve. The investigator thinks that problem solving ability is very much important in the present era which helps in the academic achievement as well as in each and every part of the life of the students. So, in this study the investigator will try to see the problem solving ability in terms of gender, residential background, course of study and different discipline on achievement of Tezpur University students.

Review of Related Literature

Singh, S.K & Paswan, S.K (2019) conducted a study on Problem solving ability among Hindu and Muslim college students. The sample selected for this study were 120. Out of 120 samples, 60 boys and 60 girls were selected. The tool used in this study was Problem solving ability test developed by Dubey. It consist of 20 items. No significant difference between Hindu and Muslim College students in problem solving ability was found in the study. Similar level of problem solving ability were showed by both college students. The result also shows that there was a significant difference between boys and girls in problem solving ability. It indicates that boys have better problem solving ability as compared to girls.

Singaravelu, S. (2017) investigated the higher secondary chemistry students 'problem solving ability'. The investigator has taken as a sample of 180 students of five higher secondary schools situated in Thanjavur district by using stratified random sampling technique. The findings of this study are that there is low level of problem solving ability of higher secondary chemistry students and found that male student are better than female students. Moreover, it was found that there is no significant difference between student residing in rural and urban area in problem solving ability.

Agnihotri, A.K (2015) investigated a study on problem solving ability among Senior secondary school students of Himachal Pradesh. The major findings revealed that no significant difference in the level of problem solving ability on gender basis but there was a significant difference between different streams. The students of science and commerce streams have the higher level of problem solving ability as compared to arts stream.

Nataraj, P.N & Manjula, M (2012) conducted a study on the topic entitled A study on problem solving ability among the matriculation school students. The objective of this study to study the problem solving ability of matriculation students and to compare with respect to gender, locality. In this study, The findings of this study reveals that problem solving ability of matriculation students at

Cuddalore district is very low and also no significant difference between boys and girls was found, rural and urban student in problem solving ability.

Langstieh, M. (2011) Conducted a Study on Problem Solving ability of higher secondary school students in Shillong. The findings of this study indicates that Problem solving ability among Higher secondary school students of Shillong is very low and it also reveals that science students shows higher level of problem solving than other streams. It also indicates male students are found to be superior in problem solving ability.

Need and Significance of the Study

It has been found that persons having problem solving ability can solve the complex problems quickly. Problem solving ability is very needed and essential skill in higher studies. It helps the individual capable of doing new things and think creatively. It helps to overcome various obstacles that come out to interfere with the attainment of a goal. Therefore, it is important to develop different problem solving ability by proper education. The researcher studies lots of research report related with this study. **Krishnan (1990)** in his study on high school students identified that the essential problem in school maths is how to teach Problem solving strategies in students so that they may become efficient problem solvers. **Singaravelu, S. (2017)** in his study no significant difference was found between student residing in rural and urban area in problem solving ability. Moreover, it also shows that male students have better problem solving ability than female students. **Shukla and Dixit (2016)** have found that the problem solving ability of adolescents female students have higher level than adolescents male students and it was also found that most of the male students used planning strategies in problem solving. **Singh and Radha Charan (1992)** in their study found that students living in urban area are more superior than students living in rural area in problem solving ability. Furthermore, it also reveals that female students problem solving ability are superior than male students. In present times problem solving ability is very necessary in higher education as well. This is very important to overcome difficulties in day to day life. It has become very relevant in the present era. The researcher found that there were many research conducted. Hence the present study try to explore the problem solving ability of undergraduate and post graduate students. Therefore, the researcher motivated to select this study.

Statement of the Problem: The problem selected for the present investigation is as follows: *“A Study on Problem solving ability of undergraduate and post graduate students of Tezpur University: A Case Study”*

Objectives of the Study: The objectives of the study are as follows:

1. To compare between the male and female students in problem solving ability.
2. To compare between the students residing in rural and urban area in problem solving ability.
3. To compare the problem solving ability between undergraduate and postgraduate students of Tezpur University.
4. To compare the problem solving ability of students in respect to its discipline i.e. Engineering, Humanities, Management and Science.

Hypothesis of the Study

1. There is no significant difference between the male and female students in problem solving ability.
2. There is no significant difference between rural and urban areas students in problem solving ability.
3. There is no significant difference between undergraduate and post graduate students in problem solving ability.
4. There is no significant difference between Engineering, Humanities, Management and Science students in problem solving ability.

Method: The present study adopted “Descriptive survey method” for data collection.

Sample & Sampling Technique: The sample of the study includes 120 students of Tezpur University. It includes both male and female, undergraduate and postgraduate students of Engineering, Humanities, Management and Science discipline. Out of which 54 male and 66 female students were there. For this study, random sampling technique was used by the researcher.

Tool Used in the Study: For measuring the problem solving ability of students, the researcher used L.N Dubey Problem Solving Ability test tool. This tool is standardized for high secondary school, college and University students with age from 12 – 25 years. It consists of 20 multiple choice items in English. The items are designed to study the problem solving ability of the respondents. Each item has four alternative responses, out of which only one answer is correct. If the respondent gives the

correct answer he/she should be given 1 mark, and he/she gives a wrong answer zero should be given and in the end all the marks should be added.

Statistical Technique Used in the Study: Descriptive statistics like mean, median, standard deviation, ANOVA were used to measure what extent two or more variable differ or close from each other. Also “t” test was used to compare the gender, area of residence and course wise scores of Problem solving ability. The collected data were analysed by using SPSS software. The questions and responses were coded and entered by using Microsoft excel software. Frequency Polygon and histogram have been used to present the data in a diagrammatic form.

Analysis & Interpretation

Objective No.1: To compare between the male and female students in problem solving ability.

H₀ 1: There is no significant difference between male and female students in problem solving ability.

Table1.1: Genderwise N, mean, standard deviation, df, t- value

Gender	Number (N)	Mean	S.D	df	t (stat)	t (tab)	Level of significance
Male	54	11.67	3.83	109	0.27	1.98	0.05
Female	66	11.85	3.51				

Table 1.1 shows that the calculated ‘t’ value was found 0.27, which is smaller than the critical ‘t’ value that is 1.98 which is not significant at 0.05 at df = 109. Therefore, the formulated null hypothesis (H₀) – “There is no significant difference between male and female student in problem solving ability” was not rejected. Hence it can be clearly interpreted that no significant difference between the male and female student in problem solving ability.

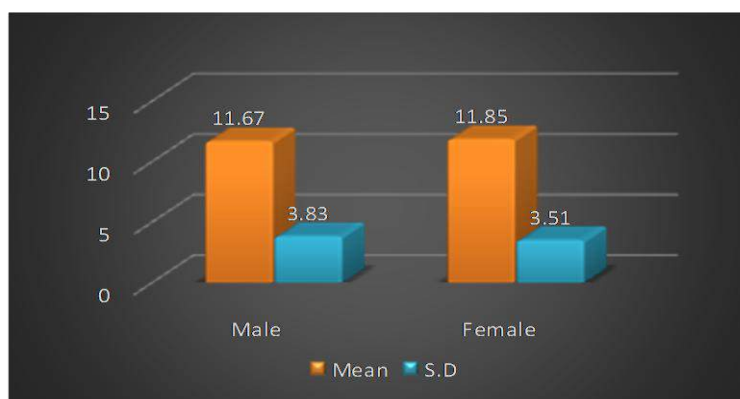


Fig 1.1 Genderwise mean & standard deviation.

Fig 1.1 depicts that no significant difference between male and female students in problem solving ability because the calculated value (-0.27) which is less than the critical value (1.98). Therefore, the formulated null hypothesis “There is no significant difference between male and female students in problem solving ability” gets accepted. Hence, the null hypothesis is not rejected.

Objective no 2: To compare between the students residing in rural and urban area in problem solving ability.

(H₀) 2: There is no significant difference between rural and urban students in problem solving ability.

Table 1.2: Area wise N, Mean, standard deviation, df & t-value

Area	Number (N)	Mean (M)	S.D	df	t (stat)	T (tab)	Level of significance
Rural	58	11.41	3.94	112	-1.021	1.98	0.05
Urban	62	12.10	3.33				

Table 2. reveals that the calculated ‘t’ value come out to be 1.021, which is smaller than the critical ‘t’ value that is 1.98 which is not significant at 0.05 level of significance for two tailed test for degree of freedom, df = 109. Therefore, the formulated null hypothesis (H₀) – “There is no significant difference between rural and urban student in problem solving ability” is not rejected. From this, it can be clearly interpreted that there is no significant difference between the rural and urban students in problem solving ability.

Fig 1.2- Areawise mean & standard deviation



Fig 1.2. depicts that there is no significant difference between rural and urban students in problem solving ability because the calculated value (-1.021) which is less than the critical value (1.98). Therefore, the formulated null hypothesis “There is no significant difference between male and female students in reasoning ability” gets accepted. Hence, the null hypothesis is not rejected.

Objective no 3 : To compare the problem solving ability between undergraduate and postgraduate students of Tezpur University.

(H₀) 3 : There is no significant difference between undergraduate and post graduate students in problem solving ability

Table1.3: Coursewise N, mean, standard deviation, df and t-value

Course	Number (N)	Mean (M)	S.D	df	t (stat)	T (tab)	Level of significance
Undergraduate	52	12.12	3.64	110	0.91	1.98	0.05
Postgraduate	68	11.50	3.65				

Table 1.3 shows that the calculated value was found 0.91, which is smaller than the critical value that is 1.98 which is not significant at 0.05 level of significance for two tailed test for degree of freedom, $df = 110$. Therefore, the formulated null hypothesis (H_0) – “There is no significant difference between undergraduate and postgraduate students in problem solving ability” is not rejected.

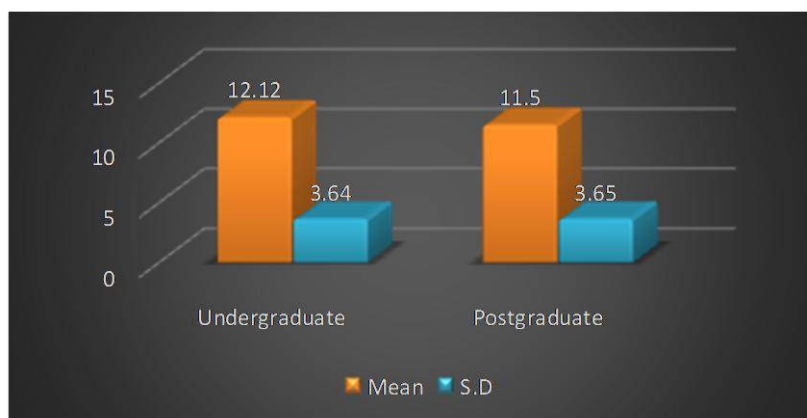


Fig 1.3-Coursewise mean & standard deviation

Fig 1.3 depicts that there is no significant difference between undergraduate and postgraduate students in problem solving ability because the calculated value (0.91) which is less than the critical value (1.98). Therefore, the formulated null hypothesis “There is no significant difference between undergraduate and post graduate students in problem solving ability” gets accepted. Hence, the null hypothesis is not rejected.

Objective no 4: To compare the problem solving ability of students in respect to its discipline i.e. Engineering, Humanities, Management and Science.

(H_0) 4: There is no significant difference between Engineering, Humanities, Management and Science students in problem solving ability.

Table 1.4The significant of difference of means of Problem solving ability

	Sum of Squares	Df	Mean Square	F
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Between Groups	287.243	3	95.748	8.595
Within Groups	1292.224	116	11.140	
Total	1579.467	119		

Table 1.4 reveals that the calculated 'F' value (8.59) is more than the tabulated 'F' value (2.70) at 0.05 level of significance with df being (variable) $v_1 = 3$ and $v_2 = 116$. Therefore, the formulated null hypothesis "There is no significant difference between Engineering, Humanities, Management and Science students in problem solving ability" gets rejected because the calculated value is more than the tabulated value. It indicates that there is significant difference between Engineering, Humanities, Management and Science students in Problem solving ability.

ig. 1.4-The significant of difference of means of Problem solving ability with respect to different schools.

Fig. 1.4 shows that Science and Engineering students performed better as compared to arts and management students. It indicates that there is significant difference between Engineering, Humanities, Management and Science students in problem solving ability. Therefore, the formulated null hypothesis "There is no significant difference between Engineering, Humanities, Management and Science students in problem solving ability" gets rejected.

FINDINGS OF THE STUDY

1. The result shows that no significant difference between male and female students was found in problem solving ability in Tezpur University. Both male and female students have same level of problem solving ability.
2. From the study it was found that there exists no difference between rural and urban students in problem solving ability in Tezpur University.
3. It has been found that no significant difference between undergraduate and postgraduate students in problem solving ability in Tezpur University.
4. It was also found that significant difference in problem solving ability between Engineering, Humanities, Management and Science students in Tezpur University. Problem solving ability was found more on Science and Engineering students.

EDUCATIONAL IMPLICATIONS

The overall results of this study have shown light on the University students problem solving ability. This study shows that male and female have equal level of problem solving ability. It also shows that rural-urban, undergraduate- postgraduate students were not differ in problem solving ability. This is an encouraging fact that they have same level of ability.

On the basis of findings of the study the following implications emerge out:

1. This study clearly reflects the ability of problem solving ability among male- female, rural-urban, undergraduate-postgraduate, different schools such as school of engineering, school of humanities, school of management, school of science students.
2. The findings of the study shows that science and engineering student performs better as compared to arts and management students. Teacher must identify students with low problem solving ability and different teaching methodologies and technique should be adopted in order to increase the problem solving and reasoning ability.
3. In many studies, the findings revealed that girls perform less as compared to boys in problem solving ability. In such case, teacher should try to identify the girls with low problem solving ability and try to modify their thinking ability and learning style through different techniques. Moreover, parents should also encourage their daughters to solve their own day to day problems through their own cognitive abilities so that they remain mentally alert and physically fit.
4. This study will help the educational institutions and teacher educators to organize various co-curricular activities to enhance problem solving ability of the students.

CONCLUSION

As we all know that problem solving is an important aspect of thinking. Because problem solving is very much important in our lives, it has become an important area of research that is both theoretical and practical importance. The persons having problem solving ability can solve the complex problems quickly. It helps to overcome difficulties that appear to interfere with the attainment of a goal. Therefore, it is necessary we must try to develop problem solving ability through proper education. From this investigation, it can be said that both male-female, rural-urban, undergraduate-postgraduate students of Tezpur University have same level of problem solving ability. It can also be inferred that science students have more problem solving ability as compared to Engineering, Humanities and Management students.

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35. Study Habits and Learning Styles among Students Pursuing in Higher Education

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Abstract

This study accomplished to explore the study habits, and learning styles, among students pursuing in Higher Education. This study majorly focuses on various methods of study habits followed by the respondents during their higher education and various modes of learning styles would prefer during higher education. By nature, each student captures their study habits were by following their self or learned attitude. Depending on their acceptance of self or learned attitude, they follow certain habits in their learning style. Study habits state to the propensity of a student to use his/her uninterrupted care to obtain knowledge via orderly routines. Some of the commonly used learning styles are: auditory, visual, logical, verbal, social, physical, social, and 'solitary'. At the outset, each student decides the study habits with a combination of two or more learning styles in it. The researcher has used a structured questionnaire in the form of a Google link to collect information from the target population. The collected information has tabulated and subjected to data analysis to draw meaningful outcomes for the study. The results of the study reveals that the order of preference of respondents towards the mode of clarification, basic interest in education, and learning styles in higher education. At the outset, this study also further focuses on the relationship of Place preference, time preference, and method preference supporting their 'study habits' and 'learning styles'. Finally, this study confirmed that there is a 'positive significant' association with study habits and learning styles among students pursuing education in higher educational institutions in the Vellore District.

Keywords: Study habits, Learning styles, Students, and Higher education.

Introduction

In modern times, the scope of students learning sources improved both online and offline methods. The students could find the available resources and subject their learning sources. The major support for their learning happens depending on their study habits. In India, the school level education system comprises four different levels, namely the primary/elementary education up to the fifth standard, the secondary school education up to 10th standards, and senior secondary school up to intermediate

education, and later they will be joining in higher education. The success of any student is initially determined by the first four levels of education. Mostly they will get equipped with their study habits 'how, where, when and how much to study'. Primarily, the students will expose to their study environment both by their teachers and parents. Secondly, their study habits were supported by their friends and classmates, and lastly by their self-supportive learning methods. However, the effectiveness of study habits can be determined by the learning style and vice-versa.

1.1 Concept of Study Habits

The Study habits signify the predisposition of a student to practice his/her incessant care to obtain knowledge via prepared procedures. Gettinger and Seibert (2002) specified that students, who found reduced study habits, are maximum likely the ones to have low academic achievement. Having decent study habits could change the let-down rate within educational institutes since learning tends to escalation a student's academic skill.

1.2 Concept of Learning Styles

The term "learning styles" expresses the accepting that every student learns inversely. Theoretically, an individual's learning style denotes to the privileged way in which the student engages progressions, understands, and retains material. This idea of adapted learning styles has extended extensive appreciation in education model and classroom management approach. Specific learning styles depend on intellectual, emotional, and conservational factors, as well as one's prior knowledge. It is significant to understand the changes in their students' education styles so that they do best exercise strategies into their daily actions, syllabus, and assessments.

1.3 'VARK' – Preferred Learning Modes

VARK is an abbreviation that refers to the four kinds of learning styles namely Auditory, Visual, Writing or Reading Favourite, and Kinesthetic. The 'VARK model' recognizes that students have different systems of how they practice information, indicated to as "preferred learning modes." Fleming and Baume (2006) in his study stated that one more pattern of style namely virtual or online, which is ahead importance in the recent enlightening era. This Study accomplishes that the students with diverse learning styles tend to do improved academically than those with just one leading style. Abidin, Rezaee, Abdullah and Singh (2011) pointed out that students' awareness of their strengths such as education style favourites and how to develop their strengths may advance their academic enactment. Students should have responsiveness of their learning style and how to exploit their

learning possible in the very early days of employment in the academic institute (Barman A, Rosniza Abdul Aziz and YusnizaMuhammedYusoff, 2014).

2 Review of Literature

The researcher used various literature to review the previous studies collected from various sources like journals, magazines, working notes, and reports were presented. Fleming, N., and Baume, D. (2006) had studied on Learning Styles Again and observed that VARK is an contraction for aural, visual, writing/Read and Kinaesthetic styles and it is additional applicable in today's background. Further, the researcher described that, VAK records had been around for more number of years. What was novel in his work was a second 'visual modality' for reading/writing learners. He noticed that some students had a different preference for the printed word while others preferred representative information as in charts, drawings, and graphs. These two types preferences were not always originate in the same person. Afsaneh and Hassanbeigi et al., (2011) had analysed the association among various study skills and educational performance among students in universities. The findings of the study showed that the study As per Timila Manandhar, Pramila Shrestha (2019), stated that the relationship between self-discipline and study habits between 'Proficiency Certificate Level' among female nursing students. Descriptive study was used. A total of 96 female nursing respondents from first year, second year and third year were selected by non-probability convenient sampling method. The final result shows that the third year students had the highest mean score value 124.43 for self-discipline, while first year students had the lowest mean score value 103.38 for self-discipline. The researcher concluded that, a moderate positive association was found between self-discipline and study habits. Haleh Jafari (2019) conducted one study on the status of study habits and their association with academic achievement in medicinal sciences students in Kermanshah, Iran. The cross-sectional study was accepted out on 380 medical sciences students at University of Medical Sciences, Kermanshah. The respondents were randomly allotted to the study. As per Palsane and Sharma, the 'Study Habit Inventory' model was used for collection of data by using descriptive and inferential statistics. The mean score value of students' average grade

data by using descriptive and inferential statistics. The mean score value of students' average grade point was 15.73 ± 1.5 out of 20 and the mean score value of total status of study habits was 45.70 ± 11.36 out of 90. The position of study habits at 81.3% of the students resulted at a moderate level. This study suggested seeing and assessing students' study habits at the time of joining into the university, also, specific exercise should be offered to students to support them learn or modify study habits to upturn their academic achievements. Gilbert C. Magulod Jr. (2019) examined the learning style of preferences, the study habits, and the level of academic achievement of the students registered in applied science degree courses of one campus of a public higher education institute in Philippines. The study deployed a descriptive correlational investigation design to a total of seventy-five students who were reasonably sampled. The research findings of the study denoted that the students of applied sciences degree courses preferred group, visual and kinaesthetic as main learning styles while they obvious a moderate level of study habits. Lastly, there were significant association between study habits, learning styles and academic performance of students in applied science degree courses.

According to Ünal Çakıroğlu from Turkey (2014) attempted to examine the relations between learning styles, study habits and learning performances with online software design language course. 'Kolb's Learning Style Inventory' (LSI) was used to identify the students' education styles. As a result, important relationships between study habits, learning styles and learning performances were exposed. Mahwish Rabia (2017) analysed the study habits and academic achievement of the students. Study habits donate significantly to the growth of knowledge and perceptual capabilities. Study habits express a person how abundant he will learn and how distant he wants to go, and how much he wants to receive could be decided with the help of one's study habits through life. Therefore, it is expected that study habits are associates of academic achievement. The output showed that there is a positive significant association between study habits and academic performance of the students. Perimala Silverrajoo and Aminuddin Hassan (2018) described the relationship among study habits and academic achievements of 'Health Science' students at University College of Health Sciences, Masterskill. This study mainly emphasises on the various study habits used by health science students and the educational achievement

of the health science students. The findings of this study pointed out that there is no significant association between summary and note taking and understanding towards the academic achievements of the health science students. Whereby, the association between 'Method of Study' shows that there is a significant association to academic achievement but it does not help the student's academic performance as it shows an adverse and weak relationship.

3. Statement of the Problem

According to the Oxford Dictionary & Thesaurus of English Language, study habit briefly refers to the propensity of a student to learn systematically and efficiently when the opportunity is given, (Shabbir Ahmad Rana & Rukhsana Kausar, 2011). It is also a model of how a student practices to study in a routine way to improve their academic performance. According to Deary, Whiteman, Starr, Whalley, & Fox (2004), academic performance is complex student behavior and measured in terms of their passing all their subject papers in the first attempt, maintaining their grades, and earning their graduation within the stipulated time. So the relationship among good study habits and academic achievement is highly important to understand how students gain results. The relevancy about understanding, what they learned, and what they studied can take as two different ways to do better grades. Numerous students fail not because of the deficiency ability, but because they do not have drivable study skills (Menzel, 1982).

In India, there is no uniform educational pattern followed. There are four levels of education up to the school level in the twelve years and later they will join for higher education in their interested field. During the time of school education, they have subjects under acquiring mode with standard texts for their learning, where, in higher education, they follow a specific field of their interest in different disciplines like Engineering, Non-Engineering.

The problem here is not entirely based on the educational system, but more towards the relationship between study habits and learning styles they follow during their study period. This research examines the relationship between study habits, learning styles associated with students pursuing their higher education. So, the researcher identified this topic to fill the research gap for the study.

4. Research Questions

1. What methods of study habits were followed by the respondents during their higher education?
2. What mode of learning styles, they would follow during their higher education.

5. Research Objectives

1. To examine various study habits Profile followed by students.
2. To study the influence of various learning styles (VARVK) preferred by students.
3. To find out the students' preference towards the mode of clarification, basic interest in education, and Learning styles in higher education.
4. To understand the relationship of Place preference, time preference, and method preference supporting their 'study habits' and 'learning styles'

6. Testing of Hypothesis

Ho1. There is no significant association between place of preference and study habits a students

Ho2. There is no significant association between preference on time and quantity of study materials with study habits among students.

Ho3. There is no significant association between methods of study influence and study habits among students.

7. The Methodology of the Study

This study intends to conduct using a descriptive research design using the respondent's profile, study habits on various criteria, learning styles applicable to students in the present context. The researcher has prepared a Google link supported questionnaire format and circulated to all the students studying various UG and PG programs' relating both engineering and non-engineering courses registered under various higher education Institutions in Vellore Tamil Nadu, India. The researcher prepared the structured questionnaire format with open and closed-end types of questions to gather the participants' profiles, study habits, learning styles, and various factors supporting their study habits. The researcher followed the judgemental sampling method by involving both and men and women students studying various Engineering and non-Engineering degree programs. After waiting for a month for data collection from the above target group, 144 good responses were collected from the target group as the sample size. After getting the required data, suitable arithmetic tools were used to infer the findings of the study. The researcher used simple descriptive statistics tools such as frequency and percentage

analysis, and mean rank used for the profiling of the respondents, followed by Non-Parametric Kendall's w test for studying about respondents' preference towards, subject clarification preference in higher education, Students' career ambition in doing higher education and the learning styles preference in higher education and non-parametric Friedman's test for studying their preference over the Place of study, time of the study, and method of study. The researcher applied the Cronbach alpha test to find out the internal consistency of the variables. The results of Cronbach's alpha showed good which more than 0.70.

8. Analysis and Interpretation

8.1.1 Study habits - Profile of the respondents

From table 1 the overall profile and study habits of the respondents with nine variables relating to the study habits and student performance were presented. From the first profile, we could infer that around 75 percent of the respondents were chosen their study program by their self-interest basis, followed by, one-fourth of them influenced by their parents, friends, and teachers. Among the total respondents say 74.3 percent of them joined their study program from outstation places and thus they preferred to stay as hostellers, the rest of them joined from local places and reminded as day scholars. More than 80 percent of them had studied more than two schools before joining their degree program and the rest of them studied only in one school. All of them (say 100 percent) had studied their schooling with English medium exposure, out of which, about 26 percent studied their schooling in both combinations of English and regional language as their medium of instruction. Among the respondents, 58.3 percent have well-trained for both combinations of 'study alone' and 'study in-group' mode of preparation, the rest of them say, 36 percent used for study alone, and around 06 percent shown their preference towards study in a group alone practice. Most of them, say, 62.5 percent for both understanding the subject and scoring marks is important, followed by about 34 percent of them given understanding the subject alone.

Among the respondents, around 38 percent of them used with any time in a day for their preparation, another 38 percent preferred only night-time for their study and the rest of them say. 24 percent preferred during day time for their study. Regarding the library usage part, around 40 percent of them use the library facility based on their class commitments only, the rest of them say, 33 percent and 25 percent, used the library most often and rarely. Now-a- days, students preferred to use the library through online/ remote facilities for their preparation, and thus physical usage patterns might reduce

substantially. Finally, around 80 percent of them were trained to use the e-gate facility available through their library, and the rest of 20 percent showed their usage pattern to nil.

8.1.2. Student Learning Style - Rank Correlation Using 'NonParametric Kendall's w test'

The researcher used 'Non Parametric Kendall's w test' for knowing the respondent's preference in perusing the higher education in various aspects like mode of subject clarification, carrier preferences, and learning styles were presented in table 2.1 to 2.3.

In table 2.1, we attempt to know various sources used by students perceiving higher education. It showed that, among the respondents, most of them using their family members as their first preference to clarify doubts in higher education, followed by their library sources of information as to their second priority, followed by their classmates, reading books, online mode of references and teachers as their priority to clarify their doubts in higher education. This clearly showed that students of higher education have preferred to use outside classroom sources like peer group learning, library, and digital tools for their priority in learning subjects.

In Table 2.2, we tried to know the respondents' carrier ambition in doing higher education. It revealed that self-employment and seeking employment stands as first and the second choice, followed by getting a degree and social respect as third and the fourth level of priority among the students perceiving higher education.

In table 2.3, we tried to understand the respondents' preference for various learning styles used by students perceiving higher education. It showed clearly, among the learning styles presented above, the oral mode of learning (prefer to listen to lectures and not take notes) and Virtual mode as their first priority than second priority, followed by, kinesthetic mode (multi-sensory) learning such as LEARN BY DOING by using visual and/or auditory study techniques as their third priority. Finally, the 'Read and Write' learning styles and visual-spatial learning style as their fourth priority.

8.2 Place of Preference, Time, Quantity, and Method of Study Habits

8.2.1 Place preference on the Study Habits using Friedman – Non-parametric test

In table number 3.1 (A), the researcher used the Non-Parametric Friedman test for understanding the preference of various parameters involved in a physical place like proper lighting, ventilation, temperature, the neatness of the place, activities associated with study place, over the study habits.

In table number 3.2 (A), the researcher used the Non-Parametric Friedman test for streamlining the preference of various parameters like time and quantity of preparation used for supporting study habits and finally,

In table 3.3 (A), the researcher used the Non-parametric Friedman test for notifying the preference of various parameters used to profile the respondent's preference over the methods used during the study time.

In table number 3.1(A) showed that, the factors related to physical study place condition and its influence on respondents' study habits. In this session, the respondents were asked to give their preference of various physical factors supporting for their study habits such as proper lighting, proper ventilation, the temperature of the place - too warm/ not too warm, neatness of the place, noise level preference, distraction at a work-related activity, and review of their subject related activity

Among the factors mentioned above, the temperature of the place of study gained more importance with the mean score of 7.3 (rank 1), followed by users to do some study-related work alone at a specific place or attempt to do multitask activity supporting for the study purpose in study area uniformly gained the same mean score of 6.23 (rank II), the activity of avoiding distraction through facing wall or corner of the place as the study area gained the mean score of 5.75 and scored the third rank. Further, the subject revision/ review the subject related work gained importance with the mean score of 5.29 (IV rank), the preference of silent place of study with the mean score of 5.12 (V Rank), Place nearness scored 5.08 as the mean score (VI rank), and finally, proper ventilation and proper lighting supported with a mean score of 4.08 (VII) and 3.85 (VIII) rank respectively. These results earmarked the importance of physical place support for study habits.

At the outset, as per the table no. 3.1 (B), the results of the Friedman test, revealed that the P-value of Chi-square is less than 0.05 shown the significance of all the Place oriented factors supporting the study habits.

✓ Hence, the Null hypothesis (Ho1) stated in this study got rejected.

8.2.2 Time and Quantity of study preference of the Respondents' Study Habits

The above table 3.2 (A), revealed the usage pattern of time management for their study habit along with the measure of study material and their preparation by the respondents. The following are the responses received from the respondents witnessing their study habits based on time management and measure of study preparation.

They review the class notes within a day on their priority with the mean score of 5.25 (Rank 1) followed by using their most available time for their study habit supported with a mean score of 4.90

(Rank II), used the most alert time for study habit gained the mean score of 4.64 (rank III), Used to set their time goal for their preparation but not fixed any measure of their preparation as their study habit gained the mean score of 4.39 (rank IV), Used to check their class notes at the time of assignment/evaluation by the teacher, with a mean score of 4.05 (rank V), used to set the learning goal during their subject preparation supported with the mean score of 4.0 (rank VI) and finally, they used to prepare the tough subjects during their alert time gained the mean score of 3.88 (rank VII).

At the outset, as per the table no. 3.2 (b), the results of the Friedman test, revealed that the P-value of Chi-square is less than 0.05 shown the statistical significance of all the factors relating to time study and measure of study materials influence of the study habits.

✓ Hence, the Null hypothesis (Ho2) stated in this study got rejected.

8.3. Methods of Study influence on the Study Habits

The above table 3.3(A) explained about various methods preferred by study respondents as their study habits. The following are the methods of the respondents' preference towards their study habits.

Thinking and list of additional questions needed to answer from reading the chapter supported with the maximum mean score of 5.57 and ranked 1; the need for additional reading to support for self-understanding the material in a more precise manner gained with a mean score 5.32 and ranked 2; learning based on exam orientation rather than a text while studying textual material supported with the mean score 5.22 and ranked 3.

Followed by, examining the study chapters headlines and subheads to list as many queries as possible about the materials to be deliberate supported with the mean score 5.04 and ranked IV; recite the readings to our understanding and try to relate the answer for the questions previously formulated gained the mean score 4.90 and ranked V; learning the concept of principles and its application than memorizing supported with a mean score of 4.88 and ranked VI; Underlining material and take notes that help us to answer the questions that previously listed gained the mean. a score of 4.74 and ranked VII; self-question and study until our understanding supported with the mean score of 4.73 and ranked VIII and finally, breaking down in small steps while working on major tasks like project stood with the support of the mean score 4.61 and ranked IX.

At the outset, as per the table no. 3.3 (B), the results of the Friedman test, revealed that the P-value of Chi-square is less than 0.05 shown the statistical significance of all the factors relating to the methods used in the study habits from the sample respondents.

✓ Hence, the Null hypothesis (Ho3) stated in this study was rejected.

9. Findings of the study

1.0 Profile of the Respondents:

In this session, the researcher presented the profile of the respondents and their preference on various parameters supported study habits and learning styles in higher education.

From the study, among the respondents, 73.6 percent of them chosen their *branch of study* by themselves and followed by 21 percent of them joined based on their parents' guidelines; 73.6 percent reported their *stay place* as hostellers, and the rest of they are staying outside in the group of day scholars or day boarders; 69.4 percent agreed they did their schooling in 2 to 4 different schools in twelve years periods, followed by 18.75 percent reported that they studied schooling in only one school.

Further, 73.6 percent reported that their *medium of instruction* as English medium during their schooling period and the rest of they say 26.4 percent used their medium of instruction through both English and regional languages; 58 percent agreed that they are used for both self-study and group study method of study preparation, followed by 36 percent shown they very much used for self-study alone.

Regarding their educational interest, 63 percent of them recorded their basic *interest in education* was both understanding the subject and scoring marks, followed by 34 percent of them recorded their interest in scoring marks only. Regarding their study time, 38 percent each, shown the preference of their *study time* at *any time during the day* and *the night-time alone* respectively and the rest of them recorded as only during day time. Further, regarding their library usage pattern, 39 percent preferred to visit the *library* only based on their class commitments, followed by 33 percent of them was visited the library very often to support for their various study obligations and finally, 80 percent of them showed their preference in using digital resources through *E-gate facility* of the library for their study purpose, out of which, 40 percent used rarely, 23 percent used often and 17 percent used very often respectively. Thus, the profile of the respondents showed highly supportive of the study purpose.

2.0 Rank correlation using Non-Parametric Kendall's w test.

In this session, the researcher studied the following three items namely;

- a) ***Subject clarifications preferences in Higher Education:*** the study results revealed that the majority of study learners of higher education preferred to use outside classroom sources like peer group learning, library, and digital tools for their priority in learning subjects.
- b) ***Students Career Ambition in doing Higher Education:*** the study results supported the preference of self-employment and seeking employment category of career ambition stands as the first and second choice, followed by getting a degree and social respect as the third and fourth level of priority among the students perceiving higher education.
- c) ***Students Learning style preference in Higher Education:*** the study results projected, among the learning styles, the '*aural mode of learning*' (prefer to listen to lectures rather than take notes) and '*Virtual mode*' as their first and second priority respectively, followed by, *Kinaesthetic mode* (multi-sensory) learning such as LEARN BY DOING by using *visual and/or auditory study techniques* as their third priority. Finally, the '*Read and Write*' learning styles and *visual-spatial learning style* as their fourth priority.

3.0: Student's preference towards Place of study, time and quantity of study, and method of study

- a) In the physical place required for the study, the temperature of the place of study gained more importance with the mean score of 7.3 (rank 1), followed by users to do study-related work at a specific place or attempt to do multitask activity supporting for the study purpose in the study area uniformly gained a mean score of 6.23 (rank II), the activity of avoiding distraction through facing wall or corner of the place as their preference of study area gained a mean score of 5.75 and scored the third rank.
- b) In the time and quantity of study, preference stated that the review of the class notes gained a maximum mean score of 5.25 and ranked I, followed by, the most available time for study purpose scored a mean score of 4.90 and ranked II, used the most alert time for study gained third place with the mean score of 4.84.

c) In the method of study preference, think and list of additional questions during the chapter preparation gained the maximum mean score of 5.57 (1 Place), reading additional material which could probably satisfy their understanding more precise, gained the mean score 5.32 (2 nd Place), learning based on the exam orientation rather reading textual material gained the mean score of 5.22 (III Place).

10. Conclusion

It is concluded that the results of this study highly supported and confirmed the significant relationships between the students' study habits and learning styles in higher education. In this study, the researcher has initially categorized the respondents' profile based on their choice of the program selected, stay place during their higher education, the number of schools studied in earlier, the medium of instruction's used at the school level, mode of self and group preparation used, the basic interest in education, time preference for their study, usage of library visits for their learning commitments and use of online sources for their extra learning. Later, the researcher used the rank correlation method to identified the preference on various modes used for their 'subject clarification', choices of their 'career ambition' and the various 'learning style' preference in their subject learning and finally, the researcher used the non-parametric test for identifying the various limits in their preference on the physical place requirements, time of study and measure of material preparation and various methods of study used in higher education. Finally, it concludes that the various learning styles preferred by students have influenced VARVK modes and there is positive significant association among study habits and learning styles between students pursuing education in higher educational institutions in Vellore District.

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36. Efficacy of Adapted Breathing Exercise on Self-Concept among Children with Intellectual Disability

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Abstract

*This paper aimed to highlight the study “Efficacy of Adapted Breathing Exercise on Self-Concept among Children with Intellectual Disability.” Therefore 60 students (30 boys and 30 girls) aged 10-15 year old of educable mild category intellectual disabled children had been selected from special unit of mental retardation, J.M. Institute of Speech & Hearing, Patna, Bihar, India, as convenience were assigned randomly in two equal (on the basis of IQ scores) groups, one experimental group and another control group, i.e. 30 in experimental group and 30 in control group. The purposive technique of sampling had been used to select the sample for the study. This study had been employed “**Pre-test Post-test Only Design**” under experimental research. To measure the amount of effect of adapted breathing exercise on Self-Concept among children with mild intellectual disability in pre-test and post-test, that tool was used, the standard Self-Description (Self-Concept) Questionnaire-I (SDQ-I; By Marsh, 1988), Individual Administration. Both the groups had been pre-tested after that the Experimental group had performed adapted breathing exercise practices 30 minutes for 30 weeks regularly (5-days in a week), while Control group didn’t perform any activities like ABE. After completion of the treatment period once again both the groups were post-tested by same tool used for pre-testing. The collected data were analyzed by applying appropriate statistical techniques such as Wilcoxon Sign Rank Test was used to test Null Hypotheses. The data had been analyzed by computer programme such as SPSS & MS Excel statistics package (free student version). The findings of the study revealed that the 30 weeks intervention programme of adapted breathing exercise practices among children with mild intellectual disability have significant impact on their Self-Concept.*

Keyword:- Adapted Breathing Exercise, Self-Concept, Intellectual Disability.

Introduction:

We all have a sense of self. Whether that sense of self is positive or negative is based upon our experiences in life and our perceptions and assessment of our self. However, the problem is that perception of self is often distorted, in the case of intellectual disabled children, physical well being

does not have a strength and calm impact on the mind resulting into low self-concept among children with Intellectual Disability.

This is the common problems of low Self-Concept in the case of intellectual disability, because they face a lot of mental cognitive limitation, high Cognitive state of anxiety, high somatic state of anxiety, self-doubts, nervousness, and weakness in physical and sensory motor skills. The children with sensory dysfunction will show many of the characteristics like poor muscle tone, poor tactile discrimination or tolerance, delayed reflex development, abnormal like or dislike of vestibular or balance activities, delays in both fine and gross motor skills, immature posture and gait pattern, poor body balance, these students have difficulty in processing sensory information, may have trouble in learning at school. The inability to understand quickly i.e. difficulties experienced in thinking and generalization and imaging this affects concept formation. These students may have a short attention span, poor grasp of the matter thought, slow reactions to environmental demands, poor problem solving skills, difficulty in staying at their seats, and problems in concentration. Dependence in movement develops low Self-Concept which affects their daily life skills as well as academic skills like reading, writing, listening, and speaking.

In other hand **Self-Concept** is also called Self-Construction, Self-Identity, or Self-Perspective which is a collection of beliefs about oneself that includes elements such as academic performance, gender roles and sexuality, and racial identity. Generally, self-Concept embodies the answer to "Who am I?" One's Self-Concept is made up of Self-Schemas, and their Past, Present, and Future Selves. Self-Concept is a cognitive or descriptive component of one's self (e.g. "I am a fast runner"), while self-esteem is evaluative and opinionated (e.g. "I feel good about the fact that I am a fast runner").Self-concept is made up of one's self-schemas, and interacts with self-esteem, self-knowledge, and the social self to form the self as whole. It includes the past, present, and future selves, where future selves (or possible selves) represent individuals' ideas of what they might become, what they would like to become, or what they are afraid of becoming. Possible selves may function as incentives for certain behaviour.

Need and Significance the Study:

This study is significant due to the following reasons:

- Use of Yogic Science is spreading very fast in all around the world.
- Use of Yogic Science method to cure the patient in medical field.

- Use of Yogic Science Technique to fight with disease, disabilities is low cost expenditure for health of the society.
- Yogic Science has proven so many cases where patient have got advantages when they faced so many disappointments from medical treatment all around the world.
- Yogic Science has benefited to patient with medical disorders, like eating disorders, asthma and irritable bowel syndrome and so many medical issues.
- The present study through light on use of adapted Breathing Exercise on Self-Concept among the children with intellectual disability.

With a view to recent development of knowledge, in the past three decades there are some studies which draw researcher attention about the problems related with disability among all groups like ID, HI, VI, VH, OH, LD, ADHD, DSI, ASD etc.. But there are very few recent studies (studies conducted by Adamson, O' Hare and graham in 2006 and by Rudel, Denka and Brown in 2004) draw our attention towards problem in sensory modulation among children with autism and LD .The studies conducted in educational setting also presents the difficulties faced by the children with MR, LD, and ASD in curricular as well as co-curricular activities due to deficit in sensory integration (DSI). By recognizing these studies, it is clear that such studies are conspicuous by its absence. Since with new developments in the area of education by *The World Declaration on Education, 1990 at Jomtien, Thailand* "Education for all", and "Salamanca Statement on Principles, Policy and Practice in Special Educational Needs", Spain, 1994 emphasized Inclusive School Model which is, inclusion of children with any disabilities is essential, in India, and also PWD (Equal Opportunity, Full Participation and Protection of Rights) Act, 1995 has been implemented in throughout India, hence there is a need for such studies, which will provide better clarity about the problems and issues, related education for children with Intellectual Disability.

OBJECTIVES:

1. To study the effect of Adapted Breathing Exercise on Self-Concept among Children with Mild Intellectual Disability.

HYPOTHESES:

H0-There is no significant difference between Pre-Test and Post-Test Scores of Self-Concept among Children with Mild Intellectual Disability.

Scope of the Study:

This study can directly help the General and Special teachers, Teacher Educators, Rehabilitation Professionals, Stakeholders, Caretakers, Vocational Trainers, Occupational Therapists, Researchers, Administrators, Curriculum Designers & Planners, Policy Makers, NGOs, Social workers, Service Providers as well as Parents of disabled children in selection of Appropriate Breathing Exercise (Yoga Practices) for Children with Intellectual Disability.

Delimitations:

This study was restricted to study the **“Efficacy of adapted breathing exercise on self-concept among children with intellectual disability”**. The study was delimited at Patna district in Bihar state, India only, and the samples had been selected on the related & concerned population in the Patna district only.

This study was restricted to 60 samples (30-boys & 30-girls) only from children with mild intellectual disability of age group 10-15 years having an IQ level of 50-70, on Wechsler Intelligence scale-IVth edition, for Indian children. Only one special school was selected in selection of samples from J.M. Institute of Speech & Hearing, Patna, Bihar, India. Only 30 minutes per day intervention programme of adapted breathing exercise was given to the children with mild intellectual disability for 30 weeks (5-days in a week) about eight months for this research.

Methodology:

The methodology for this study described here under the following heads:-

Research Design:

This study had been employed **“Pre-test Post-test Only Design”** under Experimental Research.

Sampling Design & Size:

Researcher had selected two equivalent groups Experimental Group & Control Group for the purpose of the study. The target group that had been selected which included total 60 children, 30 boys and 30 girls with mild intellectual disability within age group 10-15 years.

Table-1 Sample Distribution

Sample	Age (Years)	Experimental Group	Control Group	Total
Boys	10-15	15	15	30
Girls	10-15	15	15	30
Total	-	30	30	60

Sampling Technique of the study:-For this study **Purposive Sampling Technique** had been used to select the sample.

Table-2 Group Design

Groups	Pre-test	Treatment	Post-test
Experimental Group	O ₁	X ₁	O ₂
Control Group	O ₁	X ₂	O ₂

O₁ & O₂ are Pre-Test & Post-Test respectively.

X₁-Treatment or Intervention Programme, Adapted Breathing Exercise.

X₂-Without Treatment.

Independent Variable:-Adapted Breathing Exercise.

Dependent Variables:- Self-Concept of Intellectual Disabled Children.

Tool for Treatment:-It included Adapted Breathing Exercise (ABE) practices had been selected after consultation from Doctors (Central Dispensary, Patna University, Patna) Psychologists, Physiotherapists, Experts of Yoga and Special Educators by the investigator.

Steps of Breathing Exercises:

Step-1 Laying down in Shabasan on the floor ('like dead body structure) absolutely straight (Do not arch or bend our back)

Step-2 Keep gap between 1 to 1 ½ feet both the legs

Step-3 Keep right hand on your stomach, keep left hand on your chest

Step-4 Close you both the eyes

Step-5 Concentrate your mind on between eyebrows

Step-6 Try to see light spot at there

Step-7 Silently chant the word OM ! OM ! OM ! Three times

Step-8 Now concentrate your mind on natural breathing

Step-9 See your breathing, inhaling and exhaling at least ten times

Step-10 Now take a break for a moment

Step-11 Now, first exhale all air completely through your nose and check your right hand moving downward, but try to keep your left hand static (no any movement)

Step-12 Now inhale breath very slowly and steadily through your nose, and fill your stomach completely (without producing any sound) your right hand goes up with your stomach. This ensures that you are breathing deeply

Step-13 Now, inhale some more breath in your lungs ensuring that your chest is going up with your left hand

Step-14 Now, inhale some more breath with adjusting or shaking your solders

Step-15 Now hold or wait a moment and count the no. 1, 2, 3, 4, 5.....up to till 21 silently, so that your mind will concentrate on counting

Step-16 Now exhale some breaths slowly and steadily by relaxing your solders first.

Step-17 Now exhale total breath from your lungs ensuring that your left hand going downward

Step-18 Now exhale total breath from stomach ensuring that your right hand going downward. This is one cycle of adapted breathing exercise

Step-19 It is continued at least 20-30 cycles.

Note-Exhalation time is a little longer than inhalation time.

Validation of Tool for Treatment:-Few breathing exercise had been selected by the investigator and given to different experts in the field of Physical Health & Yoga and working with Intellectual Disability children for further suggestions/modifications. For the purpose of pilot testing the adapted breathing exercise had been treated on 05 samples and then finalized the adapted breathing exercise practices. Finally this adapted breathing exercise had been incorporated and selected for the study.

Tools for measurement:-The standard Self-Description Questionnaire-I (SDQ-I; By Marsh, 1988), Individual Administration, had been used, which was standardized.

Adaptation of the tool;

As a first step, the tool which was used for measurement, translated in the regional language (Hindi) for the easy understanding to teachers/caretakers. For that purpose, the original English version of the tool was given to an expert Bilingual Professor, Department of English, Patna University, Patna, for translation that was well versed in Hindi language as well as he possesses Ph.D. in English language. He translated the standard **Self-Description Questionnaire-I** (SDQ-I; By Marsh, 1988), Individual Administration the translated version tool in Hindi language was again re-checked by another expert for revalidation. It was found that both the versions matched about 95 to 100% of the content. The reliability and validity of

the translated tools were found sufficient split-half reliability of the test had found the content validity of the test was found optimum with the help of expert's opinion working in the related field. Then it was decided by the researcher to use the tool and also researcher made it sure that while translating the content, no part either fully or partially omitted or modified at any cost.

PROCEDURE:

Administration of Pre-Test:-The investigator administrated translated above mentioned:- The standard *Self-Description Questionnaire-I* (SDQ-I; By Marsh, 1988), Individual Administration as the two times for pre-tests after certain time intervals, the gap was 7-days between 1st & 2nd pre-tests.

Treatment:-As part of the treatment adapted breathing exercise practices had been conducted. These adapted breathing exercise practices had been carried out in a group situation. A total no. of 30 weeks (5-days in a week) of 30 minutes sessions each had been done.

Administration of Post-Test:-The investigator administrated translated:-

The standard *Self-Description Questionnaire-I* (SDQ-I; By Marsh, 1988), Individual Administration as the two times for post-tests after certain time intervals, the gap was 7-days between 1st & 2nd post-tests.

Data collection:

Written permission for data collection was taken from head of the all above mention organization. After explaining the purpose of the study the parents of the intellectual disabled students, special teachers, Physiotherapists were agreed to support in all aspects. The translated tools (Hindi Version) were supplied to the students with intellectual disability in presence of special teachers, and after marking according to their knowledge researcher had collected back from them. The researcher took the help of special teachers of their respective classes, who were very familiars with those students. Researcher had arranged the meeting with parents of the students at the time of pre-tests and post-tests.

Data Analysis:-The data collected from samples had been fitted in the data sheets of the MS-Excel computer programme and analyzed by applying appropriate statistical technique *Wilcoxon Sign Rank Test* to compare pre-test & post-test scores. The researcher had used SPSS package for data analysis.

H0 -There is no significant difference between Pre-Test and Post-Test of Self-Concept scores among Children with Mild Intellectual Disability.

Sub-Hypotheses:

H0₁: There is no significant difference between pre-test and post-test scores of Self-Concept of the children with mild intellectual disability, who were given no treatment.

(Control Group)

H0₂: There is no significant difference between pre-test and post-test scores of Self-Concept of the children with mild intellectual disability, who were treated with adapted breathing exercise for 30 weeks (5 days in a week).

(Experimental Group)

To find out if there was significant difference between pre-test and post-test of Self-Concept scores without using adapted breathing exercise treatment among **Control Group** of the children with mild intellectual disability, the researcher calculated the means, standard deviations of pre-test and post-test scores of the sample which consist of 30 children. After that, Wilcoxon sign rank test was used to compare the difference of medians between the pre-test and post-test Self-Concept scores.

Table-3.1
Descriptive Statistics of Self-Concept Scores of Control Group

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
Pre-test Self-Concept Scores-	30	79.1000	4.49022	72.00	89.00	75.00	79.00	82.25
Post-test Self-Concept Scores	30	78.9333	4.30664	70.00	88.00	75.00	79.50	82.25

Wilcoxon Signed Ranks Test

Table-3.2
Ranks

	N	Mean Rank	Sum of Ranks
Negative Ranks	10 ^a	19.90	199.00
Post-test Self-Concept - Positive Ranks	20 ^b	13.30	266.00
Pre-test Self-Concept Ties	0 ^c		
Total	30		

a. Post-test Self-Concept < Pre-test Self-Concept

b. Post-test Self-Concept > Pre-test Self-Concept

c. Post-test Self-Concept = Pre-test Self-Concept

Table-3.3
Test Statistics^a

	Post-test Self-Concept - Pre-test Self-Concept
Z	-.698 ^b
Asymp. Sig. (2-tailed)	.485
Exact Sig. (2-tailed)	.494
Exact Sig. (1-tailed)	.247
Point Probability	.002

a. Wilcoxon Signed Ranks Test

b. Based on negative ranks.

Table-3.4
Test Statistics

Total N	30
Test Statistic	266.000
Standard Error	48.029
Standardized Test Statistic	-698
Asymptotic Sig.(2-Sided test)	.485

Table-3.5**Hypothesis Test Summary of Self-Concept Scores of Control Group**

S. No	Null Hypothesis	Test	Significance	Decision
01	The median of differences between pre-test of Self-Concept and post-test of Self-Concept of Control Group ; Equals 0	Related Samples Wilcoxon Signed Rank Test	.485*	Retained the Null Hypothesis H₀

Asymptotic significances are displayed. *Level of Significance 0.05

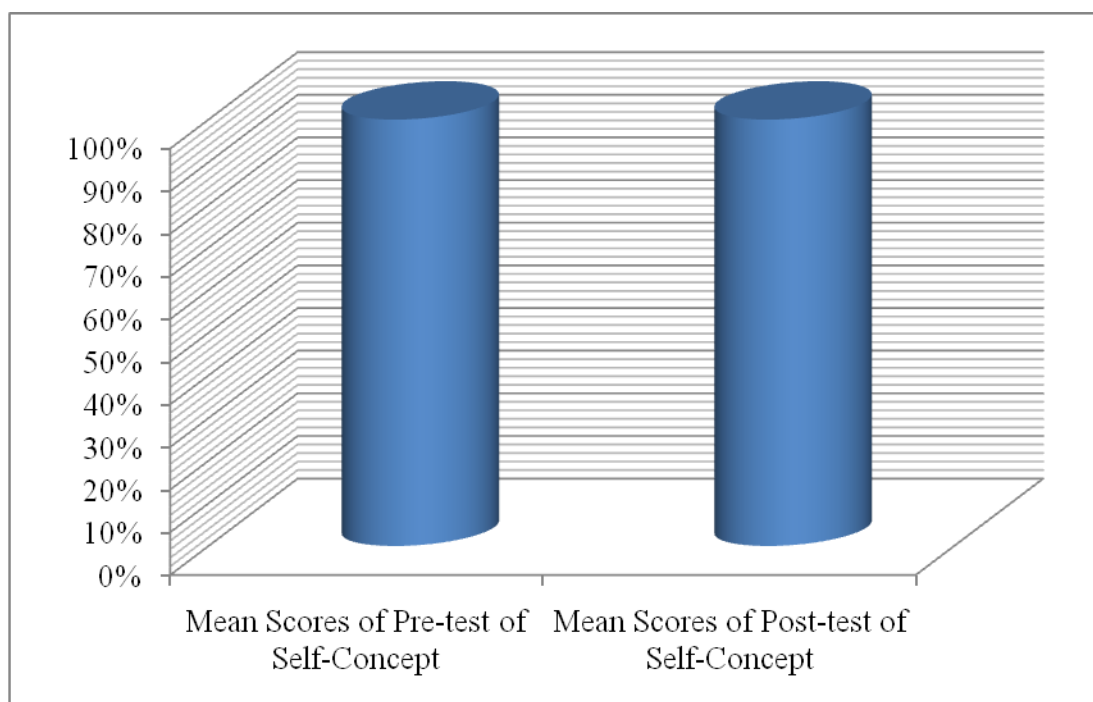
A Wilcoxon signed-rank test showed that the control group did not elicit a statistically significant change in Self-Concept scores. ($Z = -.698$, $p = 0.485$).

Table-3.1 Showed that, the mean value of pre-test was **79.1000** and the post-test was **78.9333**. Standard deviation of pre-test was **4.49022** and post-test was **4.30664**. **Table-3.3** Showed that, the Wilcoxon Z-value was **-.698**, and the significance level of Z-value was **0.05**.

Since the calculated value Z-value **.698** was lower than the critical **Z-value (2.04)** of the significance level of **0.05**, it was inferred that there was no statistically significant difference between pre-test and post-test of Self-Concept scores among control group. Hence the Null hypothesis was **H₀** retained.

And since the mean score of post-test **78.9333** was slightly smaller than the mean score of pre-test **79.1000**, which would not be considered as a great difference in mean so it might be neglected at this significant level. This result revealed that there was no change between pre-test and post-test of Self-Concept scores among the control group of children with mild intellectual disability.

Figure-1, Showed, there was no difference between pre-test and post-test in Self-Concept of Control Group Children with mild Intellectual Disability.



FINDINGS OF THIS RESULT:

There had been no change found in Self-Concept among children with Mild Intellectual Disability who did not perform adapted breathing exercise within a period of 30 weeks (5-days in a week).

To find out if there was significant difference between pre-test and post-test means of Self-Concept scores while using adapted breathing exercise treatment among **Experimental Group** of the children with mild intellectual disability, the researcher calculated the means, standard deviations of pre-test and post-test scores of the sample which consist of 30 experimental group children with mild intellectual disability. After that, Wilcoxon sign rank test was used to compare the difference in medians between the pre-test and post-test Self-Concept scores.

Table 4.1

Descriptive Statistics of Self-Concept Scores of Experimental Group

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th

Pre-test of Self-Concept Scores-	30	81.30	5.46557	72.00	94.00	77.75	80.00	85.25
Post-test of Self-Concept Scores	30	197.70	9.05596	181.00	220.00	189.75	199.00	204.00

Wilcoxon Signed Ranks Test

Table 4.2

Ranks

	N	Mean Rank	Sum of Ranks
Post-test of Self-Concept Scores - Pre-test of Self-Concept Scores			
Negative Ranks	0 ^a	.00	.00
Positive Ranks	30 ^b	15.50	465.00
Ties	0 ^c		
Total	30		

a. Post-test of Self-Concept Scores < Pre-test of Self-Concept Scores

b. Post-test of Self-Concept Scores > Pre-test of Self-Concept Scores

c. Post-test of Self-Concept Scores = Pre-test of Self-Concept Scores

Table 4.3

Test Statistics^a

	Post-test of Self-Concept Scores - Pre-test of Self-Concept Scores
Z	-4.786 ^b
Asymp. Sig. (2-tailed)	.000
Exact Sig. (2-tailed)	.000
Exact Sig. (1-tailed)	.000
Point Probability	.000

a. Wilcoxon Signed Ranks Test

b. Based on negative ranks.

Table 4.4
Test Statistics

Total N	30
Test Statistic	465.000
Standard Error	48.581
Standardized Test Statistic	-4.786
Asymptotic Sig.(2-Sided test)	.000

Table 4.5
Wilcoxon Signed Rank Test Summary Report of Self-Concept Scores of Experimental Group
(Two tailed test)

N	30
V	0.000
Expected Value	232.500
Variance	2360.125
P-Value (Two tailed)	0.0001
Alpha	0.05
Z- Value	-4.786

(An approximation method has been used to compute the p-value.)

Table 4.6
Hypothesis Test Summary of Self-Concept Scores of Experimental Group

S. No	Null Hypothesis	Test	Significance	Decision
01	The median of differences between pre-test of Self-Concept and post-test of Self-Concept ; Equals 0	Related Samples Wilcoxon Signed Rank Test	-4.786*	Rejected the Null Hypothesis H₀

Asymptotic significances are displayed. *Level of Significance 0.05

A Wilcoxon signed-rank test showed that adapted breathing exercise treatment course of 30 weeks (5-days in a week) did elicit a statistically significant change in Self-Concept scores among Children with Mild Intellectual Disability. ($Z = -4.786$, $p = 0.0001$).

Table-4.1 Showed that, the mean value of pre-test was **81.3000** and the post-test was **197.7000**. Standard deviation of pre-test was **5.46557** and post-test was **9.05596**.

Table-4.3 Showed that, the Wilcoxon Z-value was **-4.786** and $p = 0.000$ at the significance level of Z-value was **0.05**.

Since the calculated Z-value **4.786** was greater than the critical tabular **Z-value (2.04)** at the significance level of **0.05**, it was inferred that there was a statistically significant difference between pre-test and post-test of Self-Concept scores while using adapted breathing exercise among experimental group of the children with mild intellectual disability. Hence the Null hypothesis **H₀** was rejected.

And since the mean score of post-test **197.7000** was significantly greater than the mean score of pre-test **81.3000**, the significant difference was in favor of the post-test mean. This result revealed that the adapted breathing exercise had significant effect in developing Self-Concept in children with mild intellectual disability who were gone under treatment of Adapted Breathing Exercise treatment of 30 weeks (5-days in a week).

Figure-2, Showed there was difference between pre-test and post-test in Self-Concept of Experimental Group Children with mild Intellectual Disability.



FINDINGS OF THIS RESULT;

The effect of adapted breathing exercise on Self-Concept among experimental group of the children with mild Intellectual Disability had found great impact within a period of 30 weeks for enhancing the Self-Concept.

Educational Implications:

The findings of the present study revealed that the children with mild intellectual disability who had problems of low Self-Concept was changed the circumstances by the use of treatment of adapted breathing exercise for 30 weeks (5-days in a week). On the basis the results the researcher would like to confer some educational implications. If such Breathing Exercise Practices would be included in regular curriculum, it will improve the personal skills (daily life skills), social skills, occupational skills and academic skills like reading, writing, speaking, and listening, because of its low cost expenditure in nature.

Conclusion:

- Adapted breathing exercise enhanced self-concept among children with mild intellectual disability when they were subjected to intervention programme for 30 weeks (5-days in a week) in experimental conditions. The improvement was visualized in terms of changes from; low self-concept to high self-concept within 30 weeks. Therefore, it can be concluded that Adapted breathing exercise are very effective for children with mild intellectual disability.

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37. An Investigation Into Causes, Consequences And Combating Measures of School Dropout in Sikariya Village, Jehanabad District, Bihar

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Abstract

Education is the important device for development of cultural values and growth of a child in preparing them to be responsible, intelligent, strong and healthy citizen. Despite professed quality of opportunity to all citizens and various initiatives of stakeholder and government still school dropout is present in Indiansociety and particularly in case of Bihar. The present study revealed the theoretical review of literature on school dropout with various causes and factors namely personal, family, school and socioeconomic. This research work tried to find out various causes, consequences and combating measures to deal with the school dropout in Sikariya village of Bihar. For this purpose extensive primary data were collected fromthe study area which were substantiated with the secondary sources of data. The study reveals that dropout in Sikariya village is positively associated with its various social, economic and cultural factors. The tendency to not go to school is more among SC,ST and backwardcaste students. Students with SC,ST and backwardcaste leave school for social and economic reasons. Reasons for not going to school is also positively associated with sex of the children. Household work more often prevent girl students from going to school. Whereas, economic and physical reasons prevents male students from going to school.

Key Words: School Dropout, SC,ST and backwardcaste, Children, Socio-cultural.

Introduction:

Education is the driving force behind any strong economy and a prerequisite for social and economic growth. It creates opportunities and provides societies with a better educated and skilled workforce which is necessary for stimulating development. Dr. A.P.J. Abdul Kalam, former President of India, comments that we are nation of a billion brilliant people. And if each person is educated, the power of knowledge can transform India from a developing nation into a strong and economically developed nation. Despite the government's ambitious plans and projects such as SarvaShikshaAbhiyan and the Right to Education Act which aim is providing free and compulsory education to children, school dropouts rates are still a cause for concern. The dropout problem is pervasive in the Indian education

system. Many children who enter school are unable to complete secondary education and multiple factors are responsible for children dropping out of school. Risk factors begin to add up even before students enroll in school includes: poverty, low educational level of parents, weak family structure, pattern of schooling of siblings and lack of pre-school experiences. Family background and domestic problems create an environment which negatively affects the value of education. Poor socioeconomic conditions prevailing in rural area are forcing parents to send their children to work instead of school. Such parents want their children to go to work and get that extra income for the family. Also, some children are forced to look after their younger brothers and sisters when their parents go out for work. Then, girls from such families are made to sit at home once they come of age, while they learn to cook and manage other household chores. So concern like school dropout continue to remain severe bottlenecks for improvements. According to data provided by Ministry of Human Resource Development (now Ministry of Education) the dropout rate of students in class 9 and class 10 have crossed almost a fifth of the total enrolment rate in 2017-18. The dropout rate in secondary school was 32 % in Bihar. The dropout rate of SC is higher than other community.

Dropout is an act or instance of dropping out. Most commonly, dropping out refers to a student who has abandoned course of study and quitted school in an unusual way before completing a qualification or before graduation. A school dropout student is a student who leaves school before the completion of the school stage or at some intermediate or non-terminal point of a given stage of school education. High dropout rate in school education is attributed to poor socioeconomic situation of the family. School dropout is highly visible sign of a society that has failed to prepare its youth for successful transition into adulthood. School dropout is very common in Indian villages, especially in the state like Bihar which stands at the bottom in literacy and it simultaneously the condition of female literacy is at its worse in this village. Dropping out is a serious problem because it denies individual students their fundamental human right to education. This research work will helps to find out the present condition of school dropout rate in Sikariya village, causes behind such dropouts, consequences and strategies to curb this issue.

Data Sources and Methodology:

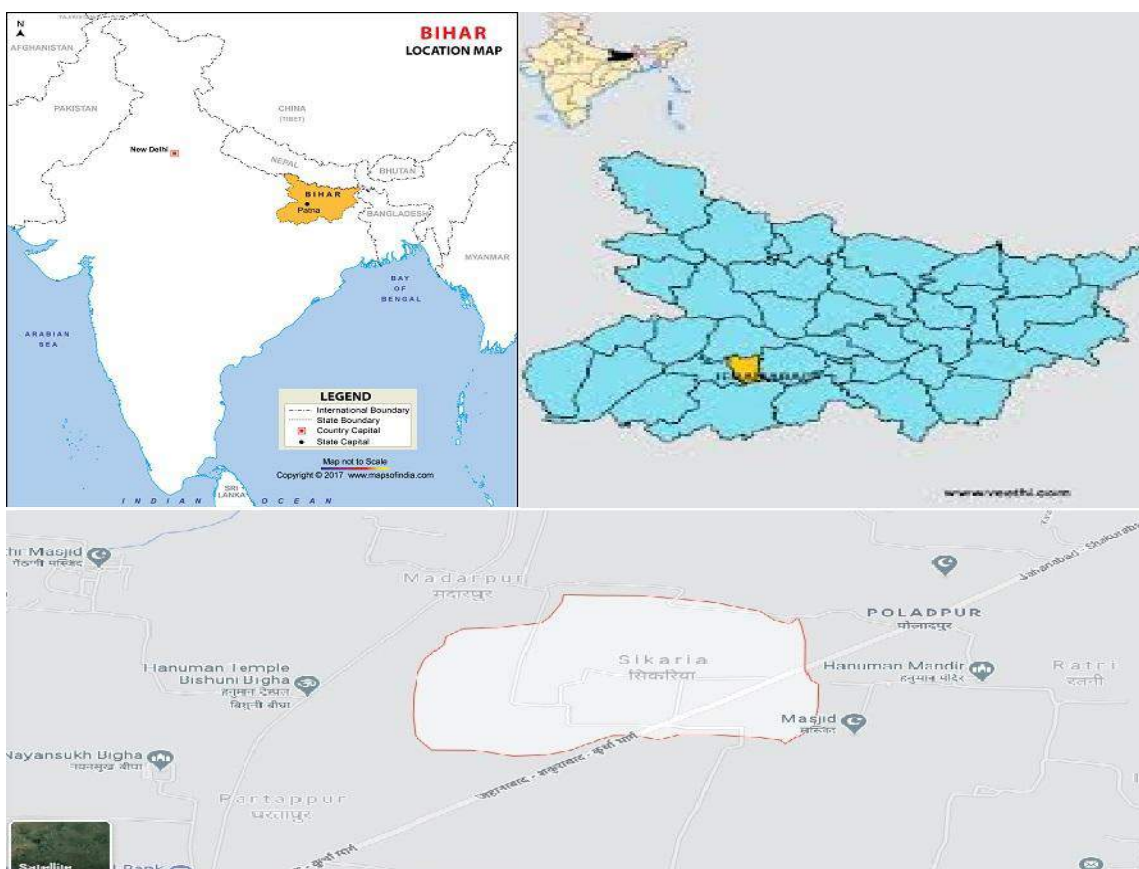
The sample design adopted for the school dropout is a systematic, multi-stage stratified sample of households. The dropout child houses were specifically searched in the field by snowball sampling method. Data were obtained in field through interview, group discussion, and questionnaire survey and classroom observation of the population. Total students enrolled in 2019-20 school years was

divided into three categories: i) students who never attended school after enrolment ii) students who dropped out and iii) the students who currently in the school. The present study aimed to find out school dropout condition of the village as a whole. It was not age specific. Major tools were household survey questionnaire, interview schedule for the teachers, school information schedule, interview schedule for parents, and interview schedule for dropout student. Primary data has been collected from 250 household and also from the school of the village. Major secondary sources of data were census of India, National Sample Survey, Ministry of Human Resource and Development, etc. The collected data were analyzed with the help of MS Excel.

Profile about the area of study:

Sikariya village is in Jehanabad Block in Jehanabad District of Bihar State, India. It belongs to Magadh Division. It is located 3 km towards west from District Head Quarters Jehanabad and 50 km from state capital Patna. Shahpur, Amain, Pandui, Kaswan are the nearby villages to Sikariya. Sikariya is surrounded by RatniFaridpur Block towards west, Makhdumpur Block towards south, Kako and Ghoshi Block towards east. Jehanabad, Makhdumpur, Masaurhi, Islampur are the nearby cities to Sikariya. Pandui is the Gram Panchayat of the village Sikariya. Total geographical area of the village is 192 hectares. Total population of the village is 1646. There are about 265 houses in Sikariya village. Male population is 864 and female population is 782. Child (0-6) population is 235 according to census 2011. Village literacy rate is 64.0 percent. Female Literacy rate is just 27.2 percent. Rajkiya Utkramit Madhya Vidyalaya. Sikariya is the middle school of the village.

Figure1: Area of Study



Sources: Mapofindia.com

Results and Discussion:

One of the major objective of the present study was to find out the drop-outs rates and its causes in the light of socio-economic profile of the respondents and their family background. Thus, an attempt is made in this section to provide socio-economic profile which is used as an important variable in the subsequent analysis chapters. The table no. 1 shows a number of measures related to the socio-economic status of the households in the village.

A study on socio-economic details of respondents and family background:

Table-1: Frequency distribution by household characteristics of respondents in the village.

Household characteristics	Categories	Frequency	Percent
Household Size	1 to 4 members	47	18.8%
	5 to 8 members	138	55.2%
	9 and above	65	26.0%

	Total	250	100%
Sex of child	Male	107	42.8%
	Female	143	57.2%
	Total	250	100%
Land holding status	No Agricultural land	38	15.2%
	Less than 1 hectare	93	37.2%
	1 to 5 hectares	72	28.8%
	5 and above hectares	47	18.8%
	Total	250	100%
Age of child	6 to 14 years	152	60.8%
	15 and above years	98	39.2%
	Total	250	100%
Level of schooling	Primary	114	45.6%
	Secondary	91	36.4%
	High	45	18%
	Total	250	100%
Sex of head of Household	Male	185	74%
	Female	65	26%
	Total	250	100%

Source: Primary Survey 2019-20

The table given above reveals that 55.2% of population maintaining their household size between 5 to 8 members followed by 26% of the population having 9 and above members and remaining 18.8% of the population having 1 to 4 members in their families. Nevertheless only small margin of population i.e., 18.8% were having ideal family size of up to 4 members which clearly indicates that there is negligible family planning norms is adopted by the people of this village. From the above table, it is clear that girl child (57.2%) is more than male child (42.8%) in the village. Out of 250 households 60.8% of children fall under the age group of 6 to 14 years of age and remaining 39.2% belong to the age group of 15 and above. In Indian villages especially in the state like Bihar birth rate is high which can be also observed by analyzing the data obtained from the survey. From the table no. 1 one can see that very little percentage of population got education up to higher level that is 18%. 45.6% of

population got education up to primary level followed by 36.4% up to secondary level. There are various reason of such fluctuations. One of them is most villagers do not consider education essential, they prefer more to earn money for their subsistence than getting degrees of higher education because most of them are poor. Household work, working on farm, working outside for employment, as well as child marriage prevents them for getting higher education. The type of family and the one who is heading the household is vital component in family management where man is supreme authority of the family. Further, in 74% of houses male was head of the family and in remaining 26 % females are heading the family. This shows the male dominating society of the village.

Table-2: Frequency distribution by parental characteristics in Sikariya village

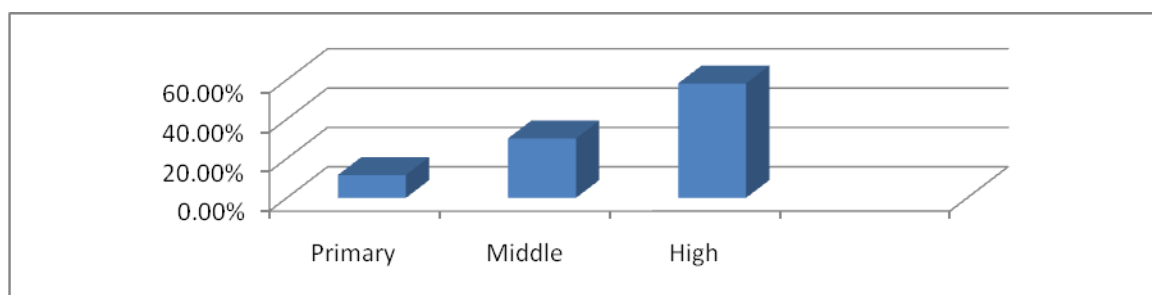
Parental characteristics	Categories	Frequency	Percent
Father occupation	Not working	09	3.6%
	Agricultural work	128	51.2%
	Non-agricultural work	113	45.2%
	Total	250	100%
Mother occupation	Housewife	166	66.4%
	Agricultural work	27	10.8%
	Non-agricultural work	57	22.8%
	Total	250	100%
Working status of parents	Father working	173	69.2%
	Both are not working	3	1.2%
	Mother working	11	4.4%
	Both are working	63	25.2%
	Total	250	100%
Number of living children to parents	2 and Below	56	22.4%
	3 to 5	135	54%
	6 and above	59	23.6%
	Total	250	100%

Education status of parents	Husband illiterate	168	67.2%
	Wife illiterate	210	84%
	Both are literate	83	33.8%
	Both are illiterate	50	20%
	Total	250	100%

Source: Primary Survey 2019-20

The above table- 2 explain about the parental background characteristics such as occupation, working status, educational status, no. of children living to parents and so forth. Eventually these background factors are very important to study from the point of its nature which has a power to impact any sort of dependent variable, especially when we undertake a research in social science sector. To begin with occupation of father it was found from the table no. 5.2 that 45.2% of parents (father) were engaged in non-agricultural sector and 51.2% in agricultural sector. Further, nearly around 4% of parents not at all working they were just remained as an ideal. It is surprise to know that these days women try to engage themselves in working sector compared to their husbands. Hence almost 66.4% of mother found to be engaged in household work followed by 22.8% in non-agricultural sector remaining 10.8% of them were found to be working in agricultural sector. After considering working status of both the parents together it was observed that 25.2% of them were working together. As far as literacy level is considered slowly but steadily one can see an upward trend. 33.2% of the parents found to be educated. But when we try to classified literacy status of parents based on the gender then we find females are lacking behind in education i.e., 84% females are illiterate. Moreover, 20% are totally illiterate that means neither of the parents is educated. With a strong ethical reason always family size was found to be large among rural community, leave apart this here too only 22.4% of the households have two or less than that children in a family which sounds ideal. And chuck the rapid growth of population whereas 23.6% of rural households have higher numbers of children that is 6 and above which goes against Indian national population policy. The reasons behind such big family size is poor establishment of family planning norms and desire to get a male child and poverty is also governing reason. The research suggest that school dropout is greatly influenced by family as well as parental background characteristics.

Figure 2: School Dropouts by level of schooling 2019-20 school years



Source: Primary Survey 2019-2020

The rate of school dropouts will probably also be related to those factors which reflect the existing level of education of the rural population and factors which measure the quality of the educational system. In this study also one can understand from the figure 2 i.e., the maximum no. of school dropout is from high schooling (58.13%) followed by middle which is 30.23% and dropout rate is quite low in primary that is 11.62%. the application of statistical value at 5% significant level also suggest that there is a strong and positive co-relation between these two variables like level of education and school dropout.

For any purpose inquiry in Indian society, caste is taken as an important variable. In the present study caste is studied for its implications for dropout. However, for the purpose of analysis castes are classified into three categories as high, intermediate and low.

Table- 3: Caste wise distribution of dropout students

Caste	No. of student dropout	Percentage
High	1	2.32%
Intermediate	13	30.24%
Low	29	67.44%
Total	43	100%

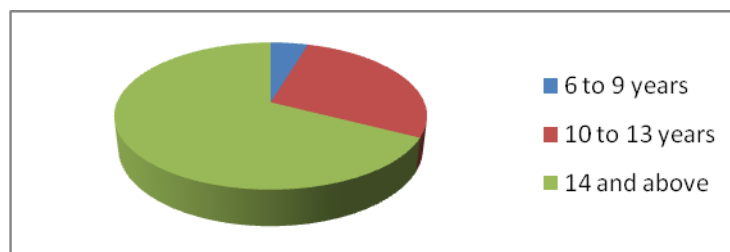
Source: Primary Survey 2019-20

The data reveal that 67.44% of the respondents come from low castes, whereas 30.24% makes the intermediate castes, whereas, the high caste constitute only 2.32% of the total sample.

Age:

It is expected that higher dropout rates are associated with higher crude fertility rate and particularly with age too. Age of the child is as important as any other socio-economic factors but in Indian context we give little importance to the age of the child at the time of enrolment in schools it may be due to lack of awareness about the educational importance and concept.

Figure 3: percentage distribution of dropouts by age of child.



Source: Primary Survey 2019-20

The above figure reveals that the dropout rate is much less in younger age group i.e., only 4.65% of the school dropout are from age group 6-9 years, and about 27.90 % of the school dropout are from the age group 10-13 years, whereas recognizable amount i.e., 67.44 % dropout are from the age group of 14 and above. It means people have realized the importance of basic education but not the necessarily of higher education.

Sex:

The wide variety of sociological and political variables have been suggested gender as major determinants of dropout rates and some time it is closely related to the cultural variables, as they reflect differential access to employment and distribution of employment opportunities based on the variables. Sex is one of the important variable to study school dropout.

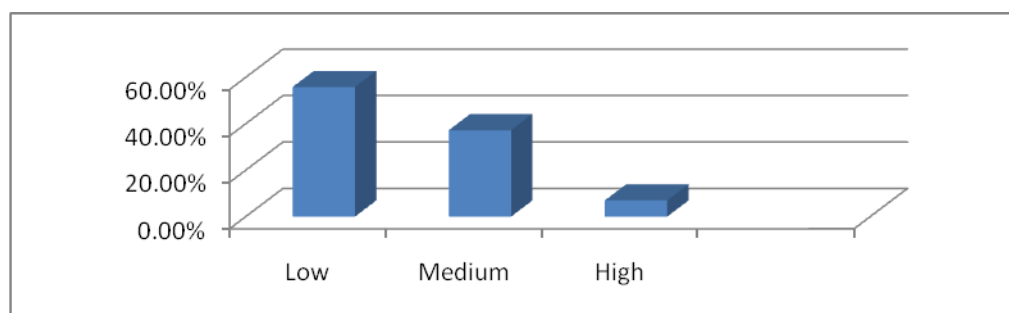
Table 4: Dropped out student distribution by sex

Sex of the dropout	No. of respondents	Percentage
Male	16	37.20%
Female	27	62.79%
Total	43	100%

Source: Primary Survey 2019-20

From the above discussion and the table no. 4 throw some light on the existing pattern of gender variation in a present research. As we see sex of the child has a strong hold in school dropout process specially 62.79% dropout found among girls, whereas 37.2% among the boys students.

Figure-4: School dropout by standard of living.



Source: Primary Survey 2019-20

The variables reflecting quality of life and differences in culture or language between the family and the school would be more closely related to early dropouts while economic costs and benefits of education would have a greater influence on later dropouts. The figure of the present study from figure 4 explains that household standard of living has a positive impact on early school dropout rates, as it was found 55.83% of the dropout was in low household income group. Further as the household income or standard of living improves there finds a steady decline in school dropout i.e. 37.2% to 6.97% respectively.

Table- 5: percentage distribution of school dropouts by parents characteristics:-

Parental characteristics	Categories	No. of dropouts	Percentage
Occupation of father	(i) Not working	22	51.16%
	(ii) Agricultural work	14	32.55%
	(iii) Non- agricultural	7	16.29%
	Total	43	100%
Occupation of mother	(i) House wife	6	13.95%
	(ii) Agricultural work	21	48.83%
	(iii) Non- agricultural	16	37.22%
	Total	43	100%
Education status of parents	(i) Both are illiterate	20	46.51%
	(ii) Wife illiterate	3	6.97%
	(iii) Husband illiterate	19	44.20%
	(iv) Both are literate	1	2.32%
	Total	43	100%

Number of living children to parents	(i) 2 and below children	2	4.65%
	(ii) 3 to 5 children	15	34.88%
	(iii) 6 and above children	26	60.47%
	Total	43	100%

Source: Primary Survey 2019-20

The occupational status of parents considered as one of the most effective component when early schooling achievement is concerned. Hence the table no. 5 unveils the fact that highest school dropouts found to be more among those students whose father are not working (51.16%) followed by 32.55% dropout whose father are working in agricultural field and drastically down ward trend was found among those students whose father are working in non-agricultural sectors. Highest school dropout was found to be more among those students either their mother are working in agricultural sector(48.83%) or non-agricultural sector(37.22%) but dropout rate found to be very less among those students whose mother preferred to remain as housewives rather than working women. The one potential link between family background and student achievement is parent involvement in the schooling process. A number of studies have found that parental commitment to education and parental education level are important determinants of dropout rates. Therefore the family atmosphere today dependent upon the educational level of parent. When we draw attention from the above table no. 6.7 that when both parents found to be literate automatically the school dropout rate is 2.32%. Never the less who's ever the literacy level is high (father or mother) the dropout level will also gone up like 44.20%, 46.51%, 6.97% respectively. High birth rates are associated with large families and the need for school age children to look after their younger siblings especially in low income families. The data for the present study clearly explains that the highest percentage of school dropout found to be from those families whose family size is more or less equal to 6 children i.e. 60.47%, 34.88%, 4.65% respectively.

Table 6: percentage distribution of causes of school dropout:-

Causes of school dropout	Frequency	Percentage (%)
school too far away	1	2.32
Further education not considered necessary	7	16.27
Required for household work	11	25.58

Required for work on farm/family business	3	6.97
Required for outside work for payment in cash or kind	5	11.63
No proper school facilities	3	6.97
Required for care of siblings	1	2.32
Not interested in studies	9	20.93
Got married	2	4.69
Other	1	2.32
Total	43	100

Source: Primary Survey 2019-20

In order to understand the concept of early school wastage, the question was asked to the parent whose children were out of school and factors responsible for happening so. Despite all governmental and non-governmental efforts put together like free education up to the age of 14 years irrespective of gender differentiated and mid-day meals, up to the high school level. Distribution of free uniform, books and by cycles to reach out therefore away distance school and specially scholarship for girl child, to undertake studies. Eventually when we critically try to review all ten reasons related to dropouts among students of the Sikariya village. As expected 25.58 % of them are required for household work. 20.93% of them stated that they are not at all interested in studies whereas for 16.27% of them further education not considered necessary followed by the statement that it's better if they go for work and earn something (13.95%) for the family expenditure sake. Never the less, other causes regarding school dropout are getting married, required for care of siblings, distance to reach out the school and lack of proper facilities in school are still emerged as the major barrier in between the schooling and dropouts.

Table 7. Percentage distribution of cluster factors responsible for school dropout:

Cluster factors	Frequency	Percentage (%)
causes related to child attitude	11	25.59
causes related to family perception	27	62.79
causes related to school environment	4	9.30
related to other reasons	1	2.32
Total	43	100

Source: Primary Survey 2019-20

Suggestions:

The study reveals that caste is positively associated with number of children not going to school. The tendency to not go to school is more among lower caste students. Students with low castes leave school for social and economic reasons. Reasons for not going to school is also positively associated with sex of the dropouts. Household work more often prevent girl students from going to school. Whereas, economic and physical reasons prevents male students from going to school. Respondents also considered lack of competence among teachers and their remaining absent from school are also identified as causes for school dropout in rural society. Some of the government programs to check the school dropouts are well appreciated by the rural people. Still there is a lot to achieve to improve the conditions of education in the village. The findings pertaining to school dropouts can be considered as quite modest and fill the scientific gap in the area of school learning children.

Based on the major findings of the study, the following suggestions are made which are divided into three sections. The first section provides suggestions to the government to address the problem of dropout children. The second sections provide suggestions to school administrations and the last section provides suggestion to the parents. These suggestions will help agencies to support children's education and to address the problem of school dropout.

The government:

Firstly Anti-child marriage efforts: one of the significant findings of the study is that prevalence of early marriage hinders the girl's child education. The govt. has already established structures to address the child marriage issue under the leadership of District Social Welfare Officer(DSWO), but the effectiveness of the functioning of the system needs to be revived; the gap need to be identified and addressed. Second, Transportation facilities: The high schools are located at distant places, meaning that the children have to travel a long distance to access high schools. Improving the public transport facilities and increasing availability of bus according to the school timings can help to reduce the transportation issue. Secondly, the state govt. is providing bicycles to the girl students from higher secondary, if they provide it to high school girls the problem of dropout at high school can be reduced a lot. Third, Improving livelihood options: From the analysis, it was clear that around half of the respondent's parents were landless laborers; monsoon failure and poor irrigation families

make people to search for job opportunities. So, the district administration should take appropriate measures to ensure minimum land ownership for all as well as better water irrigation water facilities. Secondly, the MNREGA was mainly implemented to provide rural employment .however, failure in properly implementing the program is another reason for the school dropout. Forth, Upgrading standards by upgrading the primary schools to upper primary at the village level, and the middle schools to high school, will certainly help to reduce the dropout of girl children at the high school. Fifth, Improving teacher's quality: many students said that the teaching quality is poor. In order to improve the performance of the children, it is important to improve the quality of teaching.so, on one side, the govt. should take measures to improve the quality of teacher training institutions. They should adopt more recent technology and educate teachers on how to use different methods in the classroom to make teaching more interactive to improve the quality of education.

School administration:

First, conduct regular SMC meeting: school should take care to see that SMC functions properly and should organize SMC meetings regularly. This will help to update the process of the students, and to provide concrete suggestions to help improve on their weak areas. Second, Teacher's responsibility and evaluation: late coming and living school early by the teachers and secondly the teacher's absenteeism is a major concern in the schools. So, the school administration should take appropriate measures to ensure that teachers perform their duty and should do a similar short of teacher auditing and evaluation, this will improve the responsibility of the teachers and also the learning of the children.

Parents:

First, Sensitization of parents: lack of parental interest was one of the key reasons for girl child dropout. So, parents should be sensitized, since without changing the parent attitude, gender equity cannot be achieved. Second, Child marriage: parents should be sensitized regarding the ill effects of this practice. Creative and innovative visual IEC materials should be develop to educate the people on the ill effects of child marriage which will make parents to realize the seriousness of the issue. Third, Improving parents' participation in school: parents play a vital role in motivating children to pursue education. Efforts should be made to enforce the parents to participate in the activities of the school, whereby the child will derive encouragement from parents for their education. Educational

authorities, parents and schools need to work closely together to find the reason for each student's decision to leave school and to try to do as much as possible to encourage them to stay in the system.

Conclusion:

In Sikariya village school dropout is one of the serious issue that affect the right to education of all, which are determined by various physical, economic and socio-cultural factors. Out of these factors majorly the role of parents are most important. As it is generally in our India society told that Parents are the first teacher of any children so that facts make the role of parents in education very crucial and also in dropout. For combating the primary hurdle parents should be made aware of the importance of education so that they develop a desire to educate their children. Some special programs have been active in all over India such as SSA and RTE, but it's not working effectively. So this is a high time for everyone to come forward and take a step to reduce school dropout. Though the analysis of the study provides policy insights in terms of broader educational policies aimed at improving educational equity and quality with the adoption of the significant perspectives of school dropout themselves, will further efforts to educate, the school in which they are failing to learn, the job markets they cannot enter, and the diminished communities in which they live. Thus new efforts with the improvement in existing need to emphasize for address this problem urgently and over the long term. We hope that the reflections, insights, and case study shared in this scientific research work will help transform how we view these young people not as problems to be solved, but as potential to fulfil.

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38. Teaching Strategies to Students with Visual Impairment in the Inclusive Classroom

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Abstract

The term visual impairment covers two types of disabilities namely blindness (totally blind) and low vision. A student who is totally blind is depends on his nonvisual senses, mainly hearing and touch, to access information. In the case of student with low vision, the student's everyday performance on visually dependent task will also be compromised. A student with low vision has a limited ability to see print and other objects even with correctional lenses. Due to the continuous use of senses other than sight, the senses of these students tend to be more highly developed. Therefore, for students with visual impairment all printed materials and visual tasks need to be adapted, and encouragement of the use of their sense of hearing, touch, taste, and smell should be incorporated into their learning as much as possible to ensure their accommodation and success in the inclusive education.

Inclusive education is defined as teaching students with special needs or disability in the same environment as students without special needs. Inclusion allows for the needs of students with and without disabilities to have equal access to school curricula and programs. The philosophy of inclusive education is a child-focused approach to education which requires that school adapt to the student rather than the student needing to adapt to the school. When students with disabilities are included in the inclusive classroom, both special education and general education teachers are indeed challenged to figure out ways that students with disability can learn skills and strategies in a whole class environment as opposed to a small group setting. This paper highlights the strategies to teach regular curriculum to students with visual impairment in the inclusive classroom.

Keywords: Teaching Strategies, Students with Visual Impairment, Inclusive Classroom

INTRODUCTION

All students have both similar and unique characteristics. For students with visual impairment, teacher must remember that they have the same basic needs as all students. To address the uniqueness of students with visual impairment requires supportive instruction, curricula, and environmental accommodations to minimize the effects of their disabilities and to promote their growth and development through the learning process.

Earlier it was felt the students with disability could be best nurtured and services in environments separate from students without disability. More recently, the practice has been to include children with special needs in the same environment that service children without special needs. Experience, research, and legislation have been the impetus to make this a reality (Moore, 2009). Lev Vygotsky, a developmental psychologist, affirmed that how we view ourselves is definitely influenced by how other view us as well as. It seems that Vygotsky would applaud the social benefits that students gain from learning alongside their peers in inclusive classroom. Karten (2009) stated that inclusion is an excellent way to increase student competencies when academics are skillfully taught within social environments that value the curriculum as well as how diverse student audience soaking up that knowledge interact with each other. The wonderful side effect of inclusion is that it benefits the “regular” kids too. Good practices are not exclusive to only those students with disabilities.

Student with visual impairment, when placed in inclusive settings, will need modifications, adaptations, and support, just as they would if they were in a separate setting (special school). However, the delivery system for providing these elements is different. The appropriate delivery system ensures the successful mastery of any set of teaching learning objectives. So, what constitutes appropriate learning objectives? They need to be student-based, interdisciplinary, standard-based, content driven, and challenging, thereby allowing students of all levels to secure successes (Karten, 2009). Subsequently, well designed instruction allows students to generalize, apply, and connect learning principles to their lives. Teachers not only teach, but also have the instructional power to create lifelong learners inculcated with the joy of knowledge. Teachers who give step-by-step explanations avoid the rushed instructional pace that sometimes leads to nowhere quickly. Modeling examples and frequently checking baseline knowledge is crucial. All students benefit from creative teachers whose delivery and communication system advocate and propagate the idea that learning is meaningful, including inclusive education.

TEACHING STRATEGIES IN INCLUSIVE CLASSROOM

Teachers must remember that the inclusive classroom is not a teacher-centered one, but rather one that fosters long term student learning and achievements leading towards mastery of the regular curriculum. When the classroom is student centered individual needs are addressed, aiding curriculum connections. Connecting to real life issues through problem based learning implanted the

classroom with knowledge that goes way beyond the textbooks facts and concepts. Teachers use various strategies to teach students with visual impairment in inclusive classroom, some of the strategies to teach regular curriculum to students with visual impairment in inclusive classroom are discussed below:

- **CREATION OF INCLUSIVE LEARNING ENVIRONMENT:** Defining the concept of inclusive learning environment Johnsen (2001) says, “Inclusive learning environment is one of the core element for successful inclusive education. If the environment in which learning occurs is not supportive to students with visual impairment, their learning will automatically be interrupted.” On the other hand, Fraser & Maguvhe (2008) highlighting the factors for ineffective learning and detailed, “The context in which the learning occurs; inflexible curriculum and inappropriate assessment procedures, are some of the factors leading to ineffective learning among students with visual impairment”. Further they asserted that “Inclusive learning environment is an environment that allows and supports the potential learning of all students, regardless of the learning differences and diversities these students possess in the class”. Therefore, for creating inclusive learning environment it is essential to provide some basic facilities to students with visual impairment, so that they can access quality education on par their sighted peers. These facilities include services of special education teachers, assistive devices like Braille slate, magnifying glasses, DAISY, screen reading software, use of tactile learning materials and digital technology. Therefore, it is the need of the hour to restructuring the education system for the betterment of students with visual impairment in inclusive classroom.
- **ORGANIZATION OF CO-TEACHING:** Co-teaching comprises the cooperation of two teachers who have different area of specialization. According to Dalen (1982), “Co-teaching means involvement of two teachers teaching the same class at the same time, a regular teacher taking the responsibility of the main teaching, and a special education teacher, dealing with disability specific needs of students”. Now the question arises is co-teaching beneficial in the inclusive classroom? In this regards the study conducted by Lipsky & Gartner (1997) reveals, “Co-teaching is an important feature of inclusive education, because inclusive education contains students with diverse learning needs. No single teacher can have all the skills necessary to meet students’ diverse needs in inclusive classroom”. It means a teacher specialized in one specific area of disability should be part of the teaching in an inclusive classroom having that category of students with disability. In an inclusive

classroom a special education teacher should support general education teacher in preparing TLM and also in using assistive devices for students with visual impairment.

- **ADAPTATION IN TRADITIONAL TEACHING METHODS:** Teaching methods play an important role in providing quality education to all types of students. Emphasizing this statement Webster & Roe (1998) says, “From old time, learning has been considered as a product of teaching. Teachers have been mostly using lecture method which are not effective in teaching. An effective teaching is more than merely transmission of information from teachers to students, but rather a complex interaction between the two parts”. Therefore, it is the need of the hour to move from traditional teaching methods to new one which focus child centered education and deals the need of every child. Teachers must keep in the mind individual differences during teaching in an inclusive classroom, and some time she has to teach one-to-one to students with visual impairment while engaging remaining students in some other activities. For example, after explaining process of any mathematical concept in the inclusive classroom calculation related to this concept should be given to students without disability for solving while student with visual impairment should be taught one-to-one using teaching learning materials or other concrete objects.
- **PROMOTING COOPERATIVE LEARNING:** It is the fact that cooperation is one of the essential components for success either in education or job. In cooperative learning teacher divide students in small group, and assign different task to each group. The role of teachers in cooperative learning is different, now teacher play the role of facilitator nor knowledge giver. Discussing the importance of cooperative learning Mastropieri & Scruggs (2010) say, “Cooperative learning among students of different learning capabilities and learning needs, in an inclusive classroom, has proved to be effective in promoting academic achievement, positive attitude towards the subject, and improving social interaction among students”. Further, Mitchell (2008) says, “It is a good strategy of teaching students with visual impairment, particularly in the mixed ability groups. It is especially important in third world countries where classes are very large”. “In these groups, students with visual impairments should be paired with their fellow sighted students who will help them to organize their works, find correct pages and repeat teacher’s instructions” (UNESCO, 2001). In the inclusive education teacher should encourage students to work together, in pairs and small groups. For instance, in mathematics class, a sighted student can read the questions while a student with visual impairment writes the answer using assistive device. On the other hand, in science class, a group of

students can carry out an experiment together, with the sighted students carrying out certain tasks which the student with visual impairment is unable to perform (e.g. observing a particular liquid has changed colour), but s/he can write the result of experiment. When group activities organized, it should be ensured that students with visual impairment are actively involved, not just sitting there doing nothing.

- **QUESTION - ANSWER METHOD:** A teacher in an inclusive classroom can give her lecture verbally as well writing it on blackboard, so that every student of class can understand the teaching contents. To find out whether students with visual impairment have understand the topic she taught, teacher should ask them questions and get the understanding level of these students. In this type of classroom, teacher must use students' names during asking question so that students with visual impairment can understand to whom the questions are being asked.
- **ADAPTATION OF PRINT MATERIALS:** Since students with visual impairment cannot read print materials therefore it is very essential to adapt print materials in accessible format. Mastropieri & Scruggs (2010) suggested some examples for adaption of print materials for these students, "For example, printed text can be adapted through increasing the font size, bolding the text, increasing contrast, adding colour, and adjusting spaces between characters. However, the extent of these adaptations depends solely on the severity of visual defects and the needs of the student concerned" (Mastropieri & Scruggs, 2010). Further, emphasizing the need of adaptation of printed materials for students with visual impairment Spungin (2002) says, "It is important to consult a specialist teacher on preparation of materials prior to the lesson; because different students use different materials depending on the degree of their visual impairment". In an inclusive classroom teacher should write large text on the blackboard or should use colored chalk so that students with low vision can understand the content easily (UNESCO, 2001).
- **USE OF ASSISTIVE DEVICES:** In an inclusive classroom teacher should promote student with visual impairment to use assistive devices, so that they can access the information on par with sighted peers. For example, if a teacher teaching geography in the inclusive classroom and showing the map of India to whole class, so at this time she can give a magnifier to student with low vision with a separate map. Then after giving instruction to student with low vision about using map, teacher

should elaborate this map to whole class. This way students with visual impairment can learn the map on par sighted students.

- **USE OF TACTILE TEACHING LEARNING MATERIALS:** Tactile Teaching Learning Materials (TLM) are one of the essential components for education of student with visual impairment. Highlighting the need of tactile TLM for these students Pauline (2003) says, “Teachers must be aware, that students with blindness have deficit in conceptual experiences and understanding due to absence of visual ability, therefore adaptations of teaching materials becomes paramount, if they have to learn all the things other students without visual impairments learn in the class. To help this, therefore, these students should be taught physically using concrete experiences”. Further, Salisbury (2008) emphasizes the importance of these type of TLM and says, “Tactile diagrams are very important to understand images and concepts which are difficult to explain and describe in words. Therefore, they should apparently be used when shapes and patterns are very important to understand the concept but also, when the real objects are not available to help teaching”.
- **GIVING EXTRA TIME:** Due to the limitations imposed by the disability, students with visual impairment do their assignment slowly compare to their sighted peers. Therefore, Salisbury, 2008 says, “Extra time allowance is extremely important for them to process visual information, and complete their written assignments”. For example, reading braille books need more time than print books. If students with visual impairment take the help of scribe, then they also need some extra time to complete the task. Further, Spungin (2002) specified that, “Generally, it is acceptable to add half of the time for students with low vision, and twice as much for students with blindness”. Now in India there are various rules are enforced for providing extra time for students with visual impairment in examination.
- **ENCOURAGE MULTI-SENSORY LEARNING:** Multi-sensory learning is very significant for the education of students with visual impairments. This is not only benefits to students with visual impairment but also to other students in an inclusive classroom. For example, teachers should encourage student with low vision to look at pictures in books, read print, and observe the world around them. Further, teacher should also give opportunities to students with visual impairment to learn through their senses of hearing, touch, taste and smell. For instance, if we are teaching a lesson

about different types of fruits, then we can bring fruits in the classroom so that all students, not just the students with visual impairments, can touch, taste and smell the fruits.

- **EFFECTIVE USE OF BLACKBOARD:** In an inclusive classroom using blackboard effectively is one of the option for teaching students with visual impairment. It is very important that teacher tells the class what she have written on the blackboard – the first time she is writing on the blackboard, and the second time she have finished writing on the blackboard. In order to make sure that students with low vision can read, teacher must write neatly and clearly.
- **PROVIDE ONE-TO-ONE ASSISTANCE:** In the large class of an inclusive education it could be difficult task for teacher to provide one-to-one assistance. However, there will be occasions when student with visual impairment will require this assistance, particularly if they have to learn new skill or concept. For instance, teaching the concept of map first time, teacher must give one-to- one assistance to student with visual impairment. Before elaborating the map to class, teacher should explain student with visual impairment about the way of seeing a map. Meanwhile, teacher must engage other students in some other activities like asking them to locate a particular place in the map, so that the discipline of class can be maintained.

CONCLUSION

There is a range of strategies in teaching inclusive classroom that support all students to get quality education, but there are some specific strategies that are useful in teaching students with visual impairment. This paper discussed some of the specific strategies for teaching regular curriculum to students with visual impairment in the inclusive classroom. These strategies include co-teaching, cooperative teaching, use of assistive devices, multi-sensory learning, providing extra time, one-to-one assistance etc. Peters (2003) says “Teaching in inclusive classroom is not easy, since teaching needs to be more individualized as compared to normal classrooms, where there are little diversities among students”. Although, there are some challenges for teachers to teach in the inclusive classroom; however, with the right attitude and support, they can teach easily and all students can learn and be successful.

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39.Rethinking the Role of *Ahamkara* in Performance

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Abstract

Performance is one of the dynamic concepts that can be contextualized in various personal as well as professional settings such as school, college, workplace, sports, and many others. Researchers and practitioners have continuously worked towards means and methods to improve the performance of individuals at various stages. The studies on performance suggest that most of the researches has been conducted with reference to external factors which may determine performance whereas the internal factors have got less attention. Self, an internal factor, is at the centre of all human actions and experiences. In literature, self has been differently contextualized in the western and eastern researches. In the eastern approach, self is translated as 'aham' in Sanskrit. Aham is a generic term for ahamkara, which is the mechanism for 'I-maker'. Western researchers claim that the higher the sense of self, the greater the performance whereas, eastern researchers claim that the lesser the ahamkara the greater the performance, life satisfaction, and quality of life. Contrary to these differences, there are traces of importance on the cultivation of optimal ahamkara in both the eastern as well as western researches. The studies indicate that too high or too low ahamkara can negatively affect the performance level due to impairment in perception, decision-making, and other cognitive abilities. However, the optimal level of ahamkara is essential for an effective and efficient performance level. Furthermore, the classical Indian texts provide insight on methods of lowering and controlling the development of ahamkara such as Buddhism, yoga, and others.

Keywords: Ahamkara, Performance, Individuality, Agency, Separation, Identification.

Introduction

In recent decades, performance as a concept has gained increasing attention from researchers as well as practitioners. The term 'performance' first emerged in the mid-nineteenth century in the context of sporting contests. Later, the concept evolved in the twentieth century. In linguistic form, the Oxford English dictionary describes the performance as how well or badly does an individual does something. Literature suggests that scholars continuously hold the view that no standardized definition of performance exists due to its multidimensional nature. Among the numerous definitions of performance in literature, there exist two common characteristics i.e., efficiency and effectiveness (Samsonowa, 2012). Efficiency refers to achieving maximum productivity with minimum consumed

resources, whereas effectiveness refers to the capability of goal attainment or producing desired results. Further, Samsonowa (2012) used the term performance in association with the degree of goal attainment on the organizational level instead of the individual level. Moreover, Otley (1999) stated that “the word ‘performance’ is something of a weasel word in that it appears to mean very different things to different people, thus, we use it quite freely, apparently understanding its meaning, but actually often using it to cover a lack of shared understanding”. Performance is a dynamic concept that may be applied in various settings such as sports, educational, organizational, financial, social, economic, and others.

Contextualizing performance

In educational sector: School performance deeply concerns students, teachers, parents, and nations at large. School performance is known by varied names like academic achievement, school readiness, and school performance. Attainment of educational goals is central to the school or academic performance. School thus, aims at the transformation of a given state, into a new state by integrating cognitive and structural elements. One of the determinants of school performance is the academic performance of its students. Academic performance is the indicator of student achievement across various academic subjects/courses. Conventionally, student’s results from standardized tests, classroom performance, and graduation rates are used as a measure of students’ academic performance. Performance may vary due to organic, environmental, and situational conditions that determine skills and experiences.

In organizational sector: Kenney et al. (1992) proclaimed that performance standards set by the organization may be utilized in order to measure, evaluate, and analyze employee’s performance. This helps in evaluating the progress towards goal and modifies the efforts as per the requirements. A number of measures that can be taken into consideration when measuring performance such as quality, efficiency, effectiveness, productivity, and profitability (Ahuja, 1992).

In sports context: In an athletic context, performance stands for the pursuit of excellence. Literature suggests that athletes can be categorized into two categories, one who is performance-oriented and tend to be competitive whereas another category of athletes is recreational athletes who are simply interested in participation and they do not set specific performance goals. Strength, endurance, skill, and recovery are the four major dimensions of Sports performance. A characteristic blend of these dimensions as per the sport determines the performance of athletes.

Factors affecting performance

Literature suggests that there are numerous factors that may exert an influence on human performance depending on the context. Some of the common factors are Goal Clarity (Locke & Latham, 2002), Repertoire, Knowledge of Structures, Feedback, Mental Models, Motivation, Environment, and Self. Consequently, it is established that performance is determined by the interplay of nature (internal) and nurture (external) factors. The shift from external factors to internal factors in the study of performance is recent, wherein various self-related factors, such as motivation, locus of control, self-esteem, self-efficacy, are under contemplation. Self is the centre of all experiences. Self not only initiates the process of working towards the goal but also performs goal-directed behaviour to ultimately attain goals. There are different views on self and identity between the eastern and western bodies of knowledge. Western understanding of self is very objective, quantitative, and includes experiences at a conscious level only whereas the eastern understanding of self is more transcendental and includes all kinds of experiences such as conscious, dream state, paranormal, etc. In Sanskrit, 'I' is translated to '*aham*'. '*Aham*' refers to 'being'. It can be understood at different levels. At the psychological level, *Ahamkara* refers to the mechanism of 'I-maker' (Salagame, 2013). Salagame (2014) stated that *ahamkara* is a meta-construct that incorporates various modern psychological concepts of self and identity.

Understanding Ahamkara

Psychological investigations conducted through the utilization of first-person experiences of mediators and yogis have provided in-depth insight into human nature (Salagame, 2013). Concepts and ideas illustrated in the classical Indian tradition may not find their place in modern psychology but have dynamic applications and implications for the human species. *Ahamkara* is one of such concepts rooted in Indian tradition. *Ahamkara* is a Sanskrit term that is made of two units, '*aham*' meaning 'self' and '*kara*' meaning 'doer'. When translated in English, '*ahamkara*' is equivalent to 'ego'. Herein, *Ahamkara* exemplifies the mind that links with the experiences to oneself. It constitutes the feelings and cognition of 'me' and 'mine' (Gaur 2011; Gupta & Agrawal, 2021; Reddy 2014; Salagame 2011). By personalizing experiences to one, *ahamkara* enables the soul to relate itself to perceived objects. This generates self-relevant feelings, thoughts, and mental images, which dominate the consciousness and distort the perception (Mills, 2011), which may induce exaggerated self-appreciation and arrogance (Gaur, 2011). It helps an individual identify and associate oneself with the people around them, with their own work, with things, and the world at large (Tayal & Sharma, 2020). Salagame (2011) stated that *ahamkara* is a broad concept that can subsume many modern

psychological concepts of self and identity. Range of experiences, from suffering to enjoyment can result due to collective functioning of organs of perception (*jnanendriya*), aspects of mind (*manas*), and organs of action (*karmendriya*) (Gupta & Agrawal, 2021; Rao & Paranjpe, 2016; Salagame et al., 2005).

According to the scriptures, *ahamkara* constitutes four sub-components such as separation-differentiation (*dvaita bhava*), identification (*abhimana*), agency (*kartatva*), and individuality (*vaishishtya*). First, 'Separation-differentiation' suggests the feeling of 'I' and 'I-not' or being different from others. Second, 'Identification' refers to one's association with worldly objects in form of ownership (*mamkara*), companionship (*sanga*), and attractions and attachments (*moha*). Third component 'agency' refers to the sense of 'doer ship and fourthly, 'individuality' defines the feeling of uniqueness (Salagame et al. 2005). Gaur (2011) proclaimed that individuals may adopt multiple identities due to *ahamkara*, such as the identity of a housemaker, or working woman due to a sense of doer-ship and association. Through interactions between these multiple identities, each identity and overall *ahamkara* may be enhanced.

When *ahamkara* is well cultivated, it may become a source of the power of determination. It is when the mind begins to believe that "I can do it, I will do it, I have to do it and I am going to do it!". Hence cultivating the right amount of *ahamkara* adds on survival value, it helps individuals navigate through the worldly roles and duties successfully. Rather than the complete elimination of *ahamkara*, it is wiser to make oneself aware of one's true nature of existence so that *ahamkara* can be moderated. Once aware *ahamkara* may be highly willing to be representative of one's real self. This generates immense power in an individual in form of *sankalpa-shakti* (determination). Thus, making *ahamkara* is a source of willpower or determination.

Relationship between *ahamkara* and performance:

The western concept of 'self' is well established. There exist vast differences in the understanding and functioning of ego in the literature. Literature suggests that the concept of ego exists on a continuum. At one extreme end, modern psychology advocates the establishment of a stronger ego in order to develop a high sense of self-esteem (Rao, 2014); although, elevated self-esteem has been linked with narcissism (Baumeister et al., 2003). On the other side, *Advaita Vedanta* has emphasized the complete dissolution of *ahamkara*. Between the two extremes, lies the idea of the moderate sense of self or *ahamkara*. It highlights the importance of a 'compassionate and regulated ego'. Evidence for moderated ego is also found in eastern literature as in Sri Aurobindo's 'Integral Yoga', which

emphasizes the utility of moderate ego in terms of formalizing existence and individuating an individual from the commons (Gupta & Agrawal, 2021; Reddy 1990). Furthermore, Salagame et al. (2005) highlighted the four components of *ahamkara* based on theoretical analysis of Indian classical texts. They are namely; individuality, separation, agency and identification.

The first component of *ahamkara* is individuality. It refers to the feeling of 'I-ness' or a sense of uniqueness. It is stated that the sense of uniqueness produces self-love (*abhimān*). An increase or decrease in self-esteem usually brings strong emotional reactions. Research suggests that self-esteem and academic performance are related factors. High level of self esteem increases tendency of narcissism. It is stated that narcissists perform for their interest, putting others' needs and interests at risk (Braun, 2017). Poor performance of narcissistic leaders (Braun 2017; Rosenthal 2010) is associated with their impaired decision, incomprehensible goals and disorganized functioning that may falter over time (Rosenthal & Pittinsky, 2006). In contrast, under-emphasis on individuality may lead to dependent personality characterised by excessive reliance on others, lack of consistent sense of self, fear of being alone and difficulty in making independent decision. Self-esteem refers to how much somebody appreciates and likes themselves and perceives themselves to be a valuable person (Rosenberg, 1965). This self-referential judgment has a significant impact on personal (Soto-Sanz et al, 2019) and academic adjustment (Reason, Terenzini, & Domingo, 2006), both in motivational and performance terms (Li, Han, Wang, Sun, & Cheng, 2018; Topçu & Leana-Taşcılar, 2018). In this regard, self-esteem is a powerful motivator for students to pursue academic goals (Fang, 2016). Evidence suggests that student's poor self-esteem and lack of achievement motivation mark in lack of interest strive for academic performances (Abdullah, 2000). Hence, literature exhibits that too high or too low of *ahamkara* may negatively affect an individual's performance.

The separation component of *ahamkara* refers to the self-other distinction. In our routine life, we rarely act in isolation. van der Weiden, Prikken, and van Haren (2015) explained that for efficient social interaction "people have to coordinate and *integrate* other people's thoughts, emotions, and behavioural intentions with their own". It is a prerequisite to distinguish between self and others to do this. In a confusing state between self and other, either own intentions can be projected onto others, or others' intentions and emotions can be taken over. Our brain seems to be evolved in such a way that the process of distinction and integration of self-other takes place without any conscious efforts. However, everyone is not capable to balance self-other distinction and integration. In this context, theory of mind (ToM) plays a very important role which can be defined as

the “ability to understand that others have different beliefs, desires, and knowledge than oneself” (Premack & Woodruff, 1978). Unlike children, neuro-typical adults understand that other individuals may have a different perspective than themselves (Onishi & Baillargeon, 2005), however, they may have difficulties due to arising conflicts between self and others perspective. Adult participants have been found to show “egocentrism” across different tasks, where their judgments about others' wants/thinking are slow or vulnerable to make an error when it deviates from their personal thinking, or wants (Royzman, Cassidy, & Baron, 2003). Self-disturbance is a possible explanation for the both positive (delusions of control, auditory hallucinations, grandiosity, and delusions of reference) and negative (affective flattening, apathy, anhedonia, and avolition) symptoms of schizophrenia (Sass, 2014; Sass & Parnas, 2003; van der Weiden, Prikken, & van Haren, 2015). These findings signify that self-disturbances slouch at the core of the disorder (Mishara et al., 2014, van der Weiden, Prikken, & van Haren, 2015). Studies signify that anomalous processing of self and others, not only reflects in clinical symptoms, but also in neurocognitive, social and behavioural dysfunctions as at mirror neuron level self-other disturbances may complicate an individual's cognitive abilities such as planning, regulation of actions, performance and monitoring (Nelson et al., 2014, van der Weiden, Prikken, & van Haren, 2015). Disordered individuals perform worse than healthy individuals on more complex tasks of theory of mind (Brune, 2005; van der Weiden, Prikken, & van Haren, 2015). Such evidences suggest that having an appropriate self-other integration and distinction is essential for social interactions, performance, mental and physical health and other aspects of life.

Agency is the third component of *ahamkara* that relates to the sense of doer ship. In western studies, this component has been closely correlated with the concept of locus of control. It can be observed that some individuals attribute the cause of any outcome or events either to themselves or external factors. The individuals, who attribute the cause to themselves fall in the category of internal locus of control. In contrast, those individuals, who attribute the cause of the events or outcome to outside factors, are categorized in an external locus of control. Individuals falling under the category of internal locus of control consider their behaviours are directed by their personal efforts and decisions. They have control over the events and things that they can change. Internal locus of control is closely associated with self-efficacy (Donatelle, 2011). Self-efficacy is considered as a belief in being able to do something successfully. Internal locus of causation of perceiving the negative outcomes is accounted on personal characteristics, for example, abilities, personality, and mood (Crisp & Turner, 2007). If individuals feel that the cause of peers and superiors' treatment with them resides

within them, then they may be affected negatively due to perceived accountability (Weiner, 1986). If an individual attributes the causes of negative outcomes outside of them, then their behaviour will be rationalized and severity would be less. For example, at the workplace, bullying may cause negative outcomes for employees, when individual's resources such as the expectation of control are intimidated (Hobfoll, 1989), or if the cause of bullying is attributed to themselves (Weiner, 1986). It indicates that internal locus of control (LoC) may not be favourable in case of bullying under other work stressors and circumstances. A strong internal drive of control may lead to believe that if plans do not work, one may feel responsible for the failures even if the events are beyond the control. It can induce anxiety and frustration, and in extreme situations, stress or depression. High internal LoC and Low internal LoC both were found to have negative influence on performance. It has been observed that low internal LoC heightens the conflict at the workplace and consequently, increases the psychological strain (Dijkstra et al., 2011). On the other hand, Individuals high on internal LoC can experience more stress than individuals low on internal LoC due to loss of control and social pain in the bullying situations. Empirical evidences suggest that in situation of low objective control individuals with high internal LoC may experience extreme negative consequences (Reknes et al., 2019). This indicates that extreme levels of internal locus of control may further affect an individual's performance at the workplace. There is another type of control that entails the combination of the two types of locus of control: internal and external are often referred to as Bi-locals. Individuals having bi-local type can take responsibility for their actions and the outcomes while relying upon the outside resources and circumstances. The bi-locals are known to handle stress more efficiently. The concept of bi-local is conceptually close to the idea of optimal *ahamkara*.

The fourth component of *Ahamkara* is identification. Identification is referred in associating oneself with possessions and materials such as one's body, social status, money, position, institutions, etc. Identification is understood as the bio-psycho-social Identity of an individual. It represents one's attachment with his/her social status, achievements, possession, body, mental abilities, etc. In modern psychology, identification is understood in different contexts like group identification, racial identification, organizational identification, etc. Social identity theory states that the social identity of an individual is determined by his/her group affiliations or membership (Abrams & Hogg, 1990). This group membership activates the visible characteristics of the group within an individual which further affects their behaviour. The mere categorization of the individuals into a distinctive group is ample to create their social identity (Tajfel, Flament, Billig, & Bundy, 1971).

Consequently, individuals accept assigned social category memberships on the basis of significant self-definition (e.g., ‘we doctors’). At times, they may also consider categories centred on insignificant criteria (e.g., ‘we Android users’). The overemphasis on group identity may affect its member’s thoughts, feelings, and behaviour. One such phenomenon is ‘the illusion of group effectivity’ wherein group members show the propensity to overestimate the level of in-group productivity. Other phenomena in similar lines are the ‘not invented here bias’, wherein group members overestimate their own group’s ideas and products over those of other groups (Katz & Allen, 1982). The literature discusses numerous motives underlying these biases. First, group productivity is highly accessible to members and seems quite good in comparison to a single individual’s contributions. Due to the collective participation of group members in idea generation, they fall for the illusion that their group is doing well. Secondly, on an emotional ground, group membership may become a source of positive social identity for its members. Such positive feelings naturally incline members into believing that the group is very strong and is doing well. Hence, the ‘illusion of group effectivity’ creates hindrances in group performance. Therefore, group leaders should ensure that members of the group are aware of such biases and their negative consequences; monitor group performance; motivate members to work hard in order to function effectively. Team member’s identification is another form of identification that may affect group performance. Team identity enhances the efforts of members on the task and reduces conflict between members (Tyler & Blader, 2000). This team identification may create a personalized perception that can determine an individual’s worldview and biases. Additionally, team identity gives rise to the perception of an in-group where all the team members are considered as one whole, failure to do so can result in in-group and out-group distinctions within teams. The in-group and out-group distinction may reduce the performance of the team (Williams & O’Reilly, 1998). Lacking team identification may give rise to trust issues and conflicts within teams which may reduce the level of performance of the team as a whole. Han & Harms (2010) propounded that cultivation of sense of ‘we’ rather than the sense of ‘I’ among the team members is essential for smooth functioning. Thus, suggesting that too much or too little of team identification may negatively affect the performance level of an individual and the team at large.

The review suggests that extreme of *ahamkara* either too high or too low creates barriers to performance at the individual or group level. Thus it is crucial to cultivate an optimal level of *ahamkara* among people. Various components of *ahamkara* such as individuality, agency, separation, and identification follow the same pattern. A regulated sense of self/ *ahamkara* may help

an individual in understanding and fulfilling their roles and responsibilities of the world and simultaneously accept and acknowledge their limitations and failures.

Buddhist view on cultivating appropriate *ahamkara*

In order to achieve ultimate happiness embedded in self-transcendence, classical Indian texts indicate the dissolution of *ahamkara* (Misra 2010; Salagame 2013). There are multiple worldviews that present strategies to achieve a balanced state of self. Buddhist philosophy emphasizes the teaching of the middle way or also called *Madhyamaka* school. *Madhyamaka* is a Sanskrit term that translates to the English language as middle-path. Nagarjuna, the philosopher-monk applied the middle path to existence and nonexistence. He further stated that between the two positions lies emptiness (*sunyata*). This emptiness is not simply ‘nothing’ but creative potential. Buddhist teaching is neither a path of denial nor of affirmation; attachment nor aversion, being nor non-being, subject nor object, samsara nor nirvana, part nor whole, and many other diametrically opposed pair. It reflects the paradox of the universe, within and beyond the opposites. It models the state wherein an individual is in the world but not of the world. This realization is the real application of the middle way.

To attain this state, Buddha proposed the noble eightfold path. This path emphasis on avoidance of extreme self-indulgence as well as complete self-denial. The eight steps include right view, right intention, right action, right speech, right effort, right livelihood, right mindfulness, and right concentration. The order of this eightfold path is not strictly sequential. Additionally, for the purpose of training these eight steps can be divided into three areas: a) ethical conduct (*sila*), b) concentration (*samadhi*), and c) wisdom (*prajna*). Right action, right speech, and right livelihood make the ethical conduct. The practice of concentration is comprised of right effort, right mindfulness, and right concentration. Right view and right intention are associated with the development of wisdom. Moreover, Buddhism describes sensual desire, ill will, sloth, restlessness, and doubt, as the five hindrances to the progress in meditation.

Implications

Ahamkara is an essential component in determining how an individual relates to oneself, their work, other people, and the world at large. The understanding of *ahamkara* can have broad implications in various spheres of an individual’s life such as health and illness (Tayal, Priya, & Sharma, 2020); goal attainment, life satisfaction, and many more. As reflected by the findings of the paper, the cultivation of an ‘optimal’ level of *ahamkara* may foster healthy intrapersonal and interpersonal relationships. In this regard, classical texts provide insight on how to eliminate or reduce the level of *ahamkara* among

individuals. Buddhism, as a school of thought, propounded the idea of the middle path and the eightfold path to attain it. It seems worthy to acknowledge that classical concepts like *ahamkara*, meditation, yoga, *buddhi*, and many others are in the process of re-establishing their status in the body of scientific knowledge. Thus, it is essential to acknowledge and promote them in order to utilize this knowledge and put them into practice for the large cause. Such texts not only discuss the worldly problems experienced by people but also illustrate the strategies that may be used to resolve them. Thus, it is essential a) to attain the knowledge and modify the practices accordingly such as practicing the eightfold path of Buddhism in school in order to cultivate the optimal sense of self; and b) it is well established that early experiences affect an individual's adult life experiences thus it is relevant to structurally introduce such ancient scriptures in the syllabus at early stages of life. This will help in establishing strong moral values as well as spiritual intelligence among the pupils.

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40. An Exploration of Teachers' Awareness and Understanding of the Importance of Using Psychometric Tests to Tanzanian Students during Counselling: A Case of Kinondoni Municipality

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Abstract

This paper explored the teachers' awareness and understanding of the importance of using psychometric test during counselling at Kinondoni Municipality in Tanzania. Psychometric tests as a tools to measure skills and talents of individuals without bias and link to the psychological phenomena, it was important to see if the secondary schools teachers of Kinondoni Municipality were aware of psychometric tests and also understand the importance of using psychometric tests during counselling. The study employed qualitative approach where a single case study design was used. The study involved 20 secondary school teachers in Kinondoni Municipality who were purposively selected from five secondary schools which obtained through simple random sampling. Data were collected through interview and analysed thematically by using content analysis. The study revealed that secondary school teachers from Kinondoni Municipality were not aware of the psychometric tests and also they were not understand the importance of using psychometric tests during counselling as the psychometric tests seemed to be new thing or new terminology to everyone. The study recommended that teachers should be oriented or training about the use of psychometric test and how they are important during counselling process.

Key words; Awareness, Understanding, Importance, Psychometric tests, Counselling

1.0 Introduction and Background

Psychometrics is the disciplinary home of a set of statistical models and methods that have been developed primarily to summarize, describe and draw inferences from empirical data collected in psychological research (Jones & Thissen, 2007). Psychometric tests are the tests that are designed to measure skills and talents of an individual by objectifying the measures that are non-measurable without bias, for example motivational questions, aptitude personality reasoning and ability questions. The primary stage or basic of any psychometric test is to measure the individual difference. The measurement includes the development, interpretation and evaluation of psychological tests and measures used to assess variability in behavior and link such variability to psychological

phenomena (Sattler, 2014). Psychometric tests are scientific and objective to the psychometric assessment hence they are used in a wide area by professionals, medicals, legal and even employers to the provision of therapy or rehabilitation in some domains such as ability or personality as they always tell the truth about the individual who is being assessed. Psychometric tests will be able to give out the desired outcome only when it will have the characteristics such as validity, reliability, standardization, objective and clarity (Furr & Bacharach, 2013; Sireci & Sukin, 2013; Geisinger, 2013; Sattler, 2014). These characteristics can be explained hereunder as follows.

The first characteristic is the Validity of the psychometric test. Validity is the degree to which evidence and theory support the interpretations of test scores for proposed uses of a psychometric test. In this case the validity refers not to the measure itself whether a psychometric test is valid or invalid but rather the interpretation and use of the measure's scores. To be considered valid, the interpretation of test scores must be grounded in psychological theory and empirical evidence that demonstrates a relationship between the test and what it supposed to measure. In educational psychology there are primary types of evidence related to validity as can be explained as follows. The first is *Construct evidence of validity*; this is a degree to which an individual's psychometric test scores correlate with the theoretical concept the test is designed to measure, it's the extent to which the psychometric test is meaningful to the respondent. The second evidence is *Content evidence of validity*; this is a degree to which the psychometric test content represents the targeted subject matter and supports a test's use for its intended purposes, this means the contents or questions present should be in line with subject matter of the respondents and not otherwise. The third is *Criterion-related evidence of validity*; this is the degree to which the psychometric test's score correlates with other measurable, reliable, and relevant variables thought to measure the same construct, this will help in making decision regarding the constructs that are being measured.

The second characteristic is reliability of psychometric test. Reliability is the degree to which scores from a psychometric test are stable and results are consistent. When constructs are not reliably measured, the obtained scores will not approximate a true value in relation to the psychological variable being measured. Reliability of the psychometric test can be assessed in four ways. The first way of testing reliability is the *Test-retest method*, this is the consistency of the psychometric test scores over time. With the time interval, the psychometric test results should be the same when the same test is administered. The second way of testing reliability is *Inter-rater method*, inter-rater is the consistency of test scores among independent judges. The scores should be the same when assigned

or rated by different independent people. The third way of testing reliability of a psychometric test is *Parallel or alternate forms*, this is consistency of psychometric test scores across in different forms of the test or parallel test. The fourth way of testing reliability is the *internal consistency*, this is the consistency of different items intended to measure the same thing within the test (homogeneity), for example a special case of internal consistency reliability is split-half method where scores on two halves of a single test are compared.

The third characteristics is objective of the psychometric test, being objective means the psychometric test score should not get affected by the beliefs and opinions of the tester or administer. The fourth characteristics is the Standardized of the psychometric test, psychometric test to be standardized means that the conditions and procedural details of testing are controlled and the scores are compared with the established standards of performance. These four characteristics of psychometric test are the one which differentiate the psychometric test and questionnaire. Also psychometric tests falls under normal distribution.

Counselling is process or a number of procedure where a person uses skills and principles to listen and find the best way to understand and solve the problem of the client (Oviogbodu, 2015; Nkechi, Ewomaoghene & Egenti, 2016). It is interaction or relationship between two or more individuals which involve emotional or affective in individual learning such as feelings, emotions, values and attitudes (Egbo, 2013). Therefore, it can be argued that counselling is a person to person process or interaction in which one person is getting help by another for the purpose of developing, increase in understanding and ability to solve his or her problems by choosing the right behaviours. Psychometric tests being an important tool in counselling, it has been of greater importance in a counselling in a multidimensional such as the measuring of the reasoning ability of the client, establishment of therapeutic relationship, breaking the ice, career counselling as well as couple counselling. Psychometric tests makes the counselling process to be easy as it establishes a starting point for confidence and trust building between counsellors and clients. The counselling profession emphasizes the importance of knowledge and skills acquisition by all counsellors by training in the field of psychometrics in order to protect the existing and future clients (Foster & McAdams, 2009). Teachers are encouraged to provide professional counselling services to the students with the help of psychometric tests and recommend remediation in order to fulfil their responsibilities (ACA, 2005).

1.1 Importance of Psychometric tests during Counseling

Counselling process will be difficulty with the absence of psychometric tests or instruments which save the counsellor to acquire the important information before counselling (Geisinger, 2013). The acquisition of the prior information towards counselling from tests makes psychometric tests to play an important role during counselling. The effective counselling can be achieved through the use of psychometric tests as the test play the following roles during counselling. First, psychometric tests provides the important information to the understanding of individual characteristics and capabilities through the collection, integration and interpretation of information about an individual prior to the counselling process. These important information enables counsellor to establish a good relationship with the client to be counselled due to the in-depth information of an individual. The in depth information is obtained through a variety of psychometric tests for the purpose of counselling (Groth-Marnat, 2009).

Second, the psychometric test helps to assess variability of behaviour during counselling and also help to link such variability with psychological phenomena (Weiner, 2003). With the help of psychometric test, different behaviours of the clients can be revealed during counselling process as the client has to fill the rubric. Thirdly, psychometric tests establishes the therapeutic relationship where the counselor's first task is to establish what the issues with the client and how the client sees them. This therapeutic relationship will help the counsellor to know whether the client has come for general career counseling or for a specific problem. A client may have been referred by the organization or be self-referred. Sometimes if the client is referred by the organization a client may not be in agreement with the organization as to what the issue actually is, therefore psychometric tests can be valuable to clarify and specify the situation so that both client and counsellor have the same view of it (Geisinger, 2013).

Fourth, psychometric tests breaks the ice. Breaking the ice means to get a starting point during counselling as some clients may find it difficult to talk about themselves to the counselor. Therefore, with the use of psychometric tests the information from a self-report can be used as a starting point. Tests responses can serve as a framework for the discussion as counsellor explores the picture of which has emerged from what the client has said through the psychometric tests (Groth-Marnat, 2009). Fifth, psychometric tests serves as career counseling, when counseling for the career the most of the questions to be asked will be resulted from psychometric tests hence counselling about the career choice can be conducted well. Lastly the psychometric tests can serve as couple counseling. People differ in the way they deal with the world and by identifying those differences in styles, we

can often better understand why conflicts have arisen and psychometric tests can be used as a basis for delineating the differences and save couples (Sireci and Sukin, 2013). The study conducted by Adrian and Chriss (2011) on psychometric tests on counseling revealed that all people had positive usefulness of the psychometric test due to its validity and reliability.

Teachers as counsellors in secondary schools are required to provide counselling with well knowledge and skills of psychometric tests to their students. Becoming a competent counsellor requires an individual to act ethically and professionally in fulfilling his or her responsibilities as a counsellor with the help of the use of psychometric tests. The Council for Accreditation of Counselling and Related Educational Programs (CACREP, 2009) *Standards* and the American Counselling Association (2005) *Code of Ethics* both emphasize the importance counselling with the use of quality and competent psychometric tests for the effective counselling services to the clients. Students in secondary schools being clients to the teachers, teachers have to provide counselling in different aspects including career choice among other aspects. Following the vital of counselling to the students by teachers, teachers are supposed to be knowledgeable enough on the aspect of counselling and counselling process itself which includes the use of psychometric tests for effective counselling. With these great aspect of using psychometric tests in counselling there is a need to see if the secondary schools teachers of Kinondoni Municipality in Tanzania are aware of psychometric tests and also if they understand the importance of using psychometric tests during counselling. Different literatures by different scholars in different countries have tried to explain the importance of psychometric tests for counseling in different professions including teaching profession. There is no particular study that has been conducted at Kinondoni Municipality in Tanzania specifically to explore teachers' awareness and understanding of the importance of using psychometric tests during counselling. Therefore this study aimed to fill the gap.

1.2 General and Specific Objectives

The general objective was to explore teachers' awareness and understanding of the importance of using psychometric tests in Tanzania during counselling. Specifically the study intended to explore the awareness of teachers of Kinondoni Municipality in Tanzania on the use of psychometric tests during counselling and also to explore the understanding of the importance of using psychometric tests during counselling. The specific objectives followed by the key research questions as follows hereunder.

- i. To what extent do the secondary schools teachers of Kinondoni Municipality are aware of the use of psychometric tests during counselling?
- ii. To what extent do the secondary schools teachers of Kinondoni Municipality understand the importance of using psychometric tests during counselling?

2.0 Methodology

This section covered the methodological section like study area, research design and approach, data collection as well as data analysis.

2.1 Study Area

This study was conducted at Kinondoni Municipality in Dar es Salaam, Tanzania. Twenty Teachers from five secondary schools were purposively selected. Schools were obtained through simple random sampling techniques.

2.2 Research Design, Data Collection and Data Analysis

The study employed a qualitative research approach with a single case study design. Qualitative data on the awareness and understanding of the importance of the psychometric tests during counseling were collected by using key informant interviews and documentary review. Data were analysed through content analysis. In depth descriptions of specific themes based on research objective and its related question was provided as most of the data were in form of words.

3.0 Results and Discussions

This study explored the awareness of the teachers on the use psychometric tests for counselling as well as the understanding of the importance of using psychometric tests during counselling. The findings that were revealed were discussed hereunder as follows.

3.1 Teachers' Awareness on the Use Psychometric Tests during Counselling

In this part the researcher wanted to know if the secondary schools teachers of Kinondoni Municipality were aware of the use of psychometric tests or not during counselling. During interview 16 teachers out 20 responded that they don't know what psychometric tests mean are and it was the first time to hear that word "psychometric" throughout their studies. They also insist that they may know it but they don't know if it is the one which are called psychometric tests. This implied that most of teachers in secondary schools are not aware of what psychometric tests mean. 4 respondents said that is the machine which measure brain. On insisting on this answer, one respondent said:

"...psychometric is the machine which measure brain because psycho its self is a brain and metrics is a machine..."

This quote implies that teachers were not aware of psychometric tests even to those who claim to be aware of it.

Due to the lack of awareness on the psychometric tests there is a need of the secondary schools teachers of Kinondoni Municipality to be introduced and oriented about the psychometric tests due to its great importance in teaching and learning industry including counselling. This is because teachers save as counsellors in schools and the use of psychometric test will help them to do their counselling to their students effectively.

3.2 Teachers' Understanding of the Importance of using Psychometrics Tests during Counselling

In this part the researcher wanted to know if the secondary school teachers understand the importance of using psychometric tests during counselling. During interview with teachers about the importance of using psychometric tests during counselling, all of them revealed that they cannot know the importance of using psychometric tests in counselling while they don't know what psychometrics itself mean. One of the respondent added that it is difficult to know the importance of using psychometrics tests during counselling before knowing what psychometric test is. On asking whether they used to counsel their students or not, they responded that they used to counsel them if any one ask for the advice. On insisting this one respondent said that;

"...we don't use anything or any guide when counselling, we just share idea of what he or she revealed to us..."

This quote implied that most of teachers do not understand the importance of using psychometric test during counselling since they don't know even what psychometrics instruments are. Hence it was difficult to use them for counselling though they used to counsel student but not in a professional ways of counselling.

4.0 Conclusion and Recommendations

Generally it was observed that, the secondary school teachers of Kinondoni Municipality in Tanzania were not aware of psychometrics tests and they do not understand the importance of using psychometric tests during counselling. It has shown that even if they come across with psychometric tests or using them in one way or another still they do not know if they are the one that are called psychometric test or not. The researcher recommended in-service training or orientation to teachers on psychometrics tests as well as its application in counselling. On top of that teachers should be oriented on proper way of providing counselling to their students including the proper place or special room for counselling as well as following the proper ways of providing counselling to their students.

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41. An Exploratory Study of School Curriculum: Analysing Gaps between Theory and Practice

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Abstract

This exploratory research problematises the dissonance between child-centred traditions in curriculum theory with knowledge-centred school curriculum practice. It examines prevailing curricular practices in primary grades at a private school that switched to online education during the pandemic. Data is gathered by classroom observation using the technique of virtual ethnography. Three select data vignettes are presented in this paper. The research draws from the interpretative framework offered by existentialist concepts of having, knowing and being for sense-making of the data. The concept of 'having' knowledge implies taking and keeping possession of available knowledge (information). 'Knowing' is functional and part of the process of productive thinking. 'Being' refers to human experience which in principle is not even describable. Independence, freedom and presence of critical reason are prerequisites of the mode of being. The research found school curriculum practice mainly framed in the 'having' mode. Curricular knowledge was practiced almost like an object to be possessed since prescribed texts were viewed as authoritative sources of knowledge waiting for acquisition by young primary school children. Learning consisted of an accumulation of correct answers to be reproduced in a testing situation to demonstrate children's ownership of a pre-defined body of knowledge. Such a curriculum practice ignores children's lived experiences and personal knowledge which does not foster the processes of knowing, as also brought to the fore a realization that our prescribed curriculum does not provide adequate space for children's need to know. A being curriculum that means to enjoy the delight of the knowledge and living it; is simply non-existent in school curriculum practice.

Keywords: School Curriculum, Classroom Observation, Curriculum Practice, Virtual Ethnography.

Introduction

The orienting assumption underlying our mainstream school curriculum is that education is the acquisition of a given pre-defined body of knowledge: the prescribed curriculum. The foundation for this curricular structure was laid in the early educational development of post independent India particularly after the implementation of National Policy on Education, 1968 when the 10+2+3

national system was introduced all over the country. The more than 50 school education boards of our country prescribe this pre-defined body of knowledge for the disciplinary domains of Languages, Mathematics, Science and Social Science; in the centralized examination driven education system at all levels- primary, secondary and senior secondary. This historical underpinning has given the school curriculum a tenacity that continues to regiment it tightly. The textbook culture that emerged in the colonial period continues to accord to the textbook the status of a de facto curriculum in formal schooling even in contemporary postcolonial times (Kumar, 1991:3). It is not a misnomer to say that the textbooks constitute the be-all and end-all of school curriculum. They provide all the answers to the questions that the curriculum raises and instead of probing these questions in a spirit of enquiry; the teachers and students tend to reproduce the so-called textual answers. This is notwithstanding that the information presented in the textbooks is not necessarily up-to-date or relevant. The textbook has even become a kind of cultural code and the teacher's hesitancy to depart from it is almost mainstreamed. The textbook based curriculum essentialises factual knowledge, procedural knowledge and production of standardised answers in the examination. Prevailing practices also largely ignore children's lived experiences, their need to know and to simply be. This is ironic since Indian educational thought believes that education has esoteric aims which cannot be fulfilled if schools let textbooks dominate the curriculum (Krishnamurti, 1974: 13) and that the curriculum should be prepared according to child nature as children were inherently educable (Tagore, 1933:6).

Perspective on school curriculum from education discipline

The orienting assumptions of curriculum studies in education discipline are rather different from school curriculum practice. Curriculum theory particularly its child-centered conceptions is a pivotal topic of study in pre-service teacher education. There is a gap between curriculum theory and practice which presents a challenge to a student-teacher who also undertakes school internship during pre-service teacher education. The gap is experienced as follows. In education discipline the sub field of curriculum studies has evolved as not merely as an emergent field of study but beyond the first and second generation of curriculum theory. The first refers to diverse strands of scholarship dating back to more than 100 years (Bobbitt, 1918; Dewey & Dewey, 1915). The latter, second wave of curriculum theory, was shaped by scientific method in curriculum making: its design, implementation and evaluation (Tyler, 1949). The second wave curriculum studies, emanating from behaviourism in education, since its inception continues to dominate mainstream curriculum practice in schools the world over. The current scholarship in the field of curriculum studies however has advanced to a

curriculum map located around the five curricular traditions of deliberative, systematic, pragmatic, radical and existential curriculum (Null,2011: 35) offering curriculumists an overview of the discipline of curriculum studies. An existential curriculum, for example, (Greene, 1974) does not follow a prescribed subject matter. Instead it does away with texts, instruction and teachers authority; and locates teachers and students on the same level so as to facilitate an exploration of children's immediate interests. Teachers need not present subject matter in classrooms but need to simply be keen observers of children. There is no need for teachers to be subject specialists since the aim of an existentialist curriculum is not to gain knowledge but seek personal meaning of life. A curriculum is not viewed as meaningful unless it connects with the student in a deeply personal way. It is the learner-child and not the knowledge domains that are considered the center of education. Theoretical curricular traditions, such as an existentialist curriculum, look like an impracticable utopian ideal to the curriculumist. In fact the posing of curriculum as a guided journey from theory to practice is recognized as a key curriculum problem. Graduate departments of education in some universities even offer courses with titles like Curriculum: Theory into Practice. There is a wide gap between child-centered curriculum theories versus its practice that is otherwise. The curriculum practice prevalent in mainstream school systems is neither child-centered, nor pragmatic nor existential or deliberative. It is based on the idea of education as the acquisition of a given objective body of knowledge. To a beginning student-teacher, learning to become a curriculumist it appears as if the theory and practice are almost like apples and oranges, unconnected to one another.

Statement of the Problem

There is a gap between curriculum theory and its practice in primary grades in schools.

Objectives

1. To explore prevailing school curriculum practices in primary grades.
2. To examine the spaces in the primary school curricular practices to connect children's life experiences and school curriculum as two ends of the same process.
3. To attempt child-centered curriculum development particularly incorporating children's need to know.

Methodology

The research design was exploratory, observational, descriptive and non-experimental aimed at gathering qualitative data. This research was undertaken at a time when the Covid-19 pandemic had necessitated the Government of India to suddenly close all schools across the country. The main

means of qualitative data collection was classroom observations during online education in the primary grades at a private school. The observations were undertaken directly with the researcher present in the classroom as a non-participant observer; the presence of whom was known to the children of the class. The school is a co-educational, senior secondary school in Delhi NCR region. It is permanently affiliated to the Central Board of Secondary Education, New Delhi. The observations took place in the grades I-V where the typical student strength in the online classroom was 35. The school vision seeks inspiration from the immortal lines of Rabindranath Tagore 'Where the mind is without fear and the head is held high. Where knowledge is free. Where the world has not been broken up into fragments by narrow domestic walls', but as my research found school practices belied this lofty ideal. The school had switched completely to synchronous and asynchronous modes of online education. The data for the study were gathered in October-February 2021. The observations were undertaken at a time when the students had been part of online education since several months. The researcher spent 45 days every day for at least 45 minutes each day in virtual classroom observation. The observations were semi-structured as the focus was on curriculum practice in the classroom but in a largely non predetermined fashion. The research technique used was virtual ethnography, also termed as internet ethnography since the observations were undertaken in an online classroom. While ethnography has been a prevalent method of studying schools in educational research; the technique of virtual ethnography has emerged only recently as a method of gathering qualitative research data. The technique developed from the ascendant view that research methods must reflect the changes taking place in our world. The contemporary world is one where the distinction between online and offline worlds is getting lesser (Garcia, Standlee, Bechkoff & Cui 2009). The technique has its votaries and critics. The critics argue that it eradicates 'removed social context cues such as gender, age, race, social status, facial expression and intonation resulting in disinhibiting effect upon group participants' (Williams, 2007 :7) while the votaries argue that the absence of face to face interaction facilitates reflective, participant-driven textual responses (Garcia et al, 2009). Viewing the online classroom as 'a socially constructed space' (Markham, 2004:119). The virtual ethnographic observations included: descriptions of teaching content, behaviors and activities; its transcriptions and details of the entire classroom transaction aimed at constructing a comprehensive account of the classroom curriculum practice. The physical setting of the classroom and the children's non-verbal responses could obviously not be recorded by virtual ethnography.

Data Vignettes

The detailed 40 page data recorded in the field notebook is the basis for drawing the conclusion in the final section of this research paper. Select virtual ethnographic observations are presented in three data vignette below.

1. Fragmented Curricular Experiences

The virtual ethnographic observations revealed that not only were the school curriculum domains compartmentalized in disconnected disciplinary domains; the curricular practices within domains were fragmented too. In languages in primary grades curriculum was mainly ‘rote-based’, mechanical and disjointed. This was true of both Hindi and English; the two languages taught in the school. Lessons on topics like adjectives, prepositions or pronouns were taught disjointedly by referring to exemplar sentences from puritanical grammar textbooks. The teacher would only come to the online class, teach the lesson for the day and go. In both languages the grammar topics were taught as logical instruments in complete isolation of literature, or the larger purpose of language education. The writing tasks did not go beyond the most basic, tried and tested formulaic sentences. The student’s written work showed that they are learning language more as a subject and less as a means of natural communication. Interconnections exist in real life for children; which the distinct topics of study can incorporate and practice. Even grammar is not an isolated arena as some of it is inherent to the human mind. But grammar curriculum was viewed as a set of rules, a way of getting information, or a technicality. There was a succession of topics in grammar study without putting its rules to a meaning-making social use. The main curricular focus was on explicit teaching of rules and mindless drills of decontextualized practice exercises. There was lack of opportunity for students to speak, or write, to put these rules in actual use. The virtual classrooms anyway lacked peer learning structures to facilitate meaningful communication. The curriculum was filled with definitions, grammar rules and random sentences in an unnatural language of the kind that children rarely use in their own lives. The language department of the school had prepared several powerpoint presentations that were shared with language teachers which were like a monologue that the teachers had to present in the online classroom.

There was a fair deal of curricular space in the language domain for child-centeredness as it was aspects of children’s everyday language that were taught. Yet the classroom curricular approach was one that alienated children from their own language. The researcher came up with a lesson for teaching adjectives based on a picture story titled ‘Clever Tailor’; for grade 4. The story-telling was

about a creative clothier who redesigns his old, colourful *turbans* into novel attires for his wife and children. The lesson had scope to incorporate several adjectives as children were asked about the clothes they wear, their characteristics and their personal responses to how their own old clothes could be reformed and made into something new and interesting. It was in the talk about their own clothes that children's life at school got connected with their life outside school. Story telling as a pedagogic tool allows children to re-envision characters, plots and stories of their own; in a way that is personal for them. There were often such spaces for connecting the child personally with the curriculum by linking children's life experiences with the school curriculum which the school practice largely ignored.

2. Curriculum as Production of Correct Answers

The curricular transactions in areas of mathematics and environmental studies were also mainly through teacher made video powerpoint presentations, worksheets and e-resources from internet. The grade-specific worksheets consisted of questions that students needed to answer for regurgitation in examination. Most of these sheets and questions were prepared by the teachers themselves but were based on textbook content. The emphasis in the curriculum was on production of correct answers. In mathematics students were following algorithmic procedures without necessarily developing an understanding of mathematical concepts. A grade II worksheet, for example, on operations on numbers consisted of number problems with space for writing of the final number (the answer) at the end of following the steps of performing a given operation (addition, subtraction, multiplication or addition). The students were expected to follow these stepladders for each operation and arrive at the correct answer. There was hardly any scope to encourage thinking, reflect over errors and learn by making constructions of one's own. A lesson on photosynthesis, like that for most other scientific and mathematical concepts, in grade V was based on a powerpoint presentation in which definition, examples and equation were presented to young children. The power point presentations used by the teachers were mostly information-heavy, non- interactive, text-book based and teacher centric. The language, content and presentation style used in most of them were technically correct but unnatural. This made the already textbook based curriculum even less child-centered. There was little to no scope for children to derive their own learning and construct knowledge. The topic photosynthesis, for example, offers many creative child-centered possibilities of discussion. The curricular objective however was focused on correct information (like the name of the green pigment which is necessary for the process to occur), a tight definition and examples of semi-green plants where it takes place

partially. In all of this the emphasis was on coming up with the correct answer. Consequently the children become afraid to speak in class unless they are sure of their answer. Gradually they are socialized into believing that the course curriculum is largely a production of these standard answers to textbook questions. In an comparison of observational data from primary grades ranging from grade 1I to V it was discernible that children of upper primary^h grades were more restrained in their classroom responses to teacher's questions than those of lower primary^t grades. This kind of conditioning of children begins as soon as they enter the school. By the time children come to upper primary grades they have already spent three to four years in fear-filled classrooms; centered on authoritarian knowledge where mainly correct answers are reinforced by the teacher. The burden of a heavy curriculum loaded with bookish knowledge breeds fear and conditions the mind. The Indian educational thought priorities the importance of a fear-free learning environment that draws the knower out of silence in the classroom. Philosopher J.Krishnamurti writes 'schools cause fear- fear of failure, fear of being denied uniqueness because of constant comparison with others' (Krishnamurti, 1974:13). Likewise the great visionary Tagore envisioned a mind free from fear, a head held high and filled with compassion. Children will learn only in an atmosphere where they feel they are valued, encouraged and supported. School curriculum practices observed in this research tended to neither respect child nature nor convey this value to children. School curriculum practices dominated by discipline, authority and information rather than enjoyment and satisfaction inherent in knowledge.

3. Curriculum as a Product

Some of the remarks made by a teacher during one of the observations are as follows.

Congratulations on getting 30/30.

I am sorry but I will ask questions to students who have scored below 20, this is because I know the high scorers already know the answers.

Big round of applause for 28-30 scorers.

These students are the ones who have scored lowest marks.

Stand up dear child and guide your friends about how you study and how you are able to write answers.

All those students who have scored between 12-24 will write the entire paper again.

Are marks scored on tests the only measure of learning? Is curriculum a process or just the body of knowledge the acquisition of which is to be measured solely by student scores? Don't the 'low scorers' deserve a greater share of teacher's attention, support and scaffolding? The curriculum is

obviously viewed as a product constituted of bits of linear-additive knowledge. How much of it a student possesses is assessed by marks. This product view of school curriculum resonates in the words,

Our education generally tries to train people to *have* knowledge as a possession, by and large commensurate with the amount of property or social prestige they are likely to have in later life. The minimum they receive is the amount they will need in order to function properly in their work. In addition they are given a “luxury knowledge package” to enhance their feeling of worth, with the size of each such package being in accord with the person’s probable social prestige. The schools are the factories in which these overall knowledge packages are produced- although schools usually claim they mean to bring students in touch with the highest achievements of the human mind (Fromm, 1921: 36).

Conceptual Framework

There was a need for an interpretative framework through which to make deeper sense of the virtual ethnographic classroom observational data. It was useful to turn to the existential philosophy proposed by German-American philosopher Eric Fromm (1900-1980) for a framework within which to understand, analyse and interpret the classroom observational data. He proposes that there are three different modes of human life (Ibid). These modes are ‘having’, ‘knowing’ and ‘being’. The three concepts provide the curriculum theorist a trans-disciplinary, path-breaking and novel theoretical structure for sense-making of the gap between curriculum theory and practice. The concept of having stems from owning objects, ideas and other possessions in our lives. It is therefore accompanied by concerns of losing what one owns. The loss would mean a loss of identity. In the having mode of existence one’s relationship to world is self-aggrandizing because of the need to possess objects, ideas and accomplishments. It means to be in possession of knowledge, the truth and as much more as possible. The having mode offers certainty, produces absolute knowledge and does not require independent thinking. The concept of knowing allows knowers to penetrate deeper through the surface in order to arrive roots of knowledge. The aim of knowing is beyond discovering truth. It is a self-affirmation of the value of human reason. So even the awareness of ignorance is a kind of knowledge since it is a part of the process of knowing. The concept of being has nothing to do with having. It provides meaning, delight and enrichment to human life. Being grows only when we decrease the mode of having. Being is one and same as life. Its relationship with having knowledge and knowing is as follows:

The difference between the mode of having and the mode of being in the sphere of *knowing* is expressed in two formulations: 'I have knowledge' and 'I know.' *Having* knowledge is taking and keeping possession of available knowledge (information); knowing is functional and serves only as a means in the process of productive thinking (Ibid: 34).

The three concepts provide a useful lens through which to take a philosophical look at curriculum gaps in theory and practice. It can be said that,

The American philosophical concepts of having, knowing and being; proposed more than 50 years ago; provided a useful gaze. The curriculum teaches children in school that the sum of the three angles of a triangle is 180 degrees, every sentence has a subject and a predicate and that the earth is at the center of the solar system. It never stops being in the unique moment of the now, one that was never before where each ray of light travels just once. Why must students and teachers always learn or teach? Why can't they sometimes simply be? (Raina & Verma, 2021).

Conclusion

The dissonance between the child-centered curriculum study theoretical traditions and the knowledge-centered text bookish curriculum practice was stark. Data vignette 1 trends the general lack of integration between school disciplinary domains. The gap between forward-looking perspectives from curriculum studies strands versus subject-specific curriculum practice at schools was evident. In fact the research found that a lot of classroom transaction time is spent on decontextualised grammar instruction in both Hindi and English language teaching. This does not even leave adequate opportunity for assessment of whether children even learn to use these grammar technicalities in their own spoken and written language or not. There is curricular space to put the grammar rules in context as language is part of everyday discourses of children's life which language educators have also highlighted (Sinha, 2009; Yunus, 2014). But the prescribed curriculum has so many technicalities like word identification, phonics and inert textual information that constructing meaning gets somewhat lost in the deluge of too much decontextualized edifying work. As a consequence children hardly have any opportunities for self-expression, independent writing and development of language ability. The researcher attempted curriculum planning incorporating story-telling as a technique which provoked children's interest, imagination and self-expression. The curriculum practice in the school was as far as could be from the suggestion by American curriculum theorist John Dewey,

It is true that language is a logical instrument, but it is fundamentally and primarily a social instrument. Language is the device for communication; it is the tool through which one individual comes to share the ideas and the feelings. When treated simply as a way of getting information, or as a means of showing off what one has learned, it loses its social motive. The motive should not always be the succession of studies, but also the development of new attitudes, interests and experiences (Dewey, 1897: 79).

Data vignette II reflects the trend of emphasis in school curriculum on defining knowledge as production of correct answers. This product view of curriculum, as right answers to prescribed questions, also reinforces the framing of curricular practice as largely constituted by textbooks, related materials and formal instruction. This in turn disregards children's existential realities like their interests, inclinations, imagination and out-of-school knowledge. The decontextualized textbook centered curriculum eradicates possibilities of children's meaning-making, knowledge construction and self- development by learning on their own. Such a tightly-framed prescribed curriculum practice does not facilitate teaching as a performative act. It cannot support an act that offers space for change, collaboration and spontaneity; elements that can also serve as catalysts in drawing educative elements out of every curricular experience in the classroom. School practices continue to remain indifferent to child centered conceptions of school curriculum. Children are often taught to conform to the authority of a textbook because it is expedient to do so; but the moment the children discard authority they will come into *knowing*; and only then is there cooperation, affection and *being*. School curriculum practices tend to sustain 'rote-based' or 'procedural' approach to knowledge rather than activity based teaching. This does not support learning with understanding, application and meaning-making. Developmental psychologists like Jean Piaget and educators like John Dewey have underlined the value of a curriculum based on varied modes of activity. The concrete experiences during activities facilitate the processes of children understanding the reality around them through experiential learning (Dewey, 1916; Kitchener, 1986). Physical and other non-intellectual modes of activity in teaching-learning need to therefore be an integral part of school curriculum. Virtual teaching during the pandemic could evidently not support physical activity but was not perceptive to its role in school curriculum either. Recent curriculum policy in India has recommended that moving, exploring and doing things, on one's own, with one's peers or in the company of adults, (NCF, 2005) is a method to open up rich learning possibilities. The prevailing school curriculum, with its emphasis on knowledge as a product, is adversative to a child centred curricular experience (Harrison, 2002). An analysis of

data in this research concluded that curriculum practice in primary grades curricular knowledge was practiced almost like an object, information or product to be possessed. The prescribed texts were viewed as authoritative sources of knowledge, as materials waiting for acquisition. It can be concluded that the school curriculum was framed in a *having* mode. Learning consisted of an accumulation of correct answers to be reproduced in a testing situation to demonstrate children's ownership of knowledge. Data vignette III points at how a having curriculum regards the marks scored by children as the sole markers of learning. The various remarks of teachers in the classroom about children's scored marks are akin to smothering of the knowing and being in school curriculum. Sociologists of education have argued how school achievement is contingent not only on individual but socio-cultural factors too. The having mode of curriculum ignores children's lived experiences and personal knowledge which does not foster the processes of *knowing*. The research also demonstrated that there do exist spaces of child-centred curriculum development that connect the child with the curriculum as two ends of the same process. These possibilities were seldom exploited during school curriculum practice. It can be said that school curriculum practice remained largely disengaged with child-centered curriculum theory traditions. The research confirms that prevailing school curriculum aims at acquisition of a given body of knowledge as if knowledge is a commodity to be possessed. In selecting the given body of knowledge there is hardly any recognition of children's need to know and to be. The concept of *having* resonates in tandem with such a school curriculum practice. Where *being* simply means to enjoy the delight of the knowledge and live it. Something that is professed in child-centered perspectives in curriculum theory but is not much practiced in schools. 'Optimum knowledge in the being mode is to *know more deeply*. In the having mode it is to *have more knowledge*' (Fromm, 1921:36) which curriculum practices aim at. This research confirmed a gap between child-centred curriculum theory and practice framed in the having mode; in primary grades.

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42. Online and ICT Based Education at Higher Education Level: Issues and Challenges

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Abstract

Education is a very purposeful tool for all round development of any individual and society. No nation can afford to underestimate the contribution and priority of education. Quality of education broadly depend on the availability of teaching-learning resources, training, and orientation of teachers during pre-service and in-service teacher education programmes. But the formal system of education alone cannot satisfy the needs and requirements of teachers and students at different levels. Therefore, there is an urgent need of revamping teacher education programmes in order to improve quality, standard of teaching-learning process, and prestige of teaching profession. In this context, online and digital education can play very significant role in improving the quality of education by providing access to latest knowledge, skills, values, innovative pedagogies and can also supplement the face to face mode of teaching. It proved to be the only lifeline of teaching-learning process during Covid-19 pandemic situation not only in our country but throughout the world at large. National Education Policy (2020) also recognizes the importance of integration of technology, and online/digital education can be one alternative mode of quality education for all. But there are some very crucial issues and challenges which directly or indirectly hamper the process of online/digital and ICT based education like lack of digital resources at institute level and availability of thereof at learners' level including the problem related to affordability/availability of smart phone and also the expenses of recharge thereof, lack of knowledge and orientation among teachers and students, speed and internet connectivity in rural areas, negative attitude & lack of interest of senior teachers towards online education, limitation and ineffectiveness for practical work/field based activities, feeling of monotony, besides health issues, of learners due to prolonged online teaching etc. Thus, the authors intended to make research attempt to identify the multifaceted issues and challenges regarding online and ICT based education and also to suggest some remedial measures for effective implementation of online and ICT based education at higher education level in India.

Keywords: Online/Digital Education, ICT, National Education Policy-2020

INTRODUCTION

No society can develop to the extent, it deserves without education. Therefore, education is essential for different agegroups, depending upon the required form/organizational setup- formal, informal and non-formal education. All these forms of education are closely inter-related and complimentary to each other and aims & objectives of one type of educational form cannot be achieved in absence of needed support without to other two. In the context of formal system of education, teacher education programme assumes central role, besides all other related aspects of its organization, implementation, and evaluation. It is needless to say that role of parents as first teacher in the context of informal education cannot ignored. Similarly, in case of adults/continuing and non-formal education, the role of adult educator in the context of andragogical principles, content and methodologies is of paramount importance. Further, life-long education calls for due emphasis in terms of relevant learning experience & exposures right from the child birth till the last movement of individual life.

In the present paper, the authors have delimited the scope to only formal system of education within which up to online and digital education in the context of higher education. Obviously, the online/ICT based education has already covered its journey during the past few decades, however, in very recent context of pandemic situation throughout the world, which is still prevailing, its journey has been through uneven terrain but at the same time, the educational process through online mode has been considered as only panacea by ministry of education, educational authorities/UGC/university/directorate of school education/school system. The authors thought it appropriate to carry out investigation in order to explore and identify the qualitative and quantitative aspects of online education in term of its basic requirements, available learning resources, delivery system of education institutions, educational readiness of teachers as well requirement of beneficiaries i.e. learners/students and their participation alongwith multifaceted relevant aspects related to it. Moreover, the focus has also been on issue and challenges involved in entire process of education involving teaching/learning process through online education. Apart for the aforesaid background, authors concern has also been the recommendations of National Education Policy-2020 about online education also from the point of constituting the base of educational implications of findings of this research in the present on going education scenario as well as adequate response/preparation for times to come in this regard.

ONLINE AND DIGITAL EDUCATION IN THE CONTEXT OF NEP-2020

National Education Policy-2020 also focus on the online education and digital education through ensuring equitable use of technology. It will be the alternative mode of quality education where traditional/face to facemode of education are not possible. NEP (2020) also recognizes “The importance of leveraging the advantages of technology while acknowledging its potential risks and dangers. It calls for careful designed and appropriately scaled pilot studies to determine how the benefits of online/digital education can be reaped while addressing or mitigating the downsides. In the meantime, the existing online/digital education platform and ongoing ICT-based educational initiatives must be optimized and expanded to meet the current and future challenges in providing quality education for all” (NEP-2020, P. 58).Themain provisions of NEP-2020 related to technology and its integration, online/digital education was given below:

- Education and Technology has the close relationship at all levels and can play significant role in improving quality education.
- Improving creativity of tech-savvy teachers and entrepreneurs.
- NEP (2020) Point No. 23.2- “New technologies involving artificial machine learning, block chains, smart boards, handheld computing devices, adaptive computer testing for student development, and other forms of educational software and hardware will not just change what students learn in the classroom but how they learn, and thus these areas and beyond will require extensive research both on the technological as well as educational fronts”.
- NEP (2020) Point No. 23.3- “Use and integration of technology to improve multiple aspects of education will be supported and adopted, provided these interventions are rigorously and transparently evaluated in relevant contexts before they are scaled up. An autonomous body, the National Educational Technology Forum (NETF), will be created to provide a platform for the free exchange of ideas on the use of technology to enhance learning, assessment, planning, administration, and so on, both for school and higher education”.
- NEP (2020) Point No. 23.5- “The thrust of technological interventions will be for the purposes of improving teaching-learning and evaluation processes, supporting teacher preparation and professional development, enhancing educational access, and streamlining educational planning, management, and administration including processes related to admissions, attendance, assessments, etc.”.

- NEP (2020) Point No. 23.6-“A rich variety of educational software, for all the above purposes, will be developed and made available for students and teachers at all levels. All such software will be available in all major Indian languages and will be accessible to a wide range of users including students in remote areas and Divyang students”.
- Teaching-learning e-content will be developed through SWAYAM platform in all regional languages by the State Governments and the concerned bodies.
- More focus should be on Artificial Intelligence (AI) 3D/7D Virtual Reality.
- Awareness programmes should be organised on issues related to privacy, handling, and protecting data.
- Training programmes will have to be organised for teachers/educators for effective implementation of online education.
- Improvement of digital set-up, content and technology at the institutional level.

SIGNIFICANCE OF THE STUDY

Technology is the need of the hour and it has changed our social, cultural, personal, and educational life. It is one of the most important tools of acquiring latest subject knowledge, innovative pedagogies/methodologies, life skills etc. and helps in improving the quality of education at different levels. As rightly pointed out by Mishra, et al. (2006), “In the 21st century, technology is considered to be a potential instrument for bringing revolution in social and cultural lives of our society. However, the teacher education institutions and teacher education professions in our country and in all Asian countries are yet to explore its complete advantages”. Similarly, Krishnaveni & Meenakumari (2010) have said that “Role of ICT in higher education is solicited for improving quality, widening access and enhancing operational efficiency across all functions in higher education sector and to create new dynamics in this domain both at micro and macro levels”. Dash (2014) has indicated that “Integration of Information and Communication Technologies (ICTs) and their integration in teacher education programmes are a means to improve and reinforce the quality of teacher education programmes at pre-service and in-service level”. Thus, the present investigation is quite relevant in the present educational context because of the fact that it aimed to study the online/digital, and ICT based education with special reference to issues and challenges as the proper understanding of the same is necessary for effective implementation of online/digital and ICT based

education. The findings of the study will be very useful for educational administrator and educational institutions for effective implementation of online education.

OBJECTIVES

- (i) To see the availability of online/digital and ICT tools and facilities in teaching Departments of Central University of Haryana.
- (ii) To study the level of awareness and uses of online/ICT tools among students of Central University of Haryana.
- (iii) To study the difficulties faced by students related to the process of online education and use of ICT tools.
- (iv) To suggest remedial measures in the light of findings of the study for further improvement in the online and ICT based education.

DELIMITATIONS

- (i) The present study was delimited to study the availability and use of online education at Central University of Haryana only.
- (ii) The study is delimited to only 102 students (PG & UG) studying in various teaching Departments of Central university of Haryana.

RESEARCH METHOD

Descriptive research method was adopted.

POPULATION AND SAMPLE

For the present study all the students (both UG & PG) of Central University of Haryana constituted the population. For drawing the sample, 102 students were selected on the basis of Simple Random Sampling technique.

TOOLS USED:

Self-Developed Questionnaire was used for data collection.

STATISTICAL TECHNIQUES USED

Frequencies and Percentage of responses of respondents were calculated.

RESULTS

On the basis of analysis and interpretation of the data, the results of the study are presented in the following sequence:

1. Availability of Online/Digital and ICT Tools and Facilities in Teaching Departments of Central University of Haryana

Majority of students opined that Online/Digital and ICT tools and facilities were not available in their teaching Departments of Central University of Haryana like digital set-up, computer lab, projectors, printers, interactive whiteboard, multimedia, etc.

2. Level of Awareness and Uses of Online/ICT Tools/Resources among students of Central University of Haryana

Table-1

Sl. No	Online/ICT Tools	Always (%)	Occasionally (%)	Sometimes (%)	Rarely (%)	Never (%)
1	Word Processing/ Microsoft Word	28.70	14.90	36.60	19.80	00
2	Spreadsheet/Excel	11.00	25.70	46.50	8.90	7.90
3	Power Point Presentation (PPT)	18.80	12.90	39.60	19.80	8.90
4	Interactive Whiteboard	13.80	13.90	28.70	11.90	31.70
5	Projector	5.00	23.00	44.00	8.00	20.00
6	YouTube	64.40	00	10.60	25.00	00
7	Internet	75.20	00	00	24.80	00
8	e-mail & SMS	60.40	00	17.80	21.80	00
9	e-books	24.80	8.90	45.50	20.80	00
10	e-journals	12.50	7.90	55.90	6.90	16.80
11	e-newspapers	21.80	8.90	45.50	12.90	10.90
12	e-magazines	10.80	10.90	49.50	5.00	23.80
13	Digital library	11.90	16.80	52.50	00	18.80
14	Multimedia	37.60	10.90	32.70	18.80	00
15	Video-conferencing	10.90	18.80	45.50	19.80	50.00
16	Tele-conferencing	9.90	10.90	40.60	10.90	27.70
17	Blogs	5.00	6.90	35.60	00	52.50

18	Wikipedia	34.70	7.90	38.60	18.80	00
19	Twitter	12.90	10.90	25.70	6.90	43.60
20	Facebook	27.70	6.90	29.90	21.80	13.70
21	WhatsApp	65.40	00	00	27.70	6.90
22	Telegram	18.80	12.90	25.70	8.90	33.70
23	T.V.	20.50	12.90	41.80	11.90	12.90
24	Radio	00	11.90	45.50	00	42.60
25	Discussion Form	6.90	13.90	49.50	00	29.70
26	Educational Game	5.00	15.70	44.60	5.00	29.70

Students were required to indicate the extent of use of online/ICT resources/tools on the basis of their response in terms use of frequency/extent of use under the following heads- ‘Always’, ‘Occasionally’, ‘Sometimes’, ‘Rarely’ and ‘Never’. Table-1 clearly reveals that students were using various types of online/ICT resources/tools for instance, the percentage of students who responded on ‘Always’ in case of use of Internet, WhatsApp, YouTube, e-mail & SMS, Multimedia, Word Processing/Microsoft Word, Facebook, e-books, e-newspapers, T.V., Power Point presentation and Telegram was 75.20%, 65.40%, 64.40%, 60.40%, 37.60%, 28.70%, 27.70%, 24.80%, 21.80%, 20.50%, and 18.80% respectively. Thus, the majority of students were using various online /ICT tools/resources as per their needs and requirements.

3. Uses of Online /ICT Tools /Resources among Students at Higher Education Level

Table-2

Sl. No.	Statements	SD*	D*	U*	A*	SA*
1	In modern times, learning through Online/ICT tools is very useful and relevant	0.00	0.00	15.80	39.60	44.60
2	Learning through Online/ICT tools helps in saving time and labour	5.90	10.90	9.90	52.50	20.80
3	Use of online/ICT tools are essay and communicative	8.90	12.90	21.80	43.60	12.80
4	Online/ICT tools are very useful in searching, updated course content, and	6.90	0.00	11.90	54.50	26.70

Sl. No.	Statements	SD*	D*	U*	A*	SA*
	information					
5	Exploring new information on the topic/subject through Online/ICT tools is easier	4.00	0.00	11.90	60.40	23.70
6	Learning through Online/ICT tools develops more confidence than printed books	15.80	22.70	22.70	23.00	15.80
7	Online/ICT tools enhance learning skills	0.00	6.90	25.70	48.50	18.90
8	Online/ICT tools are very helpful to improve collaborative learning	0.00	0.00	30.70	57.40	11.90
9	Online/ICT tools enhance student's motivation towards learning	6.90	15.90	26.70	38.60	11.90
10	Online/ICT tools help to find out the relevant and useful material for in-depth study and understanding of the content	0.00	10.90	29.70	43.60	15.80
11	Online/ICT tools help to interact with other students, teachers and other persons related to education	5.90	0.00	19.80	58.50	15.80
12	Online/ICT tools help in improving communication skills	5.90	7.90	21.80	48.50	15.90
13	Online/ICT tools keep constant engagement & attention of learners, and promote more innovative learning	4.00	7.90	16.80	58.40	12.90
14	Online/ICT tools help in improving efficiency and organizational skills	5.90	0.00	23.80	57.50	12.80
15	Online/ICT tools are not useful for reflective and critical learning	0.00	25.70	31.70	34.70	7.90
SD*-Strongly Disagree, D-Disagree, U-Undecided, A-Agree and SA-Strongly Agree						

It is evident on the basis of Table-2 that in the context of Statement No.1 “In modern times, learning through Online/ICT tools is very useful and relevant”, the percentage of students who responded on

‘Strongly Agree’ was 44.60% whereas in case of Statement Nos. from 2 to 15, the percentage of students who responded on ‘Strongly Agree’ was 20.80, 12.80, 26.70, 23.70, 15.80, 18.90, 11.90, 11.90, 15.80, 15.80, 15.90, 12.90, 12.80, and 7.90 respectively. In case of the percentage of students who responded on ‘Agree’ on statements from 1 to 15 was 39.60, 52.50, 43.60, 54.50, 60.40, 23.00, 48.50, 57.40, 38.60, 43.60, 58.50, 48.50, 58.40, 57.50 and 34.70 respectively. Thus, opinion of students was found varying in the context of/the extent of use of online / ICT tools/resources with regard to above given different statements. But it is evident from the table No. 2 that online/ICT tools/resources were being used by the students’ at large use to usefulness thereof.

4. Difficulties Faced By Students Related to the Process of Online Education and Use of ICT Tools

Table-3

Sr. No.	Difficulties Faced by Students	SD*	D	U	A	SA
1.	Lack of availability of Online/ICT tools in the teaching Departments	0.00	24.80	14.90	35.60	24.70
2.	Lack of awareness and orientation of learning/educational softwares	5.90	19.80	13.90	45.50	14.90
3.	Teachers do not help students about how to use Online/ICT tools	9.90	33.70	20.80	28.70	6.90
4.	Poor internet connectivity & speed in teaching departments and campus	6.90	27.70	20.80	33.70	10.90
5.	Non-availability of ICT lab in the Department	10.90	25.70	8.90	36.60	17.90
6.	Teachers do not use the latest Online/ICT tools during teaching	17.80	36.60	12.90	15.90	16.80
7.	There is no provision of period for Online/ICT lab in departmental Time-Table	10.90	20.80	15.80	39.60	12.90
8.	Lack of knowledge as to how Online/ICT can be interrelated to enhance subject knowledge	6.90	19.80	19.80	41.60	11.90
SD*-Strongly Disagree, D-Disagree, U-Undecided, A-Agree and SA-Strongly Agree						

It is evident on the basis of Table-3 that in the context of Statement No.1 “Lack of availability of Online/ICT tools in the teaching Departments”, the percentage of students who responded on ‘Strongly Agree’ was 24.70% whereas in case of statement Nos. from 2 to 08, the percentage of students who responded on ‘Strongly Agree’ was 14.90, 6.90, 10.90, 17.90, 16.80, 12.90 and 11.90 respectively. In

case of the percentage of students who responded on 'Agree' on statements from 1 to 08 was 35.60, 45.50, 28.70, 33.70, 36.60, 15.90, 39.60, and 41.60 respectively. Thus, majority of students had revealed that there were various difficulties faced by them in the context of process of online education and use of ICT tools such as lack of awareness and orientation of learning/educational softwares, lack of availability of online/ICT resources, non-availability of ICT lab and integration of Online/ICT with subject knowledge, which directly and indirectly effect the process of online education.

4. Remedial Measures in the Light of Findings of the Study for Further Improvement in the Online and ICT based education

In the light of the findings of the present study, the following remedial measures are being suggested by authors for further improvement in online education system:

- Central and State government should provide more online resources/ICT based facilities like computer labs, internet facilities&proper connectivity, power backup facilities inhigher education institutes.
- The Higher Education Institutes (HEIs)should organize orientation and time to time training programmesfor teachers as to how to use the new technologies/online resources, and tools for their capacity building for use of online/ICT tools in their teaching-learning process.
- Teachers should be more creative, collaborative and innovative in adopting new tools, methods, techniques in their teaching-learning process.
- There should also be main focus on use of online resources/Information & Communication Technology (ICT) during in-service teacher training programmes.
- Teachers should make use of available online resources in their teaching subjects.
- Teachers should help their students in their ICT enabled learning and use of online/ICT resources.
- Teachers should also use social networking sites for educational purpose to make learners more involved in subject-activities.
- The students should be motivated and encouraged to make optimum and full use of the existing ICT resources for their studies.
- Bodies of Government, and higher education institutions should give more priority on Continuing Professional Development (CPD) of the teachers across the levels.
- There should be more use of online/digital and ICT based resources and integration thereof with teaching-learning process at higher education level.
- Educational institutions should encourage teachers for effective use of ICT skills and knowledge to improve their capabilities so as to perform their duties effectively.

- Teachers should make use of new ICT tools which are technologically more interactive and effective at higher education level.

Conclusion

The main aim of the present study was to explore and identify the issues and challenges related to use of online and ICT based education. The study also highlighted the main provisions envisages in the Nation Education Policy-2020 regarding online and digital education in India. The findings of study indicate that students were using various e-resources/ICT toolsbut still there are very crucial issues and challenges which directly or indirectly hamper online and ICT based education such as lack of availibility of ICT tools/facilities; lack of orientation and training related to online/ICT based education;lack of provision for orientation of online and ICT based education in the regular time-table of the Departments. Thus, the present research study is very useful for Policy Makers, Educational Administrators, Higher Education Authorities, teachers to take necessary steps for making necessary arrangements of online and ICT based education and also orient students for use of ICT/online resources for ensuring their capacity building so that it will not only supplement and improve the traditional mode of teaching-learning but will also prove to be an effective mode of teaching/delivery of the contents to students during the absence of offline/regular mode of education, as has happened and still happening, due to the Covid-19 pandemic situation in the country.

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43. Status of Mental Health of Elementary School Teachers in Aizawl City

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Abstract

A teacher's role is very significant in society. Along with grooming the future of a student, s/he tries to shape the future of the country. Alike others, teachers also face many challenges related to the professional as well as personal, family, or social life. To deal with these challenges, strong mental health is required. Besides, it is necessary to be mentally healthy to balance intrapersonal and interpersonal relationships and to help in maintaining the mental health of the surrounding people. These all demand a teacher to be mentally healthy because mentally tender children are the responsibility of him/her. Keeping in the mind the essentiality of mental health for the teachers this study aims to find out the level of mental health among elementary school teachers. The population of this study consists of all 354 elementary schools in Aizawl city. A stratified random sampling technique was adopted for the selection of the sample. The findings of the study indicate that a majority of the elementary school teachers of Aizawl city were mentally healthy but at the same time, a large number of teachers had a low level of mental health. Female teachers and teachers who were teaching in the government schools were found to be less mentally healthy than their counterparts.

Keywords: Elementary School Teachers, Mental Health

Introduction:

The term 'Mental Health' consists of two words: The word 'mental' usually imply something more than purely cerebral functioning of a person. It includes one's emotional affective state. Similarly, 'health' refers to more than physical well-being. It also connotes the individual's intra-physic balance, the interaction of one's physic-structure with the external and social environment (Kaur, 2006). Freud has considered it as a capacity which leads us to work and love (Sebastian, 2017).

The significance of mental health can be understood for individual's life as it 'includes those behaviors, perceptions, and feelings that determine a person's overall level of personal effectiveness, success, happiness and excellence of functioning as a person. It depends on the

development and retention of goals that is neither too high nor too low to permit realistic successful maintenance of belief in one's self as a worthy, effective human being. Further, it is the condition in which the individual manifest through self-evaluation, adjustability, maturity, regular life, absence of extremism satisfactory social adjustment and satisfaction from his chief occupation' (Kornhauser, 1965) which are essential for human being's survival. It is a way of living which determines a satisfactory and effective relationship with other members of the society (Das, 2008).

Health is essential for survival as a social living being. Earlier mental health was not given much importance, but over time, the contribution of mental health for the personality of an individual was understood and got equal importance as physical health. Concretely, no part of human life is beyond the range of mental health. It includes every aspect and element of human personality and the individual's adjustment to own self, others and the environment.

The mental health not only ensures individual's adjustment but also improves the quality of life, helps to develop a healthy connections with others, helpful in making an excellent life choices, upholds physical health and well-being, makes strong to handle the natural ups and downs of life, appreciate and grow our potentials, and reduces medical costs.

The importance of teacher in the educational process is unquestionable. In every educational programmes, a teacher holds an important position with many responsibilities. These responsibilities and burden also affects the health of teachers, especially mental health. Teachers are found to be stressed (Glazzard & Rose, 2019) due to various reasons. Some reasons of the stress may be related to the professional life as; high-stakes testing; a large number of students in the class sizes; students' behavioral issue; unavailability of resources or inadequate resources, extra workload, bureaucracy in the schools, paperwork, poor physical space, high responsibility for others; inadequate recognition; and the gap between pre-service training, their expectations for work and actual work experiences (Dworkin, 2001; Fisher, 2011; Kokkinos, 2007; Travers & Cooper, 1996; Cezar-Vaz, et al. 2015; Glazzard & Rose, 2019) while some others problems may be related to the personal life as their relationship crisis, family problems as illness of family members, childcare issues etc. (Glazzard & Rose, 2019). It has been found in the researches that the mental wellbeing of teachers affects the performance of the students in the classroom (Glazzard & Rose, 2019). The mental wells being of a teacher also lower the students' mental difficulties Harding et al, 2019). Thus, it is

important to have a good mental health in the teachers. When it comes to elementary school teachers it becomes more crucial as students are going to be very young and for few children they are the first teacher, whom child comes in contact.

Keeping in mind the role of elementary school teachers and their mental health to develop appropriate social skills among students which is going to shape their future, investigator planned to find out the status of mental health of elementary level teachers of Aizawl city.

Objectives of the Study:-

The objectives of the present study were as follows-

1. To study the mental health of the elementary school teachers in Aizawl City.
2. To study the mental health of elementary school teachers under the six subscales of mental health inventory concept of life, self-concept, perception for self amongst others, perception for others, personal adjustment, and record of achievement.
3. To find out the mental health of the male and female elementary school teachers in Aizawl City.
4. To find out the mental health of the government and private elementary school teachers in Aizawl City.

Methodology:

Method:

In the present study descriptive survey method was adopted by investigators.

Population:

There were 354 elementary schools in Aizawl city and all the teachers working in these schools were the population of the present study.

Sample and Sampling Technique:

A stratified random sampling was adopted by the investigators for the selection of samples. In Aizawl city there were 354 elementary schools in which 231 were government and rest of 123 were private schools. The investigators chose 20 per cent of each type of school randomly. In the government schools there were total 335 teachers in which 129 were male and 206 were female. Same as in the private schools there were 89 male elementary school teachers and rest 196 were female teachers. More than 20 percent from each stratum was selected as sample for the study. The detail of samples is given below.

Table No.1
Number of Elementary School Teachers Selected from various Stratum

Type of school management	Gender		Total
	Male	Female	
Government	39	42	81
Private	22	46	68
Total	61	88	149

Tool and Techniques Used:

For the purpose of finding out the mental health of teachers, RCE Mental Health Scale (MHS, 2009) for teachers developed by Dr. S.P Anand was used. It is a Likert type Scale based upon six icons of mental health as; self-concept, concept of life, perception of self amongst others, perception of others, personal adjustment and record of achievement. The scale has 40 negative and 20 positive statements.

Scoring:

Five choices were given as SA (strongly agree), A (agree), UD (undecided), D (disagree) and SD (strongly disagree) in each of the statement. The respondents answers were scored as 4,3,2,1,0 for positive statements and 0,1,2,3,4 for negative statements as given in the manual (RCE Mental Health Scale, 2009).

Statistical Technique:

The gathered data were analyzed by using frequencies, percentage, mean, median and standard deviation (SD).

Analysis and Interpretation:

The findings of the present study and their interpretations are presented in the following tables.

Objective 1: To study the mental health of the elementary school teachers of Aizawl City.

Table No.– 2
Mental Health of the Elementary School Teachers of Aizawl City

Dimensions	N	Mean	Std.Dev.	Median
Concept of life	149	30.20	5.070	31
Self Concept	149	15.35	3.108	16
Perception for self amongst others	149	38.28	5.001	38
Perception for others	149	26.02	3.270	26
Personal adjustment	149	21.28	3.611	22
Record of achievement	149	27.26	4.521	28
Mental Health	149	158.30	18.252	162

The mean score, along with the median and the value of standard deviation in six different dimensions are presented in the table. In the ‘concept of life’, the mean score was 30.20 with a standard deviation (SD) 5.070. In the same area (concept of life) the median value scored by elementary school teachers was 31. The above table clearly indicates that the median value in the ‘concept of life’ was slightly higher than mean value; it means that the distribution is slightly negatively skewed which means that the majority of the elementary school teachers had scored less in the dimension ‘concept of life’.

The above table also indicates towards the ‘self-concept’ of the elementary school teachers. In this dimension, the mean score was 15.35 with SD 3.108. The median value fell to 16, which was higher than the mean value. It indicates that the distribution is slightly negatively skewed. In this dimension also the most of the teachers scored less.

The next dimension is ‘perception for self amongst others’, and in this dimension the mean value of elementary school teachers was 38.28 with SD of 5.001 and the median value was 38. Here, a positively skewed distribution is found as the mean value was slightly higher than the median value which shows that in this area majority of the elementary school teachers had scored more than average.

The fourth dimension is ‘perception for others’ and in this dimension the mean value was 26.02 with SD value 3.270 with the median value 26. A very minimal skewness has been found in this dimension.

The above table shows that in the ‘personal adjustment’ area, the mean value of the elementary school teachers is found to be 21.28 with SD of 3.611. The median value in this area is found to be 22. The table shows that the median value in ‘personal adjustment’ was higher than the mean value which means that the distribution is slightly negatively skewed.

In the dimension of ‘record of achievement’, the mean score was 27.26 with SD 4.521. The median value in ‘record of achievement’ was 28. In this area, negatively skewed distribution is found as the mean value was lesser than the median value.

The overall score for mental health of elementary school teachers is also presented in the table. The overall mean value is found to be 158.30 with SD 18.252 and median was 162. As the total mean score was lesser than the total median score, it means that the distribution is negatively skewed. The above scores show that the majority of the elementary school teachers had scored less than average. The mean score of 30.20 in the dimension of the ‘concept of life’ held the highest score among all the six dimensions which mean that elementary school teachers attained a high level in the ‘concept of life’.

According to the manual for the mentally healthy teachers, the norm value is considered to be 150. It stated that the score of 150 and above to be taken as an indicator of the subject being mentally healthy, in case of a teacher’s score is less than 150; then the teacher will not be considered as mentally healthy. As the total means score of 149 elementary school teachers was 158.30, which was higher than the given norm of mental health. It means that the teachers working in the elementary schools in Aizawl city were mentally healthy. So the research hypothesis “elementary school teachers working within Aizawl City, have a high level of mental health” is accepted. The finding is consistent with the findings of Rani et al (2012) and Srinivasan (2016); in which teachers were found to be average and good in their

mental health and inconsistent with the finding of Gohil (2016) in which she found that maximum teachers had low mental health.

Table No.- 3
Elementary School Teachers Scored below and above the Norm

	Gender		Management		Total
	Male	Female	Government	Private	
Teachers who scored below 150	19 (31.15%)	23 (26.14%)	25 (30.86)	17 (25%)	42 (28.19%)
Teachers who scored above 150	42 (68.85%)	65 (73.86%)	56 (69.14%)	51 (75%)	107 (71.81%)
Total	61	88	81	68	149

The above table shows (Table No.- 3) the number of elementary school teachers who scored below or above than 150; which is the norm of MHS. Out of 149 elementary level teachers, 61 were male and 88 were females. Out of 61 male teachers, 19 (31.15%) teachers scored less than 150 and rest 42 (68.85%) scored more than 150. Out of 88 female elementary school teachers, 23 (26.14%) female teachers scored less than 150 and 65 (73.86%) female teachers scored more than 150. It clears that 19 male teachers and 23 female teachers were not mentally healthy as per the given norm. Even though female teachers were more in numbers than the male teachers who scored below the norm; yet the percentage of male teachers scoring below 150 (31.15%) was higher than the percentage of female teachers (26.14%) scoring below 150. From the table, it is clear that more female elementary school teachers were mentally healthy than the male elementary school teachers which is inconsistent with the findings of Gorsy (2015).

The above table also indicates that out of 149 elementary school teachers, 81 elementary level teachers were from government schools and rest 68 teachers were from private schools. Table clears that from government schools 25 (30.86%) teachers scored less

than the norm. At the same time, out of 68 private school teachers, 17 teachers (25%) scored less than 150 which shows that 25 teachers from government schools and 17 teachers from private schools were not that much mentally healthy how much they should be. Table clears that at the elementary level, in comparison to government school teachers; elementary teachers of private schools were more mentally healthy. This finding is inconsistent with the finding of Rani et al. (2012) in which they found the teachers working in government and private schools do not differ in the level of mental health. While Kaur (2006) found in her study that private school teachers were more mentally healthy than government school teachers which is also consistent with the findings of this present study. At the same time the findings of Galgotra, 2013; Pachaiyappan & Raj, 2014; Gorsy et. al., 2015; are very different from findings of this study in which researchers found that government school teachers were better in mental health than the private school teachers.

Above table reveals that out of 149 elementary school teachers, there were 42 (28.19%) teachers who scored below the norm; while the majority of the teachers (total- 107 teachers) obtained the score higher than 150. Though according to the norm, it can be said that most of the teachers of the elementary level were mentally healthy, but at the same time more than 28% teachers of the elementary level were not mentally healthy.

The Major Findings of the Study:

The findings of the study has been given as per the research objectives as follows-

1. The majority (71.81%) of elementary school teachers in Aizawl city were mentally healthy.
2. Out of the 149 elementary school teachers, 42 (28.19%) were not mentally healthy and the rest 107 (71.81%) were mentally healthy.
3. Both male and female elementary school teachers in Aizawl city had good mental health.
4. Out of the 61 male elementary school teachers, 19 (31.15%) male elementary level teachers were not mentally healthy.
5. Out of the 88 female elementary school teachers, 23 (26.14%) female elementary level teachers were not mentally healthy.
6. Out of the total 81 government elementary school teachers, 25 (30.86%) were not mentally healthy.
7. Out of 68 private elementary school teachers, 17 (25%) teachers were not mentally

healthy.

Conclusion:

A good personal and professional life depends on mental health of the individual. The findings of the study show that majority of elementary school teachers in Aizawl city were mentally healthy. Although more than 70% teachers were mentally healthy but at the same time more than 28% elementary school teachers were not mentally healthy which is a huge number. Teaching profession is not an easy task. There may be multiple reasons which can be responsible for mental unhealthiness of these school teachers. The workload and stress level that the teachers' experience, dissatisfaction with salary, inefficiency of teachers in handling gadgets and different equipment, poor knowledge of infrastructural facilities or lack of infrastructural facilities in the schools etc. are some issues by which teachers go through every day. It is the only profession in which success of a person depends on the performance of others. But, workplace problems are not the only reason for poor mental health of teachers. Sometimes family or social problems also affect mental health. Therefore, it is necessary that at the time of appointment of teachers it should be ensured that the teachers getting the appointment are mentally healthy. Training and awareness programs regarding mental health can be helpful in reduction of anxiety and improving mental health of teachers. Along with it like other organizations, the authorities of the schools should organize some workshop and support programmes to reduce the stress level of the teachers before it becomes overwhelming down the line.

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44. From Theory to Praxis: Effective Inclusion of Marginalized Genders in Co-educational Institutions

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Abstract

With the references of the research works done by other researchers on gender issues, this paper offers some evidences based on a research work conducted by the authors themselves. Through open-ended questionnaire and descriptive survey method the researchers have dealt with the conception of the students about co-education and also the notion of the students about other genders with special reference to transgender people. In a nut shell this paper aims at addressing the lacuna in effective inclusion of marginalized genders. In many institutions female students are cornered in terms of gender discrimination, male supremacy, patriarchal notion about women, academic matter etc. The condition of the transgender students is more serious. Even after achieving legal rights transgender students remain in the periphery of our society. The number of transgender students is also very limited in educational institutions. Even if they secure admission, they are often humiliated and marginalized. Moreover the definition of 'co-education' also hardly mentions the term 'transgender'. The authors are concerned whether the terms like 'inclusivity', 'gender harmony', 'gender equality' etc. will remain stored in the ivory tower of intelligentsia or these progressive ideas will be actualized in the real society. However, the findings of the study are impressive in some contexts as well as the study also unlocks some rooms of concern.

Key Words: Inclusivity, Gender harmony, Gender equality, Gender discrimination, Transgender.

INTRODUCTION

Despite the proliferation of co-educational institutions, friendship between boys and girls is till now a room of concern for many people in our society and inclusion of transgender students in the field of formal education is still a reverie to most of the citizens. In the case of girls, many guardians and even sometimes teachers also subscribe into the narrow notion that boys and girls may go out of control due to co-educational set up dismantling the norms or values of the society. Moreover, such ideas also prevail among the guardians, students and some teachers also that female students are weaker in

science or sports in comparison with the male students and in single-sex education a girl can show the best of her while in Co-education she may not compete with her male friends and will lose her self-confidence, although Plato in the 4th century B.C. affirmed that there are no overriding advantages for single-sex schools on educational grounds and later Dale also seconded this thought by proving through various research works that co-education does not affect academia. However, such thoughts hinder the effective inclusion of all genders. Rather it segregates the various genders from one another. These things lead to the emergence of more single-sex institutions. In the case of transgender students, even if they are admitted, are most of the time bullied by their friends and even sometimes humiliated by the teachers and guardians. The concept of 'transgender' is still a taboo to many social members. More attack on inclusion takes place when different genders are included under the umbrella of co-education but in reality these genders are segregated inside the classroom or inside the institutional campus. Thus even in an inclusive set up inclusion of genders are not effective in such conditions. This ineffectiveness affects the desired agenda of gender harmony and betterment of the self confidence of female and transgender students.

Whatever has been stated so far is not at all a cooked up story. Research studies and circumstantial evidences including our live experiences are enough to stand by such remarks. Inside co-educational ambience there may be the twist of gender segregation or marginalization. And these contradictory things happen in the disguise of monitoring and social values. Such a scene can be observed in a research conducted in our capital New Delhi on co-education. The study finds out the contradiction in the case of inclusivity of different genders. At one point, the teachers are happy to maintain gender harmony in their co-educational institutions emphasizing the fact that their institution is free from any kind of gender discrimination. On the other hand, the same voices ensure that they keep an eye on the students so that the boys and girls cannot talk to each other freely and the boys and girls also seat separately inside the classroom. These juxtaposing views unmask the reality of gender inclusion in educational institutions (Iyer, 2018).

Women are still treated as the second sex. This concept transcends social boundaries and enters into the educational field also. Discrimination is prevalent in their academic life also. This is also a hindrance in the path of inclusion of genders. In a research work on co-education the researchers themselves have declared their belief that students, especially girls, in the single-sex education

perform well than the girls in co-educational set up because in girls' institutions the girls are free from any type of distraction and totally focus on study. Apart from this, the main reason behind this, according to them, is less competition in girls' institutions in comparison with the co-educational institutions. Their point was that in single-sex education a girl can expose her talent properly and she scores high marks (Ramteke & Ansari, 2017).

Curriculum also often plays a crucial role in the effective inclusion of marginalized genders. If the curriculum is phallogocentric in nature how would other genders find their association with that kind of knowledge? Whether the institution is single-sex or co-educational, the pedagogy or the curriculum must not encourage any gender bias. Such a picture of gender hegemony has been depicted by Kirstin Pesola McEachern in her writing. There she expressed her grief for the phallogocentric curriculum prevalent in the school. She also proposed a curriculum highlighting both males and females. But her efforts were in vain. She could not change their viewpoint which she regarded her failure as a teacher. Such atmosphere of the school refuted the concept of inclusivity; rather it believed in exclusivity by the concept of 'otherization'. They also believed in brotherhood which also marks inclusivity but this inclusion was applied on only some persons. That thing minimized the scope to add more members. Despite her repeated attempts to feed the students a proper balanced diet on gender issues, the homosocial atmosphere created a bar for the students to have it (McEachern, 2014).

So, effective inclusion of marginalized genders is challenged by the social and academic concern. But one thing in these cases is being neglected that such type of views encourages gender discrimination. If a girl is reared separately because of the notion that she will lag behind in the competition with the boys, she perhaps can never gain her self-confidence and will remain treat herself as an inferior sex, as the second sex and as the marginalized sex who can crawl to reach the center from the periphery but can never dare to think herself equal with the boys.

However, there are also some people in our society who believe that inclusivity of all genders is needed to promote gender harmony. Like, Brady being a director of co-education transition in Barker Institute, Australia, argued that there is no labour division according to gender binary and therefore, there is no longer any need of segregated institutions. She also discarded the logic behind single-sex i.e. of girls' higher academic achievement because the researches after 1995 could not provide

enough instances to prove the fact. Throughout her article her focus was on gender equity which can be easily established through co-education (Brady, 2017).

Another researcher has captured the true essence of gender equality. She was not only concerned with the inequalities between sexes but also had suggested some measures for its implementation with documents of gender inequality prevalent in the society. In this context she moved further than the traditional arguments for co-education. She went beyond the logic of compensating women's prolonged suffering. Her work aimed at establishing the right of both males and females to grow up freely. She stated that all sexes are actually living in theoretical equality. But in reality, many situations are witnessing the lack of balance regarding equal opportunity, mutual respect, equal right etc. Even in the educational field, despite there being a large number of female graduates, which is in fact more than the number of male graduates, difference is prominent in their subject selection and later in their career selection. The author urged to develop the students as human beings rather than genders. Their roles, behavioural gestures, ideologies must not be shaped by societal demand but they should discover themselves. According to the author, this task can be completed through co-education. But the author did not see co-education just merely as mingling of boys and girls. Co-education must ensure the abolition of gender discrimination practically. The author also accepted the need for separation of boys and girls in some required contexts. But, otherwise the environment should boost togetherness of both the sexes. There are many cultures that prevent this mingling. The author suggested that even in that situation also girls should be encouraged to grow self-esteem and the boys should be taught to fight against gender discrimination so that one day such differences may be erased from the society. But where co-education exists its aim must be fostering students' ability to shape their personality and to select their role in society as per their wishes. If the boys and girls are educated separately, they will not understand each other's role in the society. So, ultimately the society will remain biased. Co-education can create a healthy relationship among genders which later will better the social and mental health of the people. Finally, the author suggested that the co-educational setting must fulfill its main goal to break gender stereotypes and to grow mutual respect and friendship. For this, curriculum, gestures, language, examples are to be apt to gender equality so that through study materials also the students can make the concept of equality from the very childhood (Rosàs, 2001).

In this context Central Advisory Board of Education (CABE committee on Girls' Education and Common School System to Reduce Gender Disparity, 2005) offered a paradigm shift in the case of women education. The members noticed that even in a co-educational institution also, girls are not properly included in comparison with the boys in terms of curriculum and the behaviour of the teachers. According to the committee, the notion of women education must be changed. The aim of women education must not be limited within the logic of successful upbringing of the children by a mother, or to reduce the infant mortality rate etc. Such logic deepens the root of gender discrimination rather than uprooting it. Rejecting the view that girls should be considered separately due to their inability to compete with the boys, they also proposed that girls should get the equal education like a boy. The committee argued that if the girls are to play an active role in national development, their education cannot be allowed to be padded by soft options at any stage. They have to be taught pure mathematics, sciences rather than restricting their education within elementary mathematics, home science etc. Their revolutionary recommendation was if a girl is taught about maternity, a boy should also be taught paternity.

Transgender students face more trouble than female students. In spite of achieving their legal rights, transgender students are rarely found in most of the institutions of our country. Even if they secure admission, they are often humiliated by the students, guardians or even sometimes by the teachers. Surprisingly, till now the lexicographic definition of 'co-education' includes boys and girls only. The reference of 'transgender' is silent there. If this notion can be altered, if transgender students can be included effectively ensuring gender harmony and gender equality, they will no longer be marginalized or looked down upon.

OBJECTIVES

To observe the effectiveness of co-education as a platform of inclusivity of genders in terms of gender harmony and gender equality, a research study was conducted having the following objectives—

1. To find out the concept of the students about co-education.
2. To judge the notion of the students towards other sexes including transgender.

RESEARCH QUESTIONS

1. What is meant by co-education according to the students?
2. What are the opinions of the students regarding other genders including transgender?

3. What is the opinion of the transgender student regarding other genders?

METHODOLOGY OF THE STUDY

Descriptive survey method has been used by the researchers to carry out the present study. Total sixty undergraduate students of Kolkata from Gokhale Memorial Girls' College, Jadavpur University and Aliah University taking twenty students from each place were selected through purposive sampling technique as the sample of the present study. The purpose of selecting undergraduate students was that they had recently crossed their adolescent period which is very vital for this study. The students might come from single-sex schools or co-educational schools but at that time all of them were either in a co-educational set up or in a single-sex set up. So, they could understand the impact of the shift of the set up or the positive or negative aspects of co-education or single-sex education in their lives. The researchers might take only co-educational institutions but they mindfully took single-sex institutions to understand the situation in both set up.

TOOLS USED

Self-composed open-ended questionnaires were used by the researchers to collect data from the students. Two questionnaires were made based on the conception of students regarding co-education and the mindset of the students regarding other genders with special emphasis on transgender people. One questionnaire was constructed for the students of co-educational set up and the other for the students of single sex set up. Both of them consisted of seven questions. The researchers gave full opportunity to the respondents to express their thoughts and suggestions. Enough space was provided for such questions.

FINDINGS

Analysis with Interpretation of Objective 1:

First objective was selected to understand the concept of the students about co-education. It is customary to regard the definition of co-education as where male and female students study together. But the researchers argue that if transgender students have also the legal right to study on the same platform along with male and female students and they admit themselves filling up the transgender option in the form, why they are not included into the definition of co-education! So, the very first question in the questionnaire was:

“What do you understand by the term ‘co-education’?”

And to no utter surprise maximum students i.e. 65% described co-education as an educational platform where male and female students study together. But the researchers became pleased to observe that the rest of the students i.e. 35% had surpassed the idea by defining co-education as a space where various or different genders study together. Among them two students especially mentioned transgender students. But it was a shocking fact that the transgender student had defined co-education as a space where both genders study together. It means that the student was aware of her gender but still she had subscribed to the conventional notion about co-education.

Analysis with Interpretation of Objective 2:

There were some questions in the questionnaire to understand the notion of the students regarding various genders. One of such questions was framed to know whether the students would like to be in a homosocial group or heterosocial group. Many people still regard that homosocial group is better because they think that girls and boys should not be in the same group due to security, 'chastity' etc. That is why the researchers input this question to see what the students would think about it. More than 50% students liked both homosocial and heterosocial groups. And almost 40% students liked heterosocial groups. Only six students liked to be in a homosocial surrounding and among them four students were male and at that time were in a single-sex set up (as there were no female students in their class), although they had come from co-educational background. The two female students who liked homosocial bonding did not also remain always in a single-sex set up. One of them had come from single-sex school but at that time used to study in a co-educational set up. And the other student was then in a single-sex set up but had attended both single-sex and co-educational schools and she justified her preference based on the natural similarities of attributes of each sex. So, all of these six students had the experience of both homosocial and heterosocial classrooms. Still, they used to like homosocial group. Total twenty two students out of sixty chose heterosocial group. Among them four students were male. One of them was always in a co-educational set up. But three students were then in a single-sex set up, although coming from co-educational schools and one of them had attended single-sex school also. So, these three students had the experience of both homosocial and heterosocial set up. But they liked heterosocial group. Seventeen female students also liked heterosocial group. Among them six students were at that time in a co-educational set up and eleven students were then in a single-sex set up. Among the six girls, four students had come from single-sex background and two students had not attended single-sex set up. Six students out of the rest eleven girls had come from co-educational schools. But five girls among them had never attended co-

educational set up. Yet, they preferred heterosocial bonding. The transgender student also liked heterosocial group. Thirty two students preferred both groups. Among them fourteen students were male and eighteen were female. Three students among them had also mentioned that they used to judge a person in a group by their nature or gesture not by their sexual orientation. Out of the fourteen boys, nine were then in a single-sex set up but had come from co-educational background. Two boys were experienced of both types of schools; — one of them was at that time in a single-sex set up and another one was then in a co-educational set up. But three males, who had never experienced co-educational set up, also liked both homosocial and heterosocial groups. Among the eighteen females five students had come from single-sex school to co-educational set up and one student had come from co-educational school to single-sex set up at that time. One female studying in a single-sex set up had attended both co-educational and single-sex schools. But eleven girls had not experienced such shift; — five students were all along in a co-educational set up and six students had attended single-sex set up only. Still, they liked both homosocial and heterosocial groups.

In the questionnaire the researchers also asked the students if they were afraid of their male, female or transgender friends and the reasons behind this hesitation (if any) to befriend other genders. In a word everyone had said that they were not afraid except three boys at that time studying in a single-sex set up. Two of them mentioned transgender people in the answer. One boy had said that he used to feel nervous before transgender people but the other had shown no reason. One male student who had liked homosocial bonding said that he was afraid to befriend girls, although he had come from a co-educational school. He thought that he might be attracted towards women and be trapped in affair which he strongly wanted to avoid. However, among the fifty seven students one female student who was then in a single-sex set up but coming from co-educational background and one female student who was always in a co-educational set up had responded that they were very comfortable with other genders. Another female student, who was also at that time studying in a single-sex set up but had experienced both types of schools, especially emphasized the transgender people mentioning that she loved transgender people and she was not concerned about anyone's gender. Two female students, who had experienced single-sex set up only, were not afraid of her friends but were tensed about male persons in the society due to the reckless increasing rate of women assault and such incidents used to make her feel unsafe and unsecured when she was among male persons. They felt uncomfortable sometimes due to objectionable/obscene male gaze, their bullying and ragging attitude and sexual

abuse. So they could not easily grow friendship with boys. Another female who was always in a co-educational set up had also replied almost the same narrated above. Although she said that there was no reason to be afraid of other people unless and until they would make her afraid, still she had mentioned that males sometimes used to commit such things. She had also said that gender did not matter to her. Her attitude or impression to others depended on the behaviour of the opposite person with her and other people.

The next question discloses the impression of the students about other genders and another question was framed to know whether the students used to respect other genders also. All the students had paid respect to the other genders. And most of the students said that everyone is human-being and they did not judge a person by his/her gender identity. According to them, every gender should be treated equally; there should be no discrimination on any basis. The transgender student also demanded same acceptance of each gender to everybody in the society. One female student also added that everybody should be helpful to each other. Another female student thought that all genders are beautiful and unique in their own way and if everybody respects each other, we shall grow together towards a better world. One female student, who had already mentioned that she was sometimes afraid of male persons due to the increasing rate of women assault, was concerned about safety and security. If the male persons are non-harmful, the society will be a paradise. One female student suggested that communication with other genders could provide a different insight and perspective. One female student said that she used to show friendly attitude to other genders.

Conclusion

So, we have seen that the inclusion of transgender students into the definition of 'co-education' has still not been popularized. More astonishing is the fact that the transgender student is also not aware of this inclusion. This may be the effect of long term indifference and negative attitude of a large portion of our society towards them. If transgender students are not admitted in the educational institutions and if they are not treated well in the classroom, the dream of our inclusive society will remain a utopia. The proper inclusion of transgender students in the educational institutions is badly needed. So is required effective inclusion of female students. In a co-educational institution, their presence must not be treated as a compulsion or alternative. We should not forget that school is the miniature of the society and society is not compartmentalized with any kind of gender based division. If we want gender harmony in our society, the relationship among various genders should be based on symbiosis, love and cooperation rather than hostility or competition. If we want a tree yield good

fruits, we have to take care when it is in the form of a seed. Just like that, to ensure gender harmony in the society we have to take care of the marginalized genders of our society from the very beginning. We have to actualize the theory of gender equality. And this journey from theory to praxis should be introduced from the very beginning of our education. After all, not only the destiny of a person but the destiny of a nation also is shaped in the classrooms.

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45. Classroom Reading Strategies Programme Based on Different School Subjects for Pre-Service Teachers

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Abstract

Language has a pivotal role to play in the daily life of a person from any race, creed, and region of the world. Language is useful for expressing our feelings and desires to the world around us. Human beings communicate through written and spoken language. Besides this language is a term associated to a great extent with human learning. Thus, language is important not merely to lead a life in the society but it has a very important role to play in 'education'. Language is essential for developing human knowledge across the various branches and disciplines. We must create a learning environment in the schools which promotes importance of language learning irrespective of the subject we are learning. Researchers have highlighted the importance of 'reading' in improving the comprehension and the learning of the students. It also has an important role to play in promoting self-learning. It is useful in retrieving information, interpreting information, and reflecting and evaluating the information. It is important then that the pre-service teachers be made aware of the different reading strategies and their use in the learning of the subject. The present paper focuses on the feedback regarding a reading strategies programme for pre-service teachers in the context of its benefits for enhancing themselves as readers and writers as well as using the strategies for teaching their school subjects.

Key Words: Reading Strategies, Pre-service Teachers

Introduction

Language is a term associated to a great extent with human learning. It has a very important role to play in 'education' since it is required for developing human knowledge across the various branches. Language plays an important role in various aspects of our daily lives. Its role is not limited merely to communication, but broadens into the gaining of knowledge. Becoming a language teacher is the challenge at the initial level of career (Farrell, 2012). Therefore, Language across the Curriculums is one of the approaches towards teacher education. It refers to the integration of language learning and content learning. The basic principle is that developing language proficiency in all subjects is important in human learning. In the life of pre – service teacher, Language learning doesn't take place only in language lessons but in the learning of each and every subject, in every learning activity and

across the entire curriculum. Command over a language is related with achieving proficiency in all other subjects also.

The promotion of reading did not stop at the primary level; it should be continued with all education especially in higher education. It was also a focal point in the most recent education reform in Hong Kong which was implemented in 2009 (Mak, 2017).

Students learn a concept when they listen, talk, read and write about what they are learning. In other words, students learn 'in a language'. By expressing themselves in a language, students are able to clarify the concepts and hence understand better and consequently increase their knowledge. Verbalization through describing, explaining, concluding and evaluating helps in removing any kind of ambiguity in understanding of the students. This approach leads to an improved performance in learning. Teachers have to be aware that the student's performance in a subject may not be up to the mark, not because of subject knowledge but due to lack of language comprehension. It is hence important that all teachers take the responsibility of training the student in language skills along with making him/her learn the subject specific vocabulary. The teachers should give opportunities to the students to listen, speak, read and write while teaching their subject by organizing various activities. The students should be made to realize how the language is used in the context of the subject they are learning. Reading skill has an important role to play in education of a child since it develops a broader vocabulary, increased general knowledge and a better understanding of other cultures. Reading helps to increase our knowledge, improve vocabulary, create awareness, develop greater understanding of knowledge, in other words helps in school study (Claudia Margarita Zurek Cadena 2006). Reading is a cognitive process of decoding symbols to derive meaning from texts. Reading involves the process of Decoding, Comprehension and Retention (Rebecca, 2005). When we ask our students to read different content it is not always in the textual form.

The **NCF 2005** has also emphasized on the language learning from this context. It has recommended an integrated approach to teaching using different language skills. Special attention needs to be given to the reading and writing competence of the students. The students should learn to read and write with understanding. Language as a group of skills, thought encoders cuts across all school subjects and disciplines.

Speech and listening, reading and writing are all generalized skills and student's mastery on them is a crucial factor in their success at school. The NCF has further highlighted the fact that language

education should be everybody's concern in the school and not merely the responsibility of the language teacher. The focus of a language teacher is generally on the correctness of the language rather than the expressive and participatory functions of the language. Language needs to be developed through different activities like group talk, compare and contrast, guess and challenge, judge and evaluate. The focus should be on expanding the imagination of the students and enhancing their ability to participate and also value the other person's point of view. In fact, a culture of reading needs to be developed among the students.

Review of Related Literature

Cadena (2006) studied on effectiveness of reading strategies and improving reading comprehension in young ESL learners. The results of this study emphasize the importance of choosing the correct reading and teaching material in terms of content and length and the necessity of setting time limits for reading tasks, and last but not the least, the need to focus on the reading process and how it relates to the product of reading. Baier (2005) studied on reading strategies and comprehension for students and

results of the study showed increase in the reading comprehension of sixth grade students after using the self-questioning strategy. Brevik (2015) developed reading comprehension in English for Norwegian upper secondary school. Findings of this research also suggest little reason to claim that reading strategies are effective when taught in isolation. Instead, they have to be explicitly taught by the teachers, and then used individually and independently by the students seeing personal purposes to do so. (Khaokaew, 2012) conducted a research on explicit strategy instruction for EFL reading of undergraduate English majors in Thailand. In the study the researcher has mentioned that a course on reading strategy instruction particularly for all first-year undergraduate students should be provided as a compulsory course because it will be beneficial for these students in their academic reading. This also suggests that teacher training provision should offer more guidance on reading strategy instruction.

Need and Importance of the study:

All the research review highlights the importance of 'reading' in improving the comprehension and the learning of the students. It is imperative then that systematic training needs to be introduced at the pre-service level so that the future teachers are not only aware about the relevance of language in education, but are able to develop tasks and activities while teaching their subject and support the students in their learning.

They should understand that they might be teaching any school subject they need to be trained in using the different 'reading strategies' so as to facilitate the learning of the students in their own subjects. In other words they should not look at language learning independently but integrate it with the learning of their own subject. Language should be seen across the entire curriculum rather than in isolation. This will help them to realize that language used across the curriculum has a significant impact on the empowerment of the students.

In this context pre service teachers should be made aware regarding: -

subject specific vocabulary. (e.g. Geography- latitude, longitude, rainfall, etc., Use a *ruler* to draw the lines and He was a *great ruler*.)

subject specific word formations (prefixes, suffixes, etc.) different forms of expression. (Physical Education: - Line up properly all of you. Get ready for the warm up exercises) competencies and skills required for comprehending, identifying, selecting and integrating information in connection with specific texts, tasks, materials. (Questions like: - Summarize in your own words.)

This raised the following questions in the mind of the researchers: -

- Can some reading strategies be identified which can be used for teaching of different school subjects?
- Is it possible to train the pre-service teachers in using the different reading strategies?
- Will such a developed programme for the pre-service teachers help them to understand how to use these strategies?

Keeping these questions in mind the researchers decided to undertake the study.

For that purpose, we were stated the following objectives: -

- To develop a programme for teaching reading strategies to the pre-service teachers teaching different school subjects.
- To implement the programme and take its feedback.

Method and Procedures: -

The present research study is an Applied Research. The basic aim of applied research is to gain knowledge for the practical use. Usability of the knowledge is judged through the research work and gained knowledge is used to solve the different problems faced in the classrooms. It tries to apply the already established theories or principles to solve specific problems. In the present study the principles of language learning are applied to teaching of different school subjects.

A multiple methods study uses more than one method. There is differentiation between multi-method research and mixed method research. A multi method researches i.e. multiple qualitative or quantitative methods and mixed methods research i.e. integration of qualitative and quantitative methods (Creswell and Clark, 2007). A mono- method study uses only one type of method, one quantitative or one qualitative. In a quantitative study, the data are in numerical form and this information is analysed using quantitative techniques. In a qualitative study, the information, which is mainly in textual form, is analysed employing qualitative techniques. Drawing an initial distinction between mono-method research and multiple methods research may be helpful to determine what is understood as mixed methods. A multiple methods study uses more than one method. Moreover, a differentiation can be made within multiple method designs between multi-method researches i.e. multiple qualitative

or quantitative methods and mixed methods research i.e. integration of quantitative *and* qualitative methods

The present study was a multi method study, since two methods were used. For the first objective the researchers developed a ‘classroom reading strategies programme’ for the pre service teachers using the product development method. In this programme different reading strategies were developed based on the school subjects of the pre- service teachers. For the second objective programme evaluation method was used by conducting a feedback using a survey. In survey research, the researcher selects a sample of respondent from population and administers a standardized questionnaire to them.

In the present study the convergent parallel mixed methods research design was used. In Convergent parallel mixed method design both quantitative and qualitative data is collected. Analysis is done separately. Accordingly comparison of the results of both data is done and finally interpretation is done to see whether the results support or contradict each other. The researcher gets two databases here. These databases can be merged in different ways. The first approach is called side-by-side comparison, in which the comparison is done within the discussion. Here the researcher presents the first set of findings and then the other. In the second approach, quantitative and qualitative data is merged in a single table or a graph. The last approach is called data transformation where research can merge two databases by changing qualitative codes or themes into quantitative variables and then can combine the two quantitative databases. (Creswell, 2015).

Triangulation: - One of the important features of mixed methods research is data triangulation. It could be in the form of investigator triangulation, method triangulation, theory triangulation or data triangulation. It is useful for validating the results obtained in the research.

Theory Triangulation: - The qualitative responses given by the experts discussed above help to validate the various principles used in designing of the programme. Their responses regarding the various aspects they liked about the programme namely active involvement, application oriented, interaction, creative and variety in the activities, and innovative and interesting strategies reflect the principles of learning through activity, learning by doing, learning through creativity, learning using imagination, learning by relating to their school subjects and learning through interaction.

Data Triangulation: - In the study the data collected from different persons (experts and pre service teachers) has been compared. The quantitative analysis of the responses of the experts given in table 2 and of the pre service teachers in figure 2 and 3 and table 3 validates the results. Their responses show similarity with respect to the features like opportunity to work in pairs/groups, improve as readers and writers, variety in the texts used and useful in the future while teaching their subjects.

Reading Strategies Programme Based on Different School Subjects:

Nature of the Programme:-

In this programme the researchers tried to give exposure to the pre service teachers to a wide variety of texts/e-texts with engaging these texts/e-texts interactively individually as well as in small groups. This programme also focussed on enhancing their capacities as readers and writers. Opportunity was also given to them so as to be able to read, reflect and reproduce in writing in the form of conjectures, justifications, interpretations, arguments for or against, etc. on the read text/e-texts. Following principles of language learning were used in the programme:-

Learning through activity.

Learning by doing.

Learning through creativity.

Learning using imagination.

Learning through motivation.

Learning by relating to their school subjects.

Learning through exploration, construction and expression.

Learning through interaction.

The programme was implemented through:-

Orientation sessions in the form of lectures for developing a theoretical base.

Panel discussion for promoting learning from experts. Workshop sessions for giving hands on experiences. Various reading strategies were selected while designing the programme. They were pre-reading strategies, during – reading strategies and post-reading strategies.

The list of the selected strategies along with the respective content selected is given in the following table 1.

Table 1. Strategies selected for the study and the respective subject content

Name of the strategy	Subject content
Frayer Model	Science and Mathematics (school text)
Split reading	Science and Language (school text)
Presentation Project	Library Science (newspaper clipping)
Framing questions	Social science (e-text)
Note making	Science (school text)
Fact summary	History and Geography (school text)
Graphic organizers	Geography (school text)
Drama and Play	Language and History (school text)
Dialogue and Monologue	Literature (school text)
Summary Writing	NCF 2005 (policy document)
Anticipation Guide	Dinosaurs (Encyclopaedia)

Description of Strategies

Frayer Model: - The Frayer Model is a strategy that uses a graphic organizer. It can be used with the entire class, small groups, or for individual work. The Frayer Model draws on a student's prior knowledge to build connections among new concepts. It helps to create a visual reference by which students learn to compare features and the examples. This reading strategy helps the students to:-

- Learn new vocabulary and concepts.
- Helps in retention.

- Develops critical thinking.
- Interpret and analyze the information.
- Evaluate the information.

Go beyond the given information when thinking of examples and non-examples of the given concept.

The teacher selects a key concept from the text and gives some background related to the concept. The students then read the text. Students are asked to complete the given chart. (in pairs, groups or individually) The students after completion of the task share it with the class. This helps them to modify, clarify or correct their understanding. The teacher can model the type and quality of desired answers for the specific concept.

Split Reading: - This strategy is very useful in developing reading comprehension, retrieving information and interpreting information. As the name suggests the strategy requires dividing the reading material into sub parts. This strategy can be used for individual students or even in a group. Its focus is on the developing sequential and logical understanding of the students. The strategy is used in the following manner:-

- Students first read the entire content as a whole.
- Then the entire reading content is split into different parts by the teacher.
- The split content is then given to the students.
- The students are asked to rearrange the split content in the appropriate sequence.

It is useful for content where sequential arrangement is very important.

Presentation Project: - This activity can be given to the students individually, in pairs or in groups. Students are expected to research, read material based on the questions that are given to them related to a topic. Students then make a presentation on the information they have collected and on their understanding. This reading strategy is useful for students since it develops the capacity of:-

- Locating

- Comprehending
- Reflecting
- Synthesizing
- Expressing (spoken and written form)

The strategy also helps in developing their imagination and creative ability. The students can use pictures, cut-outs, or other ideas during their presentation.

Framing Questions: - It is the oldest and the easiest strategy that can be used in the classroom. It is used to revise, confirm, stimulate, gain attention, focus or guide.

There are mainly three types of questions. The lower order questions focus on mere recall and recognition. The middle order questions include questions on transformation, finding meaning, differentiating, giving examples, etc. The higher order questions involve analyzing, evaluating, debating, deriving conclusions, etc.

- Questions can be asked to focus on the:-
- Main idea: Global understanding
- Simple Factual details: answer available at one place
- Complex Factual details: answer available at the different places
- Inference: reading between the lines, deduction
- Interpretation: clarification
- Application: using in different context, going beyond the text
- Guessing: what comes next
- Personal response: creative, individual opinion

Note making: - To note is to observe. It means presenting big chunks in a neat and short form. The information in verbal form is presented in minimum words since it helps to remember important details. It is useful to refresh our memory about what we read sometimes in the past. The information

is presented mainly in a graphical form using Tree diagram, Tabular presentations, Flow charts, Graph, Pie- chart, etc.

Fact summary: - This strategy initially focuses on recording of the information by the students while they are reading. Gradually the students must be led towards developing their own fact sheets for summarization of learnt matter. This strategy helps the students in identifying, organizing and recording information. It develops the ability of retrieving and interpreting the information.

Introduce the strategy (by giving the fact sheet) at the beginning of the lesson.

Focus the student's attention on the key ideas and see that they are entering the information in the fact sheet as the reading is in progress.

Once the fact sheet is completed review it for accuracy.

Graphic organizers: - A graphic organizer refers to a visual or a graphic display that depicts the relationships between facts, terms, and or ideas within a learning task. They are also called as knowledge maps, story maps, cognitive organizers or concept diagrams. They are a useful tool in all subject areas. Five paragraph essay, analogy organizer, brainstorming web, Venn diagram, alphabet organizer, etc are some examples of graphic organizers. They help in organizing the thoughts and ideas and presenting them in an effective and colourful manner. They are useful in enhancing reading comprehension, retention and critical thinking.

Drama and Play: - Dramatization means putting prose or poetry into action. It helps to keep the teaching learning process activity oriented. It is also useful for developing communication skills. Putting the text into the form of a drama or role play is useful in fostering imagination and creativity. It develops the ability of interpreting, analyzing and expressing. It is also useful in integration of different subjects and inculcation of values.

Dialogue and Monologue: - Use of dialogues or monologues in teaching different subjects requires great skill. This particular strategy is linked with drama and play. Writing a dialogue or a monologue is useful in explanation of a concept. When this strategy is being used it also helps the teacher to guess what kind of questions could be asked by the students and what answers would have to be

given. Thus it is useful for a teacher to anticipate queries which could be raised by the students. It is useful for interpreting, analyzing and expressing their subject effectively.

Summary writing: - It is also one important reading strategy that is commonly used. Summarizing is separating relevant points from the irrelevant. It is a precise presentation, a restatement in shortened form of the chief ideas, points. Summarizing doesn't include commenting on the passage, adding new information, giving examples or explaining meaning. It is an academic exercise, where words, phrases and expressions from the text are avoided. What is difference between summary and note making? Summary is for general use –anyone can follow and it covers specific, highlighting points. It is more verbal in nature and is formal. In simple words it is a gist in a nutshell. In contrast Note making is for personal use and it covers all the important points in non-verbal format. It is informal and useful for revising or memorizing.

Anticipation Guide: - It helps to test comprehension. It develops the ability of prediction. It also is useful to activate prior knowledge. The teacher initially writes statements that summarize the concept. Then the students are given the list of statements. Students write if they agree or disagree with the statements. Then the students are asked to read the content. After reading, the students themselves evaluate their answers given previously.

Cubing: - This strategy involves the students in comprehending concepts from multiple perspectives. It helps to develop critical thinking and also helps in reinforcement of the concept. In this strategy students are asked to apply, describe, associate, compare analyse and argue for or against the concept.

Critique: - It is one of the important strategies which helps to understand a particular issue from various perspectives. There are three levels of criticism:-

Thematic level: subject, relevance, thought

Linguistic level: vocabulary, language style, cohesion, coherence, Structural level: presentation, layout, paragraph formation

While writing a critique we have to be constructive and hence start with the positive first yet have to be candid and specific. We need to try to understand the author's perspective and hence need to read the texts from the different angles. We should be able to provide examples to justify our point of view. We have to give our comments but never impose our ideas.

Results for Objective 1: About the Developed Programme: -

Both quantitative and qualitative feedback regarding the programme was taken from five experts.

Quantitative Results: Feedback from Experts regarding the Programme: Quantitative analysis of the ten feedback questions; which are given to the experts. They were asked various questions regarding programme. In the following table 2 the feedback received from the experts is presented.

Table 2. Feedback of the experts regarding the developed programme

Sr.No.	Item	Yes	No
1.	The selected strategies are suitable for the pre service teachers	4	1
2.	The strategies and the content are appropriately matched	5	0
3.	There is a variety in the selected texts	4	1
4.	There is ample opportunity to engage pre service teachers in pair/group activities	3	2
5.	The strategies will help the pre service teachers to become good readers	5	0
6.	The strategies will enhance the capacity of the pre service teachers to become good writers	4	1
7.	The activities will help the pre service teachers to represent the read texts in different ways	5	0
8.	The entire programme will facilitate the pre service teachers to use these strategies while teaching their subjects	4	1
9.	The programme is a good combination of theory and practice.	5	0
10.	There is a variety in the activities planned in the programme.	4	1

Table 2 shows that majority of the experts have shown their agreement with respect to all the ten aspects related with decided objectives of the designed programme. Moreover, all the experts have shown their total agreement regarding the following aspects of the programme: all strategies included in the reading programme are appropriately matched with the content, these strategies are also helpful for pre-service teachers to become good readers and that the programme is blended with theory and

practice. Hence, from TABLE 2, we can interpret that the experts were satisfied with the various aspects of the designed programme.

Qualitative Results: Feedback from Experts regarding the Programme:

For the qualitative analysis, they were asked one open ended question about the designed programme where they were asked to write their general observations regarding the designed programme. From the various answers given by the experts to the open-ended question following main themes develop regarding the programme as appreciated by the experts.

Appropriate blending of theory and practice.

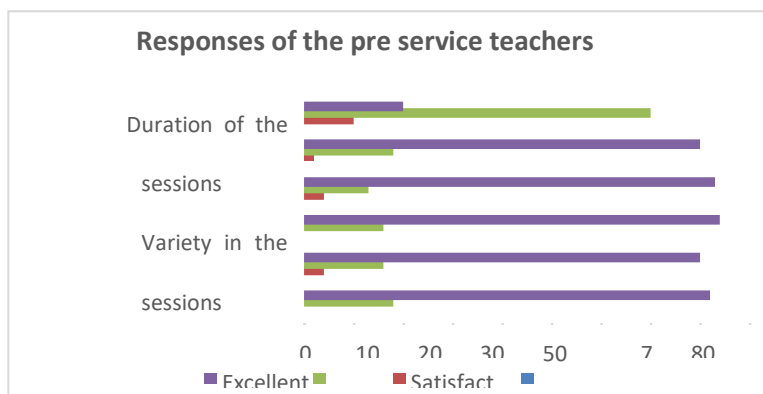
- Good combination of strategy and content.
- Active involvement.
- Application oriented by relating it to different school subjects.
- Using principles of constructivism: learning through interaction, opportunity for creativity through presentation.
- Variety of texts.
- Innovative and interesting strategies

The above responses given by the experts to the open-ended question, show their satisfaction and appreciation regarding the developed programme. Their responses not only reflect and reconfirm the principles of learning used while developing the programme but also the benefits of the different strategies used in the programme.

Results for Objective 2:- About the Implementation of the Programme: - Both quantitative and qualitative feedback regarding the programme was taken from the participating pre-service teachers.

Quantitative results: - Responses of the pre service teachers regarding the programme: - The quantitative analysis of the responses given by the pre service teachers is presented in the following Figures 2 and 3 and Table 3.

Figure 2. Responses of pre-service teachers regarding various aspects of the programme.



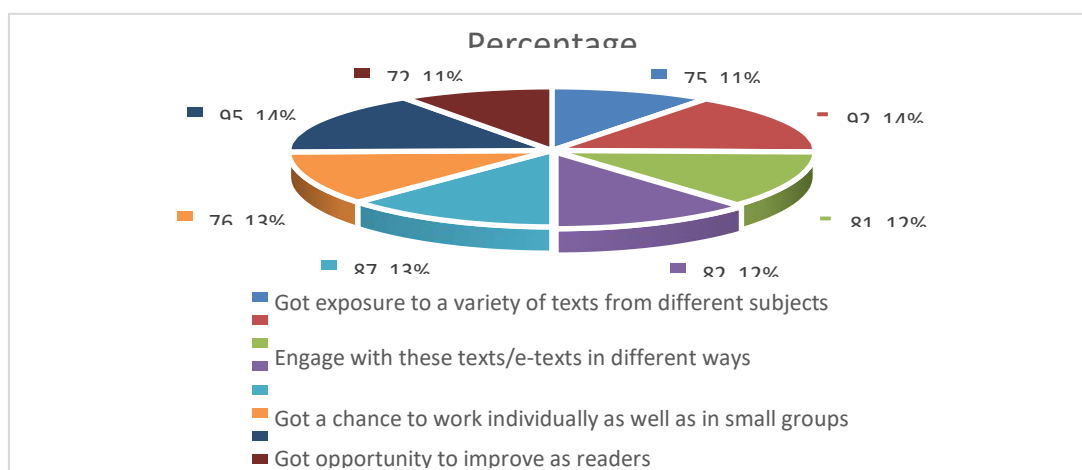
The pre service teachers were asked to respond with respect to:-

The design of the programme.

The implementation of the programme.

Statements were given and the pre service teachers had to select from the four options that were given for each statement. The analysis is presented in the above Figure 2 in the form of percentage. It shows that majority of the responses given by the pre-service teachers are in the ‘Good’ or ‘Excellent’ column with respect to the planning as well as implementation of the designed programme. It can be inferred that the pre-service teachers have a favourable opinion regarding the various aspects of the designed programme.

Figure 3. Responses of pre-service teachers regarding the benefits of the programme



The Pre-service teachers were asked about eight different aspects regarding the reading Programme. They were required to tick mark on the option they wanted. The percentage analysis shows that majority of the pre service teachers were benefitted in various ways by the programme. Maximum number of pre service teachers, have opted for the benefit of being able to present the read material in different ways. This was followed by the opportunity to engage them with different kinds of texts and e-texts in various ways. A large percentage has also mentioned that it has helped them to improve as writers and as readers. The other favourable aspects of this programme were ‘working individually as well as in small groups’, ‘read materials in different ways’ and also ‘justifying ones opinion about the texts’. Thus it can be interpreted that the pre- service teachers have a favourable opinion regarding all the above mentioned aspects of the designed programme. The responses help to understand that the developed programme was useful in fulfilling the objectives designed for the programme. They also indicate that the selected strategies and the content were appreciated by the pre service teachers.

Table 3. Benefits of the programme in planning school subject

Sr. No.	Item	Percentage
1	Using group/pair activities	82
2	Developing items to compare and contrast between concepts	67
3	Design questions related to guessing/analysing	88
4	Providing opportunities for judging and evaluating information	73
5	Selecting the strategies according to the subject content	85

Feedback regarding the designed programme was also taken in the context of its use in teaching of school subjects. The above Table 3 indicates that maximum number of the pre service teachers found that the strategy for designing questions related to guess work or to analyse would be very useful to them for planning and teaching their school subject. A large percentage of the pre-service teachers appreciated that the programme was going to be useful to them for the selection of strategies by properly relating them with their subject content. Table 3 also shows that a large percentage of pre-service teachers stated that they would now be able to use group and pair activities while teaching their subjects. Pre-service teachers also were benefitted by realizing how to provide opportunities to their students for judging and evaluating information and also for comparing and contrasting between

concepts. It can be interpreted that the pre-service teachers have found the programme useful in planning and implementing the strategies in the classroom.

From all the analysis given in the above two figures and one table we can interpret that the developed programme was:-

Useful in achieving the stated objectives:

Pre service teachers appreciated the designed programme and the way it was implemented.

Pre service teachers have found a number of benefits of the programme. It has helped them personally as well as professionally.

Pre service teachers have found that the programme has helped them to realize how to select reading strategies suitable for their subject content, design the activity and implement it in the classroom.

Qualitative Analysis: Responses of Pre-services teachers:-

For the qualitative analysis, the pre service teachers were asked one application based open ended question about the designed programme. They were asked to write the topics from their school subjects which they would teach using the discussed reading strategies. Some of the various topics (representative of different school subjects and the reading strategies) mentioned by the pre service teachers is given below in table 4.

Table 4. Analysis of the responses given to open ended question

	Fraye Model	Triangles, Amphibians, Fundamental Rights, Grammar	
	Split Reading	Water cycle, Stories, circulatory system, historical events, processes,	

Reading Strategies	Project Presentation	Poets, types of literature, social reformers, natural regions of the world, classification of plants	 All Subjects
	Note making/Graphic organizers	Structure of a cell, digestive system, administrative set up, Grammar	
	Monologue/Dialogue/Drama/Play	Environment related topics, autobiography,	
	Critique/Framing Questions	Social issues, ethical issues, economic issues, environmental issues related to the subject content	
	Summary	Literature, historical event, geographical occurrence, description of characteristics	

Pre-service teachers have stated that all reading strategies are useful for various content of subject. For the topics like grammar, fundamental rights and triangles, Frayer model could be used. The content like water cycle, stories, circulatory system, historical events, process could be taught using split reading strategy. Project presentation was a useful reading strategy for poem, types of literature from language. Moreover, this strategy was also useful especially for social sciences in the content regarding social reformers, for topics from geography like the natural regions of the world and from science like the classification of plants. Pre-service teachers stated that they would like to use note making and diagrammatic presentation for topics from science like the structure of a cell, digestive system and from history topics like the administrative set up etc. According to their responses grammar content could also be effectively presented using note making strategy. In addition, there are different reading strategies like monologue. Dialogue, drama and play which the pre service teachers found would be effective for various environment related topics; even for autobiography. The reading strategies like critique and

framing questions could be used as a teaching tool to assess knowledge and promote awareness on various issues such as social, ethical, economic and environmental. Well-crafted questions would promote insights and critiquing ability among fellows. And last but not the least the reading strategy of summary writing would be useful in teaching different topics from language, social sciences and physical and biological sciences to understand what the learner has understood at the end.

Conclusions of the study: - Reading doesn't come only in the form of text in different school subjects. It comes in different forms of Pictures, Symbols, Letters, Words, and Sentences, Group of sentences and also Diagrams, Circuits, Maps, Graphs etc. To enable our students to effectively read all kinds of material the teacher should use different reading strategies in the classroom. Research results have shown that the reading has an important role to play in self-learning. It has a significant bearing on the development of the ability of comprehending and understanding. It serves in achieving three main purposes that are important in learning namely: retrieving information, interpreting information and reflecting and evaluating the information. Hence 'Reading' has a unique relevance and importance in the life and learning of a person. It is said that the intellectual advancement and academic development of the child remains limited if he/she is unable to read effectively. There are a number of reading strategies which can be used in classroom teaching (Khaokaew, 2012). In this research, there were two major aspects: the first one being the opinion of the experts regarding the programme and the second one being the benefits of the programme to the pre-service teachers. The experts have given a positive opinion regarding the designed programme. The pre-service teachers have found the programme useful in developing their own reading, reflecting and reproducing the read content in different ways. The pre-service teachers have found the programme useful in planning and implementing the reading strategies in the classroom for all school subjects. The implementation of social constructivism and cognitive theories with embed research based instructional practices support the students and this helps in reading comprehension and reading motivation (Farkas & Jang 2019). Moreover, Kagan structures had an obvious impact on student performance and perception in reading comprehension (Shana, Lahiani & Mahmoud 2019). According to Kagan (2009), the structures present several advantages. The most important benefit of the Kagan structure is that the participants are more engaged. In the present research, we found that pre-service teachers were highly engaged in their activities.

Implications of the study: -The programme highlights the relevance of designing such programmes based on reading strategies at the pre-service level. It brings out the importance of allotting specific time for such a programme. This programme can be a reference for designing other similar programmes. Reading strategies are widely used in the classroom. These kinds of research-based strategies can be used for literacy development. However, the role of teacher educator is providing different pedagogy or strategies in the reading with richer and deeper experiences of reading in the training programme (Perkins,2013).

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